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K. Chandrasekharan

Elliptic Functions



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Dedicated to the memory of
Heinz and Anja Hopf

When the familiar scene is suddenly strange
Or the well known is what we have yet to learn,
And two worlds meet, and intersect, and change;
T. S. Eliot

Preface

This book has grown out of a course of lectures on elliptic functions, given in German, at the Swiss Federal Institute of Technology, Zürich, during the summer semester of 1982. Its aim is to give some idea of the theory of elliptic functions, and of its close connexion with theta-functions and modular functions, and to show how it provides an analytic approach to the solution of some classical problems in the theory of numbers. It comprises eleven chapters. The first seven are function-theoretic, and the next four concern arithmetical applications. There are Notes at the end of every chapter, which contain references to the literature, comments on the text, and on the ramifications, old and new, of the problems dealt with, some of them extending into cognate fields. The treatment is self-contained, and makes no special demand on the reader's knowledge beyond the elements of complex analysis in one variable, and of group theory.

Professor Raghavan Narasimhan has read the definitive English version of the text, and made illuminating comments, as a result of which I have improved the presentation in several places. Dr. Anton Good has looked through the first German version, and spared the time for many useful discussions. Dr. Peter Thurnheer, who had attended the course, helped me check the detailed calculations that lurk behind some of the statements in the text. Mr. Albert Stadler has assisted me in tracing the bibliographical references and in proof-reading. My sincere thanks go to them all for sustaining this effort during a difficult twelvemonth.

Thanks are due to the editors of the *Grundlehren*, particularly to Professor Beno Eckmann, and to Springer-Verlag, for getting this thing into print.

Zürich, October 1984 K. C.

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Chapter I

Periods of meromorphic functions

§ 1. Meromorphic functions. We assume as known the fundamentals of complex analysis, including the basic properties of *holomorphic* and of *meromorphic* functions in the *complex plane*. The meromorphic functions defined on an *open, connected*, set in the complex plane form a *field*. Unless otherwise qualified, a meromorphic function is supposed to mean a function meromorphic in the whole complex plane.

§ 2. Periodic meromorphic functions. A meromorphic function f is said to be *periodic*, if there exists a complex constant $\omega \neq 0$, such that $f(z+\omega)=f(z)$, for all $z \in \mathbb{C}$, where $\mathbb{C}$ denotes the set of all complex numbers. The number ω is said to be a *period* of f. The number zero is called the *trivial* period. If f is a constant, then every complex number is a period of f; conversely, if every complex number is a period of a meromorphic function f, then f reduces to a constant.

If ω is a period of f, and $\omega \neq 0$, then $n\omega$ is also a period of f, for $n=0, \pm 1, \pm 2, \ldots$.

The functions $\sin z$, $\cos z$, $\cot z$ are meromorphic, and periodic. In fact, $\sin z$ and $\cos z$ are *entire* functions, with $\sin(z+2\pi)=\sin z$, $\cos(z+2\pi)=\cos z$, while $\cot(z+\pi)=\cot z$, with

$$\cot z = \frac{1}{z} + \sum_{n \neq 0} \left(\frac{1}{z-n\pi} + \frac{1}{n\pi} \right).$$

If $\omega \neq 0$, the function $\exp\left(\frac{2\pi i z}{\omega}\right)$ has the period ω. Every rational function of $e^{2\frac{\pi i z}{\omega}}$, with constant coefficients, has the period ω.

(2.1) The absolute values of all non-trivial periods ω of a non-constant meromorphic function f have a positive lower bound. That is to say: $\inf_{\omega \neq 0} |\omega| = \delta > 0$.

For otherwise f will have *arbitrarily small* periods. That is to say, given $\varepsilon > 0$, there exists a period $\omega_\varepsilon \neq 0$, such that $|\omega_\varepsilon| < \varepsilon$. It follows that there exists a sequence of periods (ω_n), such that $\lim_{n \to \infty} \omega_n = 0$. If z_0 is a point at which f is holomorphic, then $f(z_0)=f(z_0+\omega_n)$, $n=1,2,\ldots$, so that $f(z)-f(z_0)$ has an infinity of zeros $z_0+\omega_n$, $n=1,2,\ldots$, which have z_0 as a finite *accumulation point*. Hence $f(z)$ is a constant, contradicting our assumption.

(2.2) The periods of a non-constant meromorphic function f have no finite accumulation point.

For otherwise let ω_0 denote such an accumulation point. Let $\varepsilon>0$, and arbitrarily small. We can then find two *different* periods ω_1, ω_2 inside the open circular disc $|z-\omega_0|<\varepsilon$. Then $\omega_1-\omega_2$ is a period with $|\omega_1-\omega_2|<2\varepsilon$. It follows that f has 'arbitrarily small' periods, contradicting (2.1).

(2.3) In consequence of (2.1) and (2.2), the absolute values of all non-trivial periods of f have a *positive minimum* δ.

Let ω_1 be a period of f, with $|\omega_1|=\delta$. Clearly $n\omega_1$ is also a period of f for $n=0, \pm 1, \pm 2, \ldots$.

Consider *any period* ω *of* f, for which the quotient $\dfrac{\omega}{\omega_1}$ is *real*. Let n be the *integer* defined by the requirement:

$$0 \leqq \frac{\omega}{\omega_1} - n < 1 .$$

The difference $\omega - n\omega_1 = \omega_0$, say, is a period, with $|\omega_0|<|\omega_1|$, hence $\omega_0=0$, and $\omega = n\omega_1$. It follows that every period ω, such that the quotient $\dfrac{\omega}{\omega_1}$ is real, is of the form $n\omega_1$, where n is an integer. *If* f has *no other* periods than $n\omega_1$, $n=0, \pm 1, \pm 2, \ldots$, f is said to be *simply periodic*, and ω_1 is said to be a *primitive*, or *reduced*, period of f. For example, $\sin z$ and $\cos z$ are simply periodic with the primitive period 2π, while $\cot z$ is simply periodic with the primitive period π.

(2.4) If the meromorphic function f *has* periods ω, such that $\dfrac{\omega}{\omega_1}$ is *not* real, let ω_2 denote one such period with the smallest absolute value. Let $\tau = \dfrac{\omega_2}{\omega_1}$. We may assume that $\operatorname{Im}\tau>0$, for otherwise we can replace ω_2 by $-\omega_2$. Further we have $|\tau| \geqq 1$. Obviously $m\omega_1 + n\omega_2$ is also a period, where m and n are integers. *Conversely, every period of* f *is of the form* $m\omega_1 + n\omega_2$. For every complex number can be written uniquely in the form $x\omega_1 + y\omega_2$, where x and y are *real* numbers, and x and y can, in turn, be written in the form $x = m + \varrho$, $y = n + \sigma$, where m and n are *integers*, and $-\frac{1}{2} \leqq \varrho < \frac{1}{2}$, $-\frac{1}{2} \leqq \sigma < \frac{1}{2}$. It follows that

$$\omega_0 = \omega - m\omega_1 - n\omega_2 = \varrho\omega_1 + \sigma\omega_2$$

is a period, where

$$\left|\frac{\omega_0}{\omega_2}\right| = \left|\frac{\varrho}{\tau} + \sigma\right| \leqq \frac{|\varrho|}{|\tau|} + |\sigma| \leqq \frac{1}{2} + \frac{1}{2} = 1 ,$$

since $|\tau| \geqq 1$. At least one of the inequalities here is *strict*, since $\dfrac{\varrho}{\tau}$ is real only if $\varrho=0$. Hence $|\omega_0|<|\omega_2|$, from which it follows that $\dfrac{\omega_0}{\omega_1}$ is *real* and $\sigma=0$ $\left(\text{since } \operatorname{Im}\dfrac{\omega_2}{\omega_1} > 0\right)$, hence also that $\varrho = \dfrac{\omega_0}{\omega_1} = 0$ (since $-\frac{1}{2} \leqq \varrho \leqq \frac{1}{2}$, and ω_1 is a period with minimum absolute value), therefore $\omega_0=0$, so that $\omega = m\omega_1 + n\omega_2$, where m and n are integers.

We call such a pair of periods ω_1, ω_2 *primitive*, or *reduced*. It is *not* unique, since the pair $(-\omega_1, -\omega_2)$ is also primitive. The meromorphic function f is then called *doubly periodic*. What we have so far proved is then subsumed under the following

Theorem 1. *If S denotes the set of all periods of a meromorphic function f, then S consists of the number zero alone, or of the numbers $\{n\omega_1\}$, where $\omega_1 \neq 0$, and $n = 0, \pm 1, \pm 2, \ldots$, or the numbers $\{m\omega_1 + n\omega_2\}$, where $m, n = 0, \pm 1, \pm 2, \ldots$, and* $\operatorname{Im} \dfrac{\omega_2}{\omega_1} > 0$.

The complex number ω_1 is a period of f with minimum absolute value, and ω_2 is a period of f with minimum absolute value, among all those periods ω of f with the property: $\operatorname{Im} \dfrac{\omega}{\omega_1} > 0$.

A non-constant meromorphic function cannot therefore have more than two periods which are linearly independent over $\mathbb{Q}$, the field of rational numbers.

This theorem can be proved in quite another way, as was first shown by Jacobi.

§ 3. Jacobi's lemma

Lemma. *Let a, b, c denote three complex numbers, $a \neq 0, b \neq 0, c \neq 0$, and l, m, n three arbitrary integers, not all of them equal to zero. Then we have*

$$\inf_{l,m,n} |la + mb + nc| = 0 .$$

Proof. Let E denote the set of numbers $\{la + mb + nc\}$, $(l, m, n) \neq (0, 0, 0)$. We consider two cases.

Case (i). There exists an element of E which is zero. If so, a, b, and c, are linearly *dependent* over $\mathbb{Q}$, and there is nothing to prove.

Case (ii). Every element of E is different from zero. If so, a, b, and c, are *linearly independent* over $\mathbb{Q}$.

If possible, let $\delta > 0$, and such that $|la + mb + nc| > 2\delta$, for every element $(la + mb + nc) \in E$.

If $(l, m, n) \neq (l_1, m_1, n_1)$, we have, in particular,

$$|(l - l_1)a + (m - m_1)b + (n - n_1)c| > 2\delta .$$

Let N be an integer, $N > 0$, and let l, m, n assume the values: $-N$, $-(N-1)$, ..., 0, 1, ..., $(N-1)$, N, such that $(l, m, n) \neq (0, 0, 0)$. We then obtain $(2N+1)^3 - 1$ elements of the set E, all of which are different from one another. We may assume without loss of generality that $|a| \geqq |b| \geqq |c|$. For each of the $(2N+1)^3 - 1$ elements we then have the inequality

$$|la + mb + nc| \leqq N|a| + N|b| + N|c| \leqq 3N|a| .$$

They may be looked upon as $(2N+1)^3-1$ points inside the circle (in the complex z-plane): $|z|<3N|a|+\delta$. With each such point as centre, and radius δ, we can describe a circle. Since the distance between the centres is greater than 2δ, these circles do *not* overlap. They are all contained in the interior of the circle with the origin as centre and radius $3N|a|+\delta$; so is the circle $|z|<\delta$. Comparing the areas, we have

$$(3N|a|+\delta)^2>(2N+1)^3\delta^2, \quad \text{or}$$
$$(3N|a|+\delta)>\delta(2N+1)^{3/2}, \quad \text{or} \quad \delta\{(2N+1)^{3/2}-1\}<3N|a|,$$

therefore

$$\delta<\frac{3N|a|}{(2N+1)^{3/2}-1},$$

where the right-hand side tends to zero as $N\to\infty$. Given $\delta>0$, we can choose N so large that this inequality is contradicted. It follows that E contains an element $la+mb+nc$, such that $|la+mb+nc|\leqq 2\delta$.

Theorem 1'. *A non-constant meromorphic function f cannot have three periods which are linearly independent over $\mathbb{Q}$, the field of rational numbers.*

Proof. Consider the meromorphic function φ defined by $\varphi(z)=f(z)-f(z_0)$, where z_0 is a point at which f is holomorphic. Obviously z_0 is a zero of φ, and since zeros are *isolated*, there exists an $\varepsilon>0$, such that φ has no other zero in the circle $|z-z_0|<\varepsilon$. Let a, b, c be three linearly independent periods of f. Then $la+mb+nc$ is also a period, where l, m, n are integers, so that $f(z_0+la+mb+nc)=f(z_0)$.

Since $la+mb+nc\neq 0$, unless $l=m=n=0$, one can choose l, m, n, such that $|la+mb+nc|<\varepsilon$, by Jacobi's lemma. In that case $\varphi(z)$ has a zero at $z_1\neq z_0$, with $|z_1-z_0|<\varepsilon$, where $z_1=z_0+la+mb+nc$, which is impossible. It follows that f cannot have three linearly independent periods.

Corollary 1. *Let α and β be real numbers, $\alpha\neq 0$, $\beta\neq 0$, and l, m integers, not both of them zero. Then we have*

$$\inf_{l,m}|l\alpha+m\beta|=0.$$

Choose $a=\alpha$, $b=\beta$, $c=i=\sqrt{-1}$. Because of Jacobi's lemma, given ε such that $0<\varepsilon<1$, there exist *integers* l', m', n', not all of them zero, such that

$$|l'a+m'b+n'c|<\varepsilon<1,$$

which implies that $n'=0$, and $|l'a+m'b|<\varepsilon$.

Corollary 2. *A non-constant meromorphic function f cannot have two real periods, which are linearly independent over $\mathbb{Q}$.*

If α and β are two real periods, which are linearly independent over $\mathbb{Q}$, with $\alpha\neq 0$, $\beta\neq 0$, then β/α is irrational. By Corollary 1, there exist integers l, m, not

both zero, such that $|l\alpha+m\beta|<\varepsilon<1$. Since β/α is irrational, $l\alpha+m\beta\neq 0$, and $l\alpha+m\beta$ is a period.

As before let z_0 be a point at which f is holomorphic, and consider the function $f(z)-f(z_0)$. There exists an $\varepsilon_0>0$, such that $f(z)-f(z_0)$ has no zero in $|z-z_0|<\varepsilon_0$, other than z_0. However, $f(z_0+l\alpha+m\beta)=f(z_0)$, so that $z_1=z_0+l\alpha+m\beta$ is a zero of $f(z)-f(z_0)$, with $z_1\neq z_0$ and $|z_1-z_0|<\varepsilon_0$, which leads to a contraction if we choose $\varepsilon_0<\varepsilon$.

Corollary 3. *A non-constant meromorphic function f cannot have two periods α, β $(\alpha\neq 0, \beta\neq 0)$ which are linearly independent over $\mathbb{Q}$, and for which the quotient α/β is real.*

For otherwise $f(\alpha z)$ will have the real periods 1 and β/α, which is impossible. Note that $f(\alpha z)=f(\alpha z+\alpha)$, and $f(\alpha(z+\beta/\alpha))=f(\alpha z+\beta)$.

Remark. The only properties of the set S (of all periods of f) used here are that S is a *sub-group* of $\mathbb{C}$, which is *discrete* (cf. (2.2)).

§ 4. Elliptic functions

Definition. A meromorphic function in the complex plane with two periods ω, ϖ, such that $\operatorname{Im}\dfrac{\varpi}{\omega}>0$, is called an *elliptic function.*

Remarks

(i) In view of Corollary 3 after Theorem 1′, the essential point in this definition is that $\operatorname{Im}\dfrac{\varpi}{\omega}\neq 0$.

(ii) The function $F(z)$ of the complex variable z defined by

$$F(z)=\sum_{m=-\omega}^{\infty}\sum_{n=-\omega}^{\infty}\frac{1}{\{z-(m+ni)\}^3}$$

is an elliptic function, with the periods 1 and i. The function $\exp\{F(z)\}$ is *not* elliptic, since it is not meromorphic in the whole z-plane.

(iii) If (ω_1, ω_2) is a pair of primitive, or reduced, periods of a non-constant meromorphic function $f(z)$, they obviously form a *basis* for the set of *all* periods of f; that is to say, every period ω of f is of the form $\omega=m\omega_1+n\omega_2$, where m, $n=0, \pm 1, \pm 2, \ldots$ (Theorem 1).

Since the points (ω) form geometrically a *lattice* in the complex plane, the set $\{m\omega_1+n\omega_2\}$ is called the *period-lattice*. The primitive, or reduced, periods form a basis for this lattice. *Any* pair of periods forming such a basis is called *a pair of basic periods.*

Given a pair of basic periods, every other pair of basic periods can be obtained from the given one by a *unimodular transformation* as shown in the following.

Theorem 2. *The complex numbers ω_1^*, ω_2^* form a pair of basic periods of an elliptic function $f(z)$, if and only if they are related to a pair of primitive (basic) periods (ω_1, ω_2) of $f(z)$ by a transformation of the type*

$$\omega_1^* = m\omega_1 + n\omega_2, \quad \omega_2^* = p\omega_1 + q\omega_2,$$

where m, n, p, q are integers, with the property $mq - np = \pm 1$.

Such a transformation of (ω_1, ω_2) into (ω_1^*, ω_2^*) is called a *unimodular* transformation. If $mq - np = +1$, it is called a *proper, unimodular* transformation.

Proof. (i) Let (ω_1^*, ω_2^*) be any pair of basic periods of the given elliptic function, while (ω_1, ω_2) is a pair of *primitive* or *reduced*, periods. We then have

$$\omega_1^* = m\omega_1 + n\omega_2, \quad \omega_2^* = p\omega_1 + q\omega_2,$$

and

$$\omega_1 = m'\omega_1^* + n'\omega_2^*, \quad \omega_2 = p'\omega_1^* + q'\omega_2^*,$$

where m, n, p, q and m', n', p', q', are all integers. If we substitute for ω_1^* and ω_2^*, we get

$$\omega_1 = (m'm + n'p)\omega_1 + (m'n + n'q)\omega_2,$$
$$\omega_2 = (p'm + q'p)\omega_1 + (p'n + q'q)\omega_2.$$

Since $\dfrac{\omega_2}{\omega_1}$ is *not rational*, it follows that

$$(m'm + n'p) = 1, \quad m'n + n'q = 0,$$
$$p'm + q'p = 0, \quad p'n + q'q = 1.$$

These equations can be expressed in matrix form by writing

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} m' & n' \\ p' & q' \end{pmatrix} \cdot \begin{pmatrix} m & n \\ p & q \end{pmatrix}.$$

By considering the corresponding determinants, we obtain

$$1 = (m'q' - n'p')(mq - np).$$

Since m, n, p, q, and m', n', p', q', are all integers, it follows that each of the terms in brackets is either $+1$ or -1. Therefore $mq - np = \pm 1$, and the transformation $(\omega_1, \omega_2) \to (\omega_1^*, \omega_2^*)$ is unimodular. So also is its inverse.

(ii) Conversely, if $\omega_1^* = m\omega_1 + n\omega_2$, $\omega_2^* = p\omega_1 + q\omega_2$, where m, n, p, q are integers such that $mq - np = \pm 1$, we get by solving for ω_1 and ω_2:

$$\omega_1 = \varepsilon(q\omega_1^* - n\omega_2^*), \quad \omega_2 = \varepsilon(-p\omega_1^* + m\omega_2^*), \quad \varepsilon = \pm 1.$$

Hence *every* number of the form $\omega = \xi\omega_1 + \eta\omega_2$, where ξ and η are integers, can also be written uniquely in the form $m'\omega_1^* + n'\omega_2^*$, where m', n' are integers, and

every number $m'\omega_1^* + n'\omega_2^*$ is of the form $m''\omega_1 + n''\omega_2$, where m'', n'' are integers. Thus (ω_1^*, ω_2^*) is a pair of basic periods.

Remarks

(i) Every two pairs of basic periods are connected by a unimodular transformation.

(ii) If $\tau = \dfrac{\omega_2}{\omega_1}$, we have the relation

$$\begin{aligned}(mq-np)\omega_1\bar{\omega}_1(\tau-\bar{\tau}) &= (mq-np)(\bar{\omega}_1\omega_2 - \bar{\omega}_2\omega_1)\\ &= \bar{\omega}_1^*\omega_2^* - \bar{\omega}_2^*\omega_1^*\\ &= \omega_1^*\bar{\omega}_1^*(\tau^*-\bar{\tau}^*),\end{aligned} \tag{4.1}$$

where $\tau^* = \dfrac{\omega_2^*}{\omega_1^*}$ (and ω_2^*, ω_1^* are as in Theorem 2). Therefore, if $(mq-np)=1$, then $(\operatorname{Im}\tau > 0)$ implies that $(\operatorname{Im}\tau^* > 0)$.

(iii) The transformation $(\omega_1, \omega_2) \to (\omega_1^*, \omega_2^*)$ given by $\omega_1^* = m\omega_1 + n\omega_2$, $\omega_2^* = p\omega_1 + q\omega_2$ induces the transformation $\tau \to \tau^*$, given by

$$\tau^* = \frac{q\tau + p}{n\tau + m}.$$

(iv) If (ω_1, ω_2) is a pair of *basic* periods, $(\omega_1, k\omega_1 + \omega_2)$ is also a pair of *basic* periods, where k is an integer. But the pair $(\omega_1, 2\omega_1 + \omega_2)$ is *not* a pair of *primitive*, or *reduced*, periods (Fig. 1).

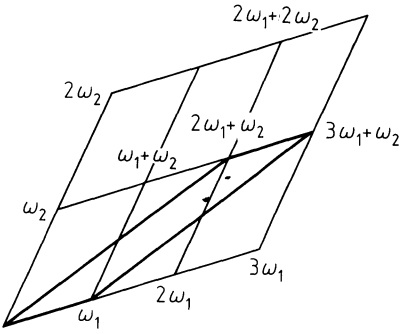


Fig. 1

(v) We have so defined a pair of primitive periods that ostensibly infinitely many conditions are involved. In fact, three of them suffice, as shown by the following

Theorem 3. *A pair of basic periods (ω_1, ω_2) of a given elliptic function is a pair of primitive, or reduced, periods, if and only if the following conditions are satisfied:*

$$|\tau| \geqq 1, \quad \operatorname{Im}\tau > 0, \quad -\tfrac{1}{2} \leqq \operatorname{Re}\tau \leqq \tfrac{1}{2},$$

where $\tau = \dfrac{\omega_2}{\omega_1}$.

Proof. Let $\tau=\xi+i\eta$, where ξ and η are real, and $i=\sqrt{-1}$. If (ω_1, ω_2) is a pair of primitive periods, then we have

$$\operatorname{Im}\tau>0, \quad \text{and} \quad |\omega_2\pm\omega_1|\geqq|\omega_2|\geqq|\omega_1|,$$

so that

$$|\tau\pm1|^2\geqq|\tau|^2\geqq1, \quad \text{or}$$
$$(\xi\pm1)^2+\eta^2=|\tau\pm1|^2\geqq|\tau|^2=\xi^2+\eta^2\geqq1,$$

or $\pm2\xi+1\geqq0$, from which follow the three conditions stated in the theorem.

Conversely, we wish to show that *if* those conditions are satisfied, then the pair (ω_1, ω_2) is primitive.

Consider any non-trivial period $\omega=m\omega_1+n\omega_2$, where m, n are integers. By hypothesis, $\left|\frac{\omega_2}{\omega_1}\right|=|\tau|\geqq1$, and $\operatorname{Im}\frac{\omega_2}{\omega_1}>0$, therefore $\frac{\omega_2}{\omega_1}$ is *non-real.*

If $n=0$, then $\frac{\omega}{\omega_1}$ is real, and equals m, where $m\neq0$, since ω is non-trivial, hence $|\omega|\geqq|\omega_1|$.

If $n\neq0$, then $\frac{\omega}{\omega_1}$ is *not* real, since $\frac{\omega_2}{\omega_1}$ is not. Set $D=|m+n\tau|^2-|\tau|^2=(m+n\xi)^2-\xi^2+(n^2-1)\eta^2$. If $n\neq\pm1$, then $n^2-1\geqq3$, and since $\eta^2=(\xi^2+\eta^2)-\xi^2\geqq1-\frac{1}{4}=\frac{3}{4}$, we have $D\geqq-\frac{1}{4}+\frac{9}{4}>0$. If $n=\pm1$, and m any integer, then we have $D=(m\pm\xi)^2-\xi^2\geqq0$, since $|\xi|\leqq\frac{1}{2}$. Hence $|\omega|\geqq|\omega_2|$, if $n\neq0$, and the pair (ω_1, ω_2) is primitive.

Theorem 4. *Given any non-constant elliptic function f, there exists a pair of basic periods (ω_1, ω_2) of f, such that the quotient $\tau=\frac{\omega_2}{\omega_1}$ satisfies the conditions:*

$$\operatorname{Im}\tau>0; \quad |\tau|\geqq1; \quad -\tfrac{1}{2}\leqq\operatorname{Re}\tau<\tfrac{1}{2},$$

with

$$\operatorname{Re}\tau\leqq0 \quad \textit{if} \quad |\tau|=1;$$

so that (after Theorem 3) (ω_1, ω_2) is a pair of primitive periods.

If (ω_1^*, ω_2^*) is another pair of primitive periods, where $\omega_1^*=m\omega_1+n\omega_2$, $\omega_2^*=p\omega_1+q\omega_2$, (m, n, p, q being integers), and the quotient $\tau^*=\frac{\omega_2^*}{\omega_1^*}$ satisfies the conditions

$$\operatorname{Im}\tau^*>0; \quad |\tau^*|\geqq1; \quad -\tfrac{1}{2}\leqq\operatorname{Re}\tau^*<\tfrac{1}{2},$$

with

$$\operatorname{Re}\tau^*\leqq0, \quad \text{if} \quad |\tau^*|=1;$$

then we have $\tau^*=\tau$.

Proof.

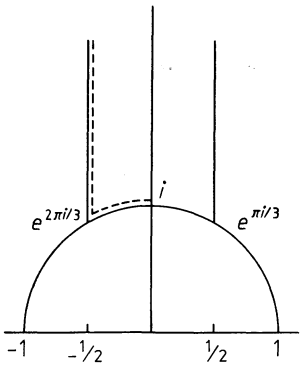


Fig. 2

Let (ω_1, ω_2) be a pair of *primitive* periods of the given elliptic function f. By Theorem 3, we have

$$\mathrm{Im}\,\tau>0; \quad -\tfrac{1}{2}\leqq \mathrm{Re}\,\tau\leqq\tfrac{1}{2}; \quad |\tau|\geqq 1, \quad \text{where} \quad \tau=\frac{\omega_2}{\omega_1}.$$

If τ lies on the half-line: $\mathrm{Re}\,\tau=\frac{1}{2}$, $\mathrm{Im}\,\tau>0$, then $\tau-1$ lies on the half-line $\mathrm{Re}\,\tau=-\frac{1}{2}$, $\mathrm{Im}\,\tau>0$. If τ lies on the semi-circular arc: $|\tau|=1$, $\frac{1}{2}>\mathrm{Re}\,\tau>0$, then $-\dfrac{1}{\tau}$ lies on the semi-circular arc: $|\tau|=1$, $-\frac{1}{2}<\mathrm{Re}\,\tau<0$. The transformation $(\omega_1,\omega_2)\to(\omega_1^*,\omega_2^*)$ given by $\omega_1^*=\omega_1$, $\omega_2^*=-\omega_1+\omega_2$ (which corresponds to $m=1$, $n=0$, $p=-1$, $q=1$) *is a proper, unimodular*, transformation, which carries $\tau=\dfrac{\omega_2}{\omega_1}$ into $\tau^*=\tau-1=\dfrac{-\omega_1+\omega_2}{\omega_1}$. Since $\mathrm{Im}\,\tau>0$, we have $\mathrm{Im}\,\tau^*>0$. Similarly the transformation $(\omega_1,\omega_2)\to(\omega_1^*,\omega_2^*)$ given by $\omega_1^*=\omega_2$, $\omega_2^*=-\omega_1$, is also a *proper, unimodular*, transformation, which carries $\tau=\dfrac{\omega_2}{\omega_1}$ into $\tau^*=-\dfrac{1}{\tau}$. Since two unimodular transformations applied in succession lead to a unimodular transformation, there is always a pair of *basic* periods (ω_1,ω_2) satisfying the conditions stated in the theorem; they are a *primitive* pair by Theorem 3.

To prove the second part, we note, first of all, that $mq-np=\pm 1$, by Theorem 2. Since $\mathrm{Im}\,\tau>0$, and $\mathrm{Im}\,\tau^*>0$, it follows (by Remark (ii) after Theorem 2) that $mq-np=+1$. Secondly, since (ω_1^*,ω_2^*) is a pair of primitive periods, we have $|\omega_1^*|\leqq|\omega_1|$, $|\omega_1^*|\leqq|\omega_2|$. Since (ω_1,ω_2) is also a pair of primitive periods, $|\omega_1|\leqq|\omega_1^*|$, $|\omega_1|\leqq|\omega_2^*|$. Hence

$$|\omega_1^*|=|\omega_1|\leqq|\omega_2|, \quad |\tau|\geqq 1.$$

We have, in general,

$$\tau^*=\frac{q\tau+p}{n\tau+m}, \quad \mathrm{Im}\,\tau^*=\frac{(mq-np)\,\mathrm{Im}\,\tau}{|n\tau+m|^2},$$

where $\omega_1^* = m\omega_1 + n\omega_2$, and $\omega_2^* = p\omega_1 + q\omega_2$. Since $|\omega_1^*| = |\omega_1|$, we have $|n\tau + m| = 1$. Since $(mq - np) = +1$, we have also $\operatorname{Im}\tau = \operatorname{Im}\tau^*$.

Let $\tau = \xi + i\eta$, $\tau^* = \xi^* + i\eta^*$, where ξ, ξ^* and η, η^* are real. If $n \neq 0$, then

$$\tau^* = \frac{1}{n^2}\left\{\frac{n^2(q\tau + p)}{n\tau + m}\right\} = \frac{1}{n^2}\frac{[qn(n\tau + m) - qmn + pn^2]}{(n\tau + m)}$$

$$= \frac{1}{n^2}\left[qn - \frac{n}{n\tau + m}\right],$$

so that

$$\xi^* + \xi = \frac{1}{n^2}[n(q - m)].$$

If $n = 0$, *then* $\tau^* = \tau$, for we have $m = q = \pm 1$, hence $\tau^* = \tau \pm p$, and since $-\frac{1}{2} \leqq \xi < \frac{1}{2}$, and $-\frac{1}{2} \leqq \xi^* < \frac{1}{2}$, it follows that $p = 0$, and $\tau^* = \tau$.

If $n \neq 0$, *then* $n = \pm 1$, *and* $|m| \leqq 1$, for $(m + n\xi)^2 + n^2\eta^2 = 1$, since $|m + n\tau|^2 = 1$, and if $|n| \geqq 2$, then $n^2 \geqq 4$, and since $\eta^2 \geqq \frac{3}{4}$, it would imply that $n^2\eta^2 \geqq 3$, which is impossible since $|m + n\tau|^2 = 1$. Thus $|\tau \pm m|^2 = 1$, and since $|\tau| \geqq 1$, we must have $m = 0$, or $m = \pm 1$.

In case $n = \pm 1$, $m = 0$, we have $|\tau| = 1$, and $-1 \leqq \xi^* + \xi = \pm q < 1$, hence $q = 0$, or -1. If $q = 0$, then $\tau^* = -1/\tau$, $\xi^* = \xi = 0$, and $\tau^* = \tau = i$. If $q = -1$, then $\xi^* = \xi = -\frac{1}{2}$, and $\tau^* = \tau = e^{2\pi i/3}$.

In case $n = \pm 1, m = \pm 1$, we have $|\tau \pm 1| = 1$, since $|m + n\tau| = 1$. The possibility $|\tau - 1| = 1$ is excluded (since $\tau = e^{\pi i/3}$ is excluded by the condition $\xi < \frac{1}{2}$). Thus $|\tau + 1| = 1$, hence $m = n = \pm 1$, and since $|\tau| \geqq 1$, we must have $\tau = e^{2\pi i/3}$, and since $\operatorname{Im}\tau^* = \operatorname{Im}\tau$, we must have $\tau^* = \tau = e^{2\pi i/3}$ (Fig. 2).

Remarks

(i) The pair (ω_1, ω_2) of primitive periods is obviously *not unique*, since the pair $(-\omega_1, -\omega_2)$ is also primitive. Both pairs define the same quotient τ.

(ii) There may be several pairs of *basic* periods with the same quotient τ, where τ satisfies the conditions:

$$\operatorname{Im}\tau > 0, \quad |\tau| \geqq 1, \quad -\tfrac{1}{2} \leqq \operatorname{Re}\tau < \tfrac{1}{2},$$

with

$$\operatorname{Re}\tau \leqq 0 \quad \text{if} \quad |\tau| = 1.$$

Since any two such pairs are related by a *proper, unimodular*, transformation, it follows that τ is a fixed point of the induced transformation on τ (Remark (iii) after Theorem 2). The proof of Theorem 4 shows that in the non-trivial case (that is, if $n \neq 0$), $\tau^* = \tau$ only when $\tau = i$, or $\tau = e^{2\pi i/3}$. Now $\tau = i$ is a fixed point only for the transformation $\tau \to -\dfrac{1}{\tau}$, and $\tau = e^{2\pi i/3}$ is a fixed point only for the transformations: $\tau \to \dfrac{-1}{\tau + 1}$ and $\tau \to -\dfrac{(\tau + 1)}{\tau}$. Hence, in case $\tau = i$, we have the four pairs: (ω_1, ω_2), $(-\omega_1, -\omega_2)$, $(-\omega_2, \omega_1)$, $(\omega_2, -\omega_1)$. In case $\tau = e^{2\pi i/3}$, we

have the six pairs: $(\omega_1, \omega_2), (-\omega_1, -\omega_2); (\omega_1+\omega_2, -\omega_1), (-(\omega_1+\omega_2), \omega_1); (\omega_2, -(\omega_1+\omega_2)), (-\omega_2, \omega_1+\omega_2)$.

§ 5. The modular group and modular functions. Given a non-constant elliptic function f, and a pair of *basic periods* (ω_1, ω_2), we can obtain another pair of basic periods (ω_1^*, ω_2^*) by an unimodular transformation: $\omega_1^* = m\omega_1 + n\omega_2$, $\omega_2^* = p\omega_1 + q\omega_2$, where m, n, p, q are integers such that $mq - np = \pm 1$. If $\tau = \dfrac{\omega_2}{\omega_1}$, $\tau^* = \dfrac{\omega_2^*}{\omega_1^*}$, then $\operatorname{Im}\tau > 0$ implies that $\operatorname{Im}\tau^* > 0$ if (and only if) $mq - np = +1$, in which case we have called the transformation *properly unimodular*. Such transformations are of prime importance in the application of the theory of elliptic functions to arithmetic.

Let z be a complex variable, and let a, b, c, d be *integers*. The transformation

$$M: z \to z' = \frac{az+b}{cz+d}, \quad \text{where} \quad ad-bc=1, \quad cz+d \neq 0,$$

is called a *modular transformation*. If $z = x+iy$, and $z' = x'+iy'$, then

$$y' = \frac{1}{2i}(z'-\bar{z}') = \frac{1}{2i}\left(\frac{az+b}{cz+d} - \frac{a\bar{z}+b}{c\bar{z}+d}\right) = \frac{1}{2i}\left(\frac{(z-\bar{z})(ad-bc)}{|cz+d|^2}\right)$$

$$= \frac{y}{|cz+d|^2} > 0,$$

if $y > 0$, since c and d cannot both be zero.

(5.1) Hence the transformation M maps the upper half-plane $\operatorname{Im} z > 0$ into itself.

(5.2) If $M: z \to z' = \dfrac{az+b}{cz+d}$, $ad-bc=1$, and $M': z' \to z'' = \dfrac{a'z'+b'}{c'z'+d'}$, $a'd'-b'c' = 1$, are two modular transformations, then $M'': z \to z''$ is again a modular transformation, since

$$z'' = \frac{\alpha z + \beta}{\gamma z + \delta},$$

with $\alpha, \beta, \gamma, \delta$ integers, *and* $\alpha\delta - \beta\gamma = 1$, as can be directly verified.

We define M'' as the *product* of M and M', and write $M'' = M'M$.

(5.3) This *product* is *associative*.

(5.4) If M is a modular transformation given by $Mz = z' = \dfrac{az+b}{cz+d}$, then the transformation M^{-1} defined by

$$M^{-1}: z \to \frac{dz-b}{-cz+a},$$

is also a modular transformation, and we have

$$M^{-1}Mz = \frac{dz'-b}{-cz'+a} = z = MM^{-1}z.$$

We call M^{-1} the *inverse* transformation of M.

(5.5) If $a=d=\pm 1$, and $b=c=0$, M reduces to the *Identity*: $Mz=z$.

(5.6) Since the integers a, b, c, d, and $-a$, $-b$, $-c$, $-d$, define the same transformation M, we may assume, without loss of generality, that $c\geqq 0$, and that $d>0$ if $c=0$. (Note that if $c=0$, and $d>0$, then $a>0$).

(5.7) *The modular transformations form a group, because of (5.2)–(5.5). This group is generated by the two transformations:*

$$A: z\to z+1, \quad \text{and} \quad B: z\to -\frac{1}{z}.$$

To prove this, we have to show that *any* modular transformation M can be expressed as a product of *powers* of A and B. Clearly $B^2=I$, the *identity*.

Let $m=\min(|c|,|d|)$. If $|c|>|d|$ in M, where $Mz=\dfrac{az+b}{cz+d}$, then the transformation

$$MB: z\to \frac{a'z+b'}{c'z+d'}$$

has the property: $|c'|<|d'|$, for $MBz=\dfrac{-\frac{a}{z}+b}{-\frac{c}{z}+d}=\dfrac{bz-a}{dz-c}$. It suffices therefore to consider a modular transformation subject to the restriction: $|c|\leqq|d|$.

Now let $|c|\leqq|d|$, and let $m=0$. Then $c=0$, hence $ad=1$, and M is given by: $z\to z\pm b$. That is to say, $M=A^{\pm b}$.

Assuming that any M for which $m\leqq n-1$, $n\geqq 1$, is generated by A and B, we shall prove that it is so generated also for $m=n$. By the definition of m, and by our hypothesis that $|c|\leqq|d|$, we have $m=|c|$. We may suppose that $c>0$, for otherwise we can multiply a, b, c, d throughout by -1 (which leaves M unchanged). Therefore $m=c$, and we have to prove that M is generated by A and B in case $c=n$ (since $m=c$). Consider the transformation

$$MA^k: z\to \frac{az+(ak+b)}{cz+(ck+d)},$$

where k is an integer such that $0\leqq ck+d<c$. (Note that k is not necessarily positive. Since c is an integer, $c>0$, d is an integer, $c\leqq|d|$, we have $-\dfrac{d}{c}\leqq k<\dfrac{c-d}{c}=1-\dfrac{d}{c}$. For example, if $c=3$, $d=4$, then $-\frac{4}{3}\leqq k<1-\frac{4}{3}$, or $-1-\frac{1}{3}\leqq k<-\frac{1}{3}$, and $k=-1$). With such a k, we have: $\min(c, ck+d)\leqq n-1$, *if* $c=n$, since $ck+d<c$. By the induction hypothesis MA^k is generated by A and B, hence also M (since $M=(\ldots)A^{-k}$).

(5.8) *Given* z, *with* $\operatorname{Im} z>0$, *there exists a modular transformation* M^*, *such that* $\operatorname{Im}(M^*z)$ *is a maximum. If* $Mz^*=z^*$, *then* $|z^*|\geqq 1$.

If $M^*z=z^*=\dfrac{az+b}{cz+d}$, and $z=x+iy$, $z^*=x^*+iy^*$, then $y^*=\dfrac{y}{|cz+d|^2}>0$. The number of pairs (c, d) with $|cz+d|<K<\infty$ is finite, since c, d are integers. Hence there exists a transformation M^*, such that $\mathrm{Im}(M^*z)$ is a maximum. And in that case, $|z^*|\geqq 1$ for otherwise $|z^*|<1$, which implies that $\mathrm{Im}\left(-\dfrac{1}{z^*}\right)=\dfrac{y^*}{x^{*2}+y^{*2}}>y^*$, which is impossible.

(5.9) Let M, A, B denote modular transformations, as follows:

$$M: z \to \frac{az+b}{cz+d}; \qquad ad-bc=1; \qquad a, b, c, d \text{ integers};$$

$$A: z \to z+1;$$

$$B: z \to -\frac{1}{z}.$$

Let F denote the set: $\{z \mid \mathrm{Im}z>0,\ |z|\geqq 1,\ |\mathrm{Re}z|\leqq \tfrac{1}{2}\}$.

(5.91) *Given z, with $\mathrm{Im}z>0$, there exists a modular transformation M, such that $Mz\in F$.*

By (5.8) there exists a modular transformation M^*, such that $\mathrm{Im}M^*z$ is a maximum, and if $M^*z=z^*$, then $|z^*|\geqq 1$. It follows that $\mathrm{Im}(A^kM^*z)$ is a maximum for all integral k, and $|A^kM^*z|\geqq 1$, for all integers k. One can choose k in such a way that $A^kM^*z\in F$, and if $M=A^kM^*$, with such a k, then M is a transformation of the kind required.

(5.92) Two points z_1, z_2, with $\mathrm{Im}z_1>0$, $\mathrm{Im}z_2>0$, are said to be *congruent* (relative to the modular group), if there exists a modular transformation M, such that $z_2=Mz_1$. We then write: $z_1\sim z_2$. The relation "$\sim$" is *reflexive*, *symmetric*, and *transitive*.

(5.93) The set F defined in (5.9) has the following property. If $z_1, z_2\in F$, $z_1\neq z_2$, $z_1\sim z_2$, then we have $\mathrm{Re}z_1=\tfrac{1}{2}$, and $z_2=z_1-1$, *or* $\mathrm{Re}z_1=-\tfrac{1}{2}$ and $z_2=z_1+1$, *or* $|z_1|=1$ and $z_2=-\dfrac{1}{z_1}$.

Proof. Let M be a modular transformation, such that $Mz_1=z_2=\dfrac{az_1+b}{cz_1+d}$, with a, b, c, d integers, and $ad-bc=1$. We may assume that $\mathrm{Im}z_2\geqq\mathrm{Im}z_1$, that is $|cz_1+d|\leqq 1$ (for otherwise, one can replace z_1 by $Mz_1=z_2$ and M by M^{-1}).

It follows then that $|c|<2$. For $1\geqq|cz_1+d|^2\geqq c^2(\mathrm{Im}z_1)^2\geqq c^2\cdot\frac{3}{4}$, since $z_1\in F$ so that $|z_1|\geqq 1$, $|\mathrm{Re}z_1|\leqq\frac{1}{2}$, $\mathrm{Im}z_1\geqq\dfrac{\sqrt{3}}{2}$.

Hence $c=0$, *or* 1, *or* -1. If $c=-1$, we can multiply a, b, c, d throughout by -1, as mentioned in (5.6), and consider the case $c=+1$. It suffices therefore to consider the cases: $c=0, 1$.

If $c=0$, then $d=\pm 1$, so that $a=\pm 1$ (in fact, $d=+1$, $a=+1$, by (5.6)), and $z_2=Mz_1=z_1\pm b=A^{\pm b}z_1$, where A is defined as in (5.9). We have, however, $|\mathrm{Re}z_1|\leqq\frac{1}{2}$, *and* $|\mathrm{Re}z_2|\leqq\frac{1}{2}$, hence *either* $b=0$, and M the Identity (which is

excluded by the assumption $z_1 \neq z_2$), or $|b|=1$. If $b=+1$, then $\mathrm{Re}\, z_1 = -\frac{1}{2}$, and $\mathrm{Re}\, z_2 = +\frac{1}{2}$, and *vice versa* if $b=-1$. *Hence* $z_2 = z_1 \pm 1$.

If $c=1$, then, first of all, we have, as before, $|cz_1+d| \geqq 1$, that is $|z_1+d| \geqq 1$. But $z_1 \in F$, hence $|z_1| \geqq 1$, and $|d| \leqq 1$ (since $|d| \geqq 2$, d integral, and $|z_1+d| \leqq 1$, imply that $z_1 \notin F$).

(5.931) *If* $d=0$, then $|z_1| \leqq 1$, hence $|z_1|=1$ and $b=-1$ (since $d=0$, and $c=1$), so that $z_2 = Mz_1 = a - \frac{1}{z_1}$, hence $z_2 + \frac{1}{z_1} = a$, $\mathrm{Re}\, z_2 + \mathrm{Re}\frac{1}{z_1} = a$, $|a| \leqq |\mathrm{Re}\, z_2| + |\mathrm{Re}\, z_1| \leqq 1$, therefore $a=0$, *or* $|a|=1$. *In case* $a=0$, we have $z_2 = -\frac{1}{z}$. *In case* $a=+1$, we have $z_2 = 1 - \frac{1}{z_1}$, with $|z_1|=1$, so that $\mathrm{Re}\, z_2 = 1 - \mathrm{Re}\, z_1$, hence $\mathrm{Re}\, z_1 = \frac{1}{2} = \mathrm{Re}\, z_2$, which implies that $z_1 = e^{\pi i/3} = z_2$, contrary to the assumption that $z_1 \neq z_2$. *In case* $a=-1$, we have $z_2 = -1 - \frac{1}{z_1}$, and $z_1 = e^{2\pi i/3} = z_2$, which is likewise excluded.

(5.932) *If* $d=1$, then $|z_1-(-d)| \leqq 1$, which defines a circular disc with centre at $-d$, and radius 1, and if $z_1 \in F$, then $z_1 = e^{2\pi i/3}$. Since $c=1$, $d=1$, and $ad-bc=1$, we have $a-b=1$, so that

$$z_2 = \frac{az_1+a-1}{z_1+1} = \frac{a(z_1+1)-1}{z_1+1} = a - \frac{1}{z_1+1} = a+z_1 .$$

Since $|\mathrm{Re}\, z_1| \leqq \frac{1}{2}$, $|\mathrm{Re}\, z_2| \leqq \frac{1}{2}$, we have *either* $a=0$, which is excluded because $z_1 \neq z_2$, *or* $a=1$, and $z_2 = z_1+1$.

(5.933) *If* $d=-1$, it follows, as in (5.932), that $a=-1$, and $z_2 = z_1 - 1$.

(5.94) By (5.91), given *any* z, with $\mathrm{Im}\, z > 0$, there exists a $z' \in F$, such that $z' \sim z$. By (5.93), if $z_1, z_2 \in \mathrm{Int}\, F$, $z_1 \neq z_2$, then $z_1 \not\sim z_2$ (z_1 is *not* congruent to z_2 relative to the modular group); if $z_1, z_2 \in F$, $z_1 \neq z_2$, and $z_1 \sim z_2$, then $z_1, z_2 \in \partial F$ (the *boundary* of F).

Let $F_0 = \{z \mid \mathrm{Im}\, z > 0,\ |z| \geqq 1,\ -\frac{1}{2} \leqq \mathrm{Re}\, z \leqq 0\}$.

$$\cup \cdot \{z \mid \mathrm{Im}\, z > 0,\ |z| > 1,\ 0 < \mathrm{Re}\, z < \tfrac{1}{2}\} .$$

Then $\mathrm{Int}\, F = \mathrm{Int}\, F_0$, so that $z_1, z_2 \in \mathrm{Int}\, F_0$, and $z_1 \neq z_2$, together imply that $z_1 \not\sim z_2$ as before.

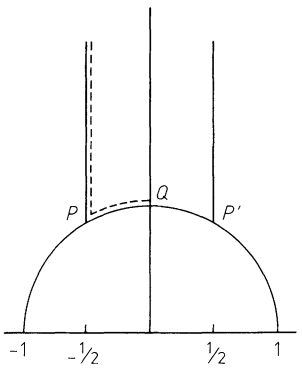


Fig. 3

Since the arc $P'Q$ ($\mathrm{Im}\, z>0$, $|z|=1$, $\frac{1}{2}\geqq \mathrm{Re}\, z\geqq 0$) is (in Fig. 3) transformed into the arc PQ ($\mathrm{Im}\, z>0$, $|z|=1$, $-\frac{1}{2}\leqq \mathrm{Re}\, z\leqq 0$) by the *modular* transformation $z \to -\frac{1}{z}$, and the line $P'\infty\left(\mathrm{Re}\, z=\frac{1}{2},\ \mathrm{Im}\, z\geqq \frac{\sqrt{3}}{2}\right)$ is transformed into the line $P\infty\left(\mathrm{Re}\, z=-\frac{1}{2},\ \mathrm{Im}\, z\geqq \frac{\sqrt{3}}{2}\right)$ by the modular transformation $z\to z-1$, we can assert the following:

(5.95) Given any z, with $\mathrm{Im}\, z>0$, there exists a $z'\in F_0$, such that $z'\sim z$. If $z_1 \neq z_2$, and $z_1, z_2\in F_0$, then $z_1 \not\sim z_2$.

The set F_0 is called *a fundamental domain* of the modular group (and has been encountered already in Theorem 4).

Definition. A complex-valued function $F(z)$ of the complex variable z is said to be a *modular function*, if it is meromorphic for $\mathrm{Im}\, z>0$, and $F(Mz)=F(z)$ for all transformations M belonging to the modular group, or for all M belonging to a sub-group of the modular group of finite index.

We shall have occasion to study two special modular functions in Chapters VI and VII, because of their close connexion with elliptic functions.

Notes on Chapter I

§ 1. As a general reference, see S. Saks and A. Zygmund, *Analytic Functions*, (3rd Edition, English translation by E. J. Scott, Elsevier, 1971), Chapters VII and VIII.

§ 2. The derivative $f'(z)$ of a periodic meromorphic function $f(z)$ is also a periodic function; every period of f is also a period of f'; but the primitive is not necessarily periodic; for example, $f'(z)=1$, $f(z)=z$.

In (2.3), if ω_1 is a primitive period of a simply periodic meromorphic function f, then no number $\alpha\omega_1$, where $0<|\alpha|<1$, is a period of f.

In (2.4), if (ω_1, ω_2) is a pair of primitive periods of a meromorphic function f, such that $|\tau|\geqq 1$, where $\tau=\mathrm{Im}\frac{\omega_2}{\omega_1}$, then $\alpha\omega_1$, where $0<|\alpha|<1$, is not a period of f, and $\beta\omega_2$, where $0<|\beta|<1$, is not a period of f (α, β real).

If $f(z)$ is a function of period 1, holomorphic in a strip R, defined by: $-\infty<b<\mathrm{Im}\, z<B<+\infty$, then in that strip it has a Fourier expansion, that is to say

$$f(z)=\sum_{n=-\infty}^{\infty} c_n e^{2\pi i n z}.$$

The series is uniformly and absolutely convergent in every strip $b'\leqq \mathrm{Im}\, z\leqq B'$ contained in R. And if the function cannot be extended holomorphically to any strip containing R but different from it, then the series diverges in the exterior of R. To different strips of holomorphy of the same function there may correspond

different Fourier expansions. For example

$$\cot \pi z = -i\left(1+2\sum_{n=1}^{\infty} e^{2\pi i n z}\right), \quad \operatorname{Im} z>0;$$

$$\cot \pi z = i\left(1+2\sum_{n=1}^{\infty} e^{-2\pi i n z}\right), \quad \operatorname{Im} z<0.$$

See Saks and Zygmund, loc. cit., p. 361.

§ 3. For Jacobi's lemma, see C. G. J. Jacobi, *J. für die reine und angewandte Math.* 13 (1835), 55–78; *Ges. Werke*, Bd. II (1882), 23–50. It is stated by E. Goursat, *A course in mathematical analysis*, Part I of Volume II (English translation by E. R. Hedrick and O. Dunkel, Boston, 1916), p. 148.

§ 4. As general references, see J. Tannery et J. Molk, *Eléments de la Théorie des Fonctions Elliptiques* (Paris, 1839–1902); P. Appell et E. Lacour, *Principes de la Théorie des Fonctions Elliptiques et Applications* (2nd edition, Paris, 1922); E. T. Whittaker and G. N. Watson, *A course of modern analysis* (4th edition, Cambridge, 1952), Chapters XX and XXII; A. Hurwitz and R. Courant, *Vorlesungen über allgemeine Funktionentheorie und elliptische Funktionen* (4th edition, Springer-Verlag, 1964). E. Goursat, loc. cit.; F. Tricomi, *Funzioni Ellittiche* (2nd edition, Bologna, 1951).

For an historical account of the theory of elliptic functions, including copious references, see Alfred Enneper, *Elliptische Funktionen (Zweite Auflage, Neu bearbeitet und herausgegeben von Felix Müller*, Halle, 1890), and the notes by Whittaker and Watson, loc. cit., §§ 20.1, 20.21, 20.312; § 21.1; §§ 22.7, 22.8.

Historically *elliptic functions* (as they are now known) arose out of *elliptic integrals* (as they are now known) by 'inversion', just as the trigonometric (or *circular*) function $\sin u$ may be said to arise by 'inversion' of the integral

$$\arcsin u = \int_0^u (1-u^2)^{-1/2} du, \quad 0 \leqq u \leqq 1.$$

By an elliptic integral is meant an integral of the form

$$\int_a^x R(x, y) dx,$$

where $R(x, y)$ is a rational function of x and y, and y^2 is a cubic or biquadratic in x, without repeated factors. A particular case is the integral

$$\int_0^t \frac{dt}{(1-t^4)^{1/2}}, \quad 0 \leqq t \leqq 1.$$

Such integrals occur in problems connected with the determination of the arc-lengths of curves such as the ellipse, the hyperbola, and the lemniscate. (The word 'elliptic' has stuck, notwithstanding the fact that such integrals are not exclusively associated with the ellipse.)

The theory of elliptic integrals may be said to have begun with the discovery by Conte G. C. di Fagnano, and later by Leonhard Euler, of two remarkable properties of an integral associated with the arc-length of a lemniscate. See Enneper, loc. cit., pp. 527, 530, 533. Fagnano's result that

$$\int_0^r \frac{dr}{(1-r^4)^{1/2}} = 2\int_0^u \frac{du}{(1-u^4)^{1/2}}, \tag{1}$$

where

$$r^2 = \frac{4u^2(1-u^4)}{(1+u^4)^2}, \tag{2}$$

is connected with the problem of doubling the arc-length of a lemniscate, and led to Euler's famous discovery, around 1761, of an *addition-theorem* for an elliptic integral, namely the relation

$$s(u)+s(v)=s(r),$$

where

$$s(u)=\int_0^u \frac{du}{(1-u^4)^{1/2}}, \qquad 0\leqq u\leqq 1, \tag{3}$$

and

$$r=\frac{u(1-v^4)^{1/2}+v(1-u^4)^{1/2}}{1+u^2v^2}.$$

See G. C. Fagnano's *Produzioni matematiche*, 2t. (Pesaro, 1750); *Opere matematiche*, 3 Vol. (Milano-Roma-Napoli, 1911–1912); Euler's *Opera* (1st series), Vol. 20, pp. 58–107; and C.L. Siegel, *Ges. Abhandlungen*, III, 249–251.

A clear and concise treatment of elliptic functions, as part of a course on the theory of functions of complex variables, is given by C. L. Siegel in his Göttingen lectures of 1953/54, later published by the Mathematical Institute of the University of Göttingen as *Ausgewählte Fragen der Funktionentheorie*, I and II. See Chapter I of Part I, where Siegel begins by explaining the work of Fagnano and Euler, and proceeds from elliptic integrals to elliptic functions. We shall here follow a different course.

Integral (3) is referred to as the *lemniscate integral*. The upper limits of integration r and u of the two lemniscate integrals in formula (1) are connected by the simple *algebraic* relation (2). By using complex substitutions, instead of purely real ones, one can obtain, instead of (1), the two following formulae:

$$\int_0^r \frac{dr}{(1-r^4)^{1/2}} = (1+i)\int_0^v \frac{dv}{(1-v^4)^{1/2}}, \tag{4}$$

where

$$r^2=\frac{2iv^2}{1-v^4}, \qquad i=\sqrt{-1};$$

and

$$\int_0^v \frac{dv}{(1-v^4)^{1/2}} = (1-i)\int_0^u \frac{du}{(1-u^4)^{1/2}}, \tag{5}$$

where

$$v^2 = \frac{-2iu^2}{1-u^4},$$

in which the upper limit v is now complex. Since $(1+i)(1-i)=2$, formulae (4) and (5) lead back to (1), and show that a lemniscate integral (with a complex limit of integration), when multiplied by a complex number, yields another lemniscate integral. They are referred to as *formulae of complex multiplication* for the lemniscate integral. This circumstance led to the theory of complex multiplication for elliptic functions, studied in the 19th century by Abel and Jacobi, and by Kronecker. See H. Weber, *Lehrbuch der Algebra*, III (zweite Auflage, 1908). See also the book by R. Fueter, *Vorlesungen über die singulären Moduln und die komplexe Multiplikation der Elliptischen Funktionen* (Teubner, 1924). A more recent reference to complex multiplication is No. 21 of the *Lecture Notes in Mathematics* (Springer Verlag, 1966), with contributions by A. Borel, S. Chowla, C. S. Herz, K. Iwasawa, and J.-P. Serre. For arithmetical applications, see C. L. Siegel's *Lectures on advanced analytic number theory* (No. 23 of the Lectures on Mathematics, Tata Institute of Fundamental Research, Bombay, 1961); K. Ramachandra, *Annals of Math.* 80 (1964), 104–148; and G. Shimura, *Introduction to the arithmetic theory of automorphic functions* (Tokyo-Princeton, 1971).

The first mathematician to deal with elliptic functions as distinct from elliptic integrals was Gauss, *Werke*, III, (1866), 404–432; 493–495; but the first results published were by Abel and Jacobi (see the comments by Whittaker and Watson, loc. cit., §§ 22.7, 22.8). See N. H. Abel, *Oeuvres*, I (1881), 263–388; and C. G. J. Jacobi, *Fundamenta Nova Theoriae Functionum Ellipticarum* (1829), *Ges. Werke* (1881), I, 49–239. For an account of the original contributions of Abel and Jacobi (and of Gauss) in this connexion, see P. G. Lejeune Dirichlet's *Gedächtnisrede* on Jacobi, printed in Jacobi's *Ges. Werke*, I, 1–28.

What we have called elliptic integrals are sometimes referred to in the classical literature as elliptic functions, e.g. A. M. Legendre, *Traité des fonctions elliptiques*.

§ 5. It has become customary to denote the modular group by Γ, which is not to be confused with the gamma-function of Leonhard Euler.

The *index* of a subgroup of Γ is the number of its distinct right cosets.

Let $SL(2,\mathbb{R})$ denote the special linear group of 2×2 matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, with coefficients a, b, c, d in $\mathbb{R}$, the set of real numbers, such that $ad-bc=1$. Let $SL(2,\mathbb{Z})$ denote the *discrete* subgroup of $SL(2,\mathbb{R})$ consisting of matrices with coefficients in $\mathbb{Z}$, the set of integers. (Discreteness here means that the Identity is

an isolated element.) Then the group Γ is isomorphic with the quotient group $\frac{SL(2,\mathbb{Z})}{\{I,-I\}}$, where $I=\begin{pmatrix}1&0\\0&1\end{pmatrix}$.

An important property of Γ is that it acts *properly discontinuously* on the complex half-plane $H=\{z\,|\,\mathrm{Im}\,z>0\}$. By that is meant that *if $a, b\in H$, then there exist neighbourhoods U of a and V of b in H such that the set $\{\gamma\in\Gamma\,|\,\gamma(U)\cap V\neq\emptyset\}$ is finite.*

This is a consequence of a more general proposition, namely: if $G=SL(2,\mathbb{R})$, and K, L are compact sets in H, then the set $E=\{g\in G\,|\,g(K)\cap L\neq\emptyset\}$ is compact in G. A short proof of this, sketched by Professor Raghavan Narasimhan, runs as follows.

Since the set E is obviously closed, it will be sufficient to prove that there exists a constant $A>0$, such that if $g=\begin{pmatrix}a&b\\c&d\end{pmatrix}\in E$, then $|a|+|b|+|c|+|d|\leqq A$.

Let $z=x+iy\in K$, and $w=\dfrac{az+b}{cz+d}=u+iv\in L$. Then there exists a constant $c_1>0$, such that

$$y+\frac{1}{y}+|x|<c_1\,,\qquad v+\frac{1}{v}+|u|<c_1\,.$$

Since $v=\dfrac{y}{|cz+d|^2}$, it follows that $\dfrac{1}{c_2}<|cz+d|^2<c_2$, with a suitable $c_2>0$, hence $\dfrac{1}{c_2}<(cx+d)^2+c^2y^2<c_2$. Since $y>\dfrac{1}{c_1}$, there exists a constant $c_3>0$, such that $c^2<c_3$, hence $|d|\leqq|cx+d|+|cx|\leqq c_4$, say. Since $g^{-1}z=\dfrac{dz-b}{-cz+a}$, and $g^{-1}w=z$, the above argument proves also that $|-c|$ and $|a|$ are bounded. Since $\dfrac{az+b}{cz+d}$ is bounded, and $|a|$, $|z|$, $\dfrac{1}{|cz+d|}$ are bounded, it follows that $\dfrac{|b|}{|cz+d|}$ is bounded, hence $|b|$ is bounded. Since a compact set which is discrete is finite, the assertion about Γ follows.

The following proposition is a Corollary: *if $a, b\in H$, and $\gamma(a)\neq b$ for all $\gamma\in\Gamma$, then there exist neighbourhoods U of a, and V of b, such that $\gamma(U)\cap V=\emptyset$ for all $\gamma\in\Gamma$.*

For let U_0, V_0 be neighbourhoods respectively of a and b, such that the set $S=\{\gamma\,|\,\gamma(U_0)\cdot\cap\cdot V_0\neq\emptyset\}$ is *finite*. Since $\gamma(a)\neq b$ for all $\gamma\in\Gamma$, there exist neighbourhoods U of a and V of b, with $U\subset U_0$, $V\subset V_0$, such that $\gamma(U)\cap V=\emptyset$, for all $\gamma\in S$. Then $\gamma(U)\cdot\cap\cdot V=\emptyset$, for all $\gamma\in\Gamma$.

For the notions of *discreteness*, *discontinuity*, and *proper discontinuity* of groups see, for instance, J. Lehner, *Discontinuous groups and automorphic functions*, Mathematical Surveys, No. VIII (American Math. Soc. 1964, Reprinted 1974). The three notions are equivalent in the case of Γ.

As general references for the theory of modular functions see F. Klein's *Vorlesungen über die Theorie der elliptischen Modulfunktionen* (Ausgearbeitet und vervollständigt von R. Fricke), (Leipzig, 1890), as well as G. Vivanti's *Fonctions polyédriques et modulaires* (French translation by A. Cahen, Paris, 1910), which seeks to provide easier access to Klein's lectures, and the book by Appell and Lacour, loc. cit., § 3 of the Notes.

To get an idea of the origins of the theory of modular functions, see also the original memoires of R. Dedekind, *J. für Math.* 83 (1877), 265–292; *Ges. Werke*, Bd. 1, 174–201, with the comments by R. Fricke about Gauss and Riemann having known and used the idea of covering the complex upper half-plane with a network of 'triangles'; F. Klein, *Math. Annalen*, 14 (1879), 111–126; 17 (1880), 62–70; K. Weierstrass, *Sitz. Preuss. Akad. Berlin*, (1883), 1271–1297; *Werke*, II, 281–309; and A. Hurwitz, *Math. Annalen*, 18 (1881), 528–592; 58 (1904), 343–360.

More recent references to modular functions, as well as modular *forms* (and automorphic functions) may be found in the following: S. Ramanujan, *Collected Papers* (Cambridge, 1927); G. H. Hardy, *Ramanujan* (Cambridge, 1940); E. Hecke, *Math. Werke* (Göttingen, 1959); C. L. Siegel, *Ges. Abhandlungen* (Springer-Verlag, 1966); R. C. Gunning, *Lectures on modular forms* (Princeton, 1962); J. Lehner, loc. cit.; H. Maass, *Lectures on modular functions of one complex variable* (Lectures on Mathematics, No. 29, Tata Institute of Fundamental Research, Bombay, 1964); Revised 1983; A. Ogg, *Modular forms and Dirichlet series* (New York, 1969); G. Shimura, *Introduction to the arithmetic theory of automorphic functions* (Tokyo-Princeton, 1971); J-P. Serre, *A course in arithmetic* (Springer-Verlag, 1973); R. A. Rankin, *Modular forms and functions* (Cambridge, 1977).

What we call elliptic functions are sometimes referred to, in the classical literature, as modular functions, e. g. K. Weierstrass, *Math. Werke* (Berlin, 1894), Bd. I, 1–49, and, in particular, the *Anmerkung*. This has led to the use, sometimes, of the double adjective 'elliptic modular', a practice which we shall try to avoid.

Chapter II

General properties of elliptic functions

§ 1. The period parallelogram. Given an elliptic function f, let (ω_1, ω_2) be a pair of *basic* periods for its period-lattice $\{m\omega_1 + n\omega_2\}$, where $m, n = 0, \pm 1, \pm 2, \ldots$. We may assume, without loss of generality, that $\operatorname{Im} \frac{\omega_2}{\omega_1} > 0$, by interchanging, if necessary, ω_1 with ω_2. Let P denote the parallelogram, in the complex z-plane, formed by the points $z = x\omega_1 + y\omega_2$, where $0 \leqq x < 1$, and $0 \leqq y < 1$. The vertices of P are $0, \omega_1, \omega_1 + \omega_2$, and ω_2, in that order (noting that $\operatorname{Im} \frac{\omega_2}{\omega_1} > 0$). In P are included the two sides extending from the origin to ω_1 and to ω_2, but not the other two sides, nor are the two end-points ω_1, ω_2 included. We call P a *fundamental period-parallelogram* associated with the given f (Fig. 4). It is *not* uniquely determined by the period-lattice, since by a proper, unimodular transformation one can pass from the given pair of basic periods (ω_1, ω_2) to another pair of basic periods (ω_1^*, ω_2^*), with $\operatorname{Im} \frac{\omega_2^*}{\omega_1^*} > 0$.

We take it that the passage from 0 to ω_1, $\omega_1 + \omega_2$, ω_2, and back to 0, in that order, defines the positive orientation of the curve ∂P which bounds P; passage in the opposite direction defines the negative orientation.

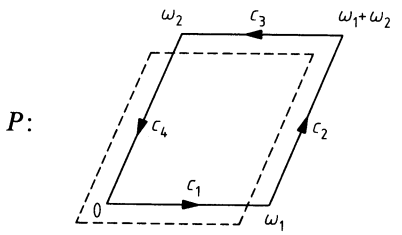


Fig. 4

If $\Omega = m\omega_1 + n\omega_2$, and m and n run through all the positive and negative integers as well as zero, Ω runs through all the periods of f, each of them occurring just once.

Let $P_{m,n}$ denote the parallelogram obtained by translating P through the vector Ω. The points of $P_{m,n}$ are given by $x\omega_1 + y\omega_2$, with $m \leqq x < m+1$, $n \leqq y < n+1$. Every point of the complex z-plane lies in exactly one such parallelogram, which we call a *period-parallelogram*, and $P_{0,0} = P$, which is a *fundamental period-parallelogram.*

Because of its periodicity, it suffices to study the given elliptic function in the period-parallelogram P defined by $z=x\omega_1+y\omega_2$, $0\leqq x<1$, $0\leqq y<1$. We denote by $\mathscr{L}$ the boundary of P, with the positive orientation, by c_1 the oriented segment extending from 0 to ω_1, by c_2, c_3, c_4 the other three oriented segments. Clearly c_2, c_3 do not belong to P, nor do the end-points of c_1, c_4 except the origin (Fig. 5).

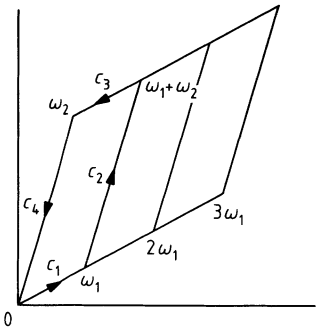


Fig. 5

§ 2. Elementary properties of elliptic functions. The class of elliptic functions has some general properties, which can be obtained by means of Cauchy's theorem of residues.

Theorem 1. *There exists no non-constant elliptic function which is entire.*

Proof. If $E(z)$ is an elliptic function, which is holomorphic on the *closed* parallelogram $\bar{P}$, then $|E(z)|<M<\infty$, for all $z\in P$, and therefore for all complex z, hence by Liouville's theorem, is a constant.

Corollary. *A non-constant elliptic function has at least one pole in any period-parallelogram.*

Theorem 2. *The sum of all the residues of an elliptic function at the poles inside a period-parallelogram is zero.*

Proof. Let $E(z)$ be an elliptic function, and $\mathscr{L}$ the boundary of a period-parallelogram of $E(z)$, positively oriented. If $E(z)$ has no pole on $\mathscr{L}$, then the sum of its residues in the period parallelogram equals

$$R=\frac{1}{2\pi i}\int_{\mathscr{L}} E(z)dz\,.$$

Because of the periodicity of $E(z)$, we have

$$\int_{c_1} E(z)dz+\int_{c_3} E(z)dz=0\,,\qquad \int_{c_4}+\int_{c_2}=0\,,$$

hence $R=0$.

If one or more poles lie on $\mathscr{L}$, then we move the boundary into $\mathscr{L}^*$, such that the sides of $\mathscr{L}^*$ are parallel to those of $\mathscr{L}$, but no poles lie on $\mathscr{L}^*$, and $\mathscr{L}^*$ encloses all the poles in P. This can be done, since poles are isolated points.

Corollary. *A non-constant elliptic function cannot have just one simple pole in a period-parallelogram. It must have therefore at least two simple poles, or at least one pole which is not simple, in any period-parallelogram.*

Theorem 3. *The number of zeros of a non-constant elliptic function in a period-parallelogram P is equal to the number of poles in P, the zeros and poles being counted according to their multiplicity.*

Proof. Let $E(z)$ be a non-constant elliptic function. Then its derivative $E'(z)$ is again an elliptic function, since it is meromorphic, and has as its periods (at least) those of $E(z)$.

The function $\dfrac{E'(z)}{E(z)}$ is likewise elliptic, and the sum of its residues in a period-parallelogram P is zero, by Theorem 2. That is to say,

$$\frac{1}{2\pi i}\int_{\mathscr{L}} \frac{E'(z)}{E(z)}\,dz = (\text{the number of zeros of } E(z) \text{ in } P) - (\text{the number of poles of } E(z) \text{ in } P) = 0,$$

assuming that none of the poles or zeros lie on $\mathscr{L}$. If not, we move the boundary as in the proof Theorem 2.

Definition. The *order* of an elliptic function is the number of its poles in a period-parallelogram, each pole being counted according to its multiplicity.

Theorem 4. *A non-constant elliptic function of order h assumes in a period-parallelogram every complex value exactly h times, taking into account the order of its multiplicity.*

Proof. Let $E(z)$ be a non-constant elliptic function of order h, and let c be a complex number. Set $F(z) = E(z) - c$. The function $F(z)$ has the same poles, and at those poles the same principal parts, as $E(z)$. Therefore $F(z)$ is a non-constant elliptic function of order h (which is independent of c). Its zeros are the roots of the equation $E(z) = c$, and with the same multiplicity in each case. Theorem 3 applied to the function F gives the required result.

Definition. If $a_1, a_2, \ldots, a_h$ denote the zeros of a non-constant elliptic function f in a period-parallelogram P, each of them repeated according to its multiplicity, *the sum of the zeros of f in P* is defined to be $a_1 + a_2 + \ldots + a_h$. If $b_1, b_2, \ldots, b_h$ denote the poles of f in P, *the sum of the poles of f in P* is similarly defined to be $b_1 + b_2 + \ldots + b_h$.

Theorem 5. *The sum of the zeros of a non-constant elliptic function in a period-parallelogram differs from the sum of its poles by a period.*

Proof. Let $E(z)$ be the given non-constant elliptic function, and let P be a period-parallelogram of $E(z)$. Let $g(z)$ denote a function which is holomorphic in $\bar{P}$, the *closure* of P. Define

$$G(z) = g(z) \cdot \frac{E'(z)}{E(z)}.$$

It is holomorphic in $\bar{P}$ except, possibly, for poles at those points where $E(z)$ has zeros or poles.

Let $a_1, a_2, \ldots, a_h$ be the zeros, and $b_1, b_2, \ldots, b_h$ the poles of $E(z)$ in P. If $a_1 = \ldots = a_k = a$ is a zero of order k, we have the Laurent expansion at a given by

$$\frac{E'(z)}{E(z)} = \frac{k}{z-a} + \ldots + \ldots = \sum_{n=1}^{k} \frac{1}{z-a_n} + \ldots,$$

and correspondingly also

$$\frac{E'(z)}{E(z)} = -\frac{l}{z-b} + \ldots = -\sum_{n=1}^{l} \frac{1}{z-b_n} + \ldots,$$

if $b_1 = \ldots = b_l = b$ is a pole of order l.

If we assume, to begin with, that no poles or zeros of $E(z)$ lie on the boundary $\mathscr{L}$ of P, then by the theorem of residues,

$$\frac{1}{2\pi i} \int_{\mathscr{L}} G(z)dz = \sum_{n=1}^{h} g(a_n) - \sum_{n=1}^{h} g(b_n).$$

If we *choose* $g(z) = z$, then the right-hand side gives the difference between the sum of the zeros of $E(z)$ and the sum of its poles. We wish to show that if

$$\frac{1}{2\pi i} \int_{\mathscr{L}} G(z)dz = \Omega,$$

then Ω is a period of $E(z)$. Because of the periodicity of $E(z)$, we obviously have (see the figure in § 1)

$$\begin{aligned} 2\pi i \cdot \Omega &= \int_{c_1} \{z-(z+\omega_2)\} \frac{E'(z)}{E(z)} dz \\ &\quad + \int_{c_4} \{z-(z+\omega_1)\} \frac{E'(z)}{E(z)} dz \\ &= \omega_1 \cdot \int_0^{\omega_2} \frac{E'(z)}{E(z)} dz - \omega_2 \int_0^{\omega_1} \frac{E'(z)}{E(z)} dz \\ &= \omega_1 [\log E(z)]_0^{\omega_2} - \omega_2 [\log E(z)]_0^{\omega_1}, \end{aligned}$$

where $[\log E(z)]_0^{\omega_1}$ denotes the variation of any fixed branch of the many-valued function $\log E(z)$ as z varies from 0 to ω_1. Since $E(0) = E(\omega_1)$, we have $\log E(0) = \log E(\omega_1) + 2\pi i n_1$, where n_1 is an integer. Similarly $\log E(0) = \log E(\omega_2) + 2\pi i n_2$, where n_2 is an integer. Hence $-\Omega = n_2\omega_1 - n_1\omega_2$, which is a period of $E(z)$.

Should zeros or poles of $E(z)$ lie on $\mathscr{L}$, we move the boundary of P as before, and then apply the above argument.

Remark. The conditions satisfied by the zeros and poles of an elliptic function, as in Theorem 3 or Theorem 5, will in fact also *suffice* for the existence

of an elliptic function with any given set of complex numbers satisfying those conditions as its zeros and poles.

We shall proceed next to *construct* a non-constant elliptic function. In seeking to do so, we note that because of Theorem 2 and its Corollary, the simplest class of elliptic functions, as far as singularities are concerned, are those of order 2. They may be of two types; those which have a double pole, with residue zero, as their only singularity in a period-parallelogram, known as the *Weierstrassian elliptic functions*, and those which have in every period-parallelogram precisely two simple poles, with residues which are equal in absolute value but opposite in sign, known as the *Jacobian elliptic functions*. We shall study the first type in Chapter III, and the second type in Chapter VII.

Notes on Chapter II

For the results of this Chapter, see J. Liouville (Lectures published by C. W. Borchardt), *J. für Math.* 88 (1880), 277–310. Liouville's method of proof is based on the use of Cauchy's theorem of residues, and differs from that of Jacobi's and Weierstrass's. Clearly this presentation is therefore not chronological.

Theorem 1 can be used to prove a number of results on elliptic functions as well as on theta-functions (cf. Chapter V).

That an entire, doubly-periodic function $f(z)$ is a constant, follows also from the Fourier expansion (cf. Notes on Chapter I, § 2)

$$f(z)=\sum_{n=-\infty}^{\infty} c_n e^{2\pi i n z/\omega}, \qquad \omega \neq 0,$$

where ω is a period of f.

Chapter III

Weierstrass's elliptic function $\wp(z)$

§ 1. The convergence of a double series. Let ω_1, ω_2 be two complex numbers, both of them different from zero, and let $\tau=\omega_2/\omega_1$, with $\operatorname{Im}\tau>0$. Let $\omega=m\omega_1+n\omega_2$, where $m, n=0, \pm 1, \pm 2, \ldots$. The points (ω) form a *lattice* $\mathfrak{L}$, say, in the complex plane. We shall consider the series

$$\sum_{\omega}{}' \frac{1}{|\omega|^{\varrho}} = \sum_{\omega\in\mathfrak{L},\,\omega\neq 0} \frac{1}{|\omega|^{\varrho}} = \sum_{\substack{m=-\infty \\ (m,n)\neq(0,0)}}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{|(m\omega_1+n\omega_2)|^{\varrho}},$$

for real values of ϱ.

Theorem 1. *The series* $\sum_{\omega}{}' |\omega|^{-\varrho}$ *converges for* $\varrho>2$, *and diverges for* $\varrho\leqq 2$.

Proof. Define the 'square' partial sums

$$S_k = \sum_{|m|\leqq k,\,|n|\leqq k}{}' |\omega|^{-\varrho}, \qquad k=1,2,3,\ldots,$$

and set $T_k=S_k-S_{k-1}$, with $S_0\equiv 0$. The series $\sum' |\omega|^{-\varrho}$ converges if and only if the series $\sum_{k=1}^{\infty} T_k$ converges, since the terms are all positive. The number of terms in T_k is $8k$, since $(2k+1)^2-1-\{(2k-2+1)^2-1\}=8k$. Each of them is either of the form

$$|k\omega_1+n\omega_2|^{-\varrho}, \quad |n|\leqq k, \quad \text{or} \quad |m\omega_1+k\omega_2|^{-\varrho}, \quad |m|\leqq k.$$

The corresponding points (ω) lie on the boundary of the parallelogram with vertices at $k\omega_1+k\omega_2$, $-k\omega_1+k\omega_2$, $-k\omega_1-k\omega_2$, and $k\omega_1-k\omega_2$. Hence there exist two numbers a and b, with $a>0$, $b>0$, and independent of k, such that if ω corresponds to a term in T_k, we have

$$a\cdot k<|\omega|<b\cdot k,$$

and hence

$$8\cdot b^{-\varrho}\cdot k^{1-\varrho}<T_k<8\cdot a^{-\varrho}\cdot k^{1-\varrho},$$

from which it follows that $\sum_{k=1}^{\infty} T_k$ converges if and only if $\sum_{k=1}^{\infty} k^{1-\varrho}$ converges, that is to say, if and only if $\varrho>2$.

Corollary 1. *For any* $R>0$, *and complex* z, *the series*

$$\sum_{\omega\in\mathfrak{L},\,|\omega|>2R} |z-\omega|^{-\varrho}, \qquad \varrho>2,$$

converges uniformly in the circle $|z| \leqq R$. *Hence the series*

$$\sum_{\omega} |z-\omega|^{-\varrho}, \quad \varrho > 2,$$

converges uniformly in every circle of finite radius, if we discard a sufficient number of terms at the beginning.

For, $|z| < \frac{1}{2}|\omega|$, so that $|z| + |\omega| \leqq \frac{3}{2}|\omega|$, and we have

$$\frac{1}{|z-\omega|} \leqq \frac{1}{|\omega| - |z|} \leqq \frac{2}{|\omega|},$$

$$\frac{1}{|z-\omega|} \geqq \frac{1}{|z| + |\omega|} \geqq \frac{2}{3} \cdot \frac{1}{|\omega|},$$

hence

$$\left(\frac{2}{3}\right)^{\varrho} \cdot \frac{1}{|\omega|^{\varrho}} \leqq \frac{1}{|z-\omega|^{\varrho}} \leqq \frac{2^{\varrho}}{|\omega|^{\varrho}}, \quad \text{for} \quad \varrho > 0.$$

The Corollary now follows from Theorem 1 and the second inequality; the first inequality shows that the assertion of the Corollary is not true for $\varrho \leqq 2$.

Corollary 2. *The series*

$$\sum_{\omega \in \mathfrak{L},\, \omega \neq 0} \left\{ \frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right\}$$

converges absolutely for $z \notin \mathfrak{L}$, *so that the sum is independent of the order of the terms. For every finite* $R > 0$, *the series converges uniformly in the circle* $|z| \leqq R$, *after the omission of a sufficient number of initial terms.*

Given $R > 0$, we have $|\omega| > 2R$ except for finitely many points ω; and if $|z| \leqq R$, we have

$$\left| \frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right| = \left| \frac{z(2\omega - z)}{\omega^2 (z-\omega)^2} \right| \leqq \frac{10 \cdot |z|}{|\omega|^3} \leqq \frac{10 \cdot R}{|\omega|^3},$$

except for finitely many points ω, since

$$\frac{|z|}{|\omega|} < \frac{1}{2}, \quad \text{and} \quad \left| \frac{z(2\omega - z)}{\omega^2 (z-\omega)^2} \right| = \left| \frac{\omega z (2 - z/\omega)}{\omega^4 (1 - z/\omega)^2} \right|,$$

where

$$\left| 2 - \frac{z}{\omega} \right| \leqq 2 + \frac{1}{2}, \quad \text{and} \quad \left| 1 - \frac{z}{\omega} \right|^2 \geqq \left(1 - \frac{1}{2} \right)^2.$$

Since the series $\sum_{\omega \in \mathfrak{L},\, \omega \neq 0} |\omega|^{-3}$ converges by Theorem 1, the Corollary follows.

§ 2. The elliptic function $\wp(z)$. After Corollary 2 of § 1, we *define*, for complex z,

$$\wp(z)=\frac{1}{z^2}+\sum_{\omega\in\mathfrak{L},\,\omega\neq 0}\left\{\frac{1}{(z-\omega)^2}-\frac{1}{\omega^2}\right\}. \tag{2.1}$$

To indicate the dependence on ω_1, ω_2 (which determine the lattice $\mathfrak{L}$), we sometimes write $\wp(z;\omega_1,\omega_2)$ instead of $\wp(z)$.

The function $\wp(z)$ so defined *is* an elliptic function, which we shall study in considerable detail. Its basic properties are given by the following

Theorem 2. *The function $\wp(z)$ is an elliptic function with the periods ω_1, ω_2. Its poles are given by $z=\omega$. It has the further properties:*

(i) *the principal part of $\wp(z)$ at $z=0$ is $1/z^2$;*

(ii) $\lim_{z\to 0}\left(\wp(z)-\frac{1}{z^2}\right)=0$;

(iii) $\wp(z)=\wp(-z)$;

(iv) $\wp'(-z)=-\wp'(z)$.

Proof. By Corollary 2, combined with Weierstrass's theorem on uniformly convergent series of holomorphic functions, it follows that $\wp(z)$ is a meromorphic function, with *double* poles at $z=\omega=m\omega_1+n\omega_2$; $m,n=0,\pm 1,\pm 2,\ldots$. By considering the difference $\wp(z)-\frac{1}{z^2}$ in a neighbourhood of $z=0$, we obtain (i) and (ii).

To prove (iii) we have only to note that

$$\wp(-z)=\frac{1}{z^2}+\sum_{\omega\in\mathfrak{L},\,\omega\neq 0}\left\{\frac{1}{(z+\omega)^2}-\frac{1}{\omega^2}\right\},$$

and that the set of points $\{-\omega\}$ is the same as the set $\{\omega\}$. Property (iv) follows from (iii).

The uniform convergence, in the sense of Corollary 2, of the (double) series defining $\wp(z)$ permits differentiation term by term, so that the derivative $\wp'(z)$ of $\wp(z)$ is, in fact, given by

$$\wp'(z)=-2\sum_{\omega\in\mathfrak{L}}\frac{1}{(z-\omega)^3},$$

the series converging absolutely for $z\notin\mathfrak{L}$. Since the set of points $\{\omega-\omega_1\}$ is the same as the set $\{\omega\}$, the series for $\wp'(z+\omega_1)$ is just a rearrangement of the series for $\wp'(z)$. Hence $\wp'(z+\omega_1)=\wp'(z)$, likewise $\wp'(z+\omega_2)=\wp'(z)$. Therefore $\wp'(z)$ is an *elliptic function*. By integration, we get the relation: $\wp(z+\omega_1)=\wp(z)+c$, where c is a constant. Setting $z=-\frac{\omega_1}{2}$, we get $\wp\left(\frac{\omega_1}{2}\right)=\wp\left(-\frac{\omega_1}{2}\right)+c$, and since $\wp$ is an even function, it follows that $c=0$, hence $\wp(z+\omega_1)=\wp(z)$, and similarly $\wp(z+\omega_2)=\wp(z)$, so that $\wp(z)$ is an *elliptic function*, with the two periods ω_1, ω_2, with $\operatorname{Im}\frac{\omega_2}{\omega_1}>0$.

Remarks

(i) The higher derivatives of $\wp(z)$ are also elliptic functions (a special case of a general property) given by the corresponding (differentiated) series.

(ii) We may assume that $\left|\frac{\omega_2}{\omega_1}\right| \geqq 1$. In the period-parallelogram defined by $z=\xi\omega_1+\eta\omega_2$, $0\leqq\xi<1$, $0\leqq\eta<1$, $\wp(z)$ has exactly one pole, of the second order, at $z=0$, with residue zero. The points of the lattice $\mathfrak{L}$, which we may call the *period-lattice*, (in as much as ω_1, ω_2 have been shown to be periods of $\wp(z)$), are all poles of $\wp(z)$; no other points are.

(iii) If λ is any complex number, $\lambda \neq 0$, then

$$\wp(\lambda z; \lambda\omega_1, \lambda\omega_2) = \lambda^{-2}\wp(z; \omega_1, \omega_2). \tag{2.2}$$

§ 3. The differential equation associated with $\wp(z)$

Theorem 3. *The elliptic function $\wp(z)$ satisfies the differential equation*

$$\wp'^2(z) = 4\wp^3(z) - g_2\wp(z) - g_3,$$

where

$$g_2 = 60 \sum_{\omega\in\mathfrak{L},\, \omega\neq 0} \omega^{-4}, \qquad g_3 = 140 \sum_{\omega\in\mathfrak{L},\, \omega\neq 0} \omega^{-6}.$$

Proof. By Corollary 2 of § 1, the series

$$\sum_{|\omega|>2R>0} \left\{\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\right\}$$

converges absolutely and uniformly in the circle $|z|\leqq R$. If $|\omega|>2R\geqq|z|$, we have

$$\left\{\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\right\} = \frac{2z}{\omega^3} + \frac{3z^2}{\omega^4} + \frac{4z^3}{\omega^5} + \frac{5z^4}{\omega^6} + \ldots.$$

Since $\wp(z)$ is an even function, its Laurent expansion at $z=0$ (i.e. for $0<|z|<\delta$) is given by

$$\wp(z) = \frac{1}{z^2} + b_1 z^2 + b_2 z^4 + \ldots + b_n z^{2n} + \ldots,$$

since $\wp(z) - \frac{1}{z^2} = 0$ for $z=0$ (Theorem 2). Here

$$b_1 = 3\sum_{\omega\neq 0}\frac{1}{\omega^4}, \qquad b_2 = 5\sum_{\omega\neq 0}\frac{1}{\omega^6}, \qquad b_n = (2n+1)\sum_{\omega\neq 0}\frac{1}{\omega^{2n+2}}.$$

It follows that

$$(\wp(z))^3 = z^{-6}(1+b_1z^4+b_2z^6+\ldots)^3 = \frac{1}{z^6} + \frac{3b_1}{z^2} + 3b_2 + \ldots,$$

and

$$(\wp'(z))^2 = z^{-6}(-2+2b_1z^4+4b_2z^6+\ldots)^2 = \frac{4}{z^6} - \frac{8b_1}{z^2} - 16b_2 + \ldots,$$

so that the difference $(\wp'(z))^2 - 4\wp^3(z)$ has a pole of the second order at $z=0$, with the expansion

$$(\wp'(z))^2 - 4\wp^3(z) = -\frac{20b_1}{z^2} - 28b_2 + z^2(\ldots).$$

On setting $g_2 = 20b_1$, $g_3 = 28b_2$, we have

$$(\wp'(z))^2 - 4\wp^3(z) = -\frac{g_2}{z^2} - g_3 + z^2(\ldots).$$

It follows that the function

$$(\wp'(z))^2 - 4\wp^3(z) + g_2\wp(z) + g_3,$$

which is obviously *elliptic*, is holomorphic at $z=0$, hence also at all points of the period-lattice. It has therefore no poles at all, so that it reduces to a constant (Theorem 1, Chapter II), which is zero, since the function vanishes for $z=0$, and the theorem follows.

Theorem 4. *The Laurent expansion of $\wp(z)$ at $z=0$ is given by*

$$\wp(z) = \frac{1}{z^2} + \sum_{n=1}^{\infty} b_n z^{2n},$$

where

$$b_1 = \frac{g_2}{20}, \quad b_2 = \frac{g_3}{28}, \quad b_n = \frac{3}{(2n+3)(n-2)} \cdot \sum_{k=1}^{n-2} b_k b_{n-k-1},$$

for $n=3, 4, \ldots$, and g_2, g_3 are defined as in Theorem 3.

Proof. In the course of proof of Theorem 3, we have seen that

$$\wp(z) = \frac{1}{z^2} + b_1z^2 + b_2z^4 + \ldots + b_nz^{2n} + \ldots.$$

By further differentiation of the equation $(\wp'(z))^2 = 4\wp^3(z) - g_2\wp(z) - g_3$, we obtain: $2\wp'(z)\wp''(z) = 12(\wp(z))^2\wp'(z) - g_2\wp'(z)$, so that $\wp''(z) = 6(\wp(z))^2 - \frac{1}{2}g_2$, since $\wp'(z)$ is not identically zero. We thus have the identity

$$\frac{6}{z^4} + \sum_{n=1}^{\infty} 2n(2n-1)b_nz^{2n-2} = \frac{6}{z^4}\left(1 + \sum_{n=1}^{\infty} b_nz^{2n+2}\right)^2 - \frac{g_2}{2}.$$

By comparing the coefficients of z^{2n-2} on either side, for $n>2$, we obtain

$$2n(2n-1)b_n = 6(2b_n + b_1b_{n-2} + b_2b_{n-3} + \ldots + b_{n-2}b_1).$$

Since $2n(2n-1)-12=2(2n+3)\,(n-2)\neq 0$, for $n>2$, we have the recurrence formula

$$b_n=\frac{6}{2(2n+3)(n-2)}\cdot\sum_{k=1}^{n-2} b_k b_{n-k-1}, \qquad n=3,4,\ldots,$$

and, in particular,

$$b_3=\frac{1}{3}b_1^2=\frac{1}{3}\left(\frac{g_2}{20}\right)^2=\frac{g_2^2}{1200}, \qquad b_4=\frac{3g_2g_3}{2^4\cdot 5\cdot 7\cdot 11}.$$

Remarks

(i) It follows from the above argument that all the coefficients b_n, $n=1,2,3,\ldots$, are polynomials in g_2, g_3, with positive, rational numbers as coefficients.

(ii) The circle of convergence of the Laurent expansion of $\wp(z)$ at $z=0$ passes through the lattice-point $z=\omega_1$, and its radius is, in fact, equal to $|\omega_1|$.

(iii) The coefficients b_n can be expressed in terms of the periods ω. By differentiating the Laurent expansion, we get

$$\wp'(z)+\frac{2}{z^3}=\sum_{n=1}^{\infty} 2nb_n z^{2n-1}.$$

On the other hand, we have, by the definition of $\wp(z)$,

$$\wp'(z)+\frac{2}{z^3}=-\sum_{\omega\neq 0}\frac{2}{(z-\omega)^3},$$

which we can differentiate, term by term, $(2n-1)$ times, and obtain, for $z=0$,

$$(2n)!\,b_n=(2n+1)!\sum_{\omega\neq 0}\frac{1}{\omega^{2n+2}}, \qquad (n=1,2,\ldots),$$

so that

$$b_{n-1}=(2n-1)\sigma_n, \qquad n=2,3,\ldots,$$

where

$$\sigma_n=\sum_{\omega\neq 0}\frac{1}{\omega^{2n}}, \qquad n=2,3,\ldots.$$

It follows that

$$\sum_{\omega\neq 0}\frac{1}{\omega^4}=\sigma_2=\frac{b_1}{3}=\frac{g_2}{60};$$

$$\sum_{\omega\neq 0}\frac{1}{\omega^6}=\sigma_3=\frac{b_2}{5}=\frac{g_3}{140};$$

$$\sum_{\omega\neq 0}\frac{1}{\omega^{2n}}=\sigma_n=\frac{b_{n-1}}{2n-1}=P_n(\sigma_2,\sigma_3), \qquad n=2,3,\ldots,$$

where $P_n(\sigma_2, \sigma_3)$ is a polynomial in σ_2 and σ_3, with positive, rational numbers as coefficients.

Given a pair of basic periods (ω_1, ω_2) of $\wp(z)$, g_2 and g_3 are uniquely defined by the formulae

$$g_2 = 60 \sum_{m,n}{}' \frac{1}{(m\omega_1 + n\omega_2)^4}, \qquad g_3 = 140 \sum_{m,n}{}' \frac{1}{(m\omega_1 + n\omega_2)^6},$$

where m and n run through all the values $0, \pm 1, \pm 2, \ldots$, except $m = n = 0$. If

$$\omega_1^* = m\omega_1 + n\omega_2, \qquad \omega_2^* = p\omega_1 + q\omega_2,$$

where m, n, p, q are integers such that $mq - np = +1$, then (ω_1^*, ω_2^*) is *also* a pair of basic periods (see Remark (i), after Theorem 2, Chapter I). Hence $g_2(\omega_1, \omega_2) = g_2(\omega_1^*, \omega_2^*)$, and $g_3(\omega_1, \omega_2) = g_3(\omega_1^*, \omega_2^*)$. For this reason, g_2 and g_3 are commonly referred to as the *invariants* of $\wp(z)$.

We note incidentally that

$$\sum_{\omega \neq 0} \frac{1}{\omega^{2n+1}} = 0, \quad \text{for} \quad n = 1, 2, 3, \ldots,$$

since the terms corresponding to periods ω of opposite signs cancel each other.

If (ω_1, ω_2) is a pair of basic periods of the $\wp$-function, we say that z_1 *is congruent to* z_2 *modulo* (ω_1, ω_2), where z_1, z_2 are complex numbers, if $z_1 - z_2 = m\omega_1 + n\omega_2$, for some integers m and n. We indicate this by writing $z_1 \equiv z_2 \pmod{\omega_1, \omega_2}$.

Theorem 5. *If g_2 and g_3 are the invariants of the function $\wp(z)$, defined by the formulae*

$$g_2 = 60 \sum_{\omega \neq 0} \omega^{-4}, \qquad g_3 = 140 \sum_{\omega \neq 0} \omega^{-6},$$

where $\omega = m\omega_1 + n\omega_2$, (ω_1, ω_2) being a pair of basic periods of the $\wp$-function, and $m, n = 0, \pm 1, \pm 2, \ldots$, then we have

$$g_2^3 - 27g_3^2 \neq 0.$$

Proof. Since the elliptic function $\wp'(z)$ is of order 3, it has only three zeros in the period-parallelogram P consisting of the points z, where $z = x\omega_1 + y\omega_2$, $0 \leqq x < 1$, $0 \leqq y < 1$. Since $\wp'(z)$ is an *odd* function of z, we have

$$\wp'\left(\frac{\omega_1}{2}\right) = -\wp'\left(-\frac{\omega_1}{2}\right) = -\wp'\left(-\frac{\omega_1}{2} + \omega_1\right) = -\wp\left(\frac{\omega_1}{2}\right),$$

hence $\wp'\left(\frac{\omega_1}{2}\right) = 0$. Similarly $\wp'\left(\frac{\omega_2}{2}\right) = 0$, and since

$$\wp'\left(\frac{\omega_1 + \omega_2}{2}\right) = \wp'\left(\frac{-\omega_1 - \omega_2}{2}\right) = -\wp'\left(\frac{\omega_1 + \omega_2}{2}\right),$$

we have also $\wp'\left(\frac{\omega_1+\omega_2}{2}\right)=0$. Thus $\frac{\omega_1}{2}$, $\frac{\omega_2}{2}$, and $\frac{\omega_1+\omega_2}{2}$ are *the* three zeros of $\wp'(z)$ in P. Since $\{\wp'(z)\}^2=4\wp^3(z)-g_2\wp(z)-g_3$, it follows that $\frac{\omega_1}{2}$, $\frac{\omega_2}{2}$, and $\frac{\omega_1+\omega_2}{2}$ are *the* three roots of the cubic equation

$$4\wp^3(z)-g_2\wp(z)-g_3=0$$

in P. Set

$$e_r=\wp\left(\frac{\omega_r}{2}\right)=\wp\left(\frac{\omega_r}{2};\omega_1,\omega_2\right),\qquad r=1,2,3;\qquad \omega_3=\omega_1+\omega_2. \tag{3.1}$$

Then e_1, e_2, e_3 are *all different*. For consider the function $\wp(z)-e_1$. It has a zero at $z=\frac{\omega_1}{2}$. But $\wp'\left(\frac{\omega_1}{2}\right)=0$, as we have just seen. Therefore it is a *double* zero. Since $\wp(z)$ has only a *double* pole in P, it follows that the zeros of $\wp(z)-e_1$ in the z-plane are *all* congruent to $\frac{\omega_1}{2}$ modulo (ω_1,ω_2), that is to say, differ from $\frac{\omega_1}{2}$ only by a period.

Similarly the (double) zeros of $\wp(z)-e_2$, and of $\wp(z)-e_3$, are *all* congruent respectively to $\frac{\omega_2}{2}$, and to $\frac{\omega_3}{2}$, modulo (ω_1,ω_2).

It follows that $e_1\neq e_2\neq e_3\neq e_1$. For if $e_1=e_2$, then $\wp(z)-e_1$ is zero for $z=\frac{\omega_2}{2}$, but $\frac{\omega_2}{2}$ is *not* congruent to $\frac{\omega_1}{2}$ modulo (ω_1,ω_2). Similarly $e_2\neq e_3$, and $e_3\neq e_1$.

Since e_1, e_2, e_3 are the roots of the cubic equation

$$4t^3-g_2t-g_3=0,$$

we have

$$e_1+e_2+e_3=0;\qquad e_1\cdot e_2\cdot e_3=\frac{g_3}{4};$$

$$e_2e_3+e_3e_1+e_1e_2=-\frac{g_2}{4};$$

and since $e_1\neq e_2\neq e_3\neq e_1$, we have $g_2^3-27g_3^2\neq 0$. (See Appendix I at the end of this chapter.)

Remarks

(i) The *discriminant of the cubic*

$$u\equiv ax^3+3bx^2+3cx+d,$$

where $a\neq 0$, and a, b, c, d are complex numbers, is *defined* to be

$$D=a^2d^2-6abcd+4ac^3+4b^3d-3b^2c^2.$$

If α, β, γ are the three roots of the equation $u=0$, then

$$a^4(\beta-\gamma)^2(\gamma-\alpha)^2(\alpha-\beta)^2=-27D.$$

If follows that a necessary and sufficient condition that the equation $u=0$ have a repeated root is that $D=0$.

(ii) If $a=4$, $b=0$, $3c=-g_2$, $d=-g_3$, then we have $D=16\left(g_3^2-\frac{g_2^3}{27}\right)$. In that case, e_1, e_2, and e_3, are the three roots. Hence

$$4^4(e_1-e_2)^2(e_2-e_3)^2(e_3-e_1)^2=27\cdot\tfrac{16}{27}(g_2^3-27g_3^2),$$

or

$$16(e_1-e_2)^2(e_2-e_3)^2(e_3-e_1)^2=(g_2^3-27g_3^2).$$

(iii) We define

$$\Delta\equiv\Delta(\omega_1,\omega_2)=g_2^3-27g_3^2.$$

The three functions $g_2(\omega_1,\omega_2)$, $g_3(\omega_1,\omega_2)$, and $\Delta(\omega_1,\omega_2)$ are known as the *elementary modular forms*. (cf. Notes on Chapter VI).

(iv) If we set $\tau=\dfrac{\omega_2}{\omega_1}$, we obtain

$$g_2(\omega_1,\omega_2)=\omega_1^{-4}g_2(1,\tau);\qquad g_3(\omega_1,\omega_2)=\omega_1^{-6}g_3(1,\tau);$$

hence $\Delta(\omega_1,\omega_2)=\omega_1^{-12}\Delta(1,\tau)$. (Note that $\operatorname{Im}\tau>0$).

§ 4. The addition-theorem. It is possible to express $\wp(u+v)$ in terms of $\wp(u)$, and $\wp(v)$, and their derivatives, as is shown by the following (which is known as the addition-theorem)

Theorem 6. *If* $z_1\not\equiv z_2$ *modulo* (ω_1,ω_2), *then we have*

$$\wp(z_1+z_2)=\frac{1}{4}\left(\frac{\wp'(z_1)-\wp'(z_2)}{\wp(z_1)-\wp(z_2)}\right)^2-\wp(z_1)-\wp(z_2). \tag{4.1}$$

Proof. Let z_2 be such that $\wp'(z_2)\neq 0$, *and* $\wp(z_2)$ is finite, while $z_1=z$ is variable. Set

$$\varphi(z)=\wp(z+z_2)+\wp(z)+\wp(z_2)-\frac{1}{4}\left(\frac{\wp'(z)-\wp'(z_2)}{\wp(z)-\wp(z_2)}\right)^2.$$

Then $\varphi(z)$ is an elliptic function. The only possible poles of $\varphi(z)$ are at those values of z which differ from $z=0$, or from $z=-z_2$, by a period of $\wp(z)$. (Note that, by assumption, $z\neq z_2$). Near $z=0$, we have the expansions

$$\wp(z)=\frac{1}{z^2}+b_1z^2+\dots,$$

$$\wp'(z)=-\frac{2}{z^3}+2b_1z+\dots,$$

and

$$\frac{1}{4}\left(\frac{\wp'(z)-\wp'(z_2)}{\wp(z)-\wp(z_2)}\right)^2-\wp(z)=\frac{1}{z^2}\left(\frac{1+\frac{1}{2}\wp'(z_2)z^3-b_1z^4+\dots}{1-\wp(z_2)z^2+b_1z^4+\dots}\right)^2-\frac{1}{z^2}-b_1z^2-\dots,$$

$$=2\wp(z_2)+\dots,$$

so that $\varphi(z)$ is holomorphic at $z=0$, with $\varphi(0)=0$.

Near $z=-z_2$, we have the expansions

$$\wp(z+z_2)=\frac{1}{(z+z_2)^2}+\dots,$$

and

$$\frac{1}{4}\left(\frac{\wp'(z)-\wp'(z_2)}{\wp(z)-\wp(z_2)}\right)^2=\left(\frac{\wp'(z_2)-\frac{1}{2}\wp''(z_2)(z+z_2)+\dots}{\wp'(z_2)(z+z_2)-\frac{1}{2}\wp''(z_2)(z+z_2)^2+\dots}\right)^2$$

$$=\frac{1}{(z+z_2)^2}-\dots,$$

so that $\varphi(z)$ is also holomorphic at $z=-z_2$.

As an elliptic function which is entire, $\varphi(z)$ reduces to a constant, which is zero, since $\varphi(0)=0$. This has been proved on the assumption that $z_2\neq 0, \frac{\omega_j}{2}$, $j=1,2,3$. But this assumption can be dropped, since we can interchange z_1 and z_2 in the assertion of the theorem.

Corollary. *If* $z_1\not\equiv z_2$ *modulo* (ω_1,ω_2), *we have the relation*

$$\wp(z_1+z_2)+\wp(z_1-z_2)=\frac{\{2\wp(z_1)\wp(z_2)-\frac{1}{2}g_2\}\{\wp(z_1)+\wp(z_2)\}-g_3}{\{\wp(z_1)-\wp(z_2)\}^2},$$

since, by Theorem 6, we have

$$\wp(z_1+z_2)+\wp(z_1-z_2)=\frac{1}{4}\left(\frac{\wp'(z_1)-\wp'(z_2)}{\wp(z_1)-\wp(z_2)}\right)^2-\wp(z_1)-\wp(z_2)$$

$$+\frac{1}{4}\left(\frac{\wp'(z_1)+\wp'(z_2)}{\wp(z_1)-\wp(z_2)}\right)^2-\wp(z_1)-\wp(z_2)$$

$$=\frac{1}{2}\left(\frac{\{\wp'(z_1)\}^2+\{\wp'(z_2)\}^2}{\{\wp(z_1)-\wp(z_2)\}^2}\right)-2\{\wp(z_1)+\wp(z_2)\},$$

while

$$\{\wp'(z_1)\}^2=4\wp^3(z_1)-g_2\wp(z_1)-g_3,\quad \textit{and}\quad \{\wp'(z_2)\}^2=4\wp^3(z_2)-g_2\wp(z_2)-g_3.$$

§ 5. The generation of elliptic functions

Theorem 7. *Let P denote a period-parallelogram associated with the function $\wp(z)$. Let a_k, b_l, where $k, l = 1, 2, \ldots, h$, be $2h$ complex numbers in P, such that $a_k \neq b_l$. If the sum $a_1 + \ldots + a_h$ differs from the sum $b_1 + b_2 + \ldots + b_h$ only by a period of $\wp(z)$, then there exists an elliptic function, which has in P just the a_k's as zeros, and the b_l's as poles, and any two such functions differ only by a non-zero, constant factor.*

Proof (Siegel). Without loss of generality, we may assume P to be the fundamental period-parallelogram in the z-plane given by $P = [z | z = x\omega_1 + y\omega_2,\ 0 \leqq x < 1,\ 0 \leqq y < 1]$, where (ω_1, ω_2) is a pair of primitive periods of $\wp(z)$.

If $E_1(z)$ and $E_2(z)$ are any two elliptic functions which have the prescribed zeros and poles in P, then the quotient $E_1(z)/E_2(z)$ is an elliptic function of order zero, without any zeros, hence equals a constant different from zero. To prove the theorem it suffices therefore to show, under the given assumptions, just the *existence* of a function $E(z)$ of the type required.

If $h = 0$, we define $E(z) = 1$. The case $h = 1$ does *not* occur, since the difference of two points in P cannot be a period. We may therefore assume that $h \geqq 2$.

Case (i). Let $b_1 = b_2 = \ldots = b_h = 0$. By hypothesis, we have

$$a_1 + a_2 + \ldots + a_h = \omega = \text{a period}. \tag{5.1}$$

We define, for $r = 1, 2, \ldots$, the elliptic functions

$$p_{2r} = (\wp(z))^r,\ p_1 = 1,\ p_{2r+1} = \wp'(z) \cdot (\wp(z))^{r-1}, \quad \text{for} \quad r \geqq 1.$$

The functions p_n, for $n = 2, 3, \ldots$, then have in P precisely one pole, namely a pole of order n at $z = 0$. On the other hand, p_1 has no pole in P.

We wish to find h constants $c_1, c_2, \ldots, c_h$, such that the elliptic function $E(z) = c_1 p_1 + c_2 p_2 + \ldots + c_h p_h$ has in P precisely the zeros $a_1, a_2, \ldots, a_h$. Since $E(z)$ is holomorphic in P except for $z = 0$, it must have (by Theorem 3, Chapter II) a pole of order h at $z = 0$, and therefore $c_h \neq 0$.

Among the h numbers $a_1, a_2, \ldots, a_h$, let $\alpha_1, \alpha_2, \ldots, \alpha_q$ be those which are all different from one another, and let n_l denote the number of times that α_l $(l = 1, \ldots, q)$ occurs, so that

$$n_1 + n_2 + \ldots + n_q = h.$$

We have now to choose $c_1, c_2, \ldots, c_h$, in such a way that all the h equations

$$E^{(k)}(\alpha_l) = 0,\ l = 1, 2, \ldots, q;\ k = 0, 1, \ldots, n_l - 1;$$

are fulfilled. If we substitute

$$E^{(k)}(z) = c_1 p_1^{(k)} + \ldots + c_h p_h^{(k)},$$

we obtain h homogeneous, linear equations for the h unknowns $c_1, c_2, \ldots, c_h$. If the determinant of this system of equations is different from zero, then there will

only be the trivial solution: $c_1 = c_2 = \ldots = c_h = 0$, which we cannot use. To get round this, we begin by omitting the last equation, namely

$$E^{(n_q - 1)}(\alpha_q) = 0, \tag{5.2}$$

which corresponds to the zero a_h. The system consisting of the remaining $h-1$ equations in h unknowns has a non-trivial solution, say $c_1, c_2, \ldots, c_h$. We shall show that the last equation is automatically satisfied with those c's.

The function $E(z)$ so constructed is not identically zero, since all the functions $p_2, p_3, \ldots, p_h$ have poles of different orders at $z = 0$, while p_1 is finite at that point. It has, in any case, the $h-1$ zeros $a_1, a_2, \ldots, a_{h-1}$. Since the only possible pole of $E(z)$ in P is at $z = 0$, at most of order h, it follows (by Theorem 3 of Chapter II) that the order j of the elliptic function $E(z)$ is either $h-1$ or h. The sum of its poles is in any case equal to zero. Suppose that $j = h-1$. Then $a_1, \ldots, a_{h-1}$ would be all the zeros of $E(z)$ in P, hence (by Theorem 5 of Chapter II) the sum of its zeros

$$a_1 + a_2 + \ldots + a_{h-1} = \omega^*, \text{ say},$$

will be a period. By (5.1), $\omega - \omega^* = a_h$ will also be a period, where a_h is a point of P, which is different from the origin. Since this is not possible, we have $j = h$, and the function $E(z)$ has besides the zeros $a_1, \ldots, a_{h-1}$, precisely one more zero in P, say at $z = a$. By Theorem 5 of Chapter II, the sum of the zeros $a_1 + a_2 + \ldots + a_{h-1} + a$ is again a period, hence $a - a_h$ is a period; but $a \in P$, $a_h \in P$, therefore $a - a_h = 0$. It follows that the equation (5.2), initially omitted from the system, is also fulfilled, and the case in which $b_1 = b_2 = \ldots = b_h = 0$ is settled.

The case in which $a_1 = a_2 = \ldots = a_h = 0$ can be similarly handled by interchanging the a_k's with the b_l's, and then replacing the elliptic function so constructed by its reciprocal.

Case (ii). We may assume that neither all the a_k's, nor all the b_k's, have the value zero.

We choose $a_0 \in P$, such that the sum $a_0 + a_1 + \ldots + a_h$ is a period. If we set $b_0 = a_0$, then, by assumption, the sum $b_0 + b_1 + \ldots + b_h$ is also a period.

Let precisely m of the $h+1$ numbers $a_0, a_1, \ldots, a_h$ be equal to zero, and precisely n of the numbers $b_0, b_1, \ldots, b_h$ be equal to zero, where $m < h+1$, and $n < h+1$. If we omit from the set of numbers a_k, $k = 0, \ldots, h$, all those which are equal to zero, the sum of the remaining $h+1-m$ numbers is a period, and as in Case (i), we can construct an elliptic function

$$E_1(z) = c_1 p_1 + \ldots + c_\mu p_\mu, \tag{5.3}$$

with constants $c_1, c_2, \ldots, c_\mu, c_\mu \neq 0$, $\mu = h+1-m > 1$, which has in P precisely the zeros $a_k \neq 0$, and a pole of order μ at the origin.

We can similarly construct an elliptic function

$$E_2(z) = d_1 p_1 + \ldots + d_\nu p_\nu, \tag{5.4}$$

with constants $d_1, d_2, \ldots, d_\nu, d_\nu \neq 0$, $\nu = h+1-n > 1$, which has in P precisely the zeros $b_l \neq 0$, and a pole of order ν at the origin.

Let $E(z)=E_1(z)/E_2(z)$. If $a_0\neq 0$, then $a_0(=b_0)$ will be a common zero of $E_1(z)$ and $E_2(z)$. Those a_k's with $a_k\neq 0$, and $k>0$, will, in any case, be zeros of $E_1(z)$. Likewise those b_l's with $b_l\neq 0$, and $l>0$, will be zeros of $E_2(z)$. And so far as the origin is concerned, since $\mu-\nu=n-m$, $E(z)$ will have a zero of order $m-n$ at the origin if $m>n$; a pole of order $n-m$ if $m<n$; and a value different from 0 and ∞ if $m=n$. The function $E(z)$ satisfies the conditions of the theorem.

Theorem 8. *Any elliptic function $E(z)$ can be written uniquely in the form*

$$E(z)=S(x)+yT(x),\ x=\wp(z),\ y=\wp'(z),$$

where $S(x)$ and $T(x)$ are rational functions of x with constant coefficients. Conversely every such expression $S(\wp)+\wp' T(\wp)$ represents an elliptic function.

Proof (Siegel). Let h be the *order* of $E(z)$. If $h=0$, then $E(z)$ is a constant, say c, in which case we define $S(x)=c$, $T(x)=0$.

If $h>0$, let $a_1,\ldots,a_h$ be the zeros, and $b_1,b_2,\ldots,b_h$ the poles of $E(z)$ in P (a fundamental period-parallelogram as in the proof of Theorem 7).

By Theorem 5 of Chapter II, the difference between the sum $a_1+a_2+\ldots+a_h$ and the sum $b_1+b_2+\ldots+b_h$ is a period. As in Theorem 7, we can construct two elliptic functions $E_1(z)$ and $E_2(z)$ of the form given in (5.3) and (5.4), such that their quotient $E_1(z)/E_2(z)$ differs from $E(z)$ only by a constant factor (different from zero). By including the constant in the definition of $E_1(z)$, we have $E(z)=\dfrac{E_1(z)}{E_2(z)}$. On using the definitions of $p_1,p_2,\ldots$, as in the proof of Theorem 7, and setting $\wp(z)=x$, $\wp'(z)=y$, we see that

$$E_1(z)=A(x)+yB(x),\ E_2(z)=C(x)+yD(x),$$

where $A(x)$, $B(x)$, $C(x)$, $D(x)$ are polynomials in x with constant coefficients. Since $E_2(z)$ is not identically zero, it cannot be that both $C(x)$ and $D(x)$ vanish identically in x. Since $\wp(z)$ is an even function, while $\wp'(z)$ is odd, it cannot happen that $E_0(z)=C(x)-yD(x)$ vanishes identically in x. The differential equation of $\wp(z)$ is given by $y^2=4x^3-g_2x-g_3$, while

$$E_0(z)E_1(z)=A_1(x)+yB_1(x),\ E_0(z)E_2(z)=C_1(x),$$

where $A_1(x)$, $B_1(x)$, $C_1(x)$ are polynomials in x, of which $C_1(x)$ is not identically zero.

On setting

$$S(x)=\frac{A_1(x)}{C_1(x)},\ T(x)=\frac{B_1(x)}{C_1(x)},$$

we obtain the required formula for $E(z)=\dfrac{E_1(z)}{E_2(z)}$. If there is another representation of $E(z)$ given by

$$E(z)=S^*(x)+yT^*(x),\ x=\wp(z),\ y=\wp'(z),$$

with rational functions S^* and T^*, we define

$$S_0(x)=S(x)-S^*(x),\ T_0(x)=T(x)-T^*(x),$$

and obtain: $S_0(x)+yT_0(x)=0$. Again, since $x=\wp(z)$ is an even function of z, $S_0(x)$ and $T_0(x)$ are even; while $y=\wp'(z)$ is an odd function of z; hence $S_0(x)=0$, $T_0(x)=0$, identically in x, thereby showing the uniqueness of the representation given in the theorem.

Since all elliptic functions with a given pair of basic periods (ω_1, ω_2) form a field, any expression of the form $S(x)+yT(x)$, where $x=\wp(z)$, $y=\wp'(z)$, and $S(x)$, $T(x)$ are rational functions of x, represents an elliptic function.

Remarks. The class $\mathbf{E}(\omega_1, \omega_2)$ of all elliptic functions with a given pair of basic periods (ω_1, ω_2) is a *field*, which contains the field of complex numbers. Theorem 8 shows that it is *generated* by the two elements $\wp$ and $\wp'$ of $\mathbf{E}(\omega_1, \omega_2)$.

Theorem 9. *Any two elliptic functions belonging to* $\mathbf{E}(\omega_1, \omega_2)$ *satisfy a non-trivial, algebraic relation.*

Proof. Let $f(u)$, $f_1(u)$ be two elliptic functions with the pair of basic periods (ω_1, ω_2). Let $a_1, a_2, \ldots, a_m$ be all the distinct points which are poles of at least one of the functions f, f_1, and which lie in a period-parallelogram P. If m_k denotes the order of the pole of f at a_k, and n_k the order of the pole of f_1 at a_k, let $\mu_k=\max(m_k, n_k)$, and $N=\mu_1+\mu_2+\ldots+\mu_m$.

Let $F(x, y)$ be a polynomial of total degree n in the two variables x and y, with constant coefficients, but *without* the constant term, say

$$F(x, y)=\sum_{\substack{k,l=0\\ 0<k+l\leqq n}}^{n} c_{kl}x^k y^l .$$

On replacing x by $f(u)$, and y by $f_1(u)$, in this polynomial, we obtain an elliptic function $\Phi(u)$, say, which cannot have in P any poles other than $a_1, a_2, \ldots, a_m$, and $\Phi(u)$ reduces to a constant, if and only if its principal parts near $a_1, a_2, \ldots, a_m$ *all* vanish, which will involve at most $n(\mu_1+\mu_2+\ldots+\mu_m)=nN$ linear, homogeneous, equations between the coefficients of the polynomial $F(x, y)$, since a_k is a pole of $\Phi(u)$ of order *at most* equal to $n\mu_k$. There are $\frac{n(n+3)}{2}$ such coefficients, and if n is chosen so large that $\frac{n(n+3)}{2}>Nn$, or $n+3>2N$, we shall have a system of linear, homogeneous equations, in which the number of unknowns is greater than the number of equations, so that a non-trivial solution exists. With the coefficients of $F(x, y)$ thus determined, we have $F(f(u), f_1(u))=c$, where c is a constant, which is the algebraic relation mentioned in the theorem; it is 'non-trivial', since the coefficients of F are not all zero.

Remarks

(i) The polynomial $F(x, y)$ in the proof of the theorem has no constant term, while the elliptic functions $f, f_1 \in \mathbf{E}(\omega_1, \omega_2)$ satisfy the relation: $F(f(u),$

$f_1(u))=c$. One can choose from among all such polynomials the one of lowest degree, and by adding the constant $-c$ to it, one can obtain an F such that $F(f(u), f_1(u))=0$.

(ii) It follows, in particular, that any non-constant elliptic function, and its derivative, satisfy an algebraic relation. For example, if $f(u)=\wp(u)$, which is of order 2, and $f_1(u)=\wp'(u)$, which is of order 3, the relation is $f_1^2=4f^2-g_2f-g_3$.

(iii) As further examples we have: $\wp''(u)=6\wp^2(u)-\frac{1}{2}g_2$, $2(\wp''(u))^3-3g_2(\wp''(u))^2+g_2^3=27(\wp'^2(u)+g_3)^2$.

Appendix I. The cubic equation

Let

$$u\equiv ax^3+3bx^2+3cx+d=0 \tag{1}$$

be a cubic equation, where a, b, c, d are complex numbers, with $a\neq 0$. The *discriminant* D of the cubic u is defined by

$$D=a^2d^2-6abcd+4ac^3+4b^3d-3b^2c^2\,. \tag{2}$$

A necessary and sufficient condition that the equation $u=0$ have a repeated root is that $D=0$.

Proof. (i) If α, β, γ are the roots of the equation

$$x^3+qx+r=0\,, \tag{3}$$

where q and r are complex numbers, with $q\neq 0$, $r\neq 0$, then $(\beta-\gamma)^2$, $(\gamma-\alpha)^2$, $(\alpha-\beta)^2$ are the roots of the equation

$$z^3+6qz^2+9q^2z+4q^3+27r^2=0\,. \tag{4}$$

For we have from (3): $\alpha+\beta+\gamma=0$, $\alpha\beta\gamma=-r$; and if we set $z=(\beta-\gamma)^2$, then $z=(\beta+\gamma)^2-4\beta\gamma=\alpha^2+\dfrac{4r}{\alpha}$, so that

$$\alpha^3-z\alpha+4r=0\,. \tag{5}$$

On the other hand, we have

$$\alpha^3+q\alpha+r=0\,, \tag{6}$$

since α satisfies (3). From (5) and (6) we get

$$(z+q)\alpha=3r\,,\quad \text{or}\quad \alpha=\frac{3r}{(z+q)}\,. \tag{7}$$

Substituting this expression for α in (6), we get $(r\neq 0)$,

$$(z+q)^3+3q(z+q)^2+27r^2=0\,,$$

which is (4).

(ii) Set $x=y-\frac{b}{a}$ in (1). Then we have

$$y^3+\frac{3H}{a^2}\cdot y+\frac{G}{a^3}=0, \tag{8}$$

where $H=ac-b^2$, $G=a^2d-3abc+2b^3$. If α, β, γ are the roots of (1), then $\alpha+\frac{b}{a}$, $\beta+\frac{b}{a}$, $\gamma+\frac{b}{a}$ are the roots of (8). Thus the differences between corresponding roots of the equations (1) and (8) are the same.

If we set $q=\frac{3H}{a^2}$, and $r=\frac{G}{a^3}$, then equation (8) takes the form

$$y^3+qy+r=0.$$

By the first part of the proof, $(\beta-\gamma)^2$, $(\gamma-\alpha)^2$, $(\alpha-\beta)^2$ are the roots of the equation

$$z^3+\frac{18H}{a^2}z^2+\frac{81H^2}{a^4}z+\frac{108H^3}{a^6}+\frac{27G^2}{a^6}=0, \tag{9}$$

hence

$$a^6(\beta-\gamma)^2(\gamma-\alpha)^2(\alpha-\beta)^2=-27(G^2+4H^3). \tag{10}$$

It can be verified directly that

$$G^2+4H^3=a^2D, \tag{11}$$

so that we have

$$a^4(\beta-\gamma)^2(\gamma-\alpha)^2(\alpha-\beta)^2=-27D, \tag{12}$$

and the proof is complete.

Appendix II. The biquadratic equation

Let

$$u\equiv ax^4+4bx^3+6cx^2+4dx+e=0, \tag{1}$$

where a, b, c, d, e are complex numbers with $a\neq 0$.

The substitution $x=y-\frac{b}{a}$ transforms equation (1) into

$$ay^4+\frac{6H}{a}y^2+\frac{4G}{a^2}\cdot y+\frac{K}{a^3}=0, \tag{2}$$

where

$$H=ac-b^2, G=a^2d-3abc+2b^3, K=a^3e-4a^2bd+6ab^2c-3b^4.$$

If we set $z=ay=ax+b$, then equation (2) is transformed into

$$z^4+6Hz^2+4Gz+K=0. \tag{3}$$

If the roots of equation (1) are denoted by $\alpha, \beta, \gamma, \delta$, then $\alpha+\frac{b}{a}, \beta+\frac{b}{a}, \gamma+\frac{b}{a}, \delta+\frac{b}{a}$ are the roots of equation (2), and $a\alpha+b, a\beta+b, a\gamma+b, a\delta+b$ the roots of equation (3).

Set

$$\lambda=\beta\gamma+\alpha\delta, \mu=\gamma\alpha+\beta\delta, \nu=\alpha\beta+\gamma\delta. \tag{4}$$

We obviously have

$$(\beta-\gamma)(\alpha-\delta)=\nu-\mu, (\gamma-\alpha)(\beta-\delta)=\lambda-\nu, (\alpha-\beta)(\gamma-\delta)=\mu-\lambda. \tag{5}$$

Therefore if *two* of the roots $\alpha, \beta, \gamma, \delta$ are equal, then *two* of the numbers λ, μ, ν are equal, *and conversely*. If *three* of the roots $\alpha, \beta, \gamma, \delta$ are equal, then $\lambda=\mu=\nu$, and *conversely*.

Now λ, μ, ν *are* the three roots of the equation

$$a^3y^3-6a^2cy^2+4a(4bd-ae)y-8(2ad^2+2eb^2-3ace)=0. \tag{6}$$

For we have

$$\sum\lambda=\sum\alpha\beta=\frac{6c}{a},$$

(where $\sum\lambda$ stands for $\lambda+\mu+\nu$, and a similar convention applies to $\sum\alpha\beta$), while

$$\sum\mu\nu=\sum\alpha\cdot\sum\alpha\beta\gamma-4\alpha\beta\gamma\delta=\frac{4(4bd-ae)}{a^2},$$

and

$$\lambda\mu\nu=\sum\alpha^2\beta^2\gamma^2+\alpha\beta\gamma\delta\sum\alpha^2=\frac{4}{a^2}(4d^2-3ce)+\frac{4e}{a^3}(4b^2-3ac)$$

$$=\frac{8}{a^3}(2ad^2+2eb^2-3ace).$$

The substitution

$$y=\frac{4t+2c}{a} \tag{7}$$

transforms equation (6) into

$$4t^3-It+J=0, \tag{8}$$

where

$$I=ae-4bd+3c^2,\ J=ace+2bcd-ad^2-c^3-eb^2. \tag{9}$$

The cubic in (8) is called the *reducing cubic* associated with the given biquadratic u in (1). It can be verified that

$$J=\begin{vmatrix} a & b & c \\ b & c & d \\ c & d & e \end{vmatrix}. \tag{10}$$

If t_1, t_2, t_3 denote the roots of equation (8), corresponding respectively to the roots λ, μ, ν of equation (6), then we have, because of (7),

$$\frac{4t_1}{a}=\lambda-\frac{2c}{a}=\lambda-\frac{1}{3}(\lambda+\mu+\nu)=\frac{1}{3}\{(\lambda-\nu)-(\mu-\lambda)\}, \tag{11}$$

with similar expressions for $\frac{4t_2}{a}$ and $\frac{4t_3}{a}$. Hence we have

$$4^2(t_1-t_2)^2=a^2(\lambda-\mu)^2, \tag{12}$$

with similar relations for the other differences. From (11) and (5) we get

$$t_1=\frac{a}{12}\{(\gamma-\alpha)(\beta-\delta)-(\alpha-\beta)(\gamma-\delta)\}, \tag{13}$$

as well as two similar expressions for t_2 and t_3.

Since the roots t_1, t_2, t_3 of equation (8) are functions of the *differences* between α, β, γ, δ, so also are I and J. Since the differences between corresponding roots are the same for equations (1) and (2), it follows that the coefficient-functions I and J are the same for both the equations. For this reason they are called the *invariants* of the given equation (1). Hence we can set $b=0$, $c=H/a$, $d=G/a^2$, $e=K/a^3$ in (9), and obtain the relations

$$a^2I=K+3H^2,\ a^3J=HK-G^2-H^3. \tag{14}$$

Now (5) and (12) give us the relation

$$\begin{aligned} &a^6(\beta-\gamma)^2(\gamma-\alpha)^2(\alpha-\beta)^2(\alpha-\delta)^2(\beta-\delta)^2(\gamma-\delta)^2 \\ &\quad=a^6(\mu-\nu)^2(\nu-\lambda)^2(\lambda-\mu)^2=4^6(t_2-t_3)^2(t_3-t_1)^2(t_1-t_2)^2. \end{aligned} \tag{15}$$

If we now apply the result of Appendix I (12) to the reducing cubic (8), by taking $a=4$, $b=0$, $c=-I/3$, $d=J$, we obtain

$$4^2(t_1-t_2)^2(t_2-t_3)^2(t_3-t_1)^2=I^3-27J^2, \tag{16}$$

so that (15) yields the relation

$$a^6(\alpha-\beta)^2(\beta-\gamma)^2(\gamma-\delta)^2(\delta-\alpha)^2(\delta-\beta)^2(\gamma-\alpha)^2=256\Delta, \tag{17}$$

where

$$\Delta=I^3-27J^2. \tag{18}$$

Hence *a necessary and sufficient condition that the equation*

$$u\equiv ax^4+4bx^3+6cx^2+4dx+e=0,\quad a\neq 0,$$

should have a repeated root is that $\Delta = 0$, *where*

$$\Delta = I^3 - 27J^2,$$

with

$$I = ae - 4bd + 3c^2,\ J = ace + 2bcd - ad^2 - c^3 - eb^2,$$

the so-called invariants of the equation $u = 0$.

The expression I^3/Δ is called the *absolute invariant* of the biquadratic u in (1).

Notes on Chapter III

Much of what is contained in this Chapter is due originally to K. Weierstrass. As general references, see his *Akademische Antrittsrede* (1857) published in his *Werke*, I (1894), 223–225, and his papers, *Sitzungsberichte der Akad. Wiss. Berlin* (1882), 443–451; (1883), 193–203; 265–275; 1271–1297, reprinted in his *Werke*, II (1895), 245–255; 257–309, and especially the account of his lectures published by H.A. Schwarz, *Formeln und Lehrsätze zum Gebrauche der elliptischen Funktionen, Nach Vorlesungen und Aufzeichnungen des Herrn Professor K. Weierstrass* (Julius Springer, Berlin, 1893). See also A. Cayley, *Journal de Math.* 10 (1845), 385–420; *Math. Papers*, I, 156–182; and G. Eisenstein, *J. für Math.* 35 (1847), 137–274; *Math. Werke*, I (Chelsea Reprint, 1975), 299–478.

It should be noted that historically the elliptic functions of Jacobi, which appear in Chapter VII, are anterior to those of Weierstrass by some thirty years.

§§ 1–2. In Schwarz's book (loc. cit.) the σ-function is introduced first, by means of an infinite product, and then the $\wp$-function by means of an infinite series. Here the procedure is reversed.

Property (2.2) implies that $\wp(z; \omega_1, \omega_2)$, considered as a function of all three variables, is *homogeneous of degree* -2. For special values of $\tau = \dfrac{\omega_2}{\omega_1}$, this leads to two formulae of complex multiplication, namely

(i) $\wp(iz) = -\wp(z)$, if $\tau = i$;
(ii) $\wp(lz) = -l\wp(z)$, if $\tau = l = e^{\pi i/3}$.

For we have $\wp(iz; i\omega_1, i\omega_2) = -\wp(z; \omega_1, \omega_2)$; and if $\omega_2 = i\omega_1$, then actually $\wp(iz; i\omega_1, i\omega_2) = \wp(iz; i\omega_1, -\omega_1)$, and the latter coincides with $\wp(iz; \omega_1, i\omega_1)$, (cf. Remark (ii) after Th. 4 of Ch. I), which proves (i). To prove (ii), we note that if $\omega_2 = l\omega_1$, then $\wp(lz; l\omega_1, l^2\omega_1) = -l\wp(z; \omega_1, l\omega_1)$, since $l^2 = -1/l$. On the other hand, we have $l^2\omega_1 = l\omega_1 - \omega_1$, so that $\wp(lz; l\omega_1, l^2\omega_1) = \wp(lz; \omega_1, l\omega_1)$. Hence $\wp(lz; \omega_1, l\omega_1) = -l\wp(z; \omega_1, l\omega_1)$.

On using the expansion of $\wp(z)$ at $z = 0$ given by Theorem 4, and comparing coefficients respectively of z^4 in the first case, and of z^2 in the second case, we see that (i) implies that $g_3 = 0$, while (ii) implies that $g_2 = 0$. (see § 4, Ch. VI).

§ 3. A number of formulae involving $\wp(z)$ are listed by Schwarz, loc. cit., §§ 9–12.

The elementary algebra needed for the remarks in § 3 is worked out in the two Appendices.

§ 4. Instead of the expression $\{\wp'(z_1)-\wp'(z_2)\}/\{\wp(z_1)-\wp(z_2)\}$ in Theorem 6, if we consider the modified expression

$$\left\{\frac{\wp'(z_1)-\wp'(z)}{z_1-z}\right\}\cdot\left\{\frac{\wp(z_1)-\wp(z)}{z_1-z}\right\}^{-1},$$

and then let $z\to z_1$, we obtain, on applying Taylor's theorem to $\wp(z)$ and $\wp'(z)$, the so-called *duplication formula:*

$$\wp(2z_1)=\frac{1}{4}\left\{\frac{\wp''(z_1)}{\wp'(z_1)}\right\}^2-2\wp(z_1),$$

unless $2z_1$ is a period.

For an analogous expression for $\wp(nz)$, where n is any positive integer, see, for instance, Tannery and Molk (cited in the Notes on Ch. I), §§ 456–460.

For a different proof of the addition-theorem by Abel's method, see Whittaker and Watson, § 20 · 312, and especially Siegel's *Funktionentheorie*, I, Kap. I, § 10 (loc. cit. Notes on Ch. I).

§ 5. Theorem 8 implies that if $E(z)$ is a non-constant elliptic function which is even, it is a rational function of $\wp(z)$; and if it is odd, it is a rational function of $\wp(z)$ multiplied by $\wp'(z)$.

If in the addition-formula (4.1) we use the differential equation of Theorem 3, which gives $\wp'(z)=(4\wp^3(z)-g_2\wp(z)-g_3)^{1/2}$, and eliminate the roots by squaring, we obtain an algebraic equation connecting the three functions $\wp(z_1+z_2)$, $\wp(z_1)$, and $\wp(z_2)$. That is to say, there exists an irreducible polynomial A, in three variables, with constant coefficients, such that

$$A\{\wp(z_1+z_2),\ \wp(z_1),\ \wp(z_2)\}=0.$$

This can be upheld for *any* elliptic function $E(z)$ with a given pair of primitive periods. For, by Theorem 8, any such function $E(z)$ can be expressed as a rational function of $\wp(z)$ and $\wp'(z)$, and on using the differential equation for $\wp(z)$, we obtain a relation of the form

$$B\{E(z),\ \wp(z)\}=0,$$

where $B(u, v)$ is an irreducible polynomial in u and v, with constant coefficients. We then have

$$B\{E(z_1),\ \wp(z_1)\}=0,\ B\{E(z_2),\ \wp(z_2)\}=0,\ B\{E(z_1+z_2),\ \wp(z_1+z_2)\}=0,$$

together with: $A\{\wp(z_1+z_2),\ \wp(z_1),\ \wp(z_2)\}=0$. By eliminating $\wp(z_1)$, $\wp(z_2)$, $\wp(z_1+z_2)$ from these four equations, we obtain an equation of the form

$$C\{E(z_1+z_2),\ E(z_1),\ E(z_2)\}=0,$$

where C is an irreducible polynomial, in three variables, with constant coefficients. This is expressed by saying that the elliptic function $E(z)$ has an *algebraic addition-theorem.*

Weierstrass proved conversely that meromorphic functions with such addition-theorems are either elliptic functions, or their *limiting cases* (referred to as the *degenerate forms*).

The *limiting case* of $\wp(z)$ arises when we fix one of its periods, say ω_1, and let $\omega_2 \to \infty$ in such a way that $\tau = \dfrac{\omega_2}{\omega_1}$ always remains in the region: $\operatorname{Im}\tau > 0$, $|\tau| \geqq 1$, $|\operatorname{Re}\tau| \leqq \frac{1}{2}$. Because of the uniform convergence of the series defining $\wp(z)$ (in the sense of Cor. 2, § 1), we then have

$$\wp(z) \to \frac{1}{z^2} + \sum_{n \neq 0} \left\{ \frac{1}{(z - n\omega_1)^2} - \frac{1}{(n\omega_1)^2} \right\},$$

and since

$$\frac{\pi^2}{\sin^2 \pi z} = \sum_{n=-\infty}^{\infty} \frac{1}{(z-n)^2}, \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6},$$

we have

$$\wp(z) \to \left(\frac{\pi}{\omega_1}\right)^2 \frac{1}{\sin^2\left(\dfrac{\pi z}{\omega_1}\right)} - \frac{1}{3}\left(\frac{\pi}{\omega_1}\right)^2 \equiv \varphi(z; \omega_1), \text{ say},$$

and similarly

$$\wp'(z) \to -2\left(\frac{\pi}{\omega_1}\right)^3 \frac{1}{\sin^3\left(\dfrac{\pi z}{\omega_1}\right)} \cdot \left(\cos \frac{\pi z}{\omega_1}\right) = \varphi'(z; \omega_1).$$

We thus have

$$\left(\frac{2\pi i}{\omega_1}\right) \frac{\left[\varphi(z; \omega_1) + \dfrac{1}{3}\left(\dfrac{\pi}{\omega_1}\right)^2\right]}{\{\varphi'(z; \omega_1)\}} = -i \tan\left(\frac{\pi z}{\omega_1}\right) = \frac{2}{1 + e^{2\pi i z/\omega_1}} - 1.$$

The *limiting cases* of $\wp(z)$ and $\wp'(z)$ are respectively $\varphi(z; \omega_1)$ and $\varphi'(z; \omega_1)$. If we set

$$\psi(z; \omega_1) = e^{(2\pi i z)/\omega_1},$$

we see that $\psi(z; \omega_1)$ is a rational function of $\varphi(z; \omega_1)$ and $\varphi'(z; \omega_1)$, and conversely $\varphi(z; \omega_1)$, $\varphi'(z; \omega_1)$ are rational functions of $\psi(z; \omega_1)$. As an example, we may take $\omega_2 = ik\omega_1$, $k = 1, 2, 3, \ldots$.

If we let not only ω_2 but also ω_1 tend to ∞ in such a way that the quotient is restricted to the region already mentioned, then the *limiting cases of* $\wp(z)$

and $\wp'(z)$ are respectively $\varphi(z)=z^{-2}$ and $\varphi'(z)=-2z^{-3}$, so that

$$-2\frac{\varphi(z)}{\varphi'(z)}=z,$$

and if we set $\psi(z)=z$, then $\psi(z)$ is a rational function of $\varphi(z)$ and $\varphi'(z)$, while obviously $\varphi(z)$ and $\varphi'(z)$ are rational functions of $\psi(z)$. As an example, we may take $\omega_1=k$, $\omega_2=ik$, $k=1,2,3,\ldots$.

Since any non-constant elliptic function $E(z)$, with the periods (ω_1,ω_2), can be expressed as a rational function of $\wp(z;\omega_1,\omega_2)$ and $\wp'(z;\omega_1,\omega_2)$, with constant coefficients (by Theorem 8), it follows that the *limiting case* of $E(z;\omega_1,\omega_2)$ is a rational function of $\psi(z;\omega_1)$ if $\omega_2\to\infty$, or of $\psi(z)$ if both ω_1 and ω_2 tend to ∞. Since $\psi(z;\omega_1)$, and $\psi(z)$, can themselves be considered to be limiting cases (in as much as they can be expressed rationally in terms of $\varphi(z;\omega_1)$, $\varphi'(z;\omega_1)$ or $\varphi(z)$, $\varphi'(z)$, as the case may be), it follows that the field of elliptic functions $\mathbf{E}(\omega_1,\omega_2)$ "degenerates" into the field of rational functions of $e^{2\pi iz/\omega_1}$, or the field of rational functions of z.

The proof of Theorem 8 makes implicit use of the fact that a polynomial in $\wp$ with constant coefficients is not identically zero unless the polynomial is zero.

In contrast to the property of doubly periodic, meromorphic functions embodied in Theorem 9, there exist non-constant, simply periodic, entire functions, say e^z and e^{e^z}, between which there is no algebraic relation.

For a study of the *real* values of $\wp(z)$, and for some applications, see Appell and Lacour (cited in § 4 of the Notes on Ch. I), and Schwarz (loc. cit.), §§ 49–50. It is possible to give a formula for the zeros of $\wp(u)$ by using elliptic integrals. See, for instance, M. Eichler and D. Zagier, *Math. Annalen*, 258 (1982), 399–407.

The $\wp$-function can be used to generate *transcendental numbers*. Siegel proved in 1932 that if the invariants g_2, g_3 of $\wp(z)$ are *algebraic*, then one at least of any pair of primitive periods of $\wp(z)$ is transcendental, *J. für Math.* 167 (1932), 62–69; *Ges. Abhandlungen*, I, 267–274. Th. Schneider carried the work forward by showing that if g_2, g_3 are algebraic, then for any algebraic $\alpha\neq 0$, $\wp(\alpha)$ is transcendental, *Math. Annalen*, 113 (1937), 1–13. See his *Einführung in die transzendenten Zahlen* (Springer Verlag, 1957), and also Alan Baker's *Transcendental Number Theory* (Cambridge, 1975), Ch. 6. For more recent work, see, for instance, G. V. Chudnovsky, *Proceedings of the International Congress of Mathematicians, Helsinki* (1978), 339–350.

Chapter IV

The zeta-function and the sigma-function of Weierstrass

§ 1. The function $\zeta(z)$. Weierstrass's ζ-function is a meromorphic function, which has *simple* poles, with residues equal to one, at all points which correspond to the periods of Weierstrass's $\wp$-function. It is *not* elliptic. But every elliptic function can be expressed in terms of ζ and its derivatives; in fact $\zeta'(z) = -\wp(z)$.

As in the case of $\wp(z)$, we shall define $\zeta(z)$ by an infinite series. Let ω_1, ω_2 be two complex numbers, with $\operatorname{Im} \frac{\omega_2}{\omega_1} > 0$. Then the series

$$\sum_{|\omega|>2R>0} \left\{ \frac{1}{z-\omega} + \frac{1}{\omega} + \frac{z}{\omega^2} \right\}, \qquad \omega = m\omega_1 + n\omega_2,$$

where m and n run through all integers, converges uniformly in the circle $|z| \leqq R$, since for $|\omega| > 2R \geqq 2|z|$, we have

$$\left| \frac{1}{z-\omega} + \frac{1}{\omega} + \frac{z}{\omega^2} \right| \leqq \frac{|z|^2}{|\omega|^3 \left(1 - \frac{|z|}{|\omega|}\right)} \leqq \frac{2|z|^2}{|\omega|^3},$$

and the series $\sum_{\omega \neq 0} |\omega|^{-3}$ converges (Theorem 1, Ch. III). It follows that the series

$$\frac{1}{z} + \sum_{\omega \neq 0} \left\{ \frac{1}{z-\omega} + \frac{1}{\omega} + \frac{z}{\omega^2} \right\}$$

converges *absolutely* for $z \neq \omega$, so that the sum is independent of the order of the terms, and that for every finite $R > 0$, the series converges uniformly in the circle $|z| \leqq R$, after the omission of a certain number of initial terms. If we *define*

$$\begin{aligned} \zeta(z) &\equiv \zeta(z; \omega) \equiv \zeta(z; \omega_1, \omega_2) \\ &= \frac{1}{z} + \sum_{\omega \neq 0} \left\{ \frac{1}{z-\omega} + \frac{1}{\omega} + \frac{z}{\omega^2} \right\}, \end{aligned} \tag{1.1}$$

for complex z, with $\omega = m\omega_1 + n\omega_2$; $\operatorname{Im} \frac{\omega_2}{\omega_1} > 0$; and $m, n = 0, \pm 1, \pm 2, \ldots$, then $\zeta(z)$ is holomorphic except for the points $z = \omega$, where it has simple poles with residue 1. It is *not* an elliptic function with periods ω_1, ω_2, since in a fundamental period-parallelogram, it has only one simple pole with residue $+1$

(Theorem 2, Chapter II). Since ω can be replaced by $-\omega$ in the defining series without altering the sum, $\zeta(z)$ is an odd function of z. By differentiation, we see that

$$\zeta'(z) = -\wp(z). \tag{1.2}$$

In a neighbourhood of the origin, we have the expansion

$$\zeta(z) = \frac{1}{z} - \frac{b_1}{3} z^3 - \frac{b_2}{5} z^5 - \dots, \tag{1.3}$$

since, by Theorem 4 of Chapter III, $\wp(z) = \frac{1}{z^2} + \sum_{n=1}^{\infty} b_n z^{2n}$ near $z=0$.

If λ is any complex number, $\lambda \neq 0$, then

$$\zeta(\lambda z; \lambda\omega_1, \lambda\omega_2) = \frac{1}{\lambda} \zeta(z; \omega_1, \omega_2). \tag{1.4}$$

By integrating the relation $\wp(z+\omega_1) = \wp(z)$, we obtain

$$\zeta(z+\omega_1) = \zeta(z) + 2\eta_1,$$

say, where $2\eta_1$ is *independent* of z. On setting $z = -\frac{\omega_1}{2}$, we obtain

$$\begin{aligned} \zeta\left(\frac{\omega_1}{2}\right) &= \zeta\left(-\frac{\omega_1}{2} + \omega_1\right) = \zeta\left(-\frac{\omega_1}{2}\right) + 2\eta_1 \\ &= -\zeta\left(\frac{\omega_1}{2}\right) + 2\eta_1, \end{aligned}$$

since $\zeta(z)$ is an odd function of z. Hence $\eta_1 = \zeta\left(\frac{\omega_1}{2}\right)$. If η_2 is defined by the relation: $\zeta(z+\omega_2) = \zeta(z) + 2\eta_2$, we similarly obtain: $\eta_2 = \zeta\left(\frac{\omega_2}{2}\right)$. If we then set $\eta_3 = \eta_2 + \eta_1$, and $\omega_3 = \omega_1 + \omega_2$, we have $\zeta(z+\omega_3) = \zeta(z) + 2(\eta_1 + \eta_2) = \zeta(z) + 2\eta_3$, hence $\eta_3 = \zeta\left(\frac{\omega_1+\omega_2}{2}\right)$. We summarize these observations in the following

Theorem 1. *The zeta-function of Weierstrass, which is denoted by $\zeta(z)$, and defined in* (1.1), *is an odd function of z, with the properties*

$$\zeta'(z) = -\wp(z);$$

$$\zeta(z+\omega_1) = \zeta(z) + 2\eta_1, \quad \eta_1 = \zeta\left(\frac{\omega_1}{2}\right);$$

$$\zeta(z+\omega_2) = \zeta(z) + 2\eta_2, \quad \eta_2 = \zeta\left(\frac{\omega_2}{2}\right);$$

$$\zeta(z+\omega_3) = \zeta(z) + 2\eta_3, \quad \eta_3 = \zeta\left(\frac{\omega_3}{2}\right),$$

where $\omega_3=\omega_1+\omega_2$, $\eta_3=\eta_1+\eta_2$, *and* (ω_1, ω_2) *is a pair of basic periods of Weierstrass's elliptic function* $\wp(z)$, *with* $\operatorname{Im}\dfrac{\omega_2}{\omega_1}>0$. *It is holomorphic except for the points* $z=\omega$, *where it has simple poles with residue 1.*

Remarks. Since $\zeta(z)$ is *not* an elliptic function, it cannot happen that $\eta_1=\eta_2=0$.

If m and n are integers, we have obviously

$$\zeta(z+m\omega_1+n\omega_2)=\zeta(z)+2m\eta_1+2n\eta_2\,.$$

The relation between the constants η_1 and η_2, and the periods ω_1 and ω_2, is given by

Theorem 2 (Legendre). *If* (ω_1, ω_2) *is a pair of basic periods of Weierstrass's* $\wp$*-function, so chosen that* $\operatorname{Im}\dfrac{\omega_2}{\omega_1}>0$, *then*

$$\eta_1\omega_2-\eta_2\omega_1=\pi i\,. \tag{1.5}$$

Proof. We choose a period-parallelogram of $\wp(z)$ with vertices A, B, C, D, in such a way that (the pole of $\wp(z)$ at) the origin is located at its centre (Fig. 6).

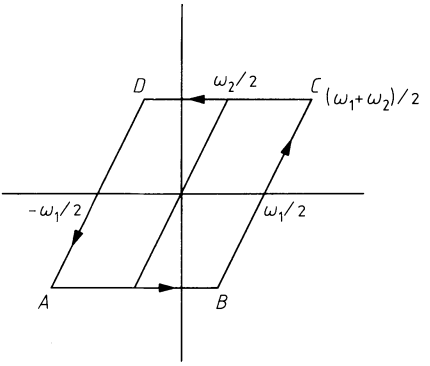


Fig. 6

By Cauchy's theorem of residues, we have

$$\int_{ABCDA}\zeta(z)dz=2\pi i\,. \tag{1.6}$$

On the other hand, we have, by Theorem 1,

$$\begin{aligned}\int_{CD}\zeta(z)dz&=\int_{BA}\zeta(z+\omega_2)dz=\int_{BA}\zeta(z)dz+\int_{BA}2\eta_2 dz\\&=\int_{BA}\zeta(z)dz-2\eta_2\omega_1\,,\end{aligned}$$

so that

$$\int_{AB}\zeta(z)dz+\int_{CD}\zeta(z)dz=-2\eta_2\omega_1\,,$$

and similarly

$$\int_{BC} \zeta(z)dz + \int_{DA} \zeta(z)dz = 2\eta_1\omega_2,$$

which taken together with (1.6) yields (1.5).

§ 2. The function $\sigma(z)$. Weierstrass's σ-function is an entire function, with its zeros, all of which are *simple*, located precisely at those points which correspond to the periods of the $\wp$-function. It is an *odd* function, which is *not* elliptic. But every elliptic function can be expressed in terms of the σ-function. To define it we need some preliminary analysis.

Lemma. *If* $E(z)=(1-z)e^{z+\frac{1}{2}z^2}$, *where* z *is complex, then*

$$|E(z)-1| \leqq 2|z|^3, \quad \textit{for} \quad |z| \leqq \tfrac{1}{2}. \tag{2.1}$$

Proof. For $|z|<1$, we have

$$1-z = e^{\mathrm{Log}(1-z)} = e^{-\left(z+\frac{z^2}{2}+\frac{z^3}{3}+\ldots\right)};$$

hence

$$E(z) = e^{-\sum_{k=3}^{\infty}\frac{z^k}{k}} = e^{g(z)}, \qquad g(z) = -\sum_{k=3}^{\infty}\frac{z^k}{k}. \tag{2.2}$$

If, in addition, $|z| \leqq \frac{1}{2}$, then

$$\begin{aligned} |g(z)| &\leqq \sum_{k=3}^{\infty}\frac{|z|^k}{k} \leqq \frac{1}{3}\sum_{k=3}^{\infty}|z|^k \\ &= \frac{|z|^3}{3}\cdot\frac{1}{1-|z|} \leqq \frac{2}{3}|z|^3. \end{aligned} \tag{2.3}$$

To estimate $|E(z)-1| = |e^{g(z)}-1|$, we need the two inequalities:
(a) $|e^z-1| \leqq e^{|z|}-1$, which follows from the expansion of e^z; and
(b) $e^x - 1 \leqq xe^x$, for any real x. This follows from the inequality

$$1+x \leqq e^x, \quad \text{for every real } x; \tag{2.4}$$

which is obvious if $x \geqq 0$; and if $x<0$, the function $e^x-(1+x)$ has a negative derivative, hence decreases in $[-\infty, 0]$ as x increases, and equals zero for $x=0$, so that it is positive for $x<0$. Replacing x by $-x$ in (2.4), we get (b).

Now

$$\begin{aligned} |E(z)-1| = |e^{g(z)}-1| &\leqq e^{|g(z)|}-1 \leqq |g(z)|\cdot e^{|g(z)|} \\ &\leqq \frac{2}{3}|z|^3 e^{|z|^3} \leqq 2|z|^3, \end{aligned}$$

since $e^{|z|^3} < e < 3$, for $|z| \leqq \frac{1}{2}$, and (2.1) follows.

We shall define $\sigma(z)$ by an infinite product, for the absolute convergence of which we shall use the above lemma.

If (ω_1, ω_2) is a pair of complex numbers, with $\operatorname{Im}\frac{\omega_2}{\omega_1} > 0$, and $\omega = m\omega_1 + n\omega_2$, with $m, n = 0, \pm 1, \pm 2, \ldots$, then for $|z| \leqq R$, we have

$$\sum_{|\omega|>2R>0} \left| E\left(\frac{z}{\omega}\right) - 1 \right| \leqq 2 \sum_{|\omega|>2R>0} \left|\frac{z}{\omega}\right|^3,$$

by the lemma, and the last series converges uniformly in the circle $|z| \leqq R$, for every finite $R > 0$. Hence the infinite product

$$z \prod_{\omega \neq 0} \left(1 - \frac{z}{\omega}\right) e^{\frac{z}{\omega} + \frac{1}{2}\left(\frac{z}{\omega}\right)^2}$$

converges absolutely, and uniformly in the circle $|z| \leqq R$, for every finite $R > 0$, and its value does not depend on the order of the factors. If we *define*

$$\sigma(z) \equiv \sigma(z; \omega) \equiv \sigma(z; \omega_1, \omega_2) = \\ = z \prod_{\omega \neq 0} \left(1 - \frac{z}{\omega}\right) e^{\frac{z}{\omega} + \frac{z^2}{2\omega^2}}, \tag{2.5}$$

then $\sigma(z)$ is an *entire* function of z, which is *not* constant, hence *not* elliptic. Since we may replace ω by $-\omega$, without altering the product, we have $\sigma(-z) = -\sigma(z)$, so that $\sigma(z)$ is an *odd* function of z. It has *simple zeros* precisely at the points $z = \omega$.

If λ is any complex number, with $\lambda \neq 0$, then

$$\sigma(\lambda z; \lambda\omega_1, \lambda\omega_2) = \lambda\sigma(z; \omega_1, \omega_2). \tag{2.6}$$

We note that

$$\zeta(z) = \frac{d}{dz}(\log\sigma(z)) = \frac{\sigma'(z)}{\sigma(z)}, \tag{2.7}$$

the many-valuedness of $\log\sigma(z)$ being removed by the differentiation.

By integrating the relation $\zeta(z+\omega_1) = \zeta(z) + 2\eta_1$, we have $\sigma(z+\omega_1) = ce^{2\eta_1 z}\sigma(z)$, where c is a constant. On setting $z = -\frac{\omega_1}{2}$, we obtain $\sigma\left(\frac{\omega_1}{2}\right) = -ce^{-\eta_1\omega_1}\sigma\left(\frac{\omega_1}{2}\right)$, since $\sigma(z)$ is an *odd* function of z, hence $c = -e^{\eta_1\omega_1}$, and

$$\left.\begin{aligned} &\sigma(z+\omega_1) = -\sigma(z) \cdot e^{2\eta_1\left(z+\frac{\omega_1}{2}\right)}; \\ &\text{similarly also} \\ &\sigma(z+\omega_2) = -\sigma(z) \cdot e^{2\eta_2\left(z+\frac{\omega_2}{2}\right)}; \\ &\text{and} \\ &\sigma(z+\omega_3) = -\sigma(z) \cdot e^{2\eta_3\left(z+\frac{\omega_3}{2}\right)}, \end{aligned}\right\} \tag{2.8}$$

where $\omega_3 = \omega_1 + \omega_2$, $\eta_3 = \eta_1 + \eta_2$.

Since $\sigma(z)$ is an *entire* function of z, which is *odd*, with simple zeros at $z=m\omega_1+n\omega_2$; $m, n=0, \pm 1, \pm 2, \ldots$, we have for $\sigma(z)$ an expansion of the form

$$\sigma(z)=a_1 z+a_3 z^3+\ldots,$$

valid in the whole z-plane, and by differentiation, a corresponding one for $\sigma'(z)$. On the other hand, $\zeta(z)$ has the expansion (1.3) near the origin:

$$\zeta(z)=\frac{1}{z}-\frac{b_1}{3}z^3-\frac{b_2}{5}z^5-\ldots,$$

where the b's are those given by the expansion

$$\wp(z)=\frac{1}{z^2}+\sum_{n=1}^{\infty} b_n z^{2n},$$

near the origin. We have seen that the coefficients b_n satisfy a recurrence relation, which enables one to calculate them, and that they are, in fact, polynomials in g_2, g_3, the invariants of $\wp(z)$, with positive, rational numbers as coefficients (Theorem 4, Chapter III). Hence we deduce that

$$\sigma(z)=z-\frac{g_2\cdot z^5}{2^4\cdot 3\cdot 5}-\frac{g_3 z^7}{2^3\cdot 3\cdot 5\cdot 7}-\frac{g_2^2 z^9}{2^9\cdot 3^2\cdot 5\cdot 7}-\frac{g_2\cdot g_3 z^{11}}{2^7\cdot 3^2\cdot 5^2\cdot 7\cdot 11}-\ldots \tag{2.9}$$

in the whole z-plane. We have thus proved

Theorem 3. *The sigma-function of Weierstrass, which is denoted by $\sigma(z)$, and defined in* (2.5), *is an entire, odd function of z, with zeros precisely at the points $z=\omega$, which correspond to the periods of $\wp(z)$, and has the properties*

$$\zeta(z)=\frac{\sigma'(z)}{\sigma(z)};$$

$$\sigma(z+\omega_1)=-\sigma(z)e^{2\eta_1\left(z+\frac{1}{2}\omega_1\right)};$$

$$\sigma(z+\omega_2)=-\sigma(z)e^{2\eta_2\left(z+\frac{1}{2}\omega_2\right)};$$

$$\sigma(z+\omega_3)=-\sigma(z)e^{2\eta_3\left(z+\frac{1}{2}\omega_3\right)},$$

where $\omega_3=\omega_1+\omega_2$, $\eta_3=\eta_1+\eta_2$, and (ω_1, ω_2) is a pair of basic periods of $\wp(z)$, with $\operatorname{Im}\frac{\omega_2}{\omega_1}>0$, *while η_1, η_2 are defined as in Theorem 1.*

Remarks. The infinite product for $\sigma(z)$ is similar to that of $\sin z$ given by

$$\sin z=z\prod_{m\neq 0}\left\{\left(1-\frac{z}{m\pi}\right)e^{z/m\pi}\right\}.$$

Now $\frac{d}{dz}(\log \sin z) = \cot z$, and $\cot z = \frac{1}{z} + \sum_{m \neq 0} \left\{ \frac{1}{z - m\pi} + \frac{1}{m\pi} \right\}$. As $\frac{d}{dz}(\cot z) =$ $= -\operatorname{cosec}^2 z$, so also we have $\frac{d}{dz}(\zeta(z)) = -\wp(z)$.

§ 3. An expression for elliptic functions. Every elliptic function can be expressed in terms of the σ-function. We can thereby furnish an alternative proof of Theorem 7 of Chapter III.

Theorem 4. *Let $f(z)$ be an elliptic function with (ω_1, ω_2) as a pair of primitive (or, reduced) periods, with $\operatorname{Im} \frac{\omega_2}{\omega_1} > 0$. Let $a_1, a_2, \ldots, a_n$ be the zeros, and $b_1, b_2, \ldots, b_n$ the poles of $f(z)$, repeated according to their degree of multiplicity, which lie in a fundamental period-parallelogram P, so that the difference between the sum of the zeros and the sum of the poles, namely*

$$\Omega = (a_1 + a_2 + \ldots + a_n) - (b_1 + b_2 + \ldots + b_n) \tag{3.1}$$

is a period of $f(z)$. Then

$$f(z) = c \cdot \frac{\sigma(z - a_1) \cdots \sigma(z - a_n)}{\sigma(z - b_1) \cdots \sigma(z - b_n - \Omega)}, \tag{3.2}$$

where c is a constant.

Conversely every function of the form (3.2), *where the a's and b's are complex numbers satisfying condition* (3.1), *is an elliptic function.*

Proof. The meromorphic function φ defined by

$$\varphi(z) = \frac{\sigma(z - a_1) \cdots \sigma(z - a_n)}{\sigma(z - b_1) \cdots \sigma(z - b_n - \Omega)}$$

has precisely the same zeros and poles as $f(z)$ inside the fundamental parallelogram P, for the only zero of the factor $\sigma(z - a_k)$ in P is at $z = a_k$ (and it has no poles, since it is entire). Further $\varphi(z)$ has ω_1 and ω_2 as periods, since

$$\begin{aligned} \varphi(z + \omega_1) &= \frac{(-1)^n e^{2\eta_1 \left(nz + n\frac{\omega_1}{2} - a_1 - a_2 - \ldots - a_n\right)} \cdot \sigma(z - a_1) \cdots \sigma(z - a_n)}{(-1)^n e^{2\eta_1 \left(nz + n\frac{\omega_1}{2} - b_1 - b_2 - \ldots - b_n - \Omega\right)} \cdot \sigma(z - b_1) \cdots \sigma(z - b_n - \Omega)} \\ &= \varphi(z) \end{aligned} \tag{3.3}$$

by (2.8), and similarly also $\varphi(z + \omega_2) = \varphi(z)$. The quotient $\frac{f(z)}{\varphi(z)}$ is therefore doubly periodic, and has no poles or zeros, hence a constant c.

Conversely, if the a's and b's satisfy (3.1), then (3.3) shows that the function given by (3.2) is elliptic.

Corollary. *We have the formulae*

$$\wp(u)-\wp(v)=-\frac{\sigma(u+v)\sigma(u-v)}{\sigma^2(u)\sigma^2(v)}, \tag{3.4}$$

$$-\frac{\wp'(v)}{\wp(u)-\wp(v)}=\zeta(u+v)-\zeta(u-v)-2\zeta(v), \tag{3.5}$$

and

$$\frac{1}{2}\cdot\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}=\zeta(u+v)-\zeta(u)-\zeta(v). \tag{3.6}$$

Proof. Consider the elliptic function $\wp(u)-\wp(v)$ as a function of u, with a fixed v, which is not equal to a period of $\wp$.

If we assume, for the moment, that v is not equal to a half-period of $\wp$, then, as in Theorem 4, with $a_1=v$, $a_2=-v$, $b_1=0$, $b_2=0$, we see that

$$\frac{\sigma(u+v)\sigma(u-v)}{\sigma^2(u)}$$

is an elliptic function, with the same zeros and poles as $\wp(u)-\wp(v)$, and with the same periods, so that

$$\wp(u)-\wp(v)=c\cdot\frac{\sigma(u+v)\sigma(u-v)}{\sigma^2(u)},$$

where c is a constant independent of u. By considering the principal parts of both sides of this formula, we deduce that $1=-c\sigma^2(v)$, hence (3.4). By continuity this holds also when v is a half-period. By logarithmic differentiation of (3.4), relative to u, we obtain

$$\frac{\wp'(u)}{\wp(u)-\wp(v)}=\zeta(u+v)+\zeta(u-v)-2\zeta(u);$$

and by interchanging u with v, we obtain

$$-\frac{\wp'(v)}{\wp(u)-\wp(v)}=\zeta(u+v)-\zeta(u-v)-2\zeta(v),$$

which is (3.5). By adding the last two formulae we get (3.6).

If u or v is a period of $\wp$, then both sides of (3.4) are infinite.

Remarks

(i) On setting $u=v+h$ in (3.4), multiplying by $\frac{1}{h}$, $h\neq 0$, and then letting $|h|\to 0$, we obtain the formula

$$\wp'(v)=-\frac{\sigma(2v)}{\{\sigma(v)\}^4}. \tag{3.7}$$

(ii) By differentiating formula (3.6) with respect to u, and then interchanging u and v, we get two new formulae. By adding them together, and using the relation $\wp''=6\wp^2-\frac{g_2}{2}$, one deduces the addition-theorem for $\wp$.

(iii) We have seen in Theorem 4 that every elliptic function can be expressed in terms of the sigma-function alone. In Theorem 8, of Chapter III, it was shown that every elliptic function can be expressed in terms of $\wp$ and $\wp'$. After Hermite, one can express every elliptic function as a linear combination of ζ and its derivatives.

(iv) Let Ω be *any* period of the $\wp$-function. By formula (3.4), we have

$$\wp(u+\Omega)-\wp(u)=0=-\frac{\sigma(2u+\Omega)\sigma(\Omega)}{\sigma^2(u+\Omega)\cdot\sigma^2(u)},$$

for all u, hence $\sigma(\Omega)=0$, so that Ω is a zero of $\sigma(u)=\sigma(u;\omega_1,\omega_2)$, which implies that $\Omega=m\omega_1+n\omega_2$, where m and n are integers (cf. Theorem 2, Chapter I).

Notes on Chapter IV

§§ 1–2. The zeta-function of Weierstrass can be defined by the equation

$$\frac{d\zeta(z)}{dz}=-\wp(z),$$

together with the condition $\lim_{z\to 0}\{\zeta(z)-z^{-1}\}=0$. See, for instance, Whittaker and Watson(cited in the Notes on Ch. I, § 4), p. 445–447. One can then introduce the sigma-function by the equation

$$\frac{d}{dz}\{\log\sigma(z)\}=\zeta(z),$$

together with the condition $\lim_{z\to 0}\{\sigma(z)/z\}=1$.

The definition of $\sigma(z)$ by an infinite product is of interest, since it is an instance of Weierstrass's general theorem on the representation of an entire function by an infinite product. See H. A. Schwarz (cited in the Notes on Ch. III), § 6; Saks and Zygmund (cited in the Notes on Ch. I), § 2.

"The first mathematician to discover that the elliptic functions could be got at starting from the infinite product $\prod\left(1-\frac{x}{n}\right)$ was Eisenstein, *Crelle's J.* Bd. 27, where he has two articles on the subject.... In *Crelle*, Bd. 35, he gives a long discussion also on the convergence of these last products. The necessity of a factor of convergence seems not to have occurred to him." So writes A. L. Daniels, *American J. Math.* 6 (1884), 253–269. See also Jacobi's remarks, *Crelle's J.* 30 (1845), 183–184; *Werke*, II, 85–86. Of interest therefore are the emendations by André Weil, *Elliptic functions according to Eisenstein and*

Kronecker, Ergebnisse der Math. 88 (Springer Verlag, 1976). See also the Notes on Chapter VII, § 5.

§ 3. For a number of formulae involving the σ-function, see the book by H. A. Schwarz (cited in the Notes on Ch. III), §§ 5–8, 13–16.

Every elliptic function can be expressed as a linear combination of zeta-functions and their derivatives. See Ch. Hermite, *Crelle's J.* 82, p. 343; *Oeuvres*, III, 420–424; Whittaker and Watson (cited in the Notes on Ch. I, § 4), § 20.52; Saks and Zygmund (cited in the Notes on Ch. I, § 1), Chapter VIII, § 8.

It is known that if $x+y+z=0$, then

$$\{\zeta(x)+\zeta(y)+\zeta(z)\}^2+\zeta'(x)+\zeta'(y)+\zeta'(z)=0\,,$$

a result due to G. Frobenius and L. Stickelberger, *J. für Math.* 83 (1877), 175–179. This is not considered an addition-theorem, since $\zeta'(x)$, $\zeta'(y)$, $\zeta'(z)$ are not algebraic functions of $\zeta(x)$, $\zeta(y)$, $\zeta(z)$. See Whittaker and Watson (loc. cit.), § 20.41.

Corresponding to the *limiting cases* $\varphi(z;\omega_1)$ and $\varphi(z)$ of $\wp(z)$ (see the Notes on Ch. III, § 5) we have the *limiting cases* of $\zeta(z)$ and $\sigma(z)$, namely

$$\zeta(z;\omega_1,\omega_2\to\infty)=\frac{1}{3}\left(\frac{\pi}{\omega_1}\right)^2 z+\frac{\pi}{\omega_1}\cot\left(\frac{\pi z}{\omega_1}\right),$$

$$\zeta(z;\omega_1\to\infty,\omega_2\to\infty)=\frac{1}{z},$$

and

$$\sigma(z;\omega_1,\omega_2\to\infty)=\frac{\omega_1}{\pi}\sin\left(\frac{\pi z}{\omega_1}\right)\cdot e^{\frac{1}{6}\left(\frac{\pi z}{\omega_1}\right)^2},$$

$$\sigma(z;\omega_1\to\infty,\omega_2\to\infty)=z\,.$$

If ω is a primitive period of a $\wp$-function, with *algebraic* invariants g_2, g_3, and $\eta=2\zeta\left(\frac{\omega}{2}\right)$, then any linear combination $\alpha\omega+\beta\eta$, where α and β are *algebraic*, not both zero, is *transcendental*. It follows that the circumference of an ellipse with *algebraic* axes-lengths is transcendental. These are consequences of a general theorem of Th. Schneider *Math. Annalen*, 121 (1949), 131–140, and are so described in Chapter 6 of the book by Alan Baker (cited in the Notes on Ch. III, § 5).

Chapter V

The theta-functions

§ 1. The function $\theta(v,\tau)$. The expansions, in infinite series, of the functions $\wp(u)$, $\zeta(u)$, and $\sigma(u)$, which we have so far considered, are not best suited to numerical computation. It is of advantage therefore to introduce another function, denoted by $\theta(v,\tau)$, which has a rapidly convergent expansion in infinite series, and which is directly connected with the σ-function of Weierstrass.

Let τ be a complex variable, with $\tau=r+is$, where r and s are real, $s>0$, $i=\sqrt{-1}$, and v a complex variable, with $v=\alpha+i\beta$, where α and β are real.

We define the function $\theta(v,\tau)$ by the relation

$$\theta(v,\tau)=\frac{1}{i}\sum_{n=-\infty}^{\infty}(-1)^n q^{(n+\frac{1}{2})^2}e^{(2n+1)\pi iv}, \qquad q=e^{\pi i\tau}. \tag{1.1}$$

Since

$$|q^{(n+\frac{1}{2})^2}e^{\pm i(2n+1)\pi v}|\leqq|q|^{(n+\frac{1}{2})^2}e^{|(2n+1)\pi v|},$$

where $|q|<1$, the series on the right-hand side of (1.1) converges absolutely, and uniformly in compact sets of the v-plane, and therefore represents an *entire* function of v, for fixed τ. Obviously we have

$$\theta(v,\tau)=2\sum_{n=0}^{\infty}(-1)^n q^{(n+\frac{1}{2})^2}\sin(2n+1)\pi v, \tag{1.2}$$

and

$$\theta(v,\tau)=-\theta(-v,\tau), \qquad \theta(0,\tau)=0, \tag{1.3}$$

$$\theta(v+1,\tau)=-\theta(v,\tau), \tag{1.4}$$

$$\theta(v+\tau,\tau)=-q^{-1}e^{-2\pi iv}\theta(v,\tau), \tag{1.5}$$

since

$$\begin{aligned}\theta(v+\tau,\tau)&=\frac{1}{i}\sum_{n=-\infty}^{\infty}(-1)^n q^{(n+\frac{1}{2})^2+(2n+1)}\cdot e^{(2n+1)\pi iv}\\&=\frac{1}{i}\sum_{m=-\infty}^{\infty}(-1)^{m-1}q^{(m-\frac{1}{2})^2+(2m-1)}\cdot e^{(2m-1)\pi iv}\qquad(n=m-1)\\&=\frac{1}{i}\sum_{m=-\infty}^{\infty}(-1)(-1)^m q^{(m+\frac{1}{2})^2}\cdot e^{(2m+1)\pi iv}\cdot q^{-1}\cdot e^{-2\pi iv}.\end{aligned}$$

By (1.3), (1.4), and (1.5), we have

$$\theta(m_1+m_2\tau,\tau)=0, \tag{1.6}$$

for all integers m_1 and m_2. In fact, the complex numbers $\{m_1+m_2\tau\}$ are the *only* zeros of $\theta(v,\tau)$.

To prove that, consider the parallelogram $ABCD$ in the v-plane with the vertices A, B, C, D located at v_0, v_0+1, $v_0+1+\tau$, $v_0+\tau$, where v_0 is so chosen that no zero of $\theta(v, \tau)$ lies on the boundary of the parallelogram. We shall see that $\theta(v, \tau)$ has *only one zero inside* the parallelogram. Consider the integral

$$\int_{ABCD} \frac{\theta'(v, \tau)}{\theta(v, \tau)}\, dv,$$

taken along the positively oriented curve $ABCD$ (since $\mathrm{Im}\,\tau > 0$, the vertices appear as in Fig. 7). By (1.4) and (1.5), we have

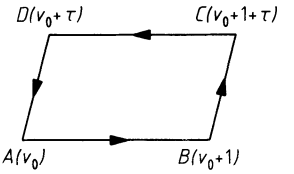


Fig. 7

$$\frac{\theta'(v+1, \tau)}{\theta(v+1, \tau)} = \frac{\theta'(v, \tau)}{\theta(v, \tau)}, \quad \frac{\theta'(v+\tau, \tau)}{\theta(v+\tau, \tau)} = \frac{\theta'(v, \tau)}{\theta(v, \tau)} - 2\pi i,$$

so that

$$\left(\int_{BC} + \int_{DA}\right) \frac{\theta'(v, \tau)}{\theta(v, \tau)}\, dv = 0, \quad \text{and} \quad \left(\int_{AB} + \int_{CD}\right) = 2\pi i,$$

hence

$$\frac{1}{2\pi i} \int_{ABCD} \frac{\theta'(v, \tau)}{\theta(v, \tau)}\, dv = 1,$$

from which it follows that $\theta(v, \tau)$, as a function of v, has *only one zero* inside $ABCD$. Here θ' is the derivative of θ with respect to v.

On the other hand, there exists only one point of the form $m_1 + m_2\tau$ inside the parallelogram $ABCD$, and every such point is a zero of $\theta(v, \tau)$. Hence $\theta(v, \tau)$ has no zeros other than those given by $\{m_1 + m_2\tau\}$, where m_1 and m_2 are arbitrary integers. We summarize our observations in the following

Theorem 1. *Let $\tau = r + is$, with r and s real, $s > 0$, $i = \sqrt{-1}$. Let $q = e^{\pi i \tau}$, and v a complex variable. Then the function $\theta(v, \tau)$, defined by*

$$\theta(v, \tau) = \frac{1}{i} \sum_{n=-\infty}^{\infty} (-1)^n q^{\left(n+\frac{1}{2}\right)^2} e^{(2n+1)\pi i v},$$

where the series converges absolutely, is an odd, entire function of v, with simple zeros at $v = m_1 + m_2\tau$, where m_1, m_2 are arbitrary integers. It has no other zeros, and satisfies the relations:

$$\theta(v+1, \tau) = -\theta(v, \tau), \qquad \theta(v+\tau, \tau) = -\frac{1}{q} e^{-2\pi i v} \theta(v, \tau).$$

The connexion between $\theta(u,\tau)$ and $\sigma(u)$ is provided by the following

Theorem 2. *We have*

$$\sigma(u)=\theta\left(\frac{u}{\omega_1},\tau\right)\cdot\frac{\omega_1}{\theta'(0,\tau)}\cdot e^{\eta_1 u^2/\omega_1},$$

where (ω_1,ω_2) *is a pair of primitive, or reduced, periods of the function* $\wp(u)$, $\tau=\dfrac{\omega_2}{\omega_1}$, *with* $\operatorname{Im}\tau>0$, $\eta_1=\zeta(\frac{1}{2}\omega_1)$, *and* $\theta'(0,\tau)$ *is the derivative of* $\theta(v,\tau)$ *with respect to* v *at* $v=0$.

Proof. Let $\varphi(u)=\theta\left(\dfrac{u}{\omega_1},\tau\right)$. Then $\varphi(u)$ is an *odd*, *entire* function of u, with *simple* zeros at $u=m\omega_1+n\omega_2$, where m and n are integers, since

$$\varphi(m\omega_1+n\omega_2)=\theta(m+n\tau,\tau)=0\,;\qquad m,n=0,\pm1,\pm2,\ldots.$$

On the other hand, we have

$$\varphi(u+\omega_1)=\theta\left(\frac{u}{\omega_1}+1,\tau\right)=-\theta\left(\frac{u}{\omega_1},\tau\right)=-\varphi(u),$$

and

$$\begin{aligned}\varphi(u+\omega_2)&=\theta\left(\frac{u}{\omega_1}+\tau,\tau\right)\\&=-e^{-\pi i\tau}\cdot e^{-2\pi i\frac{u}{\omega_1}}\cdot\theta\left(\frac{u}{\omega_1},\tau\right),\qquad\text{by (1.5)},\\&=-e^{-\pi i\frac{\omega_2}{\omega_1}-2\pi i\frac{u}{\omega_1}}\cdot\theta\left(\frac{u}{\omega_1},\tau\right)\\&=-e^{-\pi i\frac{\omega_2}{\omega_1}-2\pi i\frac{u}{\omega_1}}\cdot\varphi(u).\end{aligned}$$

On setting

$$\psi(u)=\varphi(u)\cdot e^{(\eta_1 u^2)/\omega_1}\cdot\frac{\omega_1}{\theta'(0,\tau)},$$

we note that $\psi(u)$ is again an *odd*, *entire* function of u, with the *same* zeros as $\varphi(u)$. We have, however,

$$\begin{aligned}\psi(u+\omega_1)&=\varphi(u+\omega_1)\cdot e^{\frac{\eta_1}{\omega_1}(u+\omega_1)^2}\cdot\frac{\omega_1}{\theta'(0,\tau)}\\&=-\varphi(u)\cdot\frac{\omega_1}{\theta'(0,\tau)}\cdot e^{\frac{\eta_1}{\omega_1}(u+\omega_1)^2}\\&=-\psi(u)\cdot e^{\frac{\eta_1}{\omega_1}(2u\omega_1+\omega_1^2)}=-\psi(u)\cdot e^{2\eta_1 u+\eta_1\omega_1},\end{aligned}\qquad(1.7)$$

and

$$\psi(u+\omega_2)=\varphi(u+\omega_2)\cdot e^{\frac{\eta_1}{\omega_1}(u+\omega_2)^2}\cdot\frac{\omega_1}{\theta'(0,\tau)}$$
$$=-\varphi(u)\cdot e^{-\pi i\frac{\omega_2}{\omega_1}-2\pi i\frac{u}{\omega_1}}\cdot\frac{\omega_1}{\theta'(0,\tau)}\cdot e^{\eta_1u^2/\omega_1}\cdot e^{\frac{\eta_1}{\omega_1}(2u\omega_2+\omega_2^2)}$$
$$=-\psi(u)\cdot e^{2\eta_2u}\cdot e^{\eta_2\omega_2}, \tag{1.8}$$

since

$$2\eta_1\omega_2\cdot\frac{u}{\omega_1}=(2\pi i+2\eta_2\omega_1)\frac{u}{\omega_1}=2\eta_2u+2\pi i\frac{u}{\omega_1},$$

on account of Legendre's relation: $\eta_1\omega_2-\eta_2\omega_1=\pi i$ (Theorem 2, Ch. IV), and

$$\frac{\eta_1\omega_2^2}{\omega_1}=\eta_1\omega_2\cdot\frac{\omega_2}{\omega_1}=(\pi i+\eta_2\omega_1)\frac{\omega_2}{\omega_1}=\pi i\cdot\frac{\omega_2}{\omega_1}+\eta_2\omega_2\cdot$$

Since the σ-function satisfies relations ((2.8) of Ch. IV) similar to (1.7) and (1.8), it follows that the quotient $\frac{\psi(u)}{\sigma(u)}$ is *periodic*, with the periods ω_1 and ω_2. Since $\psi(u)$ and $\sigma(u)$ have the *same* zeros, it follows that $\frac{\psi(u)}{\sigma(u)}$ is a constant. Since the coefficient of u in the expansions of both $\psi(u)$ and $\sigma(u)$, near $u=0$, equals 1, that constant is 1, and $\psi(u)=\sigma(u)$.

§ 2. The four sigma-functions. The connexion between the σ-function and the $\wp$-function was established in (3.4) of Chapter IV, namely

$$\wp(u)-\wp(v)=-\frac{\sigma(u+v)\sigma(u-v)}{\sigma^2(u)\cdot\sigma^2(v)}. \tag{2.1}$$

If we set $v=\frac{\omega_1}{2}$, so that $\wp(v)=e_1$ ((ω_1,ω_2) being a pair of primitive periods of $\wp(u)$, as in Ch. III), we have

$$\wp(u)-e_1=-\frac{\sigma\left(u+\frac{\omega_1}{2}\right)\sigma\left(u-\frac{\omega_1}{2}\right)}{\sigma^2(u)\cdot\sigma^2\left(\frac{\omega_1}{2}\right)}=\frac{\sigma\left(u+\frac{\omega_1}{2}\right)\sigma\left(\frac{\omega_1}{2}-u\right)}{\sigma^2(u)\cdot\sigma^2\left(\frac{\omega_1}{2}\right)}, \tag{2.2}$$

since $\sigma(u)$ is an *odd* function. By (2.8) of Chapter IV, however, we have

$$\sigma(u+\omega_1)=-\sigma(u)\cdot e^{2\eta_1\left(u+\frac{\omega_1}{2}\right)}\quad\text{where}\quad\eta_1=\zeta\left(\frac{\omega_1}{2}\right),$$

so that, with $u-\frac{\omega_1}{2}$ in place of u, we get

$$\sigma\left(u+\frac{\omega_1}{2}\right)=-e^{2\eta_1 u}\sigma\left(u-\frac{\omega_1}{2}\right)=e^{2\eta_1 u}\sigma\left(\frac{\omega_1}{2}-u\right). \tag{2.3}$$

If we *define*

$$\sigma_1(u)=\frac{e^{\eta_1 u}\sigma\left(\frac{\omega_1}{2}-u\right)}{\sigma\left(\frac{\omega_1}{2}\right)}=\frac{e^{-\eta_1 u}\sigma\left(u+\frac{\omega_1}{2}\right)}{\sigma\left(\frac{\omega_1}{2}\right)}, \tag{2.4}$$

the last equality resulting from (2.3), then by (2.2) and (2.3) we have

$$\wp(u)-e_1=\frac{\sigma_1^2(u)}{\sigma^2(u)}. \tag{2.5}$$

Defining

$$\sigma_2(u)=\frac{e^{\eta_2 u}\sigma\left(\frac{\omega_2}{2}-u\right)}{\sigma\left(\frac{\omega_2}{2}\right)},\quad \sigma_3(u)=\frac{e^{\eta_3 u}\sigma\left(\frac{\omega_3}{2}-u\right)}{\sigma\left(\frac{\omega_3}{2}\right)}, \tag{2.6}$$

where $\omega_3=\omega_1+\omega_2$, $\eta_3=\eta_1+\eta_2$, $\eta_1=\zeta\left(\frac{\omega_1}{2}\right)$, $\eta_2=\zeta\left(\frac{\omega_2}{2}\right)$, we obtain in a similar fashion

$$\wp(u)-e_2=\frac{\sigma_2^2(u)}{\sigma^2(u)},\quad \wp(u)-e_3=\frac{\sigma_3^2(u)}{\sigma^2(u)}. \tag{2.7}$$

Thus the three functions $\sigma_1(u)$, $\sigma_2(u)$, $\sigma_3(u)$ correspond to the three roots e_1, e_2, e_3 associated with Weierstrass's $\wp$-function (see Theorem 5, Ch. III).

Remarks

(i) We have already seen that $\sigma(u)$ is an *odd*, *entire* function of u. By (2.4) we see that $\sigma_1(u)$ is an *even*, *entire* function, so also are $\sigma_2(u)$ and $\sigma_3(u)$ *even*, and *entire*.

(ii) From the definitions of σ_1, σ_2, σ_3, we see that

$$\sigma_1(0)=\sigma_2(0)=\sigma_3(0)=1.$$

(iii) Since

$$\begin{aligned}(\wp'(u))^2&=4\wp^3(u)-g_2\wp(u)-g_3\\&=4(\wp(u)-e_1)(\wp(u)-e_2)(\wp(u)-e_3),\end{aligned}$$

we have

$$(\wp'(u))^2 = 4\left(\frac{\sigma_1(u)}{\sigma(u)}\right)^2 \cdot \left(\frac{\sigma_2(u)}{\sigma(u)}\right)^2 \cdot \left(\frac{\sigma_3(u)}{\sigma(u)}\right)^2$$
$$= 4\left(\frac{\sigma_1(u)\cdot\sigma_2(u)\cdot\sigma_3(u)}{\sigma^3(u)}\right)^2,$$

by (2.5) and (2.7).

(iv) We have

$$\sqrt{\wp(u)-e_r} = \pm\frac{\sigma_r(u)}{\sigma(u)}, \qquad r=1,2,3,$$

by (2.5) and (2.7), where $\dfrac{\sigma_r(u)}{\sigma(u)}$ is meromorphic, and if we choose the $+$sign, the residue at $u=0$ is $+1$, and correspondingly we have

$$\wp'(u) = -2\sqrt{(\wp(u)-e_1)(\wp(u)-e_2)(\wp(u)-e_3)}\,.$$

§ 3. The four theta-functions. Theorem 2 connects the σ-function with the θ-function. We can in a similar fashion connect the three other sigma-functions $\sigma_1, \sigma_2, \sigma_3$ introduced in § 2 with three other theta-functions defined as follows:

$$\theta_1(v,\tau) = 2\sum_{n=0}^{\infty} q^{\frac{1}{4}(2n+1)^2}\cos(2n+1)\pi v$$
$$= \sum_{n=-\infty}^{\infty} q^{\left(n+\frac{1}{2}\right)^2} e^{(2n+1)\pi i v}, \tag{3.1}$$

$$\theta_2(v,\tau) = 1+2\sum_{n=1}^{\infty}(-1)^n q^{n^2}\cos(2n\pi v)$$
$$= \sum_{n=-\infty}^{\infty}(-1)^n q^{n^2} e^{2n\pi i v}, \tag{3.2}$$

$$\theta_3(v,\tau) = 1+2\sum_{n=1}^{\infty} q^{n^2}\cos(2n\pi v)$$
$$= \sum_{n=-\infty}^{\infty} q^{n^2} e^{2n\pi i v}, \tag{3.3}$$

where $q=e^{\pi i\tau}$, $\operatorname{Im}\tau>0$, and v is complex. Like $\theta(v,\tau)$ these three functions are also *entire* functions of v, for fixed τ.

While $\theta(v,\tau)$ is an *odd* function of v, the other three theta-functions are *even* functions of v. They are obviously related. We have, for example, the relations

$$\theta_1(v,\tau) = \theta(\tfrac{1}{2}-v,\tau) = -\theta(v-\tfrac{1}{2},\tau),$$

$$\theta_2(v,\tau) = \theta\left(\frac{\tau}{2}-v,\tau\right) q^{1/4}e^{-\pi i v}\cdot -i,$$

$$\theta_3(v,\tau) = \theta\left(\frac{1}{2}+\frac{\tau}{2}-v,\tau\right)\cdot q^{1/4}\cdot e^{-\pi i v}, \tag{3.4}$$

which express $\theta_1(v,\tau)$, $\theta_2(v,\tau)$, and $\theta_3(v,\tau)$ in terms of $\theta(v,t)$, as well as

$$\left.\begin{aligned}\theta(v,\tau)&=\theta_3(v+\tfrac{1}{2}+\tfrac{1}{2}\tau,\tau)\left(q^{1/4}\cdot e^{\pi iv}\cdot\frac{1}{i}\right),\\ \theta_1(v,\tau)&=\theta_3(v+\tfrac{1}{2}\tau,\tau)\cdot q^{1/4}\cdot e^{\pi iv},\\ \theta_2(v,\tau)&=\theta_3(v+\tfrac{1}{2},\tau),\end{aligned}\right\}\tag{3.5}$$

which express $\theta(v,\tau)$, $\theta_1(v,\tau)$, and $\theta_2(v,\tau)$ in terms of $\theta_3(v,\tau)$.

As in the case of $\theta(v,\tau)$ in (1.4) and (1.5), the effect of replacing v by $v+1$ or by $v+\tau$ is shown by the following:

$$\left.\begin{aligned}\theta_1(v+1,\tau)&=-\theta_1(v,\tau);\quad \theta_2(v+1,\tau)=\theta_2(v,\tau);\\ \theta_3(v+1,\tau)&=\theta_3(v,\tau);\end{aligned}\right\}\tag{3.6}$$

and

$$\left.\begin{aligned}\theta_1(v+\tau,\tau)&=q^{-1}\cdot e^{-2\pi iv}\cdot\theta_1(v,\tau);\\ \theta_2(v+\tau,\tau)&=-q^{-1}\cdot e^{-2\pi iv}\cdot\theta_2(v,\tau);\\ \theta_3(v+\tau,\tau)&=q^{-1}e^{-2\pi iv}\theta_3(v,\tau).\end{aligned}\right\}\tag{3.6}$$

The connexion between the functions $\sigma_1,\sigma_2,\sigma_3$ and $\theta_1,\theta_2,\theta_3$ is provided by the following

Theorem 3. *We have the relations*

$$\left.\begin{aligned}\sigma_1(u)&=\theta_1\left(\frac{u}{\omega_1},\tau\right)e^{\eta_1u^2/\omega_1}\cdot\frac{1}{\theta_1(0,\tau)};\\ \sigma_2(u)&=\theta_2\left(\frac{u}{\omega_1},\tau\right)e^{\eta_1u^2/\omega_1}\cdot\frac{1}{\theta_2(0,\tau)};\\ \sigma_3(u)&=\theta_3\left(\frac{u}{\omega_1},\tau\right)e^{\eta_1u^2/\omega_1}\cdot\frac{1}{\theta_3(0,\tau)},\end{aligned}\right\}\tag{3.7}$$

where $\sigma_r(u)\equiv\sigma_r(u;\omega_1,\omega_2)$, $\tau=\dfrac{\omega_2}{\omega_1}$, $\operatorname{Im}\tau>0$.

Proof. By the definition of $\sigma_1(u)$ given in (2.4),

$$\sigma_1(u)=\frac{e^{\eta_1u}\sigma\left(\dfrac{\omega_1}{2}-u\right)}{\sigma\left(\dfrac{\omega_1}{2}\right)},$$

and by Theorem 2,

$$\sigma(u)=\theta\left(\frac{u}{\omega_1},\tau\right)\cdot\frac{\omega_1}{\theta'(0,\tau)}\cdot e^{\eta_1u^2/\omega_1},$$

hence

$$\sigma_1(u)=\theta\left(\frac{1}{2}-\frac{u}{\omega_1},\tau\right)e^{\eta_1 u^2/\omega_1}\cdot c_1=\theta_1\left(\frac{u}{\omega_1},\tau\right)e^{\eta_1 u^2/\omega_1}\cdot c_1,$$

with $c_1=\dfrac{\omega_1}{\theta'(0,\tau)}\cdot\dfrac{e^{\eta_1\omega_1/4}}{\sigma\left(\dfrac{\omega_1}{2}\right)}$. On letting $u\to 0$, we see that $c_1=\dfrac{1}{\theta_1(0,\tau)}$, since $\sigma_1(0)=1$. This proves the first of the three relations in (3.7). The proof of the other two is similar, if we note (3.4) as well as the relation: $\eta_1\omega_2-\eta_2\omega_1=\pi i$.

Remarks. We obviously have, for $q=e^{\pi i\tau}$, $\operatorname{Im}\tau>0$,

$$\theta_1(0,\tau)=2(q^{1/4}+q^{9/4}+q^{25/4}+\ldots);$$
$$\theta_2(0,\tau)=1-2q+2q^4-2q^9+\ldots;$$
$$\theta_3(0,\tau)=1+2q+2q^4+2q^9+\ldots;$$

while $\theta(0,\tau)=0$, and

$$\theta'(0,\tau)=2\pi(q^{1/4}-3q^{9/4}+5q^{25/4}-7q^{49/4}+\ldots).$$

These expansions, and identities involving them, form the source of a number of arithmetical results.

The four sigma-functions were of use in obtaining explicit expressions for $\wp(u)-e_r$, $r=1,2,3$ (see (2.5) and (2.7)). They are, however, connected with the four theta-functions, by Theorems 2 and 3. Combining them, we obtain

Theorem 4. *If* $e_1=\wp\left(\dfrac{\omega_1}{2}\right)$, $e_2=\wp\left(\dfrac{\omega_2}{2}\right)$, *and* $e_3=\wp\left(\dfrac{\omega_1+\omega_2}{2}\right)$, *where* (ω_1,ω_2) *is a pair of primitive periods of the* $\wp$*-function, we have*

$$e_1-e_2=\left(\frac{\theta_3(0,\tau)\cdot\theta'(0,\tau)}{\theta_1(0,\tau)\cdot\theta_2(0,\tau)\cdot\omega_1}\right)^2,\qquad \tau=\frac{\omega_2}{\omega_1},\tag{3.8}$$

$$e_1-e_3=\left(\frac{\theta_2(0,\tau)\cdot\theta'(0,\tau)}{\theta_1(0,\tau)\cdot\theta_3(0,\tau)\cdot\omega_1}\right)^2,\tag{3.9}$$

and

$$e_3-e_2=\left(\frac{\theta_1(0,\tau)\cdot\theta'(0,\tau)}{\theta_3(0,\tau)\cdot\theta_2(0,\tau)\cdot\omega_1}\right)^2.\tag{3.10}$$

Proof. By (2.5) and (2.7), we have

$$\wp(u)-e_r=\left\{\frac{\sigma_r(u)}{\sigma(u)}\right\}^2,\qquad r=1,2,3,\tag{3.11}$$

and by Theorems 2 and 3, we have

$$\left\{\frac{\sigma_r(u)}{\sigma(u)}\right\}^2 = \left(\frac{\theta_r\left(\frac{u}{\omega_1},\tau\right)}{\theta\left(\frac{u}{\omega_1},\tau\right)}\right)^2 \cdot \left\{\frac{\theta'(0,\tau)}{\omega_1 \cdot \theta_r(0,\tau)}\right\}^2, \qquad r=1,2,3. \tag{3.12}$$

Hence, with $u=\frac{\omega_1}{2}$, and $r=2$, in (3.11) we have

$$e_1-e_2=\left(\frac{\theta_2(\frac{1}{2},\tau)\cdot\theta'(0,\tau)}{\theta(\frac{1}{2},\tau)\cdot\omega_1\cdot\theta_2(0,\tau)}\right)^2=\left(\frac{\theta_3(0,\tau)\cdot\theta'(0,\tau)}{\theta_1(0,\tau)\cdot\theta_2(0,\tau)\cdot\omega_1}\right)^2,$$

because $\theta(\frac{1}{2},\tau)=\theta_1(0,\tau)$, $\theta_2(\frac{1}{2},\tau)=\theta_3(1,\tau)=\theta_3(0,\tau)$, and this is (3.8). To obtain (3.9) we proceed in a similar fashion, and use the relations: $\theta_3(\frac{1}{2},\tau)=\theta_2(0,\tau)$, $\theta(\frac{1}{2},\tau)=\theta_1(0,\tau)$. In the case of (3.10), we have to use: $\theta_2(\frac{1}{2}+\frac{1}{2}\tau,\tau)=\theta_3(1+\frac{1}{2}\tau,\tau)=$ $=\theta_3(\frac{1}{2}\tau,\tau)=q^{-1/4}\theta_1(0,\tau)$, and $\theta(\frac{1}{2}+\frac{1}{2}\tau,\tau)=\theta_3(0,\tau)\cdot q^{-1/4}$.

Remarks. We have seen in Theorem 1 that the *only* zeros of $\theta(v,\tau)$ are located at $v=m+n\tau$, where m and n are integers, and that all of them are *simple*. By (3.4) it follows that for arbitrary integers m and n,

$$\theta_1\left(m+n\tau+\frac{1}{2},\tau\right)=0, \qquad \theta_2\left(m+n\tau+\frac{\tau}{2},\tau\right)=0,$$

$$\theta_3\left(m+n\tau+\frac{1}{2}+\frac{\tau}{2},\tau\right)=0. \tag{3.13}$$

These relations give *all* the zeros of the respective theta-functions, which are *all simple*.

§ 4. The differential equation. By definition, we have

$$\theta_3(v,\tau)=\sum_{n=-\infty}^{\infty} e^{\pi i n^2\tau+2\pi i n v}, \qquad \operatorname{Im}\tau>0,$$

which, as we have seen, is an *entire* function of v. By differentiating twice, with respect to v, we get

$$\frac{\partial^2}{\partial v^2}\{\theta_3(v,\tau)\}=-4\pi^2\sum_{n=-\infty}^{\infty} n^2 e^{\pi i n^2\tau+2\pi i n v}.$$

We can also differentiate the series for $\theta_3(v,\tau)$, term by term, with respect to τ, since the differentiated series converges uniformly, so that

$$\frac{\partial}{\partial\tau}\{\theta_3(v,\tau)\}=\pi i\sum_{n=-\infty}^{\infty} n^2 e^{\pi i n^2\tau+2\pi i n v}.$$

Hence

$$\frac{\partial^2}{\partial v^2}\{\theta_3(v,\tau)\}-4\pi i\frac{\partial}{\partial\tau}\{\theta_3(v,\tau)\}=0. \tag{4.1}$$

A similar equation is satisfied by the functions θ_1, θ_2, and θ.

§ 5. Jacobi's formula for $\theta'(0,\tau)$

Theorem 5. *We have the relation*

$$\theta'(0,\tau)=\pi\theta_1(0,\tau)\cdot\theta_2(0,\tau)\cdot\theta_3(0,\tau), \tag{5.1}$$

where $\theta'(v,\tau)$ denotes the derivative of $\theta(v,\tau)$ with respect to v.

Proof. It is sufficient to prove that

$$e_r=\frac{1}{\omega_1^2}\left(\frac{1}{3}\frac{\theta'''(0,\tau)}{\theta'(0,\tau)}-\frac{\theta_r''(0,\tau)}{\theta_r(0,\tau)}\right),\qquad r=1,2,3, \tag{5.2}$$

for it would then follow, from the known relation $e_1+e_2+e_3=0$, (see the proof of Theorem 5, Ch. III) that

$$\frac{\theta'''(0,\tau)}{\theta'(0,\tau)}=\frac{\theta_1''(0,\tau)}{\theta_1(0,\tau)}+\frac{\theta_2''(0,\tau)}{\theta_2(0,\tau)}+\frac{\theta_3''(0,\tau)}{\theta_3(0,\tau)}. \tag{5.3}$$

From § 4, however, we see that

$$\frac{\partial^2}{\partial v^2}\{f(v,\tau)\}-4\pi i\frac{\partial}{\partial\tau}\{f(v,\tau)\}=0,\qquad f=\theta,\theta_1,\theta_2,\theta_3, \tag{5.4}$$

so that

$$\theta_r''(0,\tau)=4\pi i\frac{\partial\theta_r(0,\tau)}{\partial\tau},$$

$$\theta'''(0,\tau)=4\pi i\frac{\partial}{\partial v}\cdot\frac{\partial\theta(0,\tau)}{\partial\tau}=4\pi i\frac{\partial}{\partial\tau}\theta'(0,\tau).$$

Hence (5.3) gives

$$\frac{1}{\theta'(0,\tau)}\cdot\frac{\partial}{\partial\tau}\theta'(0,\tau)=\frac{1}{\theta_1(0,\tau)}\cdot\frac{\partial\theta_1(0,\tau)}{\partial\tau}+\frac{1}{\theta_2(0,\tau)}\cdot\frac{\partial\theta_2(0,\tau)}{\partial\tau}+$$
$$+\frac{1}{\theta_3(0,\tau)}\cdot\frac{\partial\theta_3(0,\tau)}{\partial\tau}.$$

By integration, we get

$$\theta'(0,\tau)=c\cdot\theta_1(0,\tau)\cdot\theta_2(0,\tau)\cdot\theta_3(0,\tau),$$

where c is a constant independent of τ. The expansions for $\theta'(0,\tau)$, $\theta_1(0,\tau)$, $\theta_2(0,\tau)$, and $\theta_3(0,\tau)$ given after Theorem 3 lead to the relation

$$2\pi(q^{1/4}-\ldots)=c(2q^{1/4}+\ldots)(1-\ldots)(1+\ldots),$$

so that $c=\pi$, and (5.1) is proved on the assumption of (5.2). To prove (5.2), we use the formula

$$\wp(u)-e_r=\left(\frac{\theta_r\left(\frac{u}{\omega_1},\tau\right)}{\theta\left(\frac{u}{\omega_1},\tau\right)}\cdot\frac{\theta'(0,\tau)}{\omega_1\theta_r(0,\tau)}\right)^2, \qquad r=1,2,3, \tag{5.5}$$

given by (3.11) and (3.12). On setting $u=v\omega_1$, we get

$$\wp(v\omega_1)-e_r=\left(\frac{1}{\omega_1}\right)^2\left(\frac{\theta_r(v,\tau)/\theta_r(0,\tau)}{\theta(v,\tau)/\theta'(0,\tau)}\right)^2, \qquad r=1,2,3.$$

In a neighbourhood of $v=0$ we have the expansions

$$\theta_r(v,\tau)=\theta_r(0,\tau)+\theta_r'(0,\tau)v+\frac{\theta_r''(0,\tau)}{2!}v^2+\ldots,$$

and

$$\theta(v,\tau)=\theta(0,\tau)+\theta'(0,\tau)v+\frac{\theta''(0,\tau)}{2!}v^2+\ldots,$$

so that

$$\frac{\theta_r(v,\tau)}{\theta_r(0,\tau)}=1+\frac{\theta_r''(0,\tau)}{2!}\frac{v^2}{\theta_r(0,\tau)}+\ldots, \qquad r=1,2,3,$$

since $\theta_r(v,\tau)$ is an *even*, entire, function of v, and

$$\frac{\theta(v,\tau)}{\theta'(0,\tau)}=v+\frac{\theta'''(0,\tau)}{3!}\frac{v^3}{\theta'(0,\tau)}+\ldots,$$

since $\theta(0,\tau)=0$, and $\theta(v,\tau)$ is an *odd*, entire, function of v. It follows that

$$\begin{aligned}\wp(v\omega_1)-e_r&=\left(\frac{1}{\omega_1}\right)^2\left(\frac{1+\frac{\theta_r''(0,\tau)}{2!}\cdot\frac{v^2}{\theta_r(0,\tau)}+\ldots}{v+\frac{\theta'''(0,\tau)}{3!}\cdot\frac{v^3}{\theta'(0,\tau)}+\ldots}\right)^2\\&=\left[\frac{1}{\omega_1 v}\left\{1+\left(\frac{\theta_r''(0,\tau)}{\theta_r(0,\tau)}-\frac{1}{3}\frac{\theta'''(0,\tau)}{\theta'(0,\tau)}\right)\frac{v^2}{2}+\ldots\right\}\right]^2,\end{aligned}$$

in a neighbourhood of $v=0$. But we know that the expansion of $\wp(v\omega_1)$ near $v=0$ has *no* constant term, hence $-e_r$ *is* the constant term in the expansion on the right-hand side, which proves (5.2), and hence the theorem.

Theorems 4 and 5 imply

Corollary 1. *We have*

$$e_1-e_2=\left(\frac{\pi}{\omega_1}\right)^2\theta_3^4(0,\tau);\qquad e_1-e_3=\left(\frac{\pi}{\omega_1}\right)^2\theta_2^4(0,\tau);$$

$$e_3-e_2=\left(\frac{\pi}{\omega_1}\right)^2\theta_1^4(0,\tau). \tag{5.6}$$

This leads us to

Corollary 2. *If*

$$\Delta\equiv\Delta(\omega_1,\omega_2)=g_2^3-27g_3^2=16(e_1-e_2)^2(e_2-e_3)^2(e_3-e_1)^2,$$

then we have

$$\Delta^{1/4}=\frac{2\pi}{\omega_1^3}\{\theta'(0,\tau)\}^2\neq 0,\quad \textit{for } \operatorname{Im}\tau>0. \tag{5.7}$$

(Theorem 5, Ch. III).

§ 6. The infinite products for the theta-functions. The four theta-functions $\theta(v,\tau)$, $\theta_1(v,\tau)$, $\theta_2(v,\tau)$, and $\theta_3(v,\tau)$ defined by infinite *series* are entire functions of v. They can be expressed also by infinite *products* which exhibit their zeros.

Theorem 6. *We have*

$$\theta_3(v,\tau)=c\cdot\prod_{n=1}^{\infty}(1+q^{2n-1}e^{2\pi iv})\cdot\prod_{n=1}^{\infty}(1+q^{2n-1}e^{-2\pi iv}), \tag{6.1}$$

$$\theta_2(v,\tau)=c\cdot\prod_{n=1}^{\infty}(1-q^{2n-1}e^{2\pi iv})\cdot\prod_{n=1}^{\infty}(1-q^{2n-1}e^{-2\pi iv}), \tag{6.2}$$

$$\theta_1(v,\tau)=c\cdot q^{1/4}\cdot e^{\pi iv}\cdot\prod_{n=1}^{\infty}(1+q^{2n}e^{2\pi iv})\cdot\prod_{n=1}^{\infty}(1+q^{2n-2}e^{-2\pi iv}), \tag{6.3}$$

$$\theta(v,\tau)=c\cdot q^{1/4}\cdot 2\sin\pi v\cdot\prod_{n=1}^{\infty}(1-q^{2n}e^{2\pi iv})\cdot\prod_{n=1}^{\infty}(1-q^{2n}e^{-2\pi iv}), \tag{6.4}$$

where

$$c=\prod_{n=1}^{\infty}(1-q^{2n}),\qquad q=e^{\pi i\tau},\qquad \operatorname{Im}\tau>0. \tag{6.5}$$

Proof. Define

$$f(v)=\prod_{n=1}^{\infty}(1+q^{2n-1}e^{2\pi iv})\cdot\prod_{n=1}^{\infty}(1+q^{2n-1}e^{-2\pi iv}).$$

Clearly both the products converge absolutely, and uniformly for $|v|\leqq\varrho<\infty$, since $\sum_{n=1}^{\infty}|q^{2n-1}|<\infty$, so that $f(v)$ is an entire function of v. It has the same zeros as $\theta_3(v,\tau)$, all of them *simple*, given by $v=m+n\tau+\frac{1}{2}+\frac{1}{2}\tau$, where m and n are

integers (positive, negative, or zero). It has the further properties

$$f(v+1)=f(v),$$

and

$$f(v+\tau)=\prod_{n=1}^{\infty}(1+q^{2n+1}e^{2\pi iv})\cdot\prod_{n=1}^{\infty}(1+q^{2n-3}e^{-2\pi iv})$$

$$=\frac{f(v)}{(1+qe^{2\pi iv})}\cdot(1+q^{-1}e^{-2\pi iv})=f(v)\cdot q^{-1}\cdot e^{-2\pi iv},$$

similar to those of $\theta_3(v,\tau)$ given by (3.6). It follows that the meromorphic function $\theta_3(v,\tau)/f(v)$ has the periods 1 and τ but *no* poles, and therefore reduces to a constant c (independent of v), hence $\theta_3(v,\tau)=c\cdot f(v)$, which is (6.1), except for the determination of c.

We have, in particular,

$$\theta_3(0,\tau)=c\cdot\prod_{n=1}^{\infty}(1+q^{2n-1})^2, \tag{6.6}$$

(where $c\neq 0$, since $\theta_3(0,\tau)\neq 0$). Since $\theta_2(v,\tau)=\theta_3(v+\frac{1}{2},\tau)$ by (3.5), we obtain (6.2), and

$$\theta_2(0,\tau)=\theta_3\left(\frac{1}{2},\tau\right)=c\cdot\prod_{n=1}^{\infty}(1-q^{2n-1})^2. \tag{6.7}$$

Since $\theta_1(v,\tau)=\theta_3(v+\frac{1}{2}\tau,\tau)\cdot q^{1/4}\cdot e^{\pi iv}$ by (3.5), we obtain (6.3), and

$$\theta_1(0,\tau)=2c\cdot q^{1/4}\cdot\prod_{n=1}^{\infty}(1+q^{2n})^2. \tag{6.8}$$

Since $\theta(v,\tau)=-\theta(-v,\tau)=-\theta_1(v+\frac{1}{2},\tau)$ by (3.4), and (1.3), we have from (6.3),

$$\theta(v,\tau)=-c\cdot q^{1/4}\cdot e^{\pi iv}\cdot i\cdot\prod_{n=1}^{\infty}(1-q^{2n}e^{2\pi iv})\times$$

$$\cdot\prod_{n=1}^{\infty}(1-q^{2n-2}e^{-2\pi iv})$$

$$=-c\cdot q^{1/4}\cdot e^{\pi iv}\cdot i\cdot\prod_{n=1}^{\infty}(1-q^{2n}e^{2\pi iv})\times$$

$$\cdot\prod_{n=1}^{\infty}(1-q^{2n}e^{-2\pi iv})(1-e^{-2\pi iv})$$

$$=-c\cdot q^{1/4}\cdot i(e^{\pi iv}-e^{-\pi iv})\cdot\prod_{n=1}^{\infty}(1-q^{2n}e^{2\pi iv})\times$$

$$\cdot\prod_{n=1}^{\infty}(1-q^{2n}e^{-2\pi iv})$$

$$=c\cdot q^{1/4}\cdot 2\sin\pi v\cdot\prod_{n=1}^{\infty}(1-q^{2n}e^{2\pi iv})\times$$

$$\cdot\prod_{n=1}^{\infty}(1-q^{2n}e^{-2\pi iv}),$$

which is (6.4), and this implies that

$$\theta'(0,\tau)=2\pi c\cdot q^{1/4}\cdot\prod_{n=1}^{\infty}(1-q^{2n})^2\,. \tag{6.9}$$

In order to determine the constant c in (6.1), we make use of Jacobi's formula (5.1), namely $\theta'(0,\tau)=\pi\theta_1(0,\tau)\cdot\theta_2(0,\tau)\cdot\theta_3(0,\tau)$. Each of the factors here has an infinite product representation given respectively by (6.9), (6.8), (6.7), and (6.6). Hence we have

$$2\pi c\cdot q^{1/4}\cdot\prod_{n=1}^{\infty}(1-q^{2n})^2=2\pi c^3 q^{1/4}\cdot\prod_{n=1}^{\infty}(1+q^{2n})^2\times$$
$$\cdot\prod_{n=1}^{\infty}(1-q^{2n-1})^2\cdot\prod_{n=1}^{\infty}(1+q^{2n-1})^2\,,$$

that is

$$\prod_{n=1}^{\infty}(1-q^{2n})^2=c^2\prod_{n=1}^{\infty}(1+q^{2n})^2\cdot\prod_{n=1}^{\infty}(1+q^{2n-1})^2\cdot\prod_{n=1}^{\infty}(1-q^{2n-1})^2\,,$$

since $c\neq 0$. We have, however,

$$\prod_{n=1}^{\infty}(1-q^{2n-1})\cdot\prod_{n=1}^{\infty}(1+q^{2n-1})\cdot\prod_{n=1}^{\infty}(1+q^{2n})=1\,,$$

since

$$\left\{\prod_{n=1}^{\infty}(1-q^{2n-1})\cdot\prod_{n=1}^{\infty}(1-q^{2n})\right\}\cdot\left\{\prod_{n=1}^{\infty}(1+q^{2n-1})\cdot(1+q^{2n})\right\}$$
$$=\prod_{n=1}^{\infty}(1-q^{n})\cdot\prod_{n=1}^{\infty}(1+q^{n})=\prod_{n=1}^{\infty}(1-q^{2n})\,,$$

all the products converging absolutely. Hence

$$c^2=\prod_{n=1}^{\infty}(1-q^{2n})^2\,,\quad\text{or}\quad c=\pm\prod_{n=1}^{\infty}(1-q^{2n})\,.$$

To determine the sign, we observe that the product on the right-hand side is holomorphic for $|q|<1$. From (6.6) we see that $\theta_3(0,\tau)\to c$ as $q\to 0$. Since $\theta_3(0,\tau)=1+2q+2q^4+\ldots$, we also see that $\theta_3(0,\tau)\to+1$ as $q\to 0$. Hence $c\to+1$ as $q\to 0$, which leads us to choose the plus sign, and (6.5) is proved.

Theorem 6 leads to an important representation of the function $\Delta(\omega_1,\omega_2)$, introduced in § 3 of Chapter III, as an infinite product.

Theorem 7. *If (ω_1,ω_2) is a pair of basic periods of Weierstrass's elliptic function $\wp(u)$, with $\tau=\dfrac{\omega_2}{\omega_1}$, $\operatorname{Im}\tau>0$, and $q=e^{\pi i\tau}$, then*

$$\Delta\equiv\Delta(\omega_1,\omega_2)=\left(\frac{2\pi}{\omega_1}\right)^{12}q^2\prod_{n=1}^{\infty}(1-q^{2n})^{24}\,. \tag{6.10}$$

This is an immediate consequence of (5.7), (6.9), and (6.5). It follows from (6.10) that $\Delta \neq 0$; something which we have already seen in (5.7).

§ 7. Theta-functions as solutions of functional equations. We have seen that $\theta(v,\tau)$, $\theta_1(v,\tau)$, and $\theta_2(v,\tau)$ can all be expressed in terms of $\theta_3(v,\tau)$, which is given by the relation

$$\theta_3(v,\tau)=\sum_{n=-\infty}^{\infty} q^{n^2}e^{2n\pi iv}=\sum_{n=-\infty}^{\infty} e^{\pi i\tau n^2+2\pi inv}, \tag{7.1}$$

where $q=e^{\pi i\tau}$, $\operatorname{Im}\tau>0$. For fixed τ, or q, θ_3 is an entire function of v. Further it satisfies the *functional equations:*

$$\theta_3(v+1,\tau)=\theta_3(v,\tau), \qquad \theta_3(v+\tau,\tau)=e^{-\pi i\tau-2\pi iv}\theta_3(v,\tau). \tag{7.2}$$

One can show conversely that any entire function of v which satisfies the functional equations (7.2) must be of the form (7.1), up to a constant factor. In this sense, the theta-function $\theta_3(v,\tau)$ is determined uniquely by its functional equations.

Theorem 8. *If the entire function $f(v)$ satisfies the equations*

$$f(v+1)=f(v), \qquad f(v+\tau)=e^{-\pi i\tau-2\pi iv}f(v), \qquad \operatorname{Im}\tau>0,\ v \text{ complex}, \tag{7.3}$$

then

$$f(v)\equiv f(v,\tau)=a_0\cdot\sum_{n=-\infty}^{\infty} e^{\pi in^2\tau+2\pi inv}, \tag{7.4}$$

where a_0 is a constant.

Proof. We introduce a new variable: $w=e^{2\pi iv}$. The strip $R: b<\operatorname{Im}v<B$ is then transformed into the annulus P formed by two circles with the origin as centre and radii $e^{-2\pi B}$, $e^{-2\pi b}$, and the function $f(v)$ into $\varphi(w)$, say, where

$$\varphi(w)=f(v)=f\left(\frac{1}{2\pi i}\log w\right). \tag{7.5}$$

If f is holomorphic in R, then φ is holomorphic in P. Because $f(v)$ is *periodic*, with period 1, $\varphi(w)$ is uniquely defined in P by (7.5). Since f is, in fact, entire, it follows that $\varphi(w)$ has a Laurent expansion for every finite $w\neq 0$, given by

$$\varphi(w)=\sum_{k=-\infty}^{\infty} a_k w^k=\sum_{k=-\infty}^{\infty} a_k e^{2\pi ikv}=f(v). \tag{7.6}$$

Hence

$$f(v+\tau)=\sum_{k=-\infty}^{\infty} a_k\cdot e^{2\pi ik(v+\tau)}=\sum_{k=-\infty}^{\infty} a_k\cdot e^{2\pi ik\tau}\cdot w^k. \tag{7.7}$$

By (7.3), however, we have

$$f(v+\tau)=e^{-\pi i\tau-2\pi iv}f(v)=\sum_{k=-\infty}^{\infty}a_k\cdot e^{-\pi i\tau}\cdot w^{k-1}=$$

$$=\sum_{k=-\infty}^{\infty}a_{k+1}e^{-\pi i\tau}w^k. \tag{7.8}$$

Because of the uniqueness of the Laurent expansion, we have from (7.7) and (7.8),

$$a_k e^{2\pi ik\tau}=a_{k+1}\cdot e^{-\pi i\tau}.$$

Since $2\pi ik\tau=-\pi i\tau k^2-\pi i\tau+\pi i\tau(k+1)^2$, it follows that

$$a_{k+1}\cdot e^{-\pi i\tau}=a_k\cdot e^{-\pi i\tau k^2}\cdot e^{-\pi i\tau}\cdot e^{\pi i\tau(k+1)^2},$$

or $a_k\cdot e^{-\pi i\tau k^2}=a_{k+1}\cdot e^{-\pi i\tau(k+1)^2}$, so that $a_{-k}=a_0e^{\pi i\tau k^2}=a_k$, for $k=\pm1, \pm2, \ldots$. Hence

$$f(v)=\sum_{k=-\infty}^{\infty}a_kw^k=a_0\sum_{k=-\infty}^{\infty}e^{\pi ik^2\tau+2\pi ikv}.$$

§ 8. The transformation formula connecting $\theta_3(v, \tau)$ and $\theta_3\left(v, -\frac{1}{\tau}\right)$. Having considered the effect of the transformations $v\to v+\tau$, $v\to v+1$, on the function $\theta_3(v, \tau)$, where τ is fixed, we shall now consider the effect of the transformations $\tau\to-\frac{1}{\tau}$, $\tau\to\tau+1$ on $\theta_3(v, \tau)$, where v is fixed. Since the other three theta-functions are expressible in terms of θ_3, analogous results hold for them as well.

Theorem 9. *Let v and τ be complex numbers, with $\operatorname{Im}\tau>0$. Then we have the formula*

$$\theta_3\left(v, -\frac{1}{\tau}\right)=\sqrt{\frac{\tau}{i}}\cdot\sum_{n=-\infty}^{\infty}e^{\pi i\tau(n+v)^2}, \tag{8.1}$$

where, as before, $\theta_3(v, \tau)=\sum_{n=-\infty}^{\infty}e^{\pi in^2\tau+2\pi inv}$, and $\sqrt{\frac{\tau}{i}}=+1$ for $\tau=i$. Hence

$$\sqrt{\frac{\tau}{i}}\,\theta_3(v, \tau)=e^{-\frac{\pi iv^2}{\tau}}\cdot\theta_3\left(\frac{v}{\tau}, -\frac{1}{\tau}\right). \tag{8.2}$$

Proof. For fixed τ, with $\operatorname{Im}\tau>0$, the series $\sum_{n=-\infty}^{\infty}e^{\pi i\tau(n+v)^2}$ converges uniformly in every compact set in the v-plane, and represents therefore an *entire* function of v. This function is *periodic in v*, with period 1, since the substitution $v\to v+1$ is equivalent to the substitution $n\to n+1$, which leaves the series unchanged because of its uniform convergence.

Let us assume at first that v is *real*, and τ *purely imaginary* (with $\mathrm{Im}\,\tau > 0$). Then we have the Fourier expansion

$$\sum_{n=-\infty}^{\infty} e^{\pi i\tau(n+v)^2} = \sum_{k=-\infty}^{\infty} c_k e^{2\pi i k v},$$

in which the coefficient c_k is given by

$$c_k = \int_0^1 \sum_{n=-\infty}^{\infty} e^{\pi i\tau(n+v)^2 - 2\pi i k v} dv.$$

We can write $n+v$ in place of v in the *second* exponent, and interchange integration and summation because of the uniform convergence, so that

$$c_k = \int_{-\infty}^{\infty} e^{\pi i\tau v^2 - 2\pi i k v} dv = e^{-\frac{\pi i k^2}{\tau}} \cdot \int_{-\infty}^{\infty} e^{\pi i\tau\left(v-\frac{k}{\tau}\right)^2} dv.$$

On setting $\tau = iy$, $y > 0$, and

$$s = \sqrt{\frac{\tau}{i}}\left(v - \frac{k}{\tau}\right) = \sqrt{y}\left(v - \frac{k}{iy}\right) = \sqrt{y}\left(v + \frac{ik}{y}\right),$$

we obtain

$$c_k = e^{-\frac{\pi k^2}{y}} \cdot \sqrt{\frac{1}{y}} \cdot \int_{-\infty+\frac{ik}{\sqrt{y}}}^{+\infty+\frac{ik}{\sqrt{y}}} e^{-\pi s^2} ds,$$

where the integration is along a line parallel to the real axis in the complex s-plane. By Cauchy's theorem of residues, this last integral is independent of the ordinate of the line of integration, therefore independent of k. Hence

$$c_k = e^{-\frac{\pi k^2}{y}} \cdot \sqrt{\frac{1}{y}} \cdot \int_{-\infty}^{\infty} e^{-\pi t^2} dt \qquad (t \text{ real})$$

$$= e^{-\frac{\pi k^2}{y}} \cdot \frac{1}{\sqrt{y}} \cdot 1.$$

Hence we obtain the formula

$$\sum_{n=-\infty}^{\infty} e^{\pi i\tau(n+v)^2} = \sum_{k=-\infty}^{\infty} e^{-\pi k^2 \cdot \frac{i}{\tau}} \cdot \sqrt{\frac{i}{\tau}} \cdot e^{2\pi i k v},$$

for $\tau = iy$, $y > 0$, and v real. By analytic continuation, this holds for *all* complex τ, with $\mathrm{Im}\,\tau > 0$, and then, further, for arbitrary complex v, hence (8.1).

If we change τ into $-\frac{1}{\tau}$ in (8.1), as we may, since $\mathrm{Im}\left(-\frac{1}{\tau}\right) > 0$, we obtain

$$\theta_3(v,\tau) = \sqrt{\frac{i}{\tau}} \cdot e^{-\frac{\pi i v^2}{\tau}} \cdot \sum_{n=-\infty}^{\infty} e^{-\frac{\pi i n^2}{\tau} - \frac{2\pi i n v}{\tau}},$$

which is (8.2).

Corollary 1. *If v is complex, $\operatorname{Im}\tau>0$, and $\sqrt{\frac{\tau}{i}}=+1$ for $\tau=i$, we have the transformation formulae:*

$$\sqrt{\frac{\tau}{i}}\,\theta_2(v,\tau)=e^{-\frac{\pi i v^2}{\tau}}\cdot\theta_1\left(\frac{v}{\tau},-\frac{1}{\tau}\right), \tag{8.3}$$

$$\sqrt{\frac{\tau}{i}}\,\theta_1(v,\tau)=e^{-\frac{\pi i v^2}{\tau}}\cdot\theta_2\left(\frac{v}{\tau},-\frac{1}{\tau}\right), \tag{8.4}$$

$$\sqrt{\frac{\tau}{i}}\,\theta(v,\tau)=i\cdot e^{-\frac{\pi i v^2}{\tau}}\cdot\theta\left(\frac{v}{\tau},-\frac{1}{\tau}\right). \tag{8.5}$$

Proof. By (3.5), we can express θ_1, θ_2, and θ in terms of θ_3 by the formulae:

$$\theta_2(v,\tau)=\theta_3(v+\tfrac{1}{2},\tau);\qquad \theta_1(v,\tau)=\theta_3(v+\tfrac{1}{2}\tau,\tau)\cdot e^{\frac{\pi i\tau}{4}}\cdot e^{\pi i v};$$

$$\theta(v,\tau)=\theta_3(v+\tfrac{1}{2}+\tfrac{1}{2}\tau,\tau)\cdot e^{\frac{\pi i\tau}{4}}\cdot e^{\pi i v}\cdot\frac{1}{i}. \tag{8.6}$$

On replacing v by $v+\frac{1}{2}$ in (8.2), we obtain

$$\sqrt{\frac{\tau}{i}}\,\theta_3\left(v+\frac{1}{2},\tau\right)=e^{-\frac{\pi i}{\tau}\left(v+\frac{1}{2}\right)^2}\cdot\theta_3\left(\frac{v+\frac{1}{2}}{\tau},-\frac{1}{\tau}\right)=$$

$$=e^{-\frac{\pi i}{\tau}\left(v+\frac{1}{2}\right)^2}\theta_3\left(\frac{-(v+\frac{1}{2})}{\tau},-\frac{1}{\tau}\right). \tag{8.7}$$

By (8.6), however, we have $\theta_3(v+\frac{1}{2},\tau)=\theta_2(v,\tau)$, and

$$\theta_1\left(-\frac{v}{\tau},-\frac{1}{\tau}\right)=\theta_3\left(-\frac{v}{\tau}-\frac{1}{2\tau},-\frac{1}{\tau}\right)\cdot e^{-\frac{\pi i}{4\tau}}\cdot e^{-\frac{\pi i v}{\tau}}=$$

$$=\theta_3\left(-\frac{(v+\frac{1}{2})}{\tau},-\frac{1}{\tau}\right)\cdot e^{-\frac{\pi i}{4\tau}}\cdot e^{-\frac{\pi i v}{\tau}},$$

so that (8.7) gives

$$\sqrt{\frac{\tau}{i}}\,\theta_2(v,\tau)=e^{-\frac{\pi i}{\tau}\left(v+\frac{1}{2}\right)^2}\cdot e^{\frac{\pi i}{4\tau}}\cdot e^{\frac{\pi i v}{\tau}}\cdot\theta_1\left(-\frac{v}{\tau},-\frac{1}{\tau}\right)=$$

$$=e^{-\frac{\pi i v^2}{\tau}}\cdot\theta_1\left(\frac{v}{\tau},-\frac{1}{\tau}\right),$$

which is (8.3). The proof of (8.4) is similar. To prove (8.5), we note that (8.7) gives

$$\sqrt{\frac{\tau}{i}}\,\theta_3\left(v+\frac{1}{2}+\frac{1}{2}\tau,\tau\right)=$$

$$=e^{-\frac{\pi i}{\tau}\left(v+\frac{1}{2}+\frac{1}{2}\tau\right)^2}\theta_3\left(\frac{v}{\tau}+\frac{1}{2}+\frac{1}{2\tau},-\frac{1}{\tau}\right). \tag{8.8}$$

Here the left-hand side equals

$$\sqrt{\frac{\tau}{i}}\, e^{-\frac{\pi i\tau}{4}} \cdot e^{-\pi i v} \cdot i \cdot \theta(v,\tau)$$

by (8.6). On the right-hand side, we use the formula (3.6)

$$\theta_3(v+\tau,\tau) = \theta_3(v,\tau) \cdot e^{-\pi i\tau} \cdot e^{-2\pi i v},$$

which yields

$$\theta_3\left(\frac{v}{\tau} + \frac{1}{2} + \frac{1}{2\tau} - \frac{1}{\tau}, -\frac{1}{\tau}\right) = \theta_3\left(\frac{v}{\tau} + \frac{1}{2} + \frac{1}{2\tau}, -\frac{1}{\tau}\right) \cdot e^{\frac{\pi i}{\tau}} \cdot e^{-2\pi i\left(\frac{v}{\tau} + \frac{1}{2} + \frac{1}{2\tau}\right)},$$

hence (8.8) gives

$$\sqrt{\frac{\tau}{i}}\, \theta(v,\tau) \cdot e^{-\frac{\pi i\tau}{4}} \cdot e^{-\pi i v} \cdot i = e^{-\frac{\pi i}{\tau}\left(v+\frac{1}{2}+\frac{1}{2}\tau\right)^2} \cdot \theta_3\left(\frac{v}{\tau} + \frac{1}{2} - \frac{1}{2\tau}, -\frac{1}{\tau}\right) \cdot e^{-\frac{\pi i}{\tau}} \cdot e^{+2\pi i\left(\frac{v}{\tau} + \frac{1}{2} + \frac{1}{2\tau}\right)}$$

or

$$\sqrt{\frac{\tau}{i}} \cdot \theta(v,\tau) \cdot i = e^{-\frac{\pi i v^2}{\tau}} \cdot e^{\frac{\pi i v}{\tau}} \cdot e^{-\frac{\pi i}{4\tau}} \cdot e^{-\frac{\pi i}{2} + \pi i} \times \cdot \theta_3\left(\frac{v}{\tau} + \frac{1}{2} - \frac{1}{2\tau}, \frac{-1}{\tau}\right) = e^{-\frac{\pi i v^2}{\tau}}\, \theta\left(\frac{v}{\tau}, -\frac{1}{\tau}\right) \cdot i^2, \quad \text{(by (8.6))}$$

which is (8.5).

Corollary 2. *We have the formulae:*

$$\theta_3(0,\tau) = \sqrt{\frac{i}{\tau}} \cdot \theta_3(0, -1/\tau); \quad \theta_2(0,\tau) = \sqrt{\frac{i}{\tau}} \cdot \theta_1\left(0, -\frac{1}{\tau}\right);$$

$$\theta_1(0,\tau) = \sqrt{\frac{i}{\tau}} \cdot \theta_2\left(0, -\frac{1}{\tau}\right); \quad \sqrt{\frac{\tau}{i}} \cdot \theta'(0,\tau) = \frac{i}{\tau}\, \theta'(0, -1/\tau), \tag{8.9}$$

where $\sqrt{\frac{\tau}{i}} = +1$ *for* $\tau = i$, *and* $\theta'(0,\tau) = \left[\frac{\partial}{\partial v}\, \theta(v,\tau)\right]_{v=0}$, *as well as the formulae:*

$$\theta_3(0,\tau+1) = \theta_2(0,\tau); \quad \theta_2(0,\tau+1) = \theta_3(0,\tau);$$

$$\theta_1(0,\tau+1) = e^{\frac{\pi i}{4}}\, \theta_1(0,\tau). \tag{8.10}$$

These are immediate from (8.2) together with the definitions of θ, θ_1, θ_2, θ_3, and Corollary 1. [Note that $\theta(0, \tau) = 0$, hence the differentiation. Note also that $\theta_r'(0, \tau) = 0$, for $r = 1, 2, 3$, *because of* (3.1)–(3.3)].

Corollary 3. *The function*

$$f(\tau) = \frac{\{\theta_1^8(0, \tau) + \theta_2^8(0, \tau) + \theta_3^8(0, \tau)\}^3}{\{\theta_1(0, \tau) \cdot \theta_2(0, \tau) \cdot \theta_3(0, \tau)\}^8} \tag{8.11}$$

is holomorphic for $\operatorname{Im}\tau > 0$, *and has the property*

$$f(\tau + 1) = f(\tau); \quad f(-1/\tau) = f(\tau). \tag{8.12}$$

Proof. The functions $\theta_1(0, \tau)$, $\theta_2(0, \tau)$, $\theta_3(0, \tau)$ are all holomorphic for $\operatorname{Im}\tau > 0$, and different from zero, since

$$\theta'(0, \tau) = \pi \cdot \theta_1(0, \tau) \cdot \theta_2(0, \tau) \cdot \theta_3(0, \tau), \tag{8.13}$$

where $\theta'(0, \tau) \neq 0$, since all the zeros of $\theta(v, \tau)$ are *simple*. It follows that $f(\tau)$ is holomorphic for $\operatorname{Im}\tau > 0$.

The invariance of $f(\tau)$ under the transformations $\tau \to -\frac{1}{\tau}$ and $\tau \to \tau + 1$ follows from (8.9) and (8.10).

Remark. From (8.12) it follows that $f(\tau)$ is invariant under *all* transformations of the *modular group* (see (5.7) of Ch. I).

There are many different approaches to the function $\frac{1}{54} f(\tau)$, as will be seen in Chapters VI and VII.

Corollary 4. *Let* a, b, c, d *be integers such that* $ad - bc = 1$. *Then we have*

$$\theta'\left(0, \frac{a\tau + b}{c\tau + d}\right) = \kappa(c\tau + d)^{3/2} \theta'(0, \tau), \quad \textit{for } \operatorname{Im}\tau > 0, \tag{8.14}$$

where κ *is some eighth root of unity depending on* a, b, c, d *(but not on* τ*), and the square-root is determined as in* (8.9).

Proof. Let M, A, B denote respectively the modular transformations

$$\tau \to \frac{a\tau + b}{c\tau + d}, \quad \tau \to \tau + 1, \quad \tau \to -\frac{1}{\tau}.$$

Then the relation

$$[\theta'(0, M\tau)]^8 = (c\tau + d)^{12} \{\theta'(0, \tau)\}^8 \tag{8.15}$$

holds for $M = A^{\pm 1}$ and $M = B$, because of (8.13), (8.10), and (8.9). If it holds for an arbitrary modular transformation M, then it holds also for $MA^{\pm 1}$ and MB,

for

$$\{\theta'(0, MA^{\pm 1}\tau)\}^8 = (c\tau \pm c + d)^{12}\{\theta'(0,\tau)\}^8, \quad \text{since}$$

$$MA^{\pm 1}\tau = \frac{a\tau \pm a + b}{c\tau \pm c + d}, \quad \text{and}$$

$$\{\theta'(0, MB\tau)\}^8 = (d\tau - c)^{12}\{\theta'(0,\tau)\}^8, \quad \text{since}$$

$$MB\tau = \frac{b\tau - a}{d\tau - c}.$$

Since M is generated by A and B, and $B^{-1} = B$, it follows that (8.15) holds in general. Taking the eighth root, and noting that $(c\tau + d)^{1/2}$, with the chosen determination of the square-root, is holomorphic for $\mathrm{Im}\,\tau > 0$, we obtain the Corollary.

Notes on Chapter V

For historical references concerning theta-functions, see A. Enneper, *Elliptische Functionen* (zweite Auflage, Neu bearbeitet und herausgegeben von Felix Müller, Halle, 1890), §§ 17–20, as well as E.T. Whittaker and G.N. Watson, *A course of modern analysis* (4-th edn., 1952), § 21.1, who say that the "first function of the Theta-function type to appear in Analysis was the Partition function $\prod_{n=1}^{\infty}(1-x^n z)^{-1}$ of Euler, *Introductio in Analysis Infinitorum*, I (Lausanne, 1748), § 304." See Euler, *Opera Omnia*, (1), VIII (Leipzig, 1912).

Theta-functions occur in J. Fourier's *La Théorie Analytique de la Chaleur* (Paris, 1822), § 241; German translation by B. Weinstein (Julius Springer, 1884), § 241, p. 176; *Oeuvres*, I (1888), § 24. See also *Joseph Fourier 1768–1830*, by I. Grattan-Guinness (M.I.T., 1972), 278.

The *theory* of theta-functions was developed from the theory of elliptic functions by Jacobi in his *Fundamenta Nova* (cf. Notes, Ch. I, § 4). The functions studied in this Chapter were introduced by him in his subsequent lectures, an account of which (1838) is given by Borchardt in his *Ges. Werke*, I, 497–538. Whittaker and Watson say that "the most important results contained in them seem to have been discovered in 1835. cf. L. Kronecker, *Sitzungsberichte der Akad. Berlin* (1891), *pp.* 653–659." See André Weil's account of Kronecker's contributions (in his book cited in the Notes on Ch. IV, §§ 1–2).

§§ 1–3. We have defined the theta-functions by infinite series, and obtained expressions for them as infinite products in § 6. This procedure, of course, can be reversed. See, for instance, H.A. Schwarz, *Formeln und Lehrsätze zum Gebrauche der elliptischen Funktionen* (Julius Springer, Berlin, 1893), § 34.

Hermite defines more general theta-functions $\theta_{\mu,\nu}$ by the series

$$\sum_{m=-\infty}^{\infty}(-1)^{m\nu}e^{\pi i\left[(2m+\mu)x+\frac{1}{4}\tau(2m+\mu)^2\right]}$$

for $\mu=0,1$ and $\nu=0,1$. See *Liouville's J.* (2) 3 (1858), 26–36; *Oeuvres*, I, 487–496; Schwarz (loc. cit.), § 34.

Most of the results of this Chapter are due to Jacobi, who obtained them by purely algebraic methods. See Enneper (loc. cit.), §§ 19–20. The systematic use of Cauchy's theorem of residues in the proofs seems to have begun with Liouville (cf. Notes on Ch. II).

The notation used here for the four theta-functions, namely $\theta(v)\equiv\theta(v,\tau)$, $\theta_1(v)$, $\theta_2(v)$, $\theta_3(v)$, is intended to relate them, in a convenient manner, to the four sigma-functions $\sigma, \sigma_1, \sigma_2, \sigma_3$. It is the same as C. Jordan's in his *Cours d'Analyse*, 3ième edn., I, II, III (1909–1915). Jacobi uses $\vartheta_1(\pi v)$, $\vartheta_2(\pi v)$, $\vartheta(\pi v)$, $\vartheta_3(\pi v)$; Weierstrass uses $\theta_1(v)$, $\theta_2(v)$, $\theta_0(v)$, $\theta_3(v)$; Tannery and Molk use $\vartheta_1(v)$, $\vartheta_2(v)$, $\vartheta_4(v)$, $\vartheta_3(v)$.

Theta-functions are more convenient for computational purposes than the sigma-functions. See Schwarz (loc. cit.), §§ 35, 40–46, 50.

§ 4. Jacobi found the differential equation not from the expansions of the theta-functions in series but by considering elliptic integrals, *Crelle's J.* 26, 80–87; *Ges. Werke*, II, 161–170. See Enneper (loc. cit.), p. 127, § 17.

§ 5. Theorem 5, which is due to Jacobi, is of fundamental importance. See his *Fundamenta Nova*, p. 90; *Ges. Werke*, I, 49–239. His original proof is different, *Ges. Werke*, I, 515–517. The proof given here is due to Hurwitz, and uses the differential equation satisfied by the theta-functions. Whittaker and Watson (loc. cit.), § 21. 41, give a proof based on the infinite products for the theta-functions, obtained in § 6, together with the differential equation.

§ 6. Jacobi gives the products in his *Fundamenta Nova*, p. 145; *Ges. Werke*, I, 49–239.

§ 7. Theorem 8 is a special case of a theorem on theta-functions of several variables proved, for instance, by C.L. Siegel in his *Ausgewählte Fragen der Funktionentheorie*, II, Kap. III, Satz 4. Theta-functions of several variables are basic to the theory of Abelian functions with which the names of Weierstrass, Riemann, and Hermite, are associated.

§ 8. The transformation formulae given in (8.2)–(8.5) were first obtained by Jacobi from the theory of elliptic functions, *J. für Math.* III (1828), 403–404; *Ges. Werke*, I, (1881), 264–265.

The case $v=0$ of (8.2) occurs in the work of Poisson, *J. de l'École polytechnique*, XII, Ch. 19, p. 420. Abel remarks, however, (*Crelle's J.* IV, p. 93; *Oeuvres*, 2-nd edn., I, p. 477) that A.L. Cauchy had given the formula in his *Exercises de mathématiques*. Enneper (loc. cit., p. 132) comments: "Die von Poisson befolgte Methode ist indess später als nicht ganz einwurfsfrei erfunden worden." (The method followed by Poisson, however, was later found to be not totally free from objections). Point is added to this by Grattan-Guinness (loc. cit.), 466–471. See A.L. Cauchy, *Bull. de la Société Philomatique* (1817), 121–124; (1818), 181–188; *Oeuvres* (2) II, 223–232. Cauchy says (p. 229) that he

had been unaware, until then, of the results presented by Fourier in 1807 and 1811, which had remained unedited; but mentions Poisson.

There exist many different proofs of Theorem 9. The first direct proof by means of contour integration seems to be due to G. Landsberg, *J. für Math.* 111 (1893), 235–258.

Whittaker and Watson (loc. cit.), § 21.51, prove (8.2) by an application of Liouville's theorem.

For transformation formulae of the type (8.14) relating $\theta(v, \tau)$ and

$$\theta\left(\frac{v}{c\tau+d}, \frac{a\tau+b}{c\tau+d}\right),$$

see, for instance, J. Tannery and J. Molk, *Éléments de la Théorie des Fonctions Elliptiques*, II, Ch. III, §§176—217.

The determination of the eighth root of unity in (8.14) is due to Hermite. See his letter to Tannery in his *Oeuvres*, I, 487–496; also Tannery and Molk (loc. cit.), IV, 294–303.

If we write $\theta'(0, \tau) = 2\pi\eta^3(\tau)$, then formula (8.14) yields the relation

$$(*) \qquad \eta\left(\frac{a\tau+b}{c\tau+d}\right) = \kappa^{1/3}(c\tau+d)^{1/2}\eta(\tau),$$

as was noted by Dedekind in *Bernhard Riemann's gesammelte mathematische Werke und wissenschaftlicher Nachlass*, 2. Aufl. (1892), 466–478; *Ges. Werke*, I, (1930), 159–173. We shall see in Chapter VIII that the function $\eta(\tau)$ can be defined by an infinite product, and the transformation formula (*) proved by Siegel's method without recourse to theta-functions (as Dedekind had wished to do, *Ges. Werke*, I, p. 195). See the Notes on Chapter VIII.

Chapter VI

The modular function $J(\tau)$

§ 1. Definition of $J(\tau)$. A complex-valued function $F(z)$ of the complex variable z is said to be a *modular function*, if it is meromorphic for $\mathrm{Im}\, z>0$, and $F(Mz)=F(z)$ for all transformations M belonging to the *modular group* (defined in § 5 of Chapter I), or for all M belonging to a sub-group of the modular group of finite index.

If $\wp(u)$ denotes Weierstrass's elliptic function, with (ω_1, ω_2) as a pair of basic periods, where $\mathrm{Im}\dfrac{\omega_2}{\omega_1}>0$, the *invariants* g_2 and g_3 associated with it are uniquely defined by the relations

$$g_2 \equiv g_2(\omega_1, \omega_2) = 60 \sum_{\substack{m,n=-\infty \\ (m,n)\neq(0,0)}}^{\infty} (m\omega_1 + n\omega_2)^{-4}, \tag{1.1}$$

and

$$g_3 \equiv g_3(\omega_1, \omega_2) = 140 \sum_{\substack{m,n=-\infty \\ (m,n)\neq(0,0)}}^{\infty} (m\omega_1 + n\omega_2)^{-6}, \tag{1.2}$$

and if

$$\Delta \equiv \Delta(\omega_1, \omega_2) = g_2^3 - 27 g_3^2, \tag{1.3}$$

then, as has been shown in Theorem 5 of Chapter III, we have $\Delta \neq 0$.

If we set $\tau = \dfrac{\omega_2}{\omega_1}$, so that $\mathrm{Im}\,\tau>0$, we see that

$$g_2(\omega_1, \omega_2) = \omega_1^{-4} g_2(1, \tau);\ g_3(\omega_1, \omega_2) = \omega_1^{-6} g_3(1, \tau);\ \Delta(\omega_1, \omega_2) = \omega_1^{-12} \Delta(1, \tau). \tag{1.4}$$

We shall write

$$\begin{gathered} g_2(\tau) = g_2(1, \tau) = 60 \sum{}'(m+n\tau)^{-4};\ g_3(\tau) = g_3(1, \tau) = 140 \sum{}'(m+n\tau)^{-6}; \\ \Delta(\tau) = \Delta(1, \tau); \end{gathered} \tag{1.5}$$

thereby *distinguishing* between g_2 and $g_2(\tau)$, g_3 and $g_3(\tau)$, and between Δ and $\Delta(\tau)$.

The function $J(\tau)$ is defined, for $\mathrm{Im}\,\tau>0$, by the relation

$$J(\tau) = \frac{g_2^3(\tau)}{\Delta(\tau)} = \frac{g_2^3(1, \tau)}{\Delta(1, \tau)} = \frac{g_2^3(\omega_1, \omega_2)}{\Delta(\omega_1, \omega_2)}. \tag{1.6}$$

We shall see that it is invariant for all transformations of the modular group Γ. If F_0 is a *fundamental domain* for Γ, (defined in § 5 of Chapter I), then the equation $J(\tau)-a=0$, where a is any given complex number, has exactly one solution in F_0. As a consequence of this fact, one can show that given any two complex numbers a_2, a_3, such that $a_2^3-27a_3^2\neq 0$, there exists a pair of complex numbers (ω_1,ω_2), with $\operatorname{Im}\dfrac{\omega_2}{\omega_1}>0$, such that

$$g_2(\omega_1,\omega_2)=60\sum{}'(m\omega_1+n\omega_2)^{-4}=a_2,$$

and

$$g_3(\omega_1,\omega_2)=140\sum{}'(m\omega_1+n\omega_2)^{-6}=a_3;$$

hence there exists an elliptic function $\wp(u;\omega_1,\omega_2)$, with (ω_1,ω_2) as a pair of basic periods, for which a_2, a_3 *are* the invariants. This proposition enables one to connect elliptic functions with elliptic integrals.

§ 2. The functions $g_2(\tau)$ and $g_3(\tau)$. In order to study the properties of $J(\tau)$ it is useful to obtain expansions, in powers of $e^{2\pi i\tau}$, for the functions $g_2(\tau)$ and $g_3(\tau)$ occurring in its definition. As a first step we prove

Lemma 1. *For any integer k, with $k\geqq 2$, we have the relation*

$$\sum_{\substack{m,n=-\infty\\(m,n)\neq(0,0)}}^{\infty}\frac{1}{(n\tau+m)^{2k}}=2\zeta(2k)+\frac{2(-2\pi i)^{2k}}{(2k-1)!}\sum_{l=1}^{\infty}\sigma_{2k-1}(l)q^{2l},\qquad(2.1)$$

where $\zeta(2k)=\sum_{n=1}^{\infty}n^{-2k}$, and $\sigma_{2k-1}(l)=\sum_{d|l}d^{2k-1}$, where d runs over all positive divisors of the positive integer l, $\operatorname{Im}\tau>0$, and $q=e^{\pi i\tau}$. The series on either side of (2.1) *converges absolutely.*

(Here ζ stands for Riemann's zeta-function, not Weierstrass's, and this arithmetical function σ has nothing to do with any of Weierstrass's σ-functions).

Proof. If $\varrho>1$, and $\operatorname{Re}z>0$, we have

$$\sum_{m=1}^{\infty}m^{\varrho-1}e^{-mz}=\Gamma(\varrho)\cdot\sum_{l=-\infty}^{\infty}(z+2\pi il)^{-\varrho},\qquad(2.2)$$

where the series on either side converges absolutely. This can be proved in many ways, one of which is as follows.

Let $f(y)=y^{\varrho-1}e^{-yz}$ for $y>0$, $\operatorname{Re}z>0$, and $f(y)=0$, for $y\leqq 0$, and $g(y)=\sum_{m=-\infty}^{\infty}f(y+m)$. Then $g(y)$ is periodic, with period 1, and its Fourier

series at $y=0$ is given by $\sum_{l=-\infty}^{\infty} a_l$, where

$$a_l = \int_0^1 g(y)e^{-2\pi lyi}dy = \int_0^1 \sum_{m=-\infty}^{\infty} f(y+m)e^{-2\pi lyi}dy =$$
$$= \int_0^1 \sum_{m=0}^{\infty} f(y+m)e^{-2\pi lyi}dy = \int_0^{\infty} y^{\varrho-1}e^{-(z+2\pi il)y}dy =$$
$$= \Gamma(\varrho)\cdot(z+2\pi il)^{-\varrho}.$$

On setting $z=-2\pi i\tau$, $\operatorname{Im}\tau>0$, $q=e^{\pi i\tau}$, $\varrho=k$, k integral, $k\geqq 2$, in (2.2), we obtain

$$\sum_{m=1}^{\infty} m^{k-1}q^{2m} = \Gamma(k)\cdot(-2\pi i)^{-k}\sum_{l=-\infty}^{\infty}(\tau-l)^{-k} = \Gamma(k)\cdot(-2\pi i)^{-k}\cdot\sum_{l=-\infty}^{\infty}(\tau+l)^{-k}. \tag{2.3}$$

Now

$$\sum_{\substack{m,n=-\infty\\(m,n)\neq(0,0)}}^{\infty}(n\tau+m)^{-2k} = 2\sum_{m=1}^{\infty}m^{-2k} + 2\sum_{n=1}^{\infty}\sum_{m=-\infty}^{\infty}(n\tau+m)^{-2k} =$$
$$= 2\zeta(2k) + \frac{2(-2\pi i)^{2k}}{\Gamma(2k)}\cdot\sum_{n=1}^{\infty}\sum_{m=1}^{\infty}m^{2k-1}q^{2mn} =$$

(by (2.3) with $n\tau$ for τ, and $2k$ for k)

$$= 2\zeta(2k) + \frac{2(-2\pi i)^{2k}}{(2k-1)!}\sum_{l=1}^{\infty}\sigma_{2k-1}(l)q^{2l},$$

with $mn=l$, and $\sigma_{2k-1}(l)=\sum_{d|l}d^{2k-1}$.

From Lemma 1 we deduce

Lemma 2. *If* $\operatorname{Im}\tau>0$, *then* $g_2(\tau)$ *and* $g_3(\tau)$ *are holomorphic, and if* $q=e^{\pi i\tau}$, *then we have*

$$g_2(\tau)\equiv g_2(1,\tau) = (2\pi)^4\left[\frac{1}{12}+20\sum_{n=1}^{\infty}\sigma_3(n)q^{2n}\right], \tag{2.4}$$

and

$$g_3(\tau)\equiv g_3(1,\tau) = (2\pi)^6\left[\frac{1}{216}-\frac{7}{3}\sum_{n=1}^{\infty}\sigma_5(n)q^{2n}\right]. \tag{2.5}$$

Proof. By Theorem 1 of Chapter III, the series of holomorphic functions (for $\operatorname{Im}\tau>0$) defining $g_2(\tau)$ and $g_3(\tau)$, namely $60\sum'(n\tau+m)^{-4}$, and $140\sum'(n\tau+m)^{-6}$, are uniformly convergent in every half-strip: $-\delta\leqq\operatorname{Re}\tau\leqq\delta$, $\operatorname{Im}\tau\geqq\varepsilon>0$, for any (finite) $\delta>0$, and $\varepsilon>0$. Hence $g_2(\tau)$ and $g_3(\tau)$ are holomorphic for $\operatorname{Im}\tau>0$.

If we take $k=2$, and $k=3$, in Lemma 1, and note that $\zeta(4)=(\pi^4/90)$, $\zeta(6)=(\pi^6/945)$, then (2.4) and (2.5) would follow. To determine $\zeta(4)$ and $\zeta(6)$, we

observe that in a neighbourhood of $z=0$, say $|z|<\pi$, we have

$$\cot z=\frac{1}{z}+\sum_{n=1}^{\infty}\left(\frac{1}{z-n\pi}+\frac{1}{z+n\pi}\right),$$

and therefore

$$\cot z=\frac{1}{z}-\frac{2}{\pi^2}\zeta(2)\cdot z-\ldots-\frac{2}{\pi^{2n}}\cdot\zeta(2n)\cdot z^{2n-1}-\ldots.$$

On the other hand, $\cot z=(\cos z)/(\sin z)$, and the expansions for $\cos z$ and $\sin z$ lead to

$$\cot z=\frac{1}{z}-\frac{z}{3}-\frac{z^3}{45}-\frac{2z^5}{945}-\frac{z^7}{4725}-\ldots,$$

where every coefficient is *rational*, so that $\dfrac{\zeta(2n)}{\pi^{2n}}$ is rational, and

$$\zeta(2)=\frac{\pi^2}{6},\ \zeta(4)=\frac{\pi^4}{90},\ \zeta(6)=\frac{\pi^6}{945},\ \zeta(8)=\frac{\pi^8}{9450},$$

and so on.

§ 3. Expansion of the function $J(\tau)$, and the connexion with theta-functions

Theorem 1. *The function $J(\tau)$, defined in* (1.6), *is holomorphic in the upper half-plane* $\operatorname{Im}\tau>0$. *If* $\tau^*=\dfrac{a\tau+b}{c\tau+d}$, *where a, b, c, d are integers, with* $ad-bc=1$, *then* $J(\tau^*)=J(\tau)$, *so that $J(\tau)$ is a modular function.*

If $q=e^{\pi i\tau}$, $\operatorname{Im}\tau>0$, *then we have the expansion*

$$J(\tau)=\frac{1}{2^6\cdot 3^3\cdot q^2}\left[1+\sum_{n=1}^{\infty}c_n q^{2n}\right], \tag{3.1}$$

where c_n is integral, with $c_n\geqq 0$, *the series converging uniformly for* $\operatorname{Im}\tau\geqq\varepsilon>0$.

Proof. Since $g_2(\tau)$ and $g_3(\tau)$ are holomorphic for $\operatorname{Im}\tau>0$, so is $\Delta(\tau)$, and since $\Delta(\tau)\neq 0$, by Theorem 5 of Chapter III, $J(\tau)$ is holomorphic for $\operatorname{Im}\tau>0$.

The functions $g_2\equiv g_2(\omega_1,\omega_2)$ and $g_3\equiv g_3(\omega_1,\omega_2)$ are left invariant when a pair of basic periods (ω_1,ω_2) is replaced by another pair (ω_1^*,ω_2^*), where $\omega_2^*=a\omega_2+b\omega_1$, $\omega_1^*=c\omega_2+d\omega_1$, with $ad-bc=1$, (a, b, c, d being integers), so that $\tau^*=\dfrac{\omega_2^*}{\omega_1^*}=\dfrac{a\tau+b}{c\tau+d}$. For g_2 and g_3 depend only on the period-lattice: $\{m\omega_1+n\omega_2\}$, and not on any particular pair of basic periods (see Ch. III, Remarks after Th. 4), and (ω_1^*,ω_2^*) *is* a pair of basic periods if (ω_1,ω_2) is (Th. 2, Ch. I). It follows that Δ is also left invariant by such a transformation, hence $J(\tau^*)=J(\tau)$.

By (2.4) and the product representation of $\Delta(\tau)$ given by Theorem 7 of Chapter V, we have

$$J(\tau)=\frac{(\frac{1}{12}+20q^2+\ldots)^3}{q^2\prod_{n=1}^{\infty}(1-q^{2n})^{24}}=\frac{1}{q^2}\cdot\frac{1}{2^6\cdot 3^3}(1+240q^2+\ldots)^3\cdot\prod_{n=1}^{\infty}(1-q^{2n})^{-24}, \tag{3.2}$$

where $q^2=e^{2\pi i\tau}$, so that $|q^2|<1$ for $\mathrm{Im}\,\tau>0$, which leads to (3.1).

The product representation of $\Delta(\tau)$ which we have just used to prove (3.1) depends on the theory of theta-functions. The function $J(\tau)$, in fact, can be expressed directly in terms of the theta-functions θ_1, θ_2, and θ_3.

Theorem 2. *We have*

$$J(\tau)=\frac{1}{54}\cdot\frac{\{\theta_1^8(0,\tau)+\theta_2^8(0,\tau)+\theta_3^8(0,\tau)\}^3}{\{\theta_1(0,\tau)\cdot\theta_2(0,\tau)\cdot\theta_3(0,\tau)\}^8}, \tag{3.3}$$

for $\mathrm{Im}\,\tau>0$.

Proof. Since e_1, e_2, e_3 are the three distinct roots of the equation $4x^3-g_2x-g_3=0$, we have

$$e_1+e_2+e_3=0, \quad e_1\cdot e_2\cdot e_3=\frac{g_3}{4}, \quad e_1e_2+e_2e_3+e_3e_1=-\frac{g_2}{4}, \tag{3.4}$$

while

$$\Delta=16(e_1-e_2)^2(e_2-e_3)^2(e_3-e_1)^2.$$

(See Ch. III, Remarks after Th. 5). Now

$$e_1^2+e_2^2+e_3^2=(e_1+e_2+e_3)^2-2(e_2e_3+e_3e_1+e_1e_2)=\frac{g_2}{2},$$

since $e_1+e_2+e_3=0$; hence

$$\begin{aligned}(e_1-e_2)^2+(e_2-e_3)^2+(e_3-e_1)^2&=2(e_1^2+e_2^2+e_3^2)-2(e_1e_2+e_2e_3+e_3e_1)\\&=\tfrac{3}{2}\cdot g_2,\end{aligned}$$

so that

$$J(\tau)=\frac{g_2^3}{\Delta}=\frac{1}{54}\cdot\frac{\{(e_1-e_2)^2+(e_2-e_3)^2+(e_3-e_1)^2\}^3}{\{(e_1-e_2)(e_2-e_3)(e_3-e_1)\}^2}. \tag{3.5}$$

By Corollary 1 of Theorem 5 of Chapter V, however, we have

$$(e_1-e_2)=\left(\frac{\pi}{\omega_1}\right)^2\theta_3^4(0,\tau); \quad (e_1-e_3)=\left(\frac{\pi}{\omega_1}\right)^2\theta_2^4(0,\tau);$$

$$(e_3-e_2)=\left(\frac{\pi}{\omega_1}\right)^2\theta_1^4(0,\tau);$$

hence formula (3.3) is a consequence of (3.5).

Remarks. One could use (3.3) as a definition of $J(\tau)$. By Corollary 3 of Theorem 9 of Chapter V, the function $J(\tau)$, so defined, is holomorphic for $\operatorname{Im}\tau>0$, and invariant under all transformations of the modular group.

If we use the product representations of the theta-functions, obtained in Theorem 6 of Chapter V, namely

$$\left.\begin{aligned} &\theta_3(0,\tau)=c\cdot\prod_{n=1}^{\infty}(1+q^{2n-1})^2; \quad \theta_2(0,\tau)=c\cdot\prod_{n=1}^{\infty}(1-q^{2n-1})^2;\\ &\theta_1(0,\tau)=c\cdot q^{1/4}\cdot 2\cdot\prod_{n=1}^{\infty}(1+q^{2n})^2; \quad \theta'(0,\tau)=2\pi c\cdot q^{1/4}\cdot\prod_{n=1}^{\infty}(1-q^{2n})^2; \end{aligned}\right\} \tag{3.6}$$

where

$$c=\prod_{n=1}^{\infty}(1-q^{2n}),\ q=e^{\pi i\tau},\ \operatorname{Im}\tau>0, \quad \text{and} \quad \theta'(0,\tau)=\pi\cdot\theta_1(0,\tau)\cdot\theta_2(0,\tau)\cdot\theta_3(0,\tau),$$

we obtain

$$J(\tau)=\frac{1}{54}\cdot\frac{1}{2^8\cdot q^2}\times$$
$$\cdot\left[\prod_{n=1}^{\infty}(1+q^{2n-1})^{16}+\prod_{n=1}^{\infty}(1-q^{2n-1})^{16}+q^2\cdot 2^8\cdot\prod_{n=1}^{\infty}(1+q^{2n})^{16}\right]^3.$$

For $|q|<1$, the third product, inside the brackets, has an expansion in even powers of q, with positive, even, integral coefficients; the first product has an expansion of which the general term is of the form $\alpha\cdot q^{j_1}\cdot q^{j_2}\ldots q^{j_n}$, where α is a positive integer, and $j_1, j_2, \ldots, j_n$ are positive, odd, integers; while the second product has an expansion of which the general term is of the form $\alpha(-q^{j_1})(-q^{j_2})\ldots(-q^{j_n})$. The terms in the first and the second product cancel if n is odd, and are equal if n is even, giving rise only to even powers of q if n is even, with positive, *even*, integral coefficients. Thus

$$J(\tau)=\frac{1}{2^9\cdot 3^3\cdot q^2}\left[2+\sum_{n=1}^{\infty}a_n q^{2n}\right]^3,$$

where $a_n\geqq 0$, a_n integral, a_n even, which yields again (3.1).

§ 4. The function $J(\tau)$ in a fundamental domain of the modular group. To study the behaviour of $J(\tau)$ in the half-plane of its definition, it obviously suffices to study it in a fundamental domain of the modular group, for given any point τ with $\operatorname{Im}\tau>0$, there exists a point τ_1 in a fundamental domain, which is 'congruent' to τ relative to the modular group. Particularly useful for such a study is the following

Theorem 3. *If $J(\tau_1)=J(\tau)$, where $\operatorname{Im}\tau_1>0, \operatorname{Im}\tau>0$, then there exist integers a, b, c, d with $ad-bc=1$, such that*

$$\tau_1=\frac{a\tau+b}{c\tau+d}.$$

Proof. Let $\wp(u; 1, \tau)$ and $\wp(u; 1, \tau_1)$ denote respectively the Weierstrassian elliptic functions constructed with $(1, \tau)$ and $(1, \tau_1)$ as pairs of primitive periods, and let g_2, g_3 and g_2^1, g_3^1 denote respectively their invariants.

Since $J(\tau_1) = J(\tau)$, we have, by (1.6), the relation

$$(g_2^1)^3 \cdot g_3^2 = g_2^3 (g_3^1)^2 .$$

Since not both g_2 and g_3 can vanish simultaneously, one can, for instance, determine a constant μ by the equation

$$g_2^1 = \mu^2 g_2 ,$$

and one of the two values of μ satisfies the condition

$$g_3^1 = \mu^3 g_3 .$$

(If $g_3 \neq 0$ one can similarly determine μ by the equation $g_3^1 = \mu^3 g_3$, and one of the three values given by it satisfies also the condition $g_2^1 = \mu^2 g_2$).

Set $\omega = \mu^{1/2}$, the choice of root being arbitrary. Let g_2', g_3' denote the invariants of the function $\wp(u; \omega, \omega\tau_1)$. Then, by (1.4), we have

$$g_2' = \mu^{-2} g_2^1 = g_2 , \qquad g_3' = \mu^{-3} g_3^1 = g_3 .$$

This implies that $\wp(u; \omega, \omega\tau_1)$ coincides with $\wp(u; 1, \tau)$, since the Laurent expansions at $u = 0$ coincide by Theorem 4 of Chapter III. It follows that the period-pairs $(\omega, \omega\tau_1)$ and $(1, \tau)$ are connected by an unimodular transformation of the form

$$\omega\tau_1 = a\tau + b , \quad \omega = c\tau + d ,$$

where a, b, c, d are integers such that $ad - bc = 1$ (Remark (ii) after Th. 2, Ch. I).

Several properties of $J(\tau)$ can be derived from Theorems 1, 2, and 3, almost directly.

(i) Theorem 1 implies that $J(\tau + 1) = J(\tau)$, and $J(-1/\tau) = J(\tau)$, for $\operatorname{Im}\tau > 0$.

(ii) The function $J(\tau)$ assumes conjugate values at points symmetrically located relative to the imaginary axis. In particular, $J(\tau)$ is real for $\operatorname{Re}\tau = 0$, $\operatorname{Im}\tau > 0$, and because of (i), $J(\tau)$ is also real for $\operatorname{Re}\tau = n$, $\operatorname{Im}\tau > 0$, where n is *any* integer.

For $g_2(-\bar{\tau}) = 60 \sum'(m - n\bar{\tau})^{-4} = 60 \sum'(\overline{m + n\tau})^{-4} = \overline{g_2(\tau)}$, and similarly also: $g_3(-\bar{\tau}) = \overline{g_3(\tau)}$. Hence $J(-\bar{\tau}) = \overline{J(\tau)}$.

(iii) Let F_0 denote the fundamental domain of the modular group, given by

$$F_0 = \{\tau \mid \operatorname{Im}\tau > 0, |\tau| \geqq 1, -\tfrac{1}{2} \leqq \operatorname{Re}\tau < \tfrac{1}{2}, \operatorname{Re}\tau \leqq 0 \quad \text{for} \quad |\tau| = 1\} .$$

If $\tau \in \partial F_0$, the boundary of F_0, then $J(\tau)$ is real. For ∂F_0 consists of the lines: $\operatorname{Re}\tau = \pm\tfrac{1}{2}$, $\operatorname{Im}\tau \geqq \dfrac{\sqrt{3}}{2}$, and the arc: $|\tau| = 1$, $-\tfrac{1}{2} \leqq \operatorname{Re}\tau \leqq \tfrac{1}{2}$. And if $\tau \in \partial F_0$, we have $J(-\bar{\tau}) = \overline{J(\tau)}$ by (ii), as well as $J(-\bar{\tau}) = J(\tau)$ by (i), so that $J(\tau)$ is real.

It follows from (i) again that $J(\tau)$ is real for $\mathrm{Re}\,\tau = n + \frac{1}{2}$, $\mathrm{Im}\,\tau > 0$, where n is any integer.

(iv) Theorem 3 implies that for any complex number a, the equation $J(\tau) - a = 0$ has *at most* one solution in the fundamental domain F_0 (cf. property (5.95) of F_0, Ch. I). If $\tau \in \mathrm{Int}\, F_0$, and $\mathrm{Re}\,\tau \neq 0$, then $J(\tau)$ is not real, because of property (ii).

(v) We see from (3.1) that $J(n + i\beta) \to +\infty$, as $\beta \to +\infty$, while $J(n + \frac{1}{2} + i\beta) \to -\infty$ as $\beta \to +\infty$, since $q^2 < 0$ for $\mathrm{Re}\,\tau = n + \frac{1}{2}$, $\mathrm{Im}\,\tau > 0$, where n is any integer.

(vi) If $\varrho = e^{2\pi i/3}$, we have $g_2(\varrho) = 0$, so that $J(\varrho) = 0$. Further $g_3(i) = 0$, so that $J(i) = 1$.

For if $\varrho = e^{2\pi i/3}$, then $\varrho + 1 = -\varrho^2 = -1/\varrho$, and by (1.4) we have

$$\frac{1}{60} g_2(\varrho) = \sum{}'(m + n\varrho)^{-4} = \frac{1}{\varrho} \sum{}'(m\varrho^2 + n)^{-4} = \frac{1}{\varrho} \sum{}'(-m\varrho - m + n)^{-4}$$
$$= \frac{1}{\varrho} \sum{}'(m + n\varrho)^{-4},$$

since $(-m + n, -m)$ and (m, n) run through the same set of pairs when m and n run through all integers except $(0, 0)$, hence $g_2(\varrho) = 0$.

Similarly we have, by (1.4),

$$\tfrac{1}{140} g_3(i) = \sum{}'(m + ni)^{-6} = -\sum{}'(-mi + n)^{-6} = -\sum (m + ni)^{-6} = -\tfrac{1}{140} g_3(i),$$

since $(-m, n)$ and (m, n) run through the same set of pairs as m and n run through all integers (the pair $(0, 0)$ excepted). Hence $g_3(i) = 0$. (cf. Notes on Ch. III, §§ 1–2).

(vii) Since $q > 0$ for $\mathrm{Re}\,\tau = 0$, $\mathrm{Im}\,\tau > 0$, $J(\tau)$ is *positive* for $\mathrm{Re}\,\tau = n$, $\mathrm{Im}\,\tau > 0$, where n is any integer. But $J(\tau)$ is *negative* for $\mathrm{Re}\,\tau = n + \frac{1}{2}$, and $\mathrm{Im}\,\tau > \frac{\sqrt{3}}{2}$.

$$\left(\text{Note that } 0 = J\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = J\left(\frac{3}{2} + i\frac{\sqrt{3}}{2}\right) = J\left(-1 \Big/ \left(\frac{3}{2} + i\frac{\sqrt{3}}{2}\right)\right) = \right.$$
$$\left. = J\left(-\frac{1}{2} + i\frac{\sqrt{3}}{6}\right)\right).$$

Of particular importance is the following

Theorem 4. *If a is any non-real number, the equation $J(\tau) - a = 0$ has exactly one solution in the interior of F_0, where*

$$F_0 = \{\tau \mid \mathrm{Im}\,\tau > 0, |\tau| \geqq 1, -\tfrac{1}{2} \leqq \mathrm{Re}\,\tau < \tfrac{1}{2}, \mathrm{Re}\,\tau \leqq 0 \quad \textit{for} \quad |\tau| = 1\}.$$

Proof. Let $\tau = \xi + i\eta$, $q = e^{\pi i \tau}$. If $\eta \geqq c > 0$, and $|\xi| \leqq \frac{1}{2}$, then $|q^2| \leqq e^{-2\pi c}$, and since

$$J(\tau) = \frac{1}{2^6 \cdot 3^3 \cdot q^2}\left[1 + \sum_{n=1}^{\infty} c_n q^{2n}\right],$$

where $c_n \geqq 0$, by (3.1), we have $|J(\tau)| > |a|$, for $\eta \geqq c$, if c is sufficiently large, and $\tau \in F_0$.

Let H denote that part of F_0 in which $\operatorname{Im}\tau=\eta\leqq c$. Then there can be no solution of $J(\tau)-a=0$ in F_0 outside H. The *number* of solutions of $J(\tau)-a=0$ in H equals the integral

$$\frac{1}{2\pi i}\int_{ABCDE}\frac{J'(\tau)}{J(\tau)-a}d\tau,$$

where J' denotes the derivative of J, and A, B, C, D, E denote respectively the points $\tau=e^{2\pi i/3}$, i, $e^{\pi i/3}$, $\frac{1}{2}+ic$, $-\frac{1}{2}+ic$, while $ABCDE$ denotes the oriented curve joining them (which bounds H) (Fig. 8).

The integral exists since a is non-real, while $J(\tau)$ is real on EA, AC, and CD; and on DE we have $|J(\tau)|>|a|$.

Since $J(\tau+1)=J(\tau)$, $J(-1/\tau)=J(\tau)$, we have

$$\frac{1}{2\pi i}\left(\int_{EA}+\int_{CD}\right)=0,\qquad \frac{1}{2\pi i}\left(\int_{AB}-\int_{CB}\right)=0.$$

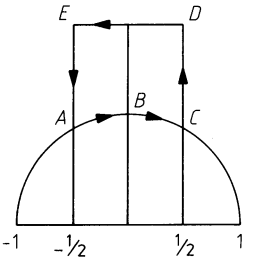


Fig. 8

Thus the number of solutions of $J(\tau)-a=0$, for a non-real, inside H, equals

$$\frac{1}{2\pi i}\int_{DE}\frac{J'(\tau)d\tau}{J(\tau)-a}.$$

If τ belongs to DE, then $\tau=\xi+ic$, with c sufficiently large, and $-\frac{1}{2}\leqq\xi\leqq\frac{1}{2}$. For $\operatorname{Im}\tau>0$, we have the expansion (3.1), namely

$$J(\tau)=e^{-2\pi i\tau}\left(2^{-6}\cdot 3^{-3}+\sum_{n=1}^{\infty}c_n'e^{2n\pi i\tau}\right),\qquad c_n'\geqq 0,\tag{4.1}$$

from which we obtain

$$\frac{J'(\xi+ic)}{J(\xi+ic)-a}=-2\pi i+O(e^{-2\pi c}).$$

Hence

$$\begin{aligned}\frac{1}{2\pi i}\int_{DE}\frac{J'(\tau)d\tau}{J(\tau)-a}&=\frac{1}{2\pi i}\int_{+\frac{1}{2}}^{-\frac{1}{2}}\frac{J'(\xi+ic)}{J(\xi+ic)-a}d\xi=\frac{1}{2\pi i}\{+2\pi i+O(e^{-2\pi c})\}\\&=1+O(e^{-2\pi c}).\end{aligned}$$

Since the integral equals an integer, and $c>0$ is at our disposal,

$$\frac{1}{2\pi i}\int_{DE}\frac{J'(\tau)d\tau}{J(\tau)-a}=1,$$

and the theorem follows.

We are now in a position to determine the image of the fundamental domain F_0 under the map: $\tau\to J(\tau)$, as in the following

Theorem 5. *Let*

$$U_1=\{\tau|\mathrm{Im}\,\tau>0,|\tau|\geqq 1,-\tfrac{1}{2}\leqq\mathrm{Re}\,\tau\leqq 0\},\quad U_2=\{\tau|\mathrm{Im}\,\tau>0,|\tau|>1,0<\mathrm{Re}\,\tau<\tfrac{1}{2}\},$$

so that $F_0=U_1\cdot\cup\cdot U_2$. *Then the function* $w=J(\tau)$ *maps the set* U_1, *one-to-one, onto the half-plane* $\mathrm{Im}\,w\geqq 0$, *and maps* $\bar{U}_2$, *the closure of* U_2, *one-to-one onto the half-plane* $\mathrm{Im}\,w\leqq 0$.

Proof. The function $J(\tau)$ is *real* for $\tau\in\partial U_1$, the boundary of U_1, because of properties (ii) and (iii) listed above. Since ∂U_1 is connected, $J(\partial U_1)$ is a real interval, and since $J(it)\to+\infty$ as $t\to+\infty$, and $J(-\frac{1}{2}+it)\to-\infty$ as $t\to+\infty$, by property (v) listed above, we have $J(\partial U_1)=(-\infty,\infty)$. Property (iv) implies that $J:\partial U_1\to(-\infty,\infty)$ is a homeomorphism and that $J(\tau)$ is non-real for $\tau\in\mathrm{Int}\,U_1\cdot\cup\cdot\mathrm{Int}\,U_2$, while $J(U_2)=\overline{J(\mathrm{Int}\,U_1)}$ by property (ii). Since U_1 and U_2 are connected, each of the sets $J(\mathrm{Int}\,U_1)$ and $J(U_2)$ must be one of the half-planes $\mathrm{Im}\,\tau>0$, or $\mathrm{Im}\,\tau<0$.

If $\tau\in\mathrm{Int}\,U_1$, $\tau=\xi+i\eta$, with η sufficiently large, then $\mathrm{Im}\,J(\tau)>0$, since (by (4.1))

$$J(\tau)=e^{-2\pi i\xi}\cdot e^{2\pi\eta}\left(2^{-6}\cdot 3^{-3}+\sum_{n=1}^{\infty}c_n' e^{2n\pi i\xi}\cdot e^{-2\pi\eta n}\right),\quad 0\geqq\xi\geqq-\tfrac{1}{2},\quad c_n'\geqq 0,$$

and

$$\mathrm{Im}\,J(\tau)=-\sin(2\pi\xi)(e^{2\pi\eta}\cdot 2^{-6}\cdot 3^{-3}+O(1)),$$

for η large enough. Hence $\mathrm{Im}\,J(\tau)>0$, for $\tau\in\mathrm{Int}\,U_1$. It follows that $\mathrm{Im}\,J(\tau)\leqq 0$, for $\tau\in\bar{U}_2$ by property (ii) of $J(\tau)$ listed above.

Thus the function $w=J(\tau)$ maps the set U_1, one-to-one, onto the closed half-plane $\mathrm{Im}\,w\geqq 0$, and the set $\bar{U}_2$, one-to-one, onto the half-plane $\mathrm{Im}\,w\leqq 0$.

Combining Theorems 4 and 5, we deduce

Theorem 5′. *The equation* $J(\tau)-a=0$ *has exactly one solution in* F_0, *whether* a *is real or non-real.*

§ 5. Relation between the periods and the invariants of $\wp(u)$. We have seen in Chapter III that given a pair of complex numbers (ω_1,ω_2) with $\mathrm{Im}\,\dfrac{\omega_2}{\omega_1}>0$, one can construct the elliptic function $\wp(u)$ with (ω_1,ω_2) as a pair of its basic periods, and $\wp(u)$ is then uniquely determined. (So are the functions $\zeta(u)$ and

$\sigma(u)$ obtained from $\wp(u)$ by integrating it once or twice). The invariants $g_2 = g_2(\omega_1, \omega_2)$, and $g_3 = g_3(\omega_1, \omega_2)$ are likewise uniquely determined (by the relations (1.1) and (1.2)) and, as has been shown in Theorem 5 of Chapter III, we have: $g_2^3 - 27g_3^2 \neq 0$. The question arises whether given any two complex numbers a_2, a_3, with the (sole) property $a_2^3 - 27a_3^2 \neq 0$, we can find a pair of complex numbers (ω_1, ω_2), with $\operatorname{Im}\frac{\omega_2}{\omega_1} > 0$, such that for the elliptic function $\wp(u)$ constructed with (ω_1, ω_2) as a pair of basic periods, the invariants are a_2 and a_3. The answer is yes, and a proof of that is facilitated by Theorems 4 and 5.

Theorem 6. *Given any two complex numbers a_2, a_3 such that $a_2^3 - 27a_3^2 \neq 0$, there exists a pair of complex numbers (ω_1, ω_2), with $\operatorname{Im}\frac{\omega_2}{\omega_1} > 0$, such that*

$$\text{and} \qquad \begin{aligned} g_2(\omega_1, \omega_2) &= 60 \sum{}' (m\omega_1 + n\omega_2)^{-4} = a_2 , \\ g_3(\omega_1, \omega_2) &= 140 \sum{}' (m\omega_1 + n\omega_2)^{-6} = a_3 ; \end{aligned} \tag{5.1}$$

hence there exists an elliptic function $\wp(u; \omega_1, \omega_2)$ with a_2 and a_3 as its invariants.

Proof (Hurwitz).

Case (i). Let $a_2 = 0$. Then $a_3 \neq 0$, since $a_2^3 - 27a_3^2 \neq 0$. Set $\tau = \frac{\omega_2}{\omega_1}$. Then (5.1) is satisfied if and only if

$$g_2(1, \tau) = 0 , \ \omega_1^{-6} g_3(1, \tau) = a_3 .$$

By property (vi) of $J(\tau)$ listed in § 4 above, $g_2(1, \varrho) = 0$, where $\varrho = e^{2\pi i/3}$. Then $g_3(1, \varrho) \neq 0$, since $g_2^3(1, \varrho) - 27g_3^2(1, \varrho) \neq 0$. We may therefore take $\tau = \varrho$, and determine ω_1 by the relation $\omega_1^{-6} g_3(1, \varrho) = a_3$, and then take $\omega_2 = \omega_1 \varrho$.

Case (ii). Let $a_3 = 0$. Then $a_2 \neq 0$. Again by property (vi) of $J(\tau)$ given in § 4, $g_3(1, i) = 0$, hence $g_2(1, i) \neq 0$. We may therefore take $\tau = i$, and determine ω_1 by the relation $\omega_1^{-4} g_2(1, i) = a_2$, and then take $\omega_2 = \omega_1 \cdot i$.

Case (iii). Let $a_2 \neq 0$, $a_3 \neq 0$. Set $a = a_2^3 - 27a_3^2$. Then $a \neq 0$, by hypothesis, and (5.1) is satisfied if and only if

$$\frac{g_2(\omega_1, \omega_2)}{g_3(\omega_1, \omega_2)} = \frac{a_2}{a_3}, \quad \text{and} \quad \frac{g_2^3(\omega_1, \omega_2)}{g_2^3(\omega_1, \omega_2) - 27g_3^2(\omega_1, \omega_2)} = \frac{a_2^3}{a} . \tag{5.2}$$

If we set $\tau = \frac{\omega_2}{\omega_1}$, the second equation in (5.2) can be written as

$$J(\tau) = \frac{g_2^3(1, \tau)}{g_2^3(1, \tau) - 27g_3^2(1, \tau)} = \frac{a_2^3}{a} , \tag{5.3}$$

and this has a unique solution $\tau \in F_0$, by Theorems 4 and 5, which implies that $\operatorname{Im}\tau > 0$. If τ is thus determined, then the first equation in (5.2), which can be

written as

$$\frac{\omega_1^2 g_2(1,\tau)}{g_3(1,\tau)} = \frac{a_2}{a_3}, \tag{5.4}$$

gives ω_1, and therefore also $\omega_2 = \omega_1 \cdot \tau$.

Thus in all three cases we have determined a pair (ω_1, ω_2), with $\operatorname{Im}\frac{\omega_2}{\omega_1} > 0$, such that $g_2(\omega_1, \omega_2) = a_2$, $g_3(\omega_1, \omega_2) = a_3$. The function $\wp(u; \omega_1, \omega_2)$ constructed with (ω_1, ω_2) as a pair of basic periods has therefore the invariants a_2 and a_3.

§ 6. Elliptic integrals of the first kind. Let $\mathscr{L}$ be a *piecewise differentiable curve* in the complex plane, which is defined by a piecewise continuously differentiable function $x = x(t)$ for $a \leqq t \leqq b$, a and b being finite (and real). Let $x_0 = x(a)$ be the *initial point* of $\mathscr{L}$, and $x_1 = x(b)$ the *terminal point* of $\mathscr{L}$.

Given any two complex *constants* g_2, g_3, such that $g_2^3 - 27g_3^2 \neq 0$, the equation

$$y^2 = 4x^3 - g_2 x - g_3 \tag{6.1}$$

defines y as a two-valued function of x. If its value is fixed at x_0, however, it is fixed all along the curve $\mathscr{L}$, if we *assume* that no zero of the polynomial $4x^3 - g_2 x - g_3$ lies on the curve $\mathscr{L}$.

The integral

$$I = \int_{\mathscr{L}(x_0, x_1)} \frac{dx}{y} \tag{6.2}$$

is known as an *elliptic integral of the first kind*. If both x_0 and x_1 are fixed, it is called a *definite* elliptic integral; if x_0 is fixed, and $x_1 = x$ varies, then the integral depends not only on x but on the curve $\mathscr{L}$. Such an integral is called an *indefinite elliptic integral.*

In order to evaluate (6.2) it is of advantage to replace both x and y by elliptic functions of a new variable u. To show that that is possible, we need to make some preliminary analysis.

We note that the assumption $g_2^3 - 27g_3^2 \neq 0$ ensures that the equation $4x^3 - g_2 x - g_3 = 0$ has *no* repeated root. Secondly, Theorem 6 ensures the existence of a pair of complex numbers (ω_1, ω_2), with $\operatorname{Im}\frac{\omega_2}{\omega_1} > 0$, such that for the elliptic function $\wp(u; \omega_1, \omega_2)$ constructed with that pair (ω_1, ω_2) as a pair of basic periods, the corresponding invariants are, in fact, g_2 and g_3.

We now prove two preliminary lemmas, which enable us to change the variables x and y with the help of the function $\wp(u)$ so constructed.

Lemma 1. *If $g_2 = g_2(\omega_1, \omega_2)$, and $g_3 = g_3(\omega_1, \omega_2)$ are the invariants of $\wp(u; \omega_1, \omega_2)$, then to every pair of complex numbers (x, y) satisfying the equation $y^2 = 4x^3 - g_2 x - g_3$, there corresponds in every period-parallelogram P of $\wp(u)$ exactly one point $\hat{u}$, such that $x = \wp(\hat{u})$, and $y = \wp'(\hat{u})$.*

Proof. Given x, there are two values of y satisfying the equation $y^2=4x^3-g_2x-g_3$, which are negatives of each other, hence two pairs (x, y_1), (x, y_2), where $y_2=-y_1$, which satisfy the equation.

The pair $(x=\wp(u),\ y=\wp'(u))$ satisfies the given equation. And the equation $\wp(u)=x$, for given x, has two roots u_1, u_2 in any given period-parallelogram P, *and* $\wp'(u_1)=-\wp'(u_2)$, whether $u_1=u_2$ (in which case $\wp'(u_1)=\wp'(u_2)=0$), or $u_1 \neq u_2$ (in which case u_1+u_2 is a period of the $\wp$-function). It follows that a given pair (x, y) can coincide with only *one* of the two pairs $(\wp(u_1),\ \wp'(u_1))$, $(\wp(u_2),\ \wp'(u_2))$, and $\hat{u}=u_1$ in the former case, while $\hat{u}=u_2$ in the latter.

Lemma 2. *Let* $g_2=g_2(\omega_1, \omega_2)$, $g_3=g_3(\omega_1, \omega_2)$ *be the invariants of* $\wp(u; \omega_1, \omega_2)$. *Let* $x=x(t)$, $a \leqq t \leqq b$, *be a piecewise differentiable curve not passing through any of the roots of the equation:* $4x^3-g_2x-g_3=0$. *Let* $y=y(t)$ *be a continuous function in* $[a, b]$, *which is related to* $x=x(t)$ *by the equation*

$$y^2=4x^3-g_2x-g_3\,.$$

Then there exists a piecewise differentiable curve $u=u(t)$, $a \leqq t \leqq b$, *such that*

$$x(t)=\wp(u(t))\,, \qquad y(t)=\wp'(u(t))\,, \qquad a \leqq t \leqq b\,.$$

Proof. Let $a \leqq t_0 \leqq b$, $x_0=x(t_0)$, $y_0=y(t_0)$. Then $y_0^2=4x_0^3-g_2x_0-g_3 \neq 0$, by hypothesis. By Lemma 1, there exists a point u_0, such that $\wp(u_0)=x_0$, and $\wp'(u_0)=y_0 \neq 0$. Therefore $\wp(u)$ is uniquely invertible in a neighbourhood of u_0, and its inverse maps the arc of the curve $x=x(t)$ in a sufficiently small interval $[t-h, t+h]$ into a piecewise differentiable arc $u=u(t)$ in the same interval. (If $t_0=a$, or $t_0=b$, we consider only the interval $[t_0, t_0+h]$, or $[t_0-h, t]$). For $t_0-h \leqq t \leqq t_0+h$, we shall therefore have

$$\wp[u(t)]=x(t),\ \{\wp'[u(t)]\}^2=[y(t)]^2\,, \tag{6.3}$$

and since $\wp'(u_0)=y_0$, and $y(t) \neq 0$, we have precisely $\wp'(u(t))=y(t)$.

To each point t_0 in $[a, b]$ we can thus associate a closed, hence also an open, interval (sufficiently small) containing it, in which $x(t)=\wp[u(t)]$, $y(t)=\wp'[u(t)]$. Because of the compactness of $[a, b]$, it is possible to subdivide the interval $[a, b]$ into a *finite* number of sub-intervals $[t_1, t_2]$, $[t_2, t_3]$, ..., $[t_{n-1}, t_n]$, with $a=t_1$, $b=t_n$, in such a way that it is possible to define in them the corresponding differentiable curves $\mathscr{C}_k$ given by the equations $u=u_k(t)$, satisfying the conditions:

$$\wp(u_k(t))=x(t)\,, \qquad \wp'(u_k(t))=y(t)\,, \quad \text{for} \quad t_k \leqq t \leqq t_{k+1}\,,$$

where $k=1, 2, \ldots, n-1$.

Since $\wp$ and $\wp'$ are periodic, we may assume that for $k=2, 3, \ldots, n-1$, the initial point of the curve $\mathscr{C}_k$ and the terminal point of the curve $\mathscr{C}_{k-1}$ belong to the *same* period-parallelogram, say, P_k. By Lemma 1, the initial point of the curve $\mathscr{C}_k$ coincides with the terminal point of the curve $\mathscr{C}_{k-1}$. We can therefore define the function $u(t)$ in $[a, b]$, by setting $u(t)=u_k(t)$ for $t_k \leqq t \leqq t_{k+1}$,

$k=1,2,\ldots,n-1$. This function then satisfies the required conditions: $x(t)=\wp(u(t))$, $y(t)=\wp'(u(t))$, for $a\leqq t\leqq b$.

Theorem 7. *Let g_2 and g_3 be two complex numbers, such that $g_2^3-27g_3^2\neq 0$, and $\wp(u)$ the elliptic function (which exists by Theorem 6) with g_2 and g_3 as its invariants.*

Let $\mathscr{L}$ be a piecewise differentiable curve defined by $x=x(t)$, $a\leqq t\leqq b$, with the initial point $x_0=x(a)$ and terminal point $x_1=x(b)$, and with the property that none of the zeros of the polynomial $4x^3-g_2x-g_3$ lie on $\mathscr{L}$.

Let $y=y(t)$ be a continuous function in $[a, b]$, which is related to $x=x(t)$ by the equation

$$y^2=4x^3-g_2x-g_3\,, \tag{6.4}$$

with y_0 and y_1 as the values of the function y at the points x_0 and x_1.

Let ℓ be a piecewise differentiable curve (which exists by Lemma 2), defined by $u=u(t)$, for $a\leqq t\leqq b$, with u_0 as its initial point and u_1 as its terminal point, such that

$$\wp(u_0)=x_0,\ \wp'(u_0)=y_0\,,\quad \text{and}\quad \wp(u_1)=x_1,\ \wp'(u_1)=y_1\,.$$

Then we have

$$I=\int_{\mathscr{L}(x_0,x_1)}\frac{dx}{y}=\int_{\mathscr{L}(x_0,x_1)}\frac{dx}{\sqrt{4x^3-g_2x-g_3}}=u_1-u_0\,. \tag{6.5}$$

If the curve $\mathscr{L}$ is replaced by $\tilde{\mathscr{L}}$, with the same end-points as $\mathscr{L}$, and

$$\tilde{I}=\int_{\tilde{\mathscr{L}}(x_0,x_1)}\frac{dx}{y}\,, \tag{6.6}$$

and if in both the integrals I and $\tilde{I}$ we assign the same value to y at the initial point x_0 of the curves of integration, then

$$\text{either}\quad \tilde{I}=I+\Omega\,,\quad \text{or}\quad \tilde{I}=-I-2u_0+\Omega'\,, \tag{6.7}$$

where Ω, Ω' stand for periods of $\wp(u)$ (constructed with the invariants g_2 and g_3), according as the function y assumes the same values at the terminal points of the curves $\mathscr{L}$ and $\tilde{\mathscr{L}}$, or those values differ in sign.

Proof. By Lemma 2, we can, in evaluating I, effect the change of variables: $x=\wp(u)$, $y=\wp'(u)$, and obtain

$$\int_{\mathscr{L}(x_0,x_1)}\frac{dx}{y}=\int_a^b\frac{x'(t)}{y(t)}\,dt=\int_a^b\frac{\wp'(u(t))\cdot u'(t)}{\wp'(u(t))}\,dt=\int_a^b u'(t)\,dt=u_1-u_0\,,$$

which gives (6.5).

To prove the second part, we note that by Lemma 2, there corresponds a curve $\tilde{\ell}$ to $\tilde{\mathscr{L}}$, just as ℓ does to $\mathscr{L}$. We may assume that $\tilde{u}_0$, the initial point of $\tilde{\ell}$ is such that $\tilde{u}_0=u_0$ (since $\wp(u_0)=x_0=\wp(\tilde{u}_0)$), and by Lemma 1 there exists in each period-parallelogram exactly one point $\hat{u}$, such that $x_0=\wp(\hat{u})$, and

$y_0 = \wp'(\hat{u})$. Let $\tilde{u}_1$ denote the terminal point of $\tilde{\ell}$. Let $\tilde{y}_1$ denote the value of the function y at the point x_1, after the continuation along $\tilde{\mathscr{P}}$. Then either $\tilde{y}_1 = y_1$, or $\tilde{y}_1 = -y_1$. In the first case, $\wp'(u_1) = y_1 = \wp'(\tilde{u}_1)$, hence $\tilde{u}_1 = u_1 + \Omega$. In the second case, we have $\wp'(\tilde{u}_1) = \tilde{y}_1 = -y_1 = -\wp'(u_1) = \wp'(-u_1)$, hence $\tilde{u}_1 = -u_1 + \Omega'$, where Ω' is a period of $\wp(u)$. Since by (6.5) we have $I = u_1 - u_0$, and $\tilde{I} = \tilde{u}_1 - u_0$, in the first case we have $\tilde{I} = I + \Omega$, while in the second case, $\tilde{I} = -u_1 - u_0 + \Omega' = -(u_1 - u_0) - 2u_0 + \Omega' = -I - 2u_0 + \Omega'$, as claimed.

Remarks. If $f(x) \equiv a_0x^4 + 4a_1x^3 + 6a_2x^2 + 4a_3x + a_4$, $a_0 \neq 0$, the solution of the biquadratic equation $f(x) = 0$, in case it has no repeated root, can be reduced to that of a *cubic* equation of the form $4t^3 - \tilde{g}_2 t - \tilde{g}_3 = 0$, where $\tilde{g}_2$ and $\tilde{g}_3$ are the invariants of the biquadratic $f(x)$, given by

$$\tilde{g}_2 = a_0a_4 - 4a_1a_3 + 3a_2^2,$$
$$\tilde{g}_3 = a_0a_2a_4 + 2a_1a_2a_3 - a_2^3 - a_0a_3^2 - a_1^2a_4.$$

A necessary and sufficient condition that the equation $f(x) = 0$ have no repeated root is that the discriminant of the biquadratic, namely $\tilde{\Delta} = \tilde{g}_2^3 - 27\tilde{g}_3^2$, be $\neq 0$. Thus the evaluation of an integral of the form

$$\int_{\mathscr{L}(x_0, x_1)} \frac{dx}{\sqrt{f(x)}}$$

can be reduced to that of an integral of the form (6.5).

Notes on Chapter VI

§ 1. Dedekind introduced the function $J(\tau)$, which he called the 'Valenz' function, by recognizing its mapping property embodied in Theorem 5. He mentions the influence of Riemann's work (§ 21 of his Inaugural Dissertation). He relates the function $J(\tau)$ to the modular function $\lambda(\tau)$ of Chapter VII ((8.1)), which had been previously studied by Abel, Jacobi, and Hermite, and to the function $\eta(\tau)$ of Chapter VIII, now named after him. See *J. für Math.* 83 (1877), 265–292; *Ges. Werke*, Bd. 1, 174–201.

Klein introduced the same function, independently of Dedekind, about a year later, in connexion with elliptic integrals, and called it the 'absolute invariant'. See *Math. Annalen*, 14 (1879), 111–126; *Ges. Abhandlungen*, Bd. 3, p. 3–9.

See also K. Weierstrass, *Sitz. Preuss. Akad. Berlin* (1883), 1271–1297; *Werke*, II, 281–309; H.A. Schwarz (cited in the Notes on Ch. III), § 55; A. Hurwitz, *Math. Annalen*, 18 (1881), 528–592; *Math. Werke*, Bd. I, 1–66.

Every modular function, which is invariant under the transformations of the modular group Γ, can be expressed as a rational function of $J(\tau)$. See, for instance, R. Fueter (cited in the Notes on Ch. I, § 4), I, (1924), Satz 12; or J. Lehner (loc. cit. § 3), p. 345.

The function $J(\tau)$ is connected with some interesting questions in arithmetic. For example, a theorem of Th. Schneider, *Math. Annalen*, 113 (1937), 1–13,

implies that $J(\tau)$ is *transcendental* for every *algebraic* number τ, with $\mathrm{Im}\,\tau > 0$, which is not a quadratic irrational. On the other hand, it is known, after Kronecker and Weber, that $J(\tau)$ is *algebraic* if $\mathrm{Im}\,\tau > 0$, and τ is a quadratic irrational. See H. Weber's *Elliptische Funktionen und algebraische Zahlen* (Braunschweig, 1891); *Lehrbuch der Algebra*, III, zweite Auflage (1908), § 115. See also R. Fueter (cited in the Notes on Ch. I, § 4), I, Satz 69, p. 63, and his Foreword, p. (iv). See also Kronecker, *J. für Math.* 57 (1860), 248–255; *Werke*, IV, 245–259; Zusätze, p. 502; H. Weber, *Acta Math.* 6 (1885), 329–416; 11 (1887/88), 333–390; G. Pick, *Math. Annalen*, 25 (1885), 433–447; 26 (1886), 219–230.

Siegel has shown that the function $J(\tau)$ can be characterized by an algebraic functional equation, *Abh. Math. Seminar der Univ. Hamburg*, 27 (1964), 32–38; *Ges. Abhandlungen*, III, 366–372; and raised a question about the transcendency of $J'(\tau)$ for a class of values of τ.

If (ω_1, ω_2) is a pair of basic periods of $\wp(u)$, then (ω_1^*, ω_2^*) is also a pair of basic periods, where $\omega_2^* = a\omega_2 + b\omega_1$, $\omega_1^* = c\omega_2 + d\omega_1$, and a, b, c, d are integers such that $ad - bc = 1$. Set $\tau = \dfrac{\omega_2}{\omega_1}$, and $\tau^* = \dfrac{\omega_2^*}{\omega_1^*}$. Since $g_2(\omega_1, \omega_2)$, $g_3(\omega_1, \omega_2)$ depend only on the period-lattice, and not on the basis, we have $g_2(\omega_1, \omega_2) = g_2(\omega_1^*, \omega_2^*)$. By (1.4) and (1.5) this implies that

$$g_2(\tau^*) = (c\tau + d)^4 g_2(\tau),$$

and similarly also

$$g_3(\tau^*) = (c\tau + d)^6 g_3(\tau). \qquad \text{(cf. Ch. III, after Th. 4)}.$$

Let G be any subgroup of finite index in the modular group Γ. A meromorphic function $f(\tau)$ on the upper half-plane $\mathrm{Im}\,\tau > 0$ is called a *modular form* of weight k for G, if

$$f(\tau^*) = (c\tau + d)^k f(\tau).$$

Series of the type

$$\sum_{\substack{m,n=-\infty \\ (m,n)\neq(0,0)}}^{\infty} (m\omega_1 + n\omega_2)^{-k},$$

where k is an even integer, $k > 2$, are known as *Eisenstein series* of weight k.

§ 2. Formula (2.2) is a special case of a well-known formula of R. Lipschitz, *Crelle's J.* 105 (1889), 127–156.

§ 3. There is an identity, due to H. Petersson, *Acta Math.* 58 (1932), p. 202, which expresses the coefficient c_n in (3.1), for $n \geqq 2$, as an infinite series of Bessel functions, with exponential sums as coefficients, namely

$$c_n = 2\pi n^{-1/2} \sum_{k=1}^{\infty} k^{-1} A_k(n) I_1\left(\frac{4\pi\sqrt{n}}{k}\right), \qquad n \geqq 2,$$

where I_1 stands for the Bessel function, and

$$A_k(n) = \sum_{h=0}^{k-1} e^{-2\pi i\left(\frac{nh+h'}{k}\right)}, \qquad hh' \equiv -1 \pmod{k}.$$

There is also an expansion of $J(\tau)$ into partial fractions, which resembles Weierstrass's expansion for $\wp(u)$, due to H. Rademacher, *American J. Math.* 60 (1938), 501–512; 61 (1939), 237–248; 64 (1942), 456.

The coefficients (c_n) grow rapidly; they are not multiplicative; and they have interesting congruence properties, as was first noticed by D.H. Lehmer, *American J. Math.* 64 (1942), 488–502. Further studies in that direction have continued. See J. Lehner, *American J. Math.* 71 (1949), 136–148; 373–386; A. van Wijngaarden, *Indagationes Math.* 15 (1953), 389–400; M. Newman, *Proc. American Math. Soc.* 9 (1958), 609–612; O. Kolberg, *Årbok Univ. Bergen*, Mat.-Natur. Ser. No. 16 (1961), 3–9; 16 (1962), 1–7; *Math. Scand.* 10 (1962), 173–181; A.O.L. Atkin and J.N. O'Brien, *Trans. American Math. Soc.* 126 (1967), 442–459; A.O.L. Atkin, *Computers in mathematical research* (North Holland, Amsterdam, 1968), 8–19. For an account of these problems and results in the context of automorphic functions, see J. Lehner, *Discontinuous groups and automorphic functions*, Math. Surveys, VIII, American Math. Soc. (1964); Reprinted (1974); *Lectures on modular forms*, Applied Math. Series, 61, U.S. Nat. Bureau of Standards (1970).

Formula (3.2) gives the expansion

$$J(\tau) = \frac{1}{12^3}\left[\frac{1}{q^2} + 744 + c_2 q^2 + \dots\right].$$

§ 4. See the Notes on Chapter III, §§ 1–2, for a different proof that $g_2(\varrho) = 0$ and $g_3(i) = 0$.

Theorem 4 can be used to give a different proof of Theorem 3, as has been remarked by Professor Raghavan Narasimhan.

Let Γ denote the modular group. If $a, b \in \bar{F}_0$, the closure of F_0, and $J(a) = J(b)$, then there exists $\gamma \in \Gamma$ such that $a = \gamma(b)$. For if $J(a) = J(b)$, and $b \neq \gamma(a)$ for every $\gamma \in \Gamma$, we can choose neighbourhoods U of a and V of b (in the half-plane $\operatorname{Im}\tau > 0$) such that $U \cdot \cap \cdot \gamma(V) = \emptyset$, for all $\gamma \in \Gamma$. This is possible because of the proper discontinuity of Γ mentioned in § 5 of the Notes on Chapter I. Since a holomorphic mapping is open, there exist points $z \in U$ and $w \in V$, and a non-real number α, such that $J(z) = J(w) = \alpha$. Let $\gamma_1, \gamma_2 \in \Gamma$ be such that $\gamma_1(z)$ and $\gamma_2(w)$ lie in F_0. Then since $J(z)$ is real for $z \in \partial F_0$, and α is not real, we have: $\gamma_1(z) \in \operatorname{Int} F_0$, $\gamma_2(w) \in \operatorname{Int} F_0$. Theorem 4 then implies that $\gamma_1(z) = \gamma_2(w)$, hence $z = \gamma(w)$, where $\gamma = \gamma_1^{-1} \cdot \gamma_2$, so that $U \cdot \cap \cdot \gamma(V) \neq \emptyset$, contrary to the assumption on U and V.

The fact that $\tau \to J(\tau)$ is *univalent* on F_0 is what led Dedekind to call it the *Valenz*.

§ 5. The problem of constructing an elliptic function $\wp(u; \omega_1, \omega_2)$ with given invariants is referred to in the classical literature as the *problem of*

inversion. A solution of that problem as in Theorem 6 is crucial for the evaluation of elliptic integrals as in § 6. The proof is due to A. Hurwitz. See Hurwitz-Courant, *Funktionentheorie* (cited in the Notes on Ch. I, § 4).

§ 6. The point $\hat{u}$, in Lemma 1, is referred to as the *parameter* of (x, y) on the cubic $y^2 = 4x^3 - g_2 x - g_3$.

The proof of Lemma 2 uses the topological argument for lifting curves to a covering space. See, for instance, Chapter II of Raghavan Narasimhan's book, *Complex analysis in one variable* (Birkhäuser, 1985).

The elliptic integral in (6.5) is known as the elliptic integral *of the first kind*, in the *normal form of Weierstrass*. The evaluation of integrals of that form is connected with the *uniformization* of plane algebraic curves of *genus* one. See the references given, for instance, by Whittaker and Watson in §§ 20·6, 20·7 of their book (cited in the Notes on Ch. I); see also Saks and Zygmund, Ch. VIII.

For the evaluation of elliptic integrals of the *first*, *second*, and *third*, kinds, after Legendre, and some applications thereof, see Whittaker and Watson (ibid. § 22·7); also Schwarz (§ 56 of his book cited in the Notes on Ch. III).

The absolute invariant of the biquadratic $f(x)$ at the end of § 6 is $\tilde{g}_2^3/\tilde{\Delta}$ (cf. Appendix 2, Ch. III), which is what led Klein to call $J(\tau)$ the *absolute invariant*.

Chapter VII

The Jacobian elliptic functions and the modular function $\lambda(\tau)$

§ 1. The functions sn u, cn u, and dn u of Jacobi. Let τ be a complex number, with $\operatorname{Im}\tau > 0$. Let ω_1 be defined as a function of τ by the relation

$$\omega_1 = \pi\theta_3^2(0,\tau) = \pi(1+2q+2q^4+\ldots)^2,\ q = e^{\pi i\tau},\ \operatorname{Im}\tau > 0, \tag{1.1}$$

and let $\omega_2 = \omega_1 \cdot \tau$. The Jacobian elliptic function $\operatorname{sn} u$ is a doubly-periodic, meromorphic function of u, with $(2\omega_1, \omega_2)$ as a pair of basic periods, with two simple poles in each period-parallelogram, the sum of the residues at those poles being zero. It satisfies the differential equation

$$\left(\frac{dy}{du}\right)^2 = (1-y^2)(1-k^2y^2), \qquad y = \operatorname{sn} u, \tag{1.2}$$

where

$$k^2 = k^2(\tau) = \frac{\theta_1^4(0,\tau)}{\theta_3^4(0,\tau)}. \tag{1.3}$$

The function defined as

$$k = k(\tau) = \frac{\theta_1^2(0,\tau)}{\theta_3^2(0,\tau)} \tag{1.4}$$

is called the *modulus* of the elliptic function $\operatorname{sn} u$, and the function defined as

$$k' = k'(\tau) = \frac{\theta_2^2(0,\tau)}{\theta_3^2(0,\tau)}, \tag{1.5}$$

the *complementary modulus*, because of the property

$$k^2 + k'^2 = 1. \tag{1.6}$$

To indicate the dependence on k, the function $\operatorname{sn} u$ is sometimes written as $\operatorname{sn}(u, k)$.

If we define $\lambda(\tau) = k^2(\tau)$, for $\operatorname{Im}\tau > 0$, then $\lambda(\tau)$ is a *modular function*, which is invariant under all transformations of the subgroup G of the modular group Γ, generated by the two transformations: $\tau \to \tau + 2$, $\tau \to \dfrac{\tau}{1-2\tau}$. If F_G denotes a fundamental domain for that subgroup, then the function $w = \lambda(\tau)$ maps F_G onto the complex w-plane without the points 0 and 1. The inverse function ϖ, defined by $\lambda(\varpi(z)) = z$, for $z \neq 0,\ 1,\ \infty$, is infinitely many-valued, but locally holomorphic, with $\operatorname{Im}\varpi(z) > 0$. Its existence enables one to obtain a short proof

of Picard's theorem, that an entire function $g(z)$, such that $g(z) \neq 0, 1$, reduces to a constant.

Associated with the function $\mathrm{sn}u$ are two other elliptic functions of Jacobi, denoted by $\mathrm{cn}u$, and $\mathrm{dn}u$, satisfying the relations

$$\mathrm{sn}^2u+\mathrm{cn}^2u=1\,,\qquad \mathrm{dn}^2u+k^2\,\mathrm{sn}^2u=1\,, \tag{1.7}$$

with $\mathrm{cn}0=\mathrm{dn}0=1$. The function $\mathrm{cn}u$ has $(2\omega_1,\ \omega_1+\omega_2)$ as a pair of basic periods, while $\mathrm{dn}u$ has $(\omega_1,\ 2\omega_2)$ as a pair of basic periods. Like $\mathrm{sn}u$, they are also elliptic functions of order 2, each of them with two simple poles in a period-parallelogram, with the sum of the residues at those poles equal to zero.

All three functions can be represented by infinite products, and an addition-theorem exists, which expresses, say, $\mathrm{sn}(u+v)$ in terms, not of $\mathrm{sn}u$ and $\mathrm{sn}v$ alone, but in terms of all the three functions sn, cn, and dn in each of the variables u and v separately.

All three functions satisfy differential equations; that of $\mathrm{sn}u$ involves k as in (1.2); that of $\mathrm{dn}u$ involves k'; and that of $\mathrm{cn}u$ involves both k and k'.

§ 2. Definition by theta-functions. (i) *The function* $\mathrm{sn}u$. If u is a complex variable, we define

$$\mathrm{sn}u=\frac{\theta\left(\dfrac{u}{\omega_1},\tau\right)}{\theta_2\left(\dfrac{u}{\omega_1},\tau\right)}\cdot\frac{\theta_3(0,\tau)}{\theta_1(0,\tau)}, \tag{2.1}$$

where τ is a complex number, with $\mathrm{Im}\,\tau>0$, $\omega_1=\pi\theta_3^2(0,\tau)$, and $\omega_2=\omega_1\cdot\tau$. If we *define* $\sqrt{k}$ by the relation

$$\sqrt{k}=\sqrt{k(\tau)}=\frac{\theta_1(0,\tau)}{\theta_3(0,\tau)}, \tag{2.2}$$

then (2.1) can be written as

$$\mathrm{sn}u=\mathrm{sn}(u,k)=\frac{1}{\sqrt{k}}\cdot\frac{\theta\left(\dfrac{u}{\omega_1},\tau\right)}{\theta_2\left(\dfrac{u}{\omega_1},\tau\right)},\qquad \omega_1=\pi\theta_3^2(0,\tau)\,. \tag{2.3}$$

We have seen [Theorem 1, and (3.6), of Ch. V] that

$$\left.\begin{aligned}&\theta(v+1,\tau)=-\theta(v,\tau)\,,\qquad &&\theta(v+\tau,\tau)=-q^{-1}e^{-2\pi iv}\theta(v,\tau)\\ &\theta_2(v+1,\tau)=\theta_2(v,\tau)\,,\qquad &&\theta_2(v+\tau,\tau)=-q^{-1}e^{-2\pi iv}\theta_2(v,\tau)\end{aligned}\right\}, \tag{2.4}$$

where $q=e^{\pi i\tau}$. Hence $\mathrm{sn}u$ has $2\omega_1$ and ω_2 as a pair of basic periods. Since $\theta(v,\tau)$ has simple zeros at $v=m+n\tau$ (by (1.6) of Ch. V), where m and n are integers, and only those zeros, $\mathrm{sn}u$ has zeros exactly at $u=m\omega_1+n\omega_2$. Since

$\theta_2(v,\tau)$ has simple zeros at $v=m+n\tau+\frac{1}{2}\tau$, and only those zeros (by (3.13) of Ch. V), $\mathrm{sn}u$ has *simple* poles at $u=m\omega_1+(n+\frac{1}{2})\omega_2$, and only those.

In a period-parallelogram spanned by the vectors $2\omega_1$ and ω_2, the function $\mathrm{sn}u$ has precisely two simple zeros, at $u=0$ and $u=\omega_1$, and precisely two simple poles at $u=\dfrac{\omega_2}{2}$ and $u=\omega_1+\dfrac{\omega_2}{2}$. Hence $\mathrm{sn}u$ is an elliptic function of order 2. The residue of $\mathrm{sn}u$ at $u=\frac{1}{2}\omega_2$ is $\dfrac{1}{k}$, and the residue at $u=\omega_1+\frac{1}{2}\omega_2$ is $-\dfrac{1}{k}$. For the residue of $\mathrm{sn}u$ at $u=\dfrac{\omega_2}{2}$ equals $\dfrac{\omega_1}{\sqrt{k}}\cdot\dfrac{\theta(\frac{1}{2}\tau,\tau)}{\theta_2'(\frac{1}{2}\tau,\tau)}$, where $\theta(\frac{1}{2}\tau,\tau)=\theta_3(\frac{1}{2}+\tau,\tau)q^{3/4}\times$ $\cdot\dfrac{1}{i}=\theta_3(\frac{1}{2},\tau)q^{-1/4}\cdot i=\theta_2(0,\tau)\cdot q^{-1/4}\cdot i$, while $\theta_2'(\frac{1}{2}\tau,\tau)=i\theta'(0,\tau)\cdot q^{-1/4}=$ $=\pi i\cdot q^{-1/4}\cdot\theta_1(0,\tau)\cdot\theta_2(0,\tau)\cdot\theta_3(0,\tau)$. Since $\omega_1=\pi\theta_3^2(0,\tau)$, the residue at $u=\frac{1}{2}\omega_2$ is $\dfrac{1}{k}$. The residue at $u=\omega_1+\frac{1}{2}\omega_2$ is $-\dfrac{1}{k}$, since $\theta(1+v,\tau)=-\theta(v,\tau)$, $\theta_2(1+v,\tau)=\theta_2(v,\tau)$. (See (3.4), (3.6), (5.1), and (1.4) of Ch. V).

Since $\theta(v,\tau)$ is an *odd* function of v, while $\theta_1(v,\tau)$, $\theta_2(v,\tau)$, and $\theta_3(v,\tau)$ are *even* functions of v, $\mathrm{sn}u$ is an *odd* function of u, and clearly $\mathrm{sn}(0)=0$. Since $\theta(v+1,\tau)=-\theta(v,\tau)$, and $\theta_2(v+1,\tau)=\theta_2(v,\tau)$ by (1.4) and (3.6) of Chapter V, we have

$$\mathrm{sn}(\omega_1-u)=\mathrm{sn}u\,.$$

(ii) *The functions* $\mathrm{cn}u$ *and* $\mathrm{dn}u$. In analogy with (2.1), we define

$$\mathrm{cn}u=\frac{\theta_1\left(\dfrac{u}{\omega_1},\tau\right)}{\theta_2\left(\dfrac{u}{\omega_1},\tau\right)}\cdot\sqrt{\frac{k'}{k}},\tag{2.5}$$

$$\mathrm{dn}u=\frac{\theta_3\left(\dfrac{u}{\omega_1},\tau\right)}{\theta_2\left(\dfrac{u}{\omega_1},\tau\right)}\cdot\sqrt{k'},\qquad \omega_1=\pi\theta_3^2(0,\tau)\,,\tag{2.6}$$

where $\sqrt{k'}$ and $\sqrt{\dfrac{k'}{k}}$ are *defined* by

$$\sqrt{k'}=\frac{\theta_2(0,\tau)}{\theta_3(0,\tau)},\qquad \sqrt{\frac{k'}{k}}=\frac{\theta_2(0,\tau)}{\theta_1(0,\tau)}.\tag{2.7}$$

By using the known properties of theta-functions, one sees that $\mathrm{cn}u$ has $(2\omega_1,\omega_1+\omega_2)$ has a pair of basic periods, while $\mathrm{dn}u$ has (ω_1,ω_2) as a pair of basic periods. The zeros of $\mathrm{cn}u$ are located at $u=(m+\frac{1}{2})\omega_1+n\omega_2$; those of $\mathrm{dn}u$ at $u=(m+\frac{1}{2})\omega_1+(n+\frac{1}{2})\omega_2$. The poles of $\mathrm{cn}u$ are located at $u=m\omega_1+$ $+(n+\frac{1}{2})\omega_2$; so are those of $\mathrm{dn}u$. Here m and n are integers. Both $\mathrm{cn}u$ and $\mathrm{dn}u$

are elliptic functions of order 2. The residue of $\mathrm{cn}u$ at $u=\frac{1}{2}\omega_2$ is $-\frac{i}{k}$, and at $u=\omega_1+\frac{1}{2}\omega_2$ it is $+\frac{i}{k}$. The residue of $\mathrm{dn}u$ at $u=\frac{1}{2}\omega_2$ is $(-i)$, and at $u=\frac{3}{2}\omega_2$ it is $(+i)$. Both $\mathrm{cn}u$ and $\mathrm{dn}u$ are *even* functions of u, since $\theta_1(v,\tau)$, $\theta_2(v,\tau)$, and $\theta_3(v,\tau)$ are all *even* functions of v. Further we have $\mathrm{cn}(0)=\mathrm{dn}(0)=1$.

§ 3. Connexion with the sigma-functions. We have seen (Theorem 2 and 3, Ch. V) that the four sigma-functions are connected with the theta-functions by the relation

$$\frac{\sigma(u)}{\sigma_r(u)}=\frac{\theta\left(\frac{u}{\omega_1},\tau\right)}{\theta_r\left(\frac{u}{\omega_1},\tau\right)}\cdot\frac{\omega_1\cdot\theta_r(0,\tau)}{\theta'(0,\tau)},\qquad r=1,2,3,\tag{3.1}$$

where $\theta'(0,\tau)=\pi\theta_1(0,\tau)\cdot\theta_2(0,\tau)\cdot\theta_3(0,\tau)$. Since ω_1 here has been so chosen that $\omega_1=\pi\theta_3^2(0,\tau)$, we have

$$\mathrm{sn}u=\frac{\sigma(u)}{\sigma_2(u)},\qquad \mathrm{cn}u=\frac{\sigma_1(u)}{\sigma_2(u)},\qquad \mathrm{dn}u=\frac{\sigma_3(u)}{\sigma_2(u)}.\tag{3.2}$$

If $\wp(u)=\wp(u;\omega_1,\omega_2)$ be Weierstrass's elliptic function, with (ω_1,ω_2) as a pair of primitive periods, we have seen in (2.5) and (2.7) of Chapter V that

$$\wp(u)-e_r=\left(\frac{\sigma_r(u)}{\sigma(u)}\right)^2,\qquad r=1,2,3,\tag{3.3}$$

where $\wp\left(\frac{\omega_1}{2}\right)=e_1$, $\wp\left(\frac{\omega_1}{2}\right)=e_2$, $\wp\left(\frac{\omega_1+\omega_2}{2}\right)=e_3$. Hence we have the relations

$$\begin{gathered}\wp(u)-e_1=\left(\frac{\mathrm{cn}u}{\mathrm{sn}u}\right)^2,\qquad \wp(u)-e_2=\left(\frac{1}{\mathrm{sn}u}\right)^2,\\ \wp(u)-e_3=\left(\frac{\mathrm{dn}u}{\mathrm{sn}u}\right)^2,\end{gathered}\tag{3.4}$$

where $\omega_1=\pi\theta_3^2(0,\tau)$, from which we obtain:

$$e_1-e_2=\frac{1-\mathrm{cn}^2u}{\mathrm{sn}^2u},\quad (e_1-e_2)\mathrm{sn}^2u+\mathrm{cn}^2u=1,\tag{3.5}$$

and

$$e_3-e_2=\frac{1-\mathrm{dn}^2u}{\mathrm{sn}^2u},\quad (\mathrm{dn}u)^2+(e_3-e_2)\mathrm{sn}^2u=1.\tag{3.6}$$

We have, however, by (5.6) of Chapter V, the general relations

$$(e_1-e_2)=\left(\frac{\pi}{\omega_1}\right)^2\theta_3^4(0,\tau);\quad e_1-e_3=\left(\frac{\pi}{\omega_1}\right)^2\theta_2^4(0,\tau);\quad e_3-e_2=\left(\frac{\pi}{\omega_1}\right)^2\theta_1^4(0,\tau).\tag{3.7}$$

Since $\omega_1 = \pi\theta_3^2(0,\tau)$, by choice, we have, in fact

$$e_1 - e_2 = 1, \quad e_1 - e_3 = \frac{\theta_2^4(0,\tau)}{\theta_3^4(0,\tau)}, \quad e_3 - e_2 = \frac{\theta_1^4(0,\tau)}{\theta_3^4(0,\tau)}, \tag{3.8}$$

and equations (3.5) and (3.6) yield the relations

$$\mathrm{cn}^2 u + \mathrm{sn}^2 u = 1, \quad \mathrm{dn}^2 u + k^2 \mathrm{sn}^2 u = 1, \tag{3.9}$$

where $k = \dfrac{\theta_1^2(0,\tau)}{\theta_3^2(0,\tau)}$. Since $k' = \dfrac{\theta_2^2(0,\tau)}{\theta_3^2(0,\tau)}$, we have

$$k^2 + k'^2 = \frac{\theta_1^4(0,\tau) + \theta_2^4(0,\tau)}{\theta_3^4(0,\tau)} = (e_3 - e_2) + (e_1 - e_3) = e_1 - e_2 = 1. \tag{3.10}$$

§ 4. The differential equation. We have proved in Chapter III that $\{\wp'(u)\}^2 = 4\{\wp(u) - e_1\}\{\wp(u) - e_2\}\{\wp(u) - e_3\}$. On the other hand, we have defined in § 2 of Chapter V

$$\sqrt{\wp(u) - e_r} = + \frac{\sigma_r(u)}{\sigma(u)}, \quad r = 1, 2, 3,$$

so that

$$\wp'(u) = -2\sqrt{\wp(u) - e_1} \cdot \sqrt{\wp(u) - e_2} \cdot \sqrt{\wp(u) - e_3}.$$

Hence, by (3.4),

$$\wp'(u) = -2\frac{\mathrm{cn}\, u \cdot \mathrm{dn}\, u}{(\mathrm{sn}\, u)^3}.$$

By (3.4), however, we have $\wp(u) - e_2 = \dfrac{1}{\mathrm{sn}^2 u}$, so that $\wp'(u) = -\dfrac{2}{\mathrm{sn}^3(u)} \times \left(\dfrac{d}{du}\,\mathrm{sn}\, u\right)$, hence

$$\frac{d}{du}(\mathrm{sn}\, u) = (\mathrm{cn}\, u) \cdot (\mathrm{dn}\, u). \tag{4.1}$$

Since $\mathrm{cn}^2 u = 1 - \mathrm{sn}^2 u$, we have $2\mathrm{cn}\, u \cdot \dfrac{d}{du}(\mathrm{cn}\, u) = -2\mathrm{sn}\, u(\mathrm{cn}\, u \cdot \mathrm{dn}\, u)$, hence

$$\frac{d}{du}(\mathrm{cn}\, u) = -(\mathrm{sn}\, u) \cdot (\mathrm{dn}\, u). \tag{4.2}$$

On the other hand, $\mathrm{dn}^2 u = 1 - k^2 \mathrm{sn}^2 u$ by (3.9), hence

$$\frac{d}{du}(\mathrm{dn}\, u) = -k^2(\mathrm{sn}\, u)(\mathrm{cn}\, u). \tag{4.3}$$

From (4.1)–(4.3) we obtain the three differential equations satisfied by the three functions $\mathrm{sn}u$, $\mathrm{cn}u$, and $\mathrm{dn}u$, namely

$$\left\{\frac{d}{du}(\mathrm{sn}u)\right\}^2 = (1-\mathrm{sn}^2u)(1-k^2\mathrm{sn}^2u), \qquad k^2 = \frac{\theta_1^4(0,\tau)}{\theta_3^4(0,\tau)}; \tag{4.4}$$

$$\left\{\frac{d}{du}(\mathrm{cn}u)\right\}^2 = (1-\mathrm{cn}^2u)(k'^2+k^2\mathrm{cn}^2u), \tag{4.5}$$

and

$$\left\{\frac{d}{du}(\mathrm{dn}u)\right\}^2 = -(1-\mathrm{dn}^2u)(k'^2-\mathrm{dn}^2u), \qquad k'^2 = \frac{\theta_2^4(0,\tau)}{\theta_3^4(0,\tau)}. \tag{4.6}$$

We note that because of (3.7), resulting from (5.6) of Chapter V, we also have

$$k^2 = \frac{e_3-e_2}{e_1-e_2}, \qquad k'^2 = \frac{e_1-e_3}{e_1-e_2}, \tag{4.7}$$

where, as usual, $\wp\left(\frac{\omega_1}{2}\right)=e_1$, $\wp\left(\frac{\omega_2}{2}\right)=e_2$, and $\wp\left(\frac{\omega_1+\omega_2}{2}\right)=e_3$. Here ω_1, ω_2 are defined as in (2.1). Since e_1, e_2, e_3 are all different (Proof of Theorem 5 of Chapter III), we have $k^2 \neq 0, 1$ (and ∞). Thus to each value of τ, with $\mathrm{Im}\,\tau > 0$, there corresponds a value of $k^2(\tau)$ which is different from 0 and 1, and a function $\mathrm{sn}u = \mathrm{sn}(u,k)$ which satisfies (4.4).

Conversely, let a be a given complex number, with $a \neq 0, 1$. Then there exists a τ, with $\mathrm{Im}\,\tau > 0$, such that

$$k^2(\tau) = a, \tag{4.8}$$

where, as before, $k^2(\tau) = \frac{\theta_1^4(0,\tau)}{\theta_3^4(0,\tau)}$. To prove this, *define* e_1, e_2, e_3 as follows: $e_2 = -\frac{1}{3}(a+1)$, $e_1 = e_2+1$, and $e_3 = e_2+a$. Then we have $e_1+e_2+e_3=0$. *Define* $g_2 = -4(e_1e_2+e_2e_3+e_3e_1)$, and $g_3 = 4(e_1e_2e_3)$. Since $a \neq 0, 1$, the numbers e_1, e_2, e_3 are *all* different, and we have: $g_2^3 - 27g_3^2 \neq 0$. By Theorem 5 of Chapter VI we can determine a pair of complex numbers ω_1, ω_2, with $\mathrm{Im}\frac{\omega_2}{\omega_1} > 0$, such that g_2 and g_3 *are* the invariants of the $\wp$-function constructed with (ω_1, ω_2) as a pair of basic periods. On setting $\tau = \frac{\omega_2}{\omega_1}$, we find that this τ fulfils the condition: $k^2(\tau) = a$, since

$$k^2(\tau) = \frac{\theta_1^4(0,\tau)}{\theta_3^4(0,\tau)} = \frac{e_3-e_2}{e_1-e_2} = \frac{a}{1} = a.$$

It follows that given any *constant* k (not τ), such that $k \neq 0, 1, -1$ (and ∞), there exists an elliptic function $\mathrm{sn}(u,k)$ which satisfies the differential equation (4.4). We summarize our observations concerning the function $\mathrm{sn}u$ in the following

Theorem 1. *Given* τ, *with* $\operatorname{Im}\tau>0$, *let* $\omega_1=\pi\theta_3^2(0,\tau)$, *and* $\omega_2=\omega_1\cdot\tau$. *The function* $\operatorname{sn}u$, *of the complex variable* u, *defined by the relation*

$$\operatorname{sn}u=\frac{\theta\left(\dfrac{u}{\omega_1},\tau\right)}{\theta_2\left(\dfrac{u}{\omega_1},\tau\right)}\cdot\frac{\theta_3(0,\tau)}{\theta_1(0,\tau)} \tag{2.1}$$

is an elliptic function of order 2, with $(2\omega_1,\omega_2)$ *as a pair of basic periods. It is an odd function of* u, *with the property*

$$\operatorname{sn}(\omega_1-u)=\operatorname{sn}u\,. \tag{4.9}$$

It satisfies the differential equation

$$\left(\frac{dy}{du}\right)^2=(1-y^2)(1-k^2y^2),\qquad y=\operatorname{sn}u, \tag{4.4}$$

where k^2 *is a function of* τ, *defined for* $\operatorname{Im}\tau>0$, *by the relation*

$$k^2=k^2(\tau)=\frac{\theta_1^4(0,\tau)}{\theta_3^4(0,\tau)}=\frac{e_3-e_2}{e_1-e_2}, \tag{4.7}$$

which has the property: $k^2\neq0,1$. *Here* e_1, e_2, e_3 *are respectively equal to* $\wp\left(\dfrac{\omega_1}{2}\right)$, $\wp\left(\dfrac{\omega_2}{2}\right)$, *and* $\wp\left(\dfrac{\omega_1+\omega_2}{2}\right)$, $\wp$ *being the Weierstrass* $\wp$*-function with* (ω_1,ω_2) *as a pair of basic periods.*

Given any constant k, *such that* $k\neq0,\ +1,\ -1$ (*and* ∞), *there exists an elliptic function* $\operatorname{sn}(u,k)\equiv\operatorname{sn}u$, *which satisfies the differential equation* (4.4).

Remarks

(i) Analogous statements can be made about $\operatorname{cn}u$, and $\operatorname{dn}u$.

(ii) Theorem 1 is sometimes expressed loosely by saying that if $u=\int_0^y\dfrac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}$, then $y=\operatorname{sn}(u,k)$. But the inversion of the integral has to be considered with care (Ch. VI, § 6).

§ 5. Infinite products for the Jacobian elliptic functions. The infinite products which we have obtained for the theta-functions yield the infinite products for the Jacobian elliptic functions.

If we set $u=\omega_1x=\pi\theta_3^2(0,\tau)\cdot x$, and write $q=e^{\pi i\tau}$, where $\operatorname{Im}\tau>0$, then from § 6 of Chapter V, we obtain the formulae:

$$\operatorname{sn}u=k^{-1/2}\cdot2q^{1/4}\cdot\sin(\pi x)\cdot\frac{\prod_{n=1}^{\infty}(1-2q^{2n}\cos2\pi x+q^{4n})}{\prod_{n=1}^{\infty}(1-2q^{2n-1}\cos2\pi x+q^{4n-2})}, \tag{5.1}$$

$$\operatorname{cn}u=2q^{1/4}\cdot k'^{1/2}\cdot k^{-1/2}\cdot\cos\pi x\cdot\frac{\prod_{n=1}^{\infty}(1+2q^{2n}\cos2\pi x+q^{4n})}{\prod_{n=1}^{\infty}(1-2q^{2n-1}\cos2\pi x+q^{4n-2})}, \tag{5.2}$$

and

$$\mathrm{dn}u = k'^{1/2} \cdot \frac{\prod_{n=1}^{\infty} (1+2q^{2n-1} \cdot \cos 2\pi x + q^{4n-2})}{\prod_{n=1}^{\infty} (1-2q^{2n-1} \cdot \cos 2\pi x + q^{4n-2})}, \tag{5.3}$$

where

$$u = \pi\theta_3^2(0,\tau) \cdot x,\ q = e^{\pi i\tau},\ \mathrm{Im}\,\tau > 0.$$

§ 6. Addition-theorems for sn*u*, cn*u*, dn*u*. The addition theorem for the function sn expresses $\mathrm{sn}(u+v)$ in terms of sn, cn, dn in each of the variables u and v separately. There are corresponding addition-theorems also for cn and dn. We take them up together.

Let α be a complex number, such that $\alpha \neq \frac{m}{2}\omega_1 + \frac{n}{2}\omega_2$, where m and n are integers (zero included), and ω_1, ω_2 are defined as in (1.1). Then α is neither a pole nor a zero of any of the functions sn, cn, or dn. We consider the three auxiliary functions

$$\mathrm{sn}u \cdot \mathrm{sn}(u-\alpha), \quad \mathrm{cn}u \cdot \mathrm{cn}(u-\alpha), \quad \mathrm{dn}u \cdot \mathrm{dn}(u-\alpha). \tag{6.1}$$

By (2.4), we have

$$\mathrm{sn}(u+\omega_1) \cdot \mathrm{sn}(u+\omega_1-\alpha) = \mathrm{sn}u \cdot \mathrm{sn}(u-\alpha),$$

and

$$\mathrm{sn}(u+\omega_2) \cdot \mathrm{sn}(u+\omega_2-\alpha) = \mathrm{sn}u \cdot \mathrm{sn}(u-\alpha).$$

Hence (ω_1, ω_2) is a pair of periods for the first of the three functions in (6.1), and similarly also for the second and the third. They are all elliptic functions of order 2. Each of them has a simple pole at $u = \frac{\omega_2}{2}$, and at $u = \frac{\omega_2}{2} + \alpha$. In a period-parallelogram spanned by the vectors ω_1 and ω_2, each of them has a simple pole at a point u_1 congruent to $\frac{\omega_2}{2}$ modulo (ω_1, ω_2) and another simple pole at a point u_2 congruent to $\frac{\omega_2}{2} + \alpha$ modulo (ω_1, ω_2), the sum of the residues at u_1 and u_2 being equal to zero.

It follows from § 2 that the residues of the three functions in (6.1) at $u = \frac{\omega_2}{2}$ are respectively equal to

$$\frac{1}{k}\mathrm{sn}\left(\frac{\omega_2}{2}-\alpha\right), \frac{-i}{k}\mathrm{cn}\left(\frac{\omega_2}{2}-\alpha\right), \quad \text{and} \quad -i \cdot \mathrm{dn}\left(\frac{\omega_2}{2}-\alpha\right). \tag{6.2}$$

Since $\theta_2(v,\tau) = \theta\left(\frac{\tau}{2}-v,\tau\right) \cdot q^{1/4} \cdot e^{-\pi iv} \cdot -i$, (by (3.4) of Ch. V), we have

$$\mathrm{sn}\left(u-\frac{\omega_2}{2}\right) = \frac{1}{k\,\mathrm{sn}u},$$

and similarly also

$$\operatorname{cn}\left(u-\frac{\omega_2}{2}\right)=\frac{i\operatorname{dn}u}{k\operatorname{sn}u},\qquad \operatorname{dn}\left(u-\frac{\omega_2}{2}\right)=i\cdot\frac{\operatorname{cn}u}{\operatorname{sn}u}.$$

Since sn$u$ is an *odd* function of u, while cn$u$ and dn$u$ are even functions of u, the residues given in (6.2) of the three functions in (6.1) at $u=\frac{\omega_2}{2}$ are respectively equal to

$$-\frac{1}{k^2\operatorname{sn}\alpha},\qquad \frac{\operatorname{dn}\alpha}{k^2\operatorname{sn}\alpha},\quad \text{and}\quad \frac{\operatorname{cn}\alpha}{\operatorname{sn}\alpha}. \tag{6.3}$$

Hence *each* of the functions

$$-k^2\operatorname{sn}\alpha\cdot\operatorname{sn}u\cdot\operatorname{sn}(u-\alpha),\qquad \frac{k^2\operatorname{sn}\alpha}{\operatorname{dn}\alpha}\operatorname{cn}u\cdot\operatorname{cn}(u-\alpha),\qquad \frac{\operatorname{sn}\alpha}{\operatorname{cn}\alpha}\cdot\operatorname{dn}u\cdot\operatorname{dn}(u-\alpha), \tag{6.4}$$

considered as a function of u, has a simple pole at $u=\frac{\omega_2}{2}$ with residue 1, and in consequence a simple pole at $u=\frac{\omega_2}{2}+\alpha$ with residue -1. Thus the difference between any two of the functions in (6.4) is an elliptic function *without* poles, hence a constant. This constant can be determined by setting $u=0$, and using the fact that $\operatorname{sn}(0)=0$, $\operatorname{cn}(0)=1=\operatorname{dn}(0)$. And we obtain the relation

$$\operatorname{dn}\alpha\cdot\operatorname{sn}u\cdot\operatorname{sn}(u-\alpha)+\operatorname{cn}u\cdot\operatorname{cn}(u-\alpha)=\operatorname{cn}\alpha, \tag{6.5}$$

by subtracting the second of the functions in (6.4) from the first, and similarly also the relation

$$k^2\operatorname{cn}\alpha\cdot\operatorname{sn}u\cdot\operatorname{sn}(u-\alpha)+\operatorname{dn}u\cdot\operatorname{dn}(u-\alpha)=\operatorname{dn}\alpha, \tag{6.6}$$

which results from subtracting the third of the functions in (6.4) from the first.

If we put $\alpha=u+v$ in (6.5) and (6.6), we obtain

$$\operatorname{cn}(u+v)=\operatorname{cn}u\cdot\operatorname{cn}v-\operatorname{sn}u\cdot\operatorname{sn}v\cdot\operatorname{dn}(u+v), \tag{6.7}$$

and

$$\operatorname{dn}(u+v)=\operatorname{dn}u\cdot\operatorname{dn}v-k^2\operatorname{sn}u\cdot\operatorname{sn}v\cdot\operatorname{cn}(u+v), \tag{6.8}$$

provided that $u+v\neq\frac{m}{2}\omega_1+\frac{n}{2}\omega_2$.

On substituting for $\operatorname{dn}(u+v)$ in (6.7) the expression given by (6.8), we obtain

$$\operatorname{cn}(u+v)=\frac{\operatorname{cn}u\cdot\operatorname{cn}v-\operatorname{dn}u\cdot\operatorname{dn}v\cdot\operatorname{sn}u\cdot\operatorname{sn}v}{1-k^2\operatorname{sn}^2u\cdot\operatorname{sn}^2v}. \tag{6.9}$$

By continuity, this holds without the restriction $u+v\neq\frac{m}{2}\omega_1+\frac{n}{2}\omega_2$, and is known as the *addition-theorem for* cn.

On substituting in (6.8) the expression for $\mathrm{cn}(u+v)$ given by (6.7), we similarly obtain

$$\mathrm{dn}(u+v)=\frac{\mathrm{dn}u\cdot\mathrm{dn}v-k^2\mathrm{cn}u\cdot\mathrm{cn}v\cdot\mathrm{sn}u\cdot\mathrm{sn}v}{1-k^2\,\mathrm{sn}^2u\cdot\mathrm{sn}^2v}, \tag{6.10}$$

which is *the addition-theorem for* dn.

If we put $\alpha=-v$ in (6.5), however, we get the relation

$$\mathrm{dn}v\cdot\mathrm{sn}u\cdot\mathrm{sn}(u+v)=\mathrm{cn}v-\mathrm{cn}u\cdot\mathrm{cn}(u+v),$$

provided that $v\neq\frac{m}{2}\omega_1+\frac{n}{2}\omega_2$. If in this we now use the addition-theorem for cn given by (6.9), we get

$$\mathrm{sn}(u+v)=\frac{\mathrm{sn}u\cdot\mathrm{cn}v\cdot\mathrm{dn}v+\mathrm{sn}v\cdot\mathrm{cn}u\cdot\mathrm{dn}u}{1-k^2\,\mathrm{sn}^2u\,\mathrm{sn}^2v}, \tag{6.11}$$

on using the known relations: $\mathrm{sn}^2u=1-\mathrm{cn}^2u$, and $\mathrm{dn}^2v=1-k^2\,\mathrm{sn}^2v$. This is the *addition-theorem for* sn, which holds without the restriction on v, by continuity.

§ 7. The modular function $\lambda(\tau)$. If τ is a complex number, with $\mathrm{Im}\,\tau>0$, we define the function $\lambda(\tau)$ by the relation

$$\lambda(\tau)=k^2(\tau)=\frac{\theta_1^4(0,\tau)}{\theta_3^4(0,\tau)}. \tag{7.1}$$

Since $\theta_1(0,\tau)$ and $\theta_3(0,\tau)$ are holomorphic for $\mathrm{Im}\,\tau>0$, and different from zero (cf. § 8, Corollary 3, Chapter V), $\lambda(\tau)$ is *holomorphic* for $\mathrm{Im}\,\tau>0$.

The connexion between $\lambda(\tau)$ and Weierstrass's function $\wp(u;\omega_1,\omega_2)$, with (ω_1,ω_2) as a pair of basic periods, and $\omega_2=\omega_1\tau$, $\mathrm{Im}\,\tau>0$, is provided by the relation (Corollary 1, Theorem 5, of Chapter V)

$$\lambda(\tau)=\frac{e_3-e_2}{e_1-e_2}, \tag{7.2}$$

where $e_1=\wp\left(\frac{\omega_1}{2}\right)$, $e_2=\wp\left(\frac{\omega_2}{2}\right)$, and $e_3=\wp\left(\frac{\omega_1+\omega_2}{2}\right)$. (No special choice of ω_1 is involved here). Since e_1, e_2, e_3 are all different (Proof of Theorem 5 in Chapter III), we have

$$\lambda(\tau)\neq 0,1. \tag{7.3}$$

Since $\theta_3(0,\tau+2)=\theta_3(0,\tau)$, and $\theta_1(0,\tau+1)=e^{\frac{\pi i}{4}}\theta_1(0,\tau)$, (see (8.10) in Chapter V), we have

$$\lambda(\tau+2)=\lambda(\tau). \tag{7.4}$$

This, as well as other properties of λ, can be proved from (7.2). For the transformation of a pair of basic periods (ω_1, ω_2) into another pair of basic periods (ω_1^*, ω_2^*) by means of the relations:

$$\omega_2^* = a\omega_2 + b\omega_1, \qquad \omega_1^* = c\omega_2 + d\omega_1,$$

where a, b, c, d are integers such that $ad - bc = 1$, induces a transformation of the quotient $\tau = \dfrac{\omega_2}{\omega_1}$ into $\tau^* = \dfrac{\omega_2^*}{\omega_1^*}$ by the relation

$$\tau^* = \frac{a\tau + b}{c\tau + d}.$$

Thus $\tau \to \tau^*$ is a *modular transformation* (as in Chapter I). It is generated by the two transformations

$$\tau \to \tau + 1, \quad \text{and} \quad \tau \to -\frac{1}{\tau}.$$

To determine the effect of these transformations on $\lambda(\tau)$, let

$$e_1^* = \wp\left(\frac{\omega_1^*}{2}\right), \qquad e_2^* = \wp\left(\frac{\omega_2^*}{2}\right), \qquad e_3^* = \wp\left(\frac{\omega_1^* + \omega_2^*}{2}\right).$$

If $a = 0, d = 0, c = 1, b = -1$, we have $\tau^* = -\dfrac{1}{\tau}, \dfrac{\omega_1^*}{2} = \dfrac{\omega_2}{2}, \dfrac{\omega_2^*}{2} = -\dfrac{\omega_1}{2}$, hence $e_1^* = e_2$, $e_2^* = e_1$, $e_3^* = e_3$, and

$$\lambda\left(\frac{-1}{\tau}\right) = \lambda(\tau^*) = \frac{e_3^* - e_2^*}{e_1^* - e_2^*} = \frac{e_3 - e_1}{e_2 - e_1} = 1 - \frac{e_3 - e_2}{e_1 - e_2} = 1 - \lambda(\tau). \tag{7.5}$$

If $a = b = 1, c = 0, d = 1$, we have $\tau^* = \tau + 1, \dfrac{\omega_2^*}{2} = \dfrac{\omega_2 + \omega_1}{2}$, and $\dfrac{\omega_1^*}{2} = \dfrac{\omega_1}{2}$, so that $e_1^* = e_1$, $e_2^* = e_3$, $e_3^* = e_2$, hence

$$\lambda(\tau + 1) = \lambda(\tau^*) = \frac{e_2 - e_3}{e_1 - e_3} = \frac{\lambda(\tau)}{\lambda(\tau) - 1}, \tag{7.6}$$

which also implies (7.4).

We note that each of the two transformations $\tau \to -\dfrac{1}{\tau}$ and $\tau \to \tau + 1$ just permutes the three numbers e_1, e_2, e_3 among themselves. It follows that *any* modular transformation does so.

Particular modular transformations $\tau \to \tau^*$ may leave the individual numbers e_1, e_2, e_3 unchanged. Such transformations obviously leave $\lambda(\tau)$ invariant.

If a, b, c, d are integers, such that $ad - bc = 1$, *and* b and c are *even*, then both a and d are *odd*. Then $\dfrac{\omega_1^*}{2} - \dfrac{\omega_1}{2}$ is a period of $\wp(u; \omega_1, \omega_2)$; so also $\dfrac{\omega_2^*}{2} - \dfrac{\omega_2}{2}$.

Hence $e_1^*=e_1$, $e_2^*=e_2$, $e_3^*=e_3$. Therefore

$$\lambda\left(\frac{a\tau+b}{c\tau+d}\right)=\lambda(\tau^*)=\frac{e_3^*-e_2^*}{e_1^*-e_2^*}=\frac{e_3-e_2}{e_1-e_2}=\lambda(\tau). \tag{7.7}$$

Conversely, if

$$\lambda\left(\frac{a\tau+b}{c\tau+d}\right)=\lambda(\tau), \tag{7.7$'$}$$

where a, b, c, d are integers with $ad-bc=1$, then b and c must be even, so that a and d are odd.

To see this, let

$$\omega_2'=a'\omega_2+b'\omega_1, \qquad \omega_1'=c'\omega_2+d'\omega_1,$$

where a', b', c', d' are integers such that $a'd'-b'c'=1$, and

$$\tau'=\frac{\omega_2'}{\omega_1'}=\frac{a'\tau+b'}{c'\tau+d'}, \quad \text{with} \quad \operatorname{Im}\tau'>0.$$

If

$$a\equiv a',\ b\equiv b',\ c\equiv c',\ d\equiv d' \pmod 2,$$

then $\lambda(\tau')=\lambda(\tau^*)$, since

$$\wp\left(\frac{\omega_2'}{2}\right)=\wp\left(\frac{a'\omega_2+b'\omega_1}{2}\right)=\wp\left(\frac{a\omega_2+b\omega_1}{2}\right)=\wp\left(\frac{\omega_2^*}{2}\right),$$

and similarly also

$$\wp\left(\frac{\omega_1'}{2}\right)=\wp\left(\frac{\omega_1^*}{2}\right), \qquad \wp\left(\frac{\omega_1'+\omega_2'}{2}\right)=\wp\left(\frac{\omega_1^*+\omega_2^*}{2}\right).$$

To determine the effect on $\lambda(\tau)$ of any modular transformation

$$\tau\to\tau^*=\frac{a\tau+b}{c\tau+d},\ ad-bc=1,$$

we may therefore suppose that a, b, c, d assume the values 0, $+1$, -1. Since $ad-bc=1$, one of the products ad and bc is necessarily zero, while the other is ± 1. The six different possibilities are given in the following table, which shows that the effect of the modular group acting on $\lambda(\tau)$ in the τ-plane is the same as that of the *anharmonic group*

$$\left\{\lambda, \frac{1}{1-\lambda}, \frac{\lambda-1}{\lambda}, \frac{1}{\lambda}, \frac{\lambda}{\lambda-1}, 1-\lambda\right\},$$

where $\lambda\neq 0, 1$, acting on λ in the λ-plane. The table also shows that the only modular transformations $\tau\to\tau^*$ which leave λ invariant are those which leave e_1, e_2, e_3 unchanged, and they are precisely those in which b and c are even, so that a and d are odd.

a	b	c	d	$\tau^* = \frac{a\tau+b}{c\tau+d}$	$e_1^* = \wp\left(\frac{\omega_1^*}{2}\right)$	$e_2^* = \wp\left(\frac{\omega_2^*}{2}\right)$	$e_3^* = \wp\left(\frac{\omega_1^*+\omega_2^*}{2}\right)$	$\lambda(\tau^*) = \frac{e_3^*-e_2^*}{e_1^*-e_2^*}$
1	0	0	1	τ	e_1	e_2	e_3	$\lambda(\tau)$
1	1	0	1	$\tau+1$	e_1	e_3	e_2	$\frac{\lambda(\tau)}{\lambda(\tau)-1}$
0	−1	1	0	$-\frac{1}{\tau}$	e_2	e_1	e_3	$1-\lambda(\tau)$
1	−1	1	0	$\frac{\tau-1}{\tau}$	e_2	e_3	e_1	$\frac{\lambda(\tau)-1}{\lambda(\tau)}$
1	0	1	1	$\frac{\tau}{\tau+1}$	e_3	e_2	e_1	$\frac{1}{\lambda(\tau)}$
0	1	−1	1	$\frac{1}{1-\tau}$	e_3	e_1	e_2	$\frac{1}{1-\lambda(\tau)}$

It follows that *if a, b, c, d are integers such that $ad-bc=1$, then*

$$\lambda\left(\frac{a\tau+b}{c\tau+d}\right)=\lambda(\tau), \qquad \operatorname{Im}\tau>0,$$

if and only if b and c are even, so that a and d are odd.

All transformations

$$\tau\to\tau^*=\frac{a\tau+b}{c\tau+d}, \qquad \operatorname{Im}\tau>0, \qquad ad-bc=1,$$

where b and c are even, while a and d are odd, form a group, which we shall denote by G. *Then G is a sub-group of the modular group Γ, of finite index.* By (7.7), the function $\lambda(\tau)$ is invariant under all transformations belonging to G. *Hence $\lambda(\tau)$ is a modular function.*

If A denotes the transformation $\tau\to\tau+1$, and B denotes the transformation $\tau\to-\frac{1}{\tau}$, then the transformations: $A^2:\tau\to\tau+2$, and $(BA^2B):\tau\to\frac{\tau}{1-2\tau}$ belong to G. In fact we have

Lemma 1. *The group G is generated by the transformations*

$$\tau\to\tau+2, \qquad \tau\to\frac{\tau}{1-2\tau}. \tag{7.8}$$

Proof. Let $M\in G$, where

$$M\tau=\frac{a\tau+b}{c\tau+d},\ ad-bc=1,\ \left.\begin{matrix} b\equiv c\equiv 0 \pmod 2 \\ a\equiv d\equiv 1 \pmod 2\end{matrix}\right\}.$$

We shall show that M is a product of powers of A^2 and (BA^2B), by using induction on $m=\min(|c|, |d|)$.

We may assume that $|c|\leqq|d|$, for if $|c|>|d|$, then the transformation $M'=M(BA^2B)^l$, l integral, is given by

$$M'\tau=\frac{a'\tau+b'}{c'\tau+d'}, \quad \text{say, with} \quad c'=c-2ld,$$

and belongs to G, since (BA^2B) and M do. Here we choose l such that

$$|c-2ld|\leqq|d|;$$

and if M' is generated by A^2 and (BA^2B), then so is M.

Let $m=0$. Then $c=0$, so that $ad=1$, and b is even, hence $M\tau=\tau\pm b=A^{\pm b}\tau$, and M is of the form claimed.

Assuming that any M for which $m\leqq n-1$, $n\geqq 1$, is generated by A^2 and (BA^2B), we shall show that M is so generated also when $m=n$. By the definition of m, and by our assumption on $|c|$, we have $|c|=m$. We may assume that $c>0$, for otherwise one can multiply a, b, c, d by -1 without changing M, or the *parity* of a and d, or that of b and c. [Two integers are said to have the same parity if their difference is even; they are said to be of opposite parity if their difference is odd.] Then $m=c$, and we have to show that M is generated by A^2 and (BA^2B) for $c=n$.

Since $\dfrac{d}{c}$ is not an integer, there exists an *even* integer k, such that $-1-(d/c)<k<1-(d/c)$. For such a k, we have $0<|ck+d|<c$, so that $\min(|c|, |ck+d|)\leqq n-1$, if $c=n$. Now

$$MA^k\tau=\frac{a\tau+(ak+b)}{c\tau+(ck+d)}, \qquad MA^k\in G,$$

and by the induction hypothesis, MA^k is generated by A^2 and (BA^2B), hence also M.

To determine a fundamental domain for G, as we did in the case of the modular group Γ in Chapter I, it is convenient to consider yet another subgroup, say Δ, of the group Γ, which may be defined in either of two ways, as in the following

Lemma 2. *Let Δ denote the subgroup of the modular group Γ, generated by the transformations:*

$$\tau\to\tau+2, \quad \textit{and} \quad \tau\to-\frac{1}{\tau}, \tag{7.9}$$

of the half-plane $\operatorname{Im}\tau>0$ into itself. Then a modular transformation M defined by $M\tau=\dfrac{a\tau+b}{c\tau+d}$, where a, b, c, d are integers with $ad-bc=1$, belongs to Δ if and only if both a and d are even, while b and c are odd, or both a and d are odd, while b and

c are even; in other words, a and d have the same parity, and b and c have the same parity, which is opposite to that of a and d.

Proof. Transformations with the properties of parity just stated form a group Δ', say. The assertion of the lemma is that $\Delta=\Delta'$.

Let A denote the transformation $\tau\to\tau+1$ (with $a=1, d=1, b=1, c=0$), and B denote: $\tau\to-\frac{1}{\tau}$ (with $a=0$, $d=0$, $c=1$, $b=-1$), and C denote A^2: $\tau\to\tau+2$ (with $a=1$, $d=1$, $b=2$, $c=0$).

Now $A^2\in\Delta'$, $B\in\Delta'$, hence $\Delta\subset\Delta'$. We have only to prove that $\Delta'\subset\Delta$. (Clearly $A\notin\Delta'$).

If $M\in\Delta'$, we have to show that M can be represented as a product of powers of A^2 and B. This can be done in the same way as in the proof of Lemma 1, by induction on $m=\min(|c|, |d|)$, where

$$M\tau=\frac{a\tau+b}{c\tau+d},\quad ad-bc=1,$$

and the parity of a and d is the opposite of the parity of b and c.

As before, it suffices to consider such transformations $M\in\Delta'$ as satisfy the condition: $|c|\leqq|d|$. For if $|c|>|d|$, then the transformation

$$MB:\tau\to\frac{-\frac{a}{\tau}+b}{-\frac{c}{\tau}+d}=\frac{b\tau-a}{d\tau-c}=\frac{a'\tau+b'}{c'\tau+d'}$$

has the property: $|c'|<|d'|$, and $MB\in\Delta'$.

Let $m=0$. Then $c=0$, so that $ad=1$, and b *even*, hence $M\tau=\tau\pm b=A^{\pm b}\tau$, with b even, and the lemma holds. It remains therefore only to consider the case $m>0$.

Let us assume that any M for which $m\leqq n-1$, where $n\geqq1$, is generated by A^2 and B. We shall show that M is so generated also when $m=n$. By the definition of m, and by the assumption that $|c|\leqq|d|$, we have $m=|c|$. As in Lemma 1, we may assume that $c>0$. Thus $m=c$, and we wish to show that M is generated by A^2 and B in case $c=n$.

Let k be an *even* integer, such that $|ck+d|<c$. Such a k exists, since $-(d/c)$ is not an integer, so that the open interval $(-1-(d/c), 1-(d/c))$ always contains an even integer. For such a k, we have $\min(|c|, |ck+d|)\leqq n-1$, *if* $c=n$. Consider the transformation

$$MA^k\tau=\frac{a\tau+(ak+b)}{c\tau+(ck+d)}.$$

Since k is even, $MA^k\in\Delta'$. By the induction hypothesis, together with the inequality: $\min(|c|, |ck+d|)\leqq n-1$, the transformation MA^k is generated by A^2 and B, hence also M, since k is even, in case $m=n$. It follows by induction that $\Delta=\Delta'$.

Lemma 3. *We have the disjoint decomposition*

$$\Gamma = \Delta \cdot \cup \cdot \Delta M_0 \cdot \cup \cdot \Delta M_0^2,$$

where Γ denotes the modular group, and Δ its subgroup generated by the two transformations $A^2 : \tau \to \tau + 2$, and $B : \tau \to -1/\tau$, while $M_0\tau = 1/(1-\tau)$.

Proof. Obviously we have $M_0^2\tau = (\tau - 1)/\tau$, and $M_0 \notin \Delta$, $M_0^2 \notin \Delta$, $M_0^3 = 1$ (the identity transformation). Further $M_0 = BA^{-1}$, $M_0^2 = AB$, while $B^2 = 1$, $BM_0 = A^{-1}$, and $A^2BM_0 = A$. If $M \in \Delta$, then $MM_0 \notin \Delta$, and $MM_0^2 \notin \Delta$, so that Δ and ΔM_0 are disjoint, as are Δ and ΔM_0^2. Further ΔM_0 and ΔM_0^2 are disjoint, for if $MM_0 = M'M_0^2$, for some M and M' belonging to Δ, then $M_0 = (M')^{-1} \cdot M \in \Delta$.

If we denote by Γ' the set $\Delta \cdot \cup \cdot \Delta M_0 \cdot \cup \cdot \Delta M_0^2$, obviously we have $\Gamma' \subset \Gamma$. To show that $\Gamma \subset \Gamma'$, it is sufficient to show that if $M \in \Gamma'$, then $A^{\pm 1}M \in \Gamma'$, $B^{\pm 1}M \in \Gamma'$, for Γ is generated by A and B (as we have seen in Chapter I).

Now $A \in \Gamma'$, since $A = A^2BM_0 \in \Delta M_0$; and $A^{-1} \in \Gamma'$, since $A^{-1} = BM_0 \in \Delta M_0$, while $B^{\pm 1} \in \Gamma'$, since $B^{\pm 1} \in \Delta$. We next show that $M_0\Delta \subset \Delta M_0 \cdot \cup \cdot \Delta M_0^2$. For that purpose, let $S \in \Delta$, so that $S\tau = \dfrac{a\tau + b}{c\tau + d}$, where a, b, c, d are integers such that $ad - bc = 1$, and a and d have the same parity, while b and c have the same parity, which is the opposite of that of the former pair. Then we have

$$M_0SM_0\tau = \frac{-d\tau + (c+d)}{(b-d)\tau + \{(c+d) - (a+b)\}},$$

$$M_0SM_0^{-1}\tau = \frac{(c+d)\tau - c}{\{(c+d) - (a+b)\}\tau - (c-a)}.$$

If d is *even*, then $M_0SM_0 \in \Delta$, which implies that $M_0S \in \Delta M_0^2$. If d is *odd*, then $M_0SM_0^{-1} \in \Delta$, which implies that $M_0S \in \Delta M_0$. It follows that $M_0\Delta \subset \Delta M_0 \cdot \cup \cdot \Delta M_0^2$ in both cases. We use this fact to show that if $M \in \Gamma'$, then $B^{\pm 1}M \in \Gamma'$, *and* $A^{\pm 1}M \in \Gamma'$. The former follows from the fact that Δ is a group. To show the latter we consider the three cases in which $M \in \Delta$, or ΔM_0, or ΔM_0^2.

If $M \in \Delta$, then $A^{\pm 1}M \in \Delta M_0\Delta \subset \Delta M_0 \cdot \cup \cdot \Delta M_0^2 \subset \Gamma'$. If $M \in \Delta M_0$, then $A^{\pm 1}M \in \Delta M_0 \cdot \Delta M_0 \subset \Delta M_0^2 \cdot \cup \cdot \Delta M_0^3 = \Delta M_0^2 \cdot \cup \cdot \Delta \subset \Gamma'$. If $M \in \Delta M_0^2$, then $A^{\pm 1}M \in \Delta M_0 \cdot \Delta M_0^2 \subset \Delta M_0^3 \cdot \cup \cdot \Delta M_0^4 = \Delta \cdot \cup \cdot \Delta M_0 \subset \Gamma'$. This completes the proof of Lemma 3.

A fundamental domain for G.

Let F_Γ be a fundamental domain for the modular group Γ, given by

$$F_\Gamma = U_1 \cdot \cup \cdot U_2,$$

where

$$U_1 = \{\tau \mid \operatorname{Im}\tau > 0,\ |\tau| \geqq 1,\ -\tfrac{1}{2} \leqq \operatorname{Re}\tau \leqq 0\},$$

and

$$U_2 = \{\tau \mid \operatorname{Im}\tau > 0,\ |\tau| > 1,\ 0 < \operatorname{Re}\tau < \tfrac{1}{2}\}.$$

For convenience, we obtain another fundamental domain F'_Γ for the same group Γ, by subjecting U_1 to the transformation $\tau \to \tau + 1$, so that

$$F'_\Gamma = U'_1 \cdot \cup \cdot U_2, \qquad U'_1 = \{\tau | \operatorname{Im}\tau > 0, |\tau - 1| \geqq 1, \tfrac{1}{2} \leqq \operatorname{Re}\tau \leqq 1\}.$$

By subjecting F'_Γ to the transformations M_0, M_0^2, and M_0^3, we obtain F_Δ, a fundamental domain for the group Δ, (because of Lemma 3), which is given by

$$\begin{aligned} F_\Delta &= F'_\Gamma \cdot \cup \cdot M_0^2 F'_\Gamma \cdot \cup \cdot M_0 F'_\Gamma \\ &= \{\tau | \operatorname{Im}\tau > 0, |\tau - \tfrac{1}{2}| \geqq \tfrac{1}{2}, 0 \leqq \operatorname{Re}\tau \leqq 1, \quad \text{with} \quad \tfrac{1}{2} \leqq \operatorname{Re}\tau \leqq 1 \quad \text{if} \quad |\tau - \tfrac{1}{2}| = \tfrac{1}{2}, \\ &\quad \text{and} \quad \operatorname{Im}\tau \geqq 1 \quad \text{if} \quad \operatorname{Re}\tau = 1, \quad \text{and} \quad \operatorname{Im}\tau \leqq 1 \quad \text{if} \quad \operatorname{Re}\tau = 0\}. \end{aligned}$$

Every point in the half-plane $\operatorname{Im}\tau > 0$ is congruent, with respect to Δ, to exactly one point in F_Δ.

Now let $M \in G$, where $M\tau = \dfrac{a\tau + b}{c\tau + d}$, $ad - bc = 1$, $a \equiv d \equiv 1 \pmod 2$, and $b \equiv c \equiv 0 \pmod 2$. Then $MB\tau = \dfrac{b\tau - a}{d\tau - c}$, so that we have

$$\Delta = GB \cdot \cup \cdot G, \quad \text{since} \quad B^2 = 1,$$

and *the set*

$$\tilde{F}_G = F_\Delta \cdot \cup \cdot BF_\Delta$$

is a fundamental domain for the group G, since F_Δ is a fundamental domain for Δ. Clearly (see Fig. 9)

$$F_G = \{\tau | \operatorname{Im}\tau > 0, -1 \leqq \operatorname{Re}\tau < 1, |\tau + \tfrac{1}{2}| \geqq \tfrac{1}{2}, |\tau - \tfrac{1}{2}| > \tfrac{1}{2}\}$$

is also a fundamental domain for the group G, so that every point in the half-plane $\operatorname{Im}\tau > 0$ is congruent, with respect to G, to exactly one point in F_G. (The line $\operatorname{Re}\tau = 1$, $\operatorname{Im}\tau \geqq 1$ goes into $\operatorname{Re}\tau = -1$, $\operatorname{Im}\tau \geqq 1$, by the transformation $A^{-2} \in G$, and the arc $|\tau - \tfrac{1}{2}| = \tfrac{1}{2}$, $\tfrac{1}{2} \leqq \operatorname{Re}\tau \leqq 1$, goes into the arc $|\tau + \tfrac{1}{2}| = \tfrac{1}{2}$, $-1 \leqq \operatorname{Re}\tau \leqq -\tfrac{1}{2}$, by the transformation $\tau \to \tau/(1 - 2\tau)$.) To study the behaviour of $\lambda(\tau)$, we may restrict ourselves to the domain F_G.

Summarizing our observations concerning the function $\lambda(\tau)$, we have

Theorem 2. *The function $\lambda(\tau)$, defined for $\operatorname{Im}\tau > 0$, by the relation*

$$\lambda(\tau) = \frac{\theta_1^4(0, \tau)}{\theta_3^4(0, \tau)},$$

is holomorphic, and invariant under all transformations of the (proper) subgroup G of the modular group Γ, which is generated by the transformations

$$\tau \to \tau + 2, \quad \text{and} \quad \tau \to \frac{\tau}{1 - 2\tau},$$

so that $\lambda(\tau)$ is a modular function, and a fundamental domain for the group G is the set

$$F_G = F_G^1 \cdot \cup \cdot F_G^2,$$

where

$$F_G^1 = \{\tau | \operatorname{Im}\tau > 0, |\tau - \tfrac{1}{2}| > \tfrac{1}{2}, 0 < \operatorname{Re}\tau < 1\},$$

and (7.10)

$$F_G^2 = \{\tau | \operatorname{Im}\tau > 0, |\tau + \tfrac{1}{2}| \geqq \tfrac{1}{2}, -1 \leqq \operatorname{Re}\tau \leqq 0\}.$$

Remark. If a is any complex number, the equation $J(\tau) - a = 0$ has six roots in F_G, because of the way F_G has been obtained from F_Γ, a fundamental domain for the modular group Γ, for every transformation of which $J(\tau)$ remains invariant.

§ 8. Mapping properties of $\lambda(\tau)$ and Picard's theorem. Some general properties of λ are easily derived from the definition itself.

(i) The function $\lambda(\tau)$ is real for $\tau \in \partial F_G$, the boundary of the fundamental domain F_G, as well as on the boundary of F_G^1 and of F_G^2 separately (defined as in (7.10)). For $\lambda(\tau)$ assumes conjugate complex values at points symmetric relative to the imaginary axis in the half-plane $\operatorname{Im}\tau > 0$, since $\theta_1(0,\tau)$ and $\theta_3(0,\tau)$ have that property, so that $\lambda(iy)$ is real for $y > 0$; and since $\lambda(\tau+1) = \dfrac{\lambda(\tau)}{\lambda(\tau)-1}$, by (7.6), $\lambda(\tau)$ is also real on the line $\operatorname{Re}\tau = 1$, $\operatorname{Im}\tau > 0$, and on the line $\operatorname{Re}\tau = -1$, $\operatorname{Im}\tau > 0$. The transformation $\tau \to 1 - \dfrac{1}{\tau}$ maps the line $\operatorname{Re}\tau = 1$, $\operatorname{Im}\tau > 0$, on the semi-circle $|\tau - \tfrac{1}{2}| = \tfrac{1}{2}$, $\operatorname{Im}\tau > 0$, and by (7.5) and (7.6), $\lambda(\tau)$ is real on that semi-circle, hence also on the semi-circle $|\tau + \tfrac{1}{2}| = \tfrac{1}{2}$, $\operatorname{Im}\tau > 0$.

(ii) The behaviour of $\lambda(\tau)$ at the 'corners' of F_G can be seen as follows. The function $\lambda(\tau) \to 0$ as $\operatorname{Im}\tau \to +\infty$, uniformly for $-1 \leqq \operatorname{Re}\tau \leqq 1$, since $\theta_1^4(0,\tau) \to 0$, and $\theta_3^4(0,\tau) \to 1$, as $\operatorname{Im}\tau \to +\infty$, in that strip. In particular, if $\tau = i\xi$, $\xi > 0$, then $\lambda(\tau) \to 0$, as $\xi \to +\infty$, and by (7.5), $\lambda(\tau) \to 1$, as $\xi \downarrow 0$. Further $\lambda(\tau) \to +\infty$, as $\tau \to 1$ along the semi-circle $|\tau - \tfrac{1}{2}| = \tfrac{1}{2}$ in the half-plane $\operatorname{Im}\tau > 0$. If $\tau = \pm 1 + i\xi$, $\xi > 0$, then $\lambda(\tau) \to 0$, as $\xi \to +\infty$, while $\lambda(\tau) \to -\infty$ as $\xi \downarrow 0$. $\lambda(\tau)$ is positive for $\tau = i\xi$, $\xi > 0$, and negative for $\tau = \pm 1 + i\xi$, $\xi > 0$.

(iii) Since

$$\lambda(\tau) = \frac{\theta_1^4(0,\tau)}{\theta_3^4(0,\tau)} = \frac{[2\{q^{1/4} + q^{9/4} + \ldots\}]^4}{[1 + 2q + 2q^4 + \ldots]^4}, \qquad q = e^{\pi i \tau}, \operatorname{Im}\tau > 0,$$

$$= \frac{16\{q + \ldots\}}{\{1 + \ldots\}},$$

we have

$$\lambda(\tau)\cdot e^{-\pi i\tau}\to 16, \quad \text{as} \quad \operatorname{Im}\tau\to+\infty. \tag{8.1}$$

(iv) By using the transformation formulae (7.4), (7.5), and (7.6), one can evaluate $\lambda(\tau)$ for special values of τ. Thus

$$\lambda(i)=\frac{1}{2}, \quad \lambda(1+i)=\lambda(-1+i)=-1, \quad \lambda\left(\frac{1}{2}+\frac{i}{2}\right)=2. \tag{8.2}$$

The first follows from (7.5) with $\tau=i$; the second follows on using (7.6) with $\tau=i$, together with the first; the third results from

$$\lambda\left(\frac{1}{2}+\frac{i}{2}\right)=\lambda\left(1+\frac{-1+i}{2}\right)=\frac{\lambda\left(\dfrac{-1+i}{2}\right)}{\lambda\left(\dfrac{-1+i}{2}\right)-1}=\frac{1-\lambda(1+i)}{-\lambda(1+i)}=2,$$

by (7.6) with $\tau=\dfrac{-1+i}{2}$, and (7.5) with $\tau=1+i$.

(v) There is a direct connexion between the functions $\lambda(\tau)$ and $J(\tau)$. By Theorem 2 of Chapter VI, we have

$$J(\tau)=\frac{1}{54}\cdot\frac{\{\theta_3^8(0,\tau)+\theta_2^8(0,\tau)+\theta_1^8(0,\tau)\}^3}{\{\theta_1(0,\tau)\cdot\theta_2(0,\tau)\cdot\theta_3(0,\tau)\}^8}, \quad \operatorname{Im}\tau>0.$$

From Corollary 1 of Theorem 5 of Chapter V, we have

$$\theta_1^4(0,\tau)+\theta_2^4(0,\tau)=\theta_3^4(0,\tau),$$

and since $\lambda(\tau)=\theta_1^4(0,\tau)/\theta_3^4(0,\tau)$, we obtain

$$(\theta_3^8+\theta_2^8+\theta_1^8)=(1+(1-\lambda)^2+\lambda^2)\theta_3^8=2(1-\lambda+\lambda^2)\theta_3^8,$$

where $\theta_r=\theta_r(0,\tau)$, for $r=1, 2, 3$, and $\lambda=\lambda(\tau)$. On the other hand,

$$(\theta_1\cdot\theta_2\cdot\theta_3)^8=(\lambda^2\theta_3^8)\cdot((1-\lambda)^2\theta_3^8)\cdot\theta_3^8=\lambda^2(1-\lambda)^2\theta_3^{24},$$

hence we have

$$J(\tau)=\frac{1}{54}\frac{\{2(1-\lambda+\lambda^2)\theta_3^8\}^3}{\lambda^2(1-\lambda)^2\theta_3^{24}}=\frac{4}{27}\cdot\frac{(1-\lambda+\lambda^2)^3}{\lambda^2(1-\lambda)^2}, \tag{8.3}$$

where $\lambda=\lambda(\tau)$, $\operatorname{Im}\tau>0$.

This formula can be used to prove the following

Theorem 3. *If a is any complex number, where $a\neq 0, 1$, then the equation $\lambda(\tau)-a=0$ has one and only one root in F_G, a fundamental domain for the group G, which leaves $\lambda(\tau)$ invariant.*

Proof. We have seen in the proof of Theorem 1 that there exists a τ, with $\operatorname{Im}\tau>0$, such that $\lambda(\tau)=a$. Since the function λ is left invariant by the transformations of the group G, and since F_G is a fundamental domain for G, there exists a point τ in F_G, such that $\lambda(\tau)=a$.

To show that this point is unique, suppose that for a given number a, there exist two points τ_1, τ_2, $\tau_1 \neq \tau_2$, in F_G, such that $\lambda(\tau_1)=\lambda(\tau_2)=a$. Then $J(\tau_1)=J(\tau_2)$ by (8.3). Hence there exists a modular transformation $M \in \Gamma$, such that $\tau_2 = M\tau_1$ (cf. Theorem 3, Chapter VI). By (7.7)′ it follows that $M \in G$, which is impossible since F_G is a fundamental domain for G, and $\tau_1, \tau_2 \in F_G$, with $\tau_1 \neq \tau_2$.

Theorem 4. *Let* $F_G = F_G^1 \cdot \cup \cdot F_G^2$,
where

$$F_G^1 = \{\tau \mid \operatorname{Im}\tau>0,\ |\tau-\tfrac{1}{2}|>\tfrac{1}{2},\ 0<\operatorname{Re}\tau<1\},$$

and

$$F_G^2 = \{\tau \mid \operatorname{Im}\tau>0,\ |\tau+\tfrac{1}{2}|\geqq\tfrac{1}{2},\ -1\leqq\operatorname{Re}\tau\leqq 0\},$$

so that F_G is a fundamental domain for the (proper) subgroup G of the modular group. Then the function $\lambda(\tau)=w$ maps the set F_G^1, one-to-one, onto the half-plane $\operatorname{Im}w>0$, and maps the set F_G^2, one-to-one, onto the half-plane $\operatorname{Im}w\leqq 0$, with the exception of the points $w=0,1$.

Proof. We have seen that $\lambda(\tau)\neq 0, 1$, for $\operatorname{Im}\tau>0$ (Theorem 1), and that $\lambda(\tau)$ is real for $\tau \in \partial F_G^2$. If $\tau = i\xi$, with $\xi>0$, then $\lambda(\tau)\to 0$ as $\xi\to+\infty$, and $\lambda(\tau)\to 1$ as $\xi\downarrow 0$. Further $\lambda(\tau)\to+\infty$ as $\tau\to 1$ along the semi-circle $|\tau-\frac{1}{2}|=\frac{1}{2}$ in the half-plane $\operatorname{Im}\tau>0$. If $\tau=\pm 1+i\xi$, then $\lambda(\tau)\to 0$ as $\xi\to+\infty$, and $\lambda(\tau)\to-\infty$, as $\xi\downarrow 0$. (Properties (i) and (ii) listed above). By Theorem 3, $\lambda(\tau)$ assumes no real value more than once in ∂F_G^2.

By the same reasoning as in the case of $J(\tau)$ in Chapter VI, we prove that $\lambda(\tau)$ is *not* real for $\tau \in \operatorname{Int}F_G^1 \cdot \cup \cdot \operatorname{Int}F_G^2$. By Theorem 3, every non-real value is assumed only once in $\operatorname{Int}F_G^1 \cdot \cup \cdot \operatorname{Int}F_G^2$.

It follows that $\operatorname{Im}\lambda(\tau)$ is always of the same sign for $\tau \in F_G^1$. Since $\lambda(\tau)\cdot e^{-\pi i\tau} \to 16$, as $\operatorname{Im}\tau\to+\infty$, it follows that $\operatorname{Im}\lambda(\tau)>0$, for $\tau \in F_G^1$, hence $\operatorname{Im}\lambda(\tau)\leqq 0$, for $\tau \in F_G^2$.

Picard's theorem. *If $g(z)$ is an entire function of the complex variable z, such that $g(z)\neq 0, 1$, then $g(z)$ reduces to a constant.*

Proof. The function $\lambda(\tau)=w$ maps F_G onto the complex w-plane without the points $w=0, 1$. The inverse function ϖ, defined by $\lambda(\varpi(z))=z$, for $z\neq 0, 1, \infty$, is infinitely many-valued, but locally holomorphic, with $\operatorname{Im}\varpi(z)>0$. By the principle of monodromy, $\varpi(g(z))$ is holomorphic, and since $\operatorname{Im}\varpi(g(z))>0$, the function $e^{i\varpi(g(z))}$ is a bounded, entire function, hence a constant (by Liouville's theorem), so that $g(z)$ must reduce to a constant.

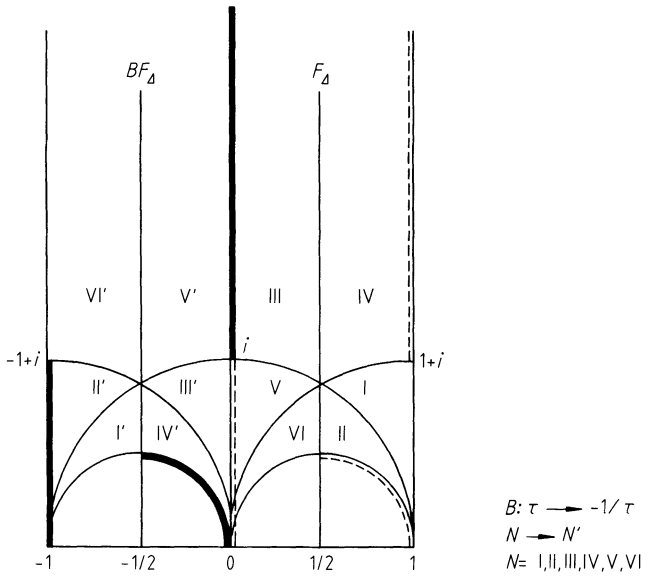


Fig. 9

Notes on Chapter VII

§§ 1–3. There are many different ways of defining the Jacobian elliptic functions. As mentioned already in the Notes on Chapters I and III, they were originally discovered by the inversion of integrals of the type

$$u=\int_0^y \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}, \qquad k^2 \neq 0, 1,$$

(cf. Remarks, § 4), some thirty years before the Weierstrassian elliptic functions. One can define them, after Hurwitz, by means of the sigma-functions as in (3.2). Defining them as quotients of theta-functions makes for simpler proofs.

The function $k^2(\tau)$ – the square of the modulus – was studied by Abel, Jacobi, and Hermite. See the references to Hermite in Dedekind's paper cited in the Notes on Chapter VI. See also § 54 of H.A. Schwarz's book (cited in the Notes on Ch. III).

§ 4. The problem of solving the equation $k^2(\tau)=a$ in (4.8) for a given a, $a \neq 0, 1$, is referred to as the *problem of inversion* for k^2. The solution here effected, by an appeal to Theorem 6 of Chapter VI, is due to Hurwitz. Jacobi's original proof (*Ges. Werke*, I, p. 520) holds only for $0 < a < 1$, as was noted by Weierstrass (ibid. p. 545), who filled the gap, *Werke*, II, 257–309.

§ 5. The infinite products for $\mathrm{sn}u$, $\mathrm{cn}u$, $\mathrm{dn}u$, are here derived from the infinite products for theta-functions obtained in Chapter V. This reverses, of course, the historical development.

It may be of interest here to note the contributions of G. Eisenstein, *Math. Werke*, Bd. I (Chelsea Reprint, 1975), 299–478. Unlike Weierstrass, Eisenstein was unaware of the notion of uniform convergence. Nevertheless he was aware that the quotients of infinite products gave rise to elliptic functions. His method of proof elicited critical comments from Jacobi (*Crelle's J.* 30 (1845), 183–184; *Werke*, II, 85–86), A.L. Daniels (*American J. Math.* 6 (1884), 253–269; see pp. 261–262), and A. Enneper (in § 11 of his book cited in the Notes on Ch. I). As Enneper says, the results found by Eisenstein can be verified by relatively elementary means, if one follows the method of H.E. Heine, *Crelle's J.* 34, pp. 122–137. (cf. André Weil, *Elliptic functions according to Eisenstein and Kronecker*, *Ergebnisse der Math.* 88 (1976)).

§ 6. This proof of the addition-theorem seems to go back to Liouville. See, for instance, § 89 of the book by Appell and Lacour, cited in the Notes on Chapter I. Four other methods of proving the addition-theorem are mentioned by Whittaker and Watson (in § 22.2 of their book cited in § 4 of the Notes on Ch. I), due respectively to Euler, Abel, Jacobi, and Cayley.

§ 7. As has been mentioned already (in § 1 of the Notes on Ch. VI), the function $k^2(\tau)$ was studied by Abel, Jacobi, and Hermite, *before* Dedekind, and Klein, introduced the function $J(\tau)$.

Eisenstein and Weierstrass use the notation $\lambda(\tau)$ instead of $k^2(\tau)$.

The *anharmonic group* is generated by the transformations $S: \lambda \to \frac{1}{\lambda}$ and T: $\lambda \to 1-\lambda$. We have $S^2 = T^2 = (ST)^3 = (TS)^3 = 1$, while $ST\lambda = \frac{1}{1-\lambda}$, $TS\lambda = \frac{\lambda-1}{\lambda}$, and $STS\lambda = TST\lambda = \frac{\lambda}{\lambda-1}$.

The table shows that G is of index 6 in Γ; G is known as the *principal congruence subgroup of level* 2 (of Γ). Lemma 3 shows that Δ is of index 3 in Γ. The argument preceding Theorem 2 shows that G is of index 2 in Δ.

§ 8. Dedekind (*Ges. Werke*, I, p. 193) remarks on formula (8.3): "In dieser Form erscheint die Funktion $v(=J(\tau))$ an mehreren Stellen der berühmten Abhandlung von Hermite über die Theorie der Modulargleichungen; ich bemerke zugleich, daß auch Gauss (*Werke*, III, s. 386) die Absicht gehabt hat, eine solche Funktion einzuführen." (The function v appears in this form at several places in the famous paper of Hermite on the theory of modular equations. I note at the same time that Gauss also had the intent to introduce such a function).

Dedekind proves that if $\lambda(\tau_1) = \lambda(\tau_2)$, $\operatorname{Im}\tau_1 > 0$, $\operatorname{Im}\tau_2 > 0$, then

$$\tau_2 = \frac{a\tau_1 + b}{c\tau_1 + d}, \qquad ad - bc = 1,$$

in which b and c are *even*, while a and d are *odd*, integers. His proof consists in comparing the zeros of $\operatorname{sn}^2(u, k(\tau_1))$ and of $\operatorname{sn}^2(u, k(\tau_2))$. It runs as follows.

The function $y=\mathrm{sn}(u,k)=\mathrm{sn}(u,k(\tau))=\mathrm{sn}(u,\tau)$ satisfies the differential equation

$$\left(\frac{dy}{du}\right)^2=(1-y^2)(1-k^2(\tau)y^2),\qquad k^2(\tau)=\lambda(\tau).$$

The function $z=\mathrm{sn}^2(u,\tau)$ satisfies the equation

$$\left(\frac{dz}{du}\right)^2=4z(1-z)(1-k^2(\tau)z),$$

by (3.9) and (4.1). If $\lambda(\tau_1)=\lambda(\tau_2)$, then $\mathrm{sn}^2(u,\tau_1)$ and and $\mathrm{sn}^2(u,\tau_2)$ satisfy the same differential equation; hence they coincide, since $\mathrm{sn}^2(u,\tau)$ has an expansion of the form

$$Au^2+Bu^4+Cu^6+\ldots,$$

and the differential equation determines the coefficients A, B, C, ..., of this expansion as functions of k. Thus $\mathrm{sn}^2(u,\tau_1)$ and $\mathrm{sn}^2(u,\tau_2)$ have the same zeros. Since $\mathrm{sn}(u,\tau_1)$ has the periods $(2\omega_1,\omega_2)$, where $\omega_2=\omega_1\tau_1$, and has its zeros precisely at the points $\{m\omega_1+n\omega_2\}$, where m,n are integers, the set $\{m\omega_1+n\omega_2\}$ is the same as the set $\{m'\omega_1'+n'\omega_2'\}$, where $\omega_2'=\omega_1'\tau_2$, and m', n' are integers. This implies that

$$\omega_2'=a\omega_2+b\omega_1,\qquad \omega_1'=c\omega_2+d\omega_1,$$

where a,b,c,d are integers, with $ad-bc=\pm 1$. Since $\mathrm{Im}\,\tau_1>0$, and $\mathrm{Im}\,\tau_2>0$, we have: $ad-bc=1$. Now $\mathrm{sn}^2(u,\tau_2)$ has a pole at $u=\frac{1}{2}\omega_2'=\dfrac{a\omega_2+b\omega_1}{2}$, which implies that b must be *even*, hence both a and d must be odd. Further $\mathrm{sn}^2\left(\dfrac{\omega_1'}{2},\tau_2\right)=1$, hence $\mathrm{cn}(u,\tau_2)=0$, for $u=\dfrac{c\omega_2+d\omega_1}{2}$, which implies that c is *even* (cf. § 2).

For Picard's theorem, see E. Picard, *Comptes Rendus*, Paris, 88 (1879), 1024–1027; *Oeuvres*, I, pp. 19–22 (*CNRS*, Paris, 1978). As Lars Ahlfors stresses in his *Complex Analysis*, 3rd edn., (1979), Ch. 8, the rôle of the principle of monodromy in the proof is just as important as that of the modular function λ. It is also the reason, as Raghavan Narasimhan has remarked, that the Second Theorem of Picard (which states that if $f(z)$ is holomorphic in the punctured circular disc $D_\varepsilon: 0<|z|<\varepsilon$, and has an essential singularity at $z=0$, then $f(D_\varepsilon)$ contains every complex number with at most one exception) requires a little more work – while $\mathbb{C}$ is simply connected, the set $\{0<|z|<\varepsilon\}$ is not. See Chapter IV of Raghavan Narasimhan's *Complex analysis in one variable* (Birkhäuser, 1985); the method there used has generalizations to several complex variables.

Chapter VIII

Dedekind's η-function and Euler's theorem on pentagonal numbers

§ 1. Connexion with the invariants of the $\wp$-function, and with the theta-functions. Associated with Weierstrass's function $\wp(u;\omega_1,\omega_2)$, with (ω_1,ω_2) as a pair of primitive periods, are the *invariants* $g_2=g_2(\omega_1,\omega_2)$ and $g_3=g_3(\omega_1,\omega_2)$, defined by

$$g_2(\omega_1,\omega_2)=60\sum{}'(m\omega_1+n\omega_2)^{-4},\qquad g_3(\omega_1,\omega_2)=140\sum{}'(m\omega_1+n\omega_2)^{-6},$$

as well as the function

$$\Delta=\Delta(\omega_1,\omega_2)=g_2^3(\omega_1,\omega_2)-27g_3^2(\omega_1,\omega_2),$$

(§ 1 of Chapter VI) for which we have (by Theorem 7, Chapter V) the product-representation

$$\Delta(\omega_1,\omega_2)=\left(\frac{2\pi}{\omega_1}\right)^{12}\cdot q^2\cdot\prod_{n=1}^{\infty}(1-q^{2n})^{24},\qquad q=e^{\pi i\tau},\qquad \operatorname{Im}\tau>0,\tau=\frac{\omega_2}{\omega_1}. \tag{1.1}$$

If $\omega_1=1$, then $\tau=\omega_2$, with $\operatorname{Im}\tau=\operatorname{Im}\omega_2>0$, and

$$\Delta(\tau)\equiv\Delta(1,\tau)=(2\pi)^{12}\cdot e^{2\pi i\tau}\cdot\prod_{n=1}^{\infty}(1-e^{2\pi in\tau})^{24}. \tag{1.2}$$

As already observed, $\Delta(\tau)$ is holomorphic for $\operatorname{Im}\tau>0$, and $\Delta(\tau)\neq 0$ (Theorem 1, Chapter VI).

Dedekind's η-function is defined by the relation

$$\eta(z)=e^{\pi iz/12}\cdot\prod_{n=1}^{\infty}(1-e^{2\pi inz}),\qquad \operatorname{Im}z>0, \tag{1.3}$$

so that $\eta(z)$ is holomorphic for $\operatorname{Im}z>0$, and

$$\eta^{24}(z)=(2\pi)^{-12}\Delta(1,z)=(2\pi)^{-12}\Delta(z),\qquad \operatorname{Im}z>0. \tag{1.4}$$

The η-function is obviously connected with the theta-functions by the formula

$$2\pi\cdot\eta^3(z)=\theta'(0,z),\qquad \operatorname{Im}z>0, \tag{1.5}$$

which results from (6.9) and (6.5) of Chapter V. Of particular interest is another connexion, given by the relation

$$\eta(z)=e^{\pi iz/12}\theta_3\left(\frac{1}{2}+\frac{z}{2},3z\right),\qquad \operatorname{Im}z>0, \tag{1.6}$$

which yields Euler's theorem on pentagonal numbers (Corollary 2, Theorem 1).

We shall prove (1.6) in two different ways. The first proof, due to Jacobi, consists in using the product-representation of $\theta_3\left(\frac{1}{2}+\frac{z}{2}, 3z\right)$, and showing directly that it coincides with the defining product for $e^{-\pi i z/12}\eta(z)$. The second proof, due to Siegel, is more sophisticated but of wider scope (as will be clear from the following chapters), and consists in showing that the function $K(z)$, defined for $\mathrm{Im}\, z>0$, by the relation

$$K(z)=\frac{\eta(z)-g(z)}{\eta(z)},\quad g(z)=e^{\pi i z/12}\theta_3\left(\frac{1}{2}+\frac{z}{2}, 3z\right)$$

is a *modular function* which vanishes identically.

§ 2. Euler's theorem and Jacobi's proof

Theorem 1. *For* $\mathrm{Im}\, z>0$, *we have*

$$\eta(z)=e^{\frac{\pi i z}{12}}\theta_3\left(\frac{1}{2}+\frac{z}{2}, 3z\right)$$

Proof (Jacobi). By Theorem 6 of Chapter V, we have

$$\theta_3(v, z)=\prod_{n=1}^{\infty}(1-e^{2n\pi i z})\cdot\prod_{n=1}^{\infty}(1+e^{(2n-1)\pi i z}\cdot e^{2\pi i v})\cdot\prod_{n=1}^{\infty}(1+e^{(2n-1)\pi i z}\cdot e^{-2\pi i v}), \tag{2.1}$$

where v is a complex variable, $\mathrm{Im}\, z>0$, so that

$$\theta_3\left(\frac{1}{2}+\frac{z}{2}, 3z\right)=\prod_{n=1}^{\infty}(1-e^{6n\pi i z})\cdot\prod_{n=1}^{\infty}(1+e^{(2n-1)3\pi i z}\cdot e^{2\pi i\left(\frac{1}{2}+\frac{z}{2}\right)})\times$$
$$\cdot\prod_{n=1}^{\infty}(1+e^{(2n-1)3\pi i z}\cdot e^{-2\pi i\left(\frac{1}{2}+\frac{z}{2}\right)}).$$

If we set $Q=e^{2\pi i z}$, we can rewrite this as

$$\begin{aligned}\theta_3\left(\frac{1}{2}+\frac{z}{2}, 3z\right)&=\prod_{n=1}^{\infty}(1-Q^{3n})\cdot\prod_{n=1}^{\infty}(1+Q^{3\left(n-\frac{1}{2}\right)}\cdot(-1)Q^{1/2})\times\\&\cdot\prod_{n=1}^{\infty}(1+Q^{3\left(n-\frac{1}{2}\right)}\cdot(-1)\cdot Q^{-1/2})\\&=\prod_{n=1}^{\infty}(1-Q^{3n})\cdot\prod_{n=1}^{\infty}(1-Q^{3n-1})\cdot\prod_{n=1}^{\infty}(1-Q^{3n-2})\\&=\prod_{n=0}^{\infty}(1-Q^{3n+3})\cdot\prod_{n=0}^{\infty}(1-Q^{3n+2})\cdot\prod_{n=0}^{\infty}(1-Q^{3n+1})\\&=\prod_{m=1}^{\infty}(1-Q^{m}),\end{aligned}$$

and because of (1.3), the theorem follows.

Corollary 1 (L. Euler). *If* $|x|<1$, *then*

$$\prod_{n=1}^{\infty}(1-x^n)=\sum_{n=-\infty}^{\infty}(-1)^n x^{\frac{1}{2}n(3n+1)}. \tag{2.2}$$

This is an immediate consequence of Theorem 1, if we note that by definition ((3.2) of Chapter V), we have

$$\theta_3\left(\frac{1}{2}+\frac{z}{2}, 3z\right)=\sum_{n=-\infty}^{\infty} e^{3\pi i n^2 z+2\pi i n\left(\frac{1}{2}+\frac{z}{2}\right)}.$$

Corollary 2 (L. Euler). *Let* b_n *denote the number of representations of a given positive integer* n *as a sum of* k *different integers, namely*

$$n=n_1+\ldots+n_k,\ 0<n_1<n_2<\ldots<n_k, \tag{2.3}$$

in which the number of summands k *is even, minus the number of representations of* n *in which* k *is odd. Then*

$$b_n=\begin{cases}(-1)^m, & \text{for } n=\frac{1}{2}(3m+1)m,\\ 0, & n\neq\frac{1}{2}(3m+1)m,\end{cases} \tag{2.4}$$

where m *is integral,* $m\gtrless 0$.

This follows from Corollary 1, if we note that

$$\prod_{n=1}^{\infty}(1-x^n)=\sum_{n=0}^{\infty}b_n x^n, \qquad |x|<1, \qquad b_0=1,$$

for the infinite product on the left-hand side contributes terms of the form

$$(-x^{n_1})(-x^{n_2})\ldots(-x^{n_k})=(-1)^k x^{n_1+n_2+\ldots+n_k}=(-1)^k x^n.$$

Remarks

(i) It is easy to check that $b_1=-1$, $b_2=-1$, $b_{11}=0$, $b_{12}=-1$, ..., and so on, and that

$$\prod_{n=1}^{\infty}(1-x^n)=1-x-x^2+x^5+x^7-x^{12}-x^{15}+x^{22}+x^{26}-x^{35}-x^{40}+\ldots. \tag{2.5}$$

The numbers $(\frac{1}{2}m(3m+1))$, $m=-1, -2, -3, \ldots$, are known as the *pentagonal numbers*. These are: 1, 5, 12, 22, ... The numbers 1, 2, 5, 7, 12, 15, 22, ..., occurring in the sequence of exponents in the series (2.5) are known as *generalized pentagonal numbers*, and Corollary 2 is known as Euler's theorem on pentagonal numbers.

If $E(n)$ denotes the number of representations of n of the form (2.3) in which k is *even*, and $U(n)$ denotes the number of representations of n of the form (2.3) in which k is *odd*, then, by Corollary 2, we have $E(n)=U(n)$ for all n *except* those n of the form $\frac{1}{2}m(3m\pm 1)$; for *such* n, we have $E(n)-U(n)=(-1)^m$.

(ii) Let $p(n)$ denote the number of unrestricted *partitions* of a positive integer n as a sum of *integers*, that is to say the number of representations of n of

the form

$$n=t_1+2t_2+3t_3+\ldots+kt_k+\ldots,$$

where $t_k\geqq 0$, for $k=1,2,\ldots$, (where, of course, only finitely many integers (t_k) can be different from zero). Then t_k is just the number of times that the positive integer k occurs in the representation

$$n=y_1+y_2+\ldots+y_r,$$

where $y_l>0$, y_l integral, for $l=1,2,\ldots,r$. Obviously we have

$$1+\sum_{n=1}^{\infty}p(n)x^n=\frac{1}{(1-x)(1-x^2)(1-x^3)\ldots}, \quad \text{for} \quad |x|<1. \tag{2.6}$$

On the other hand, by Corollary 1, we have

$$\begin{aligned}(1-x)(1-x^2)(1-x^3)\ldots&=1+\sum_{n=1}^{\infty}(-1)^n\{x^{\frac{1}{2}n(3n-1)}+x^{\frac{1}{2}n(3n+1)}\}\\&=1-x-x^2+x^5+x^7-x^{12}-\ldots.\end{aligned}$$

Hence

$$\left(1+\sum_{n=1}^{\infty}p(n)x^n\right)(1-x-x^2+x^5+x^7-\ldots)=1, \quad \text{for} \quad |x|<1,$$

so that we have

$$\begin{aligned}p(n)-p(n-1)-p(n-2)+p(n-5)+\ldots+(-1)^k p(n-\tfrac{1}{2}k(3k-1))+\\+(-1)^k p(n-\tfrac{1}{2}k(3k+1))+\ldots=0,\end{aligned} \tag{2.7}$$

a formula which can be used to compute $p(n)$.

The infinite product for $\theta_3(v,z)$ given in (2.1) can be used to derive an identity due to Gauss, and another due to Jacobi, which are closely allied to Euler's, and which have arithmetical applications.

Corollary 3 (Gauss). *If* $|x|<1$, *then*

$$\frac{\prod_{n=0}^{\infty}(1-x^{2n+2})}{\prod_{n=0}^{\infty}(1-x^{2n+1})}=\sum_{n=-\infty}^{\infty}x^{\frac{1}{2}n(n+1)}. \tag{2.8}$$

This is an immediate consequence of (2.1), if we use the definition

$$\theta_3(v,z)=\sum_{n=-\infty}^{\infty}e^{\pi i n^2 z+2\pi i n v}, \quad \operatorname{Im} z>0,$$

for $v=z/2$, and set $e^{2\pi i z}=x$.

Corollary 4 (Jacobi). *If* $|x|<1$, *then*

$$\prod_{n=1}^{\infty}(1-x^n)^3=\sum_{n=0}^{\infty}(-1)^n(2n+1)x^{\frac{1}{2}n(n+1)}. \tag{2.9}$$

For, by definition, we have (from (1.1) of Chapter V)

$$\theta(v,z)=\frac{1}{i}\cdot\sum_{n=-\infty}^{\infty}(-1)^n q^{\left(n+\frac{1}{2}\right)^2}\cdot e^{(2n+1)\pi iv},\qquad q=e^{\pi iz},\qquad \mathrm{Im}\,z>0,$$

so that

$$\theta'(0,z)\equiv\left(\frac{\partial}{\partial v}\cdot\theta(v,z)\right)_{v=0}=\pi\sum_{n=-\infty}^{\infty}(-1)^n(2n+1)q^{\left(n+\frac{1}{2}\right)^2},$$

while Jacobi's relation (6.9) of Chapter V gives

$$\theta'(0,z)=2\pi\cdot q^{1/4}\cdot\prod_{n=1}^{\infty}(1-q^{2n})^3.$$

On writing $q^2=x$, we get

$$\begin{aligned}2\prod_{n=1}^{\infty}(1-x^n)^3&=\sum_{n=-\infty}^{\infty}(-1)^n(2n+1)x^{\frac{1}{2}n(n+1)}\\&=2\sum_{n=0}^{\infty}(-1)^n(2n+1)x^{\frac{1}{2}n(n+1)}.\end{aligned}$$

§ 3. The transformation formula connecting $\eta(z)$ and $\eta\left(-\frac{1}{z}\right)$. From the definition of $\eta(z)$ in (1.3) it is obvious that

$$\eta(z\pm 1)=e^{\pm\frac{\pi i}{12}}\eta(z). \tag{3.1}$$

Less obvious is the transformation formula contained in the following

Theorem 2. *If* $\mathrm{Im}\,z>0$, *we have*

$$\eta\left(-\frac{1}{z}\right)=\sqrt{\frac{z}{i}}\cdot\eta(z), \tag{3.2}$$

where $\sqrt{\frac{z}{i}}=+1$ *for* $z=+i$.

Proof (Siegel). Let $z=x+iy$, with $y>0$. Since $\mathrm{Im}\left(-\frac{1}{z}\right)=\frac{y}{x^2+y^2}>0$, if $y>0$, and $\eta(z)$ is holomorphic for $y>0$, it suffices to prove (3.2) for purely imaginary z, say $z=iy$, $y>0$. It would then follow, by analytic continuation, that (3.2) holds for all complex z, with $\mathrm{Im}\,z>0$. We therefore suppose, from now on, that $z=iy$, $y>0$.

Since $\eta(z)\neq 0$ for $y>0$, we can take logarithms in (1.3), and use the expansion

$$\log(1-e^{2\pi imz})=-\sum_{k=1}^{\infty}\frac{1}{k}\cdot e^{2\pi ikmz},\qquad \mathrm{Im}\,z>0,\qquad m=1,2,\ldots,$$

which defines the logarithm uniquely, so that

$$\log\eta(z)=\frac{\pi iz}{12}+\sum_{m=1}^{\infty}\log(1-e^{2\pi imz})=\frac{\pi iz}{12}-\sum_{m=1}^{\infty}\sum_{k=1}^{\infty}\frac{1}{k}\cdot e^{2\pi ikmz}, \tag{3.3}$$

and the right-hand side defines the branch of $\log\eta(z)$ uniquely, since the double series converges absolutely, because $|e^{2\pi ikmz}|=e^{-2\pi kmy}$, $y>0$.

Replacing z by $-\frac{1}{z}$ in (3.3), which we may since $\mathrm{Im}\left(-\frac{1}{z}\right)>0$ if $\mathrm{Im}\,z>0$, we obtain

$$\log\eta\left(-\frac{1}{z}\right)=-\frac{\pi i}{12z}-\sum_{m=1}^{\infty}\sum_{k=1}^{\infty}\frac{1}{k}\cdot e^{-2\pi ikm/z}. \tag{3.4}$$

In order to prove the theorem, it suffices, in view of (3.4) and (3.3), to prove that

$$-\frac{\pi iz}{12}-\frac{\pi i}{12z}+\sum_{k=1}^{\infty}\frac{1}{k}\left(\frac{e^{2\pi ikz}}{1-e^{2\pi ikz}}-\frac{e^{-2\pi ik/z}}{1-e^{-2\pi ik/z}}\right)=\frac{1}{2}\log\frac{z}{i}, \tag{3.5}$$

since

$$\sum_{m=1}^{\infty}e^{2\pi ikmz}=\frac{e^{2\pi ikz}}{1-e^{2\pi ikz}},$$

and the same is true with $-\frac{1}{z}$ in place of z.

The nth partial sum of the infinite series in (3.5) can be written as

$$\begin{aligned}
&\sum_{k=1}^{n}\frac{1}{2k}\left(\frac{2e^{2\pi ikz}}{1-e^{2\pi ikz}}+1-\frac{2e^{-2\pi ik/z}}{1-e^{-2\pi ik/z}}-1\right)\\
&=\sum_{k=1}^{n}\frac{i}{2k}\left(-i\cdot\frac{(1+e^{2\pi ikz})}{(1-e^{2\pi ikz})}+i\cdot\frac{(1+e^{-2\pi ik/z})}{(1-e^{-2\pi ik/z})}\right)\\
&=\sum_{k=1}^{n}\frac{i}{2k}\left(\cot\pi kz+\cot\frac{\pi k}{z}\right)\\
&=\sum_{\substack{k=-n\\k\neq 0}}^{n}\frac{i}{4k}\left(\cot\pi kz+\cot\frac{\pi k}{z}\right).
\end{aligned}$$

It suffices therefore to prove that

$$-\frac{\pi i}{12}\left(z+\frac{1}{z}\right)+\sum_{\substack{k=-n\\k\neq 0}}^{n}\frac{i}{4k}\left(\cot\pi kz+\cot\frac{\pi k}{z}\right)\to\frac{1}{2}\log\frac{z}{i},\quad\text{as}\quad n\to+\infty, \tag{3.6}$$

for $\mathrm{Im}\,z>0$. This can be done by applying Cauchy's theorem of residues to a suitably chosen integral.

Let

$$\varphi(s)=\cot s\cdot\cot\frac{s}{z},\qquad v=\pi(n+\tfrac{1}{2}),\qquad n=0,1,2,\ldots, \tag{3.7}$$

where s is a complex variable, which is independent of z. The function $\frac{\varphi(vs)}{s}$ is meromorphic in the s-plane, with *simple* poles (since $y>0$) at $s=\frac{\pi k}{v}$ and $s=\frac{\pi k}{v}z$, where k is any integer different from zero, and a *triple* pole at $s=0$.

To calculate the residues at these poles, we note that in a neighbourhood of $s=0$,

$$\cot s=\frac{\cos s}{\sin s}=\frac{\left(1-\frac{s^2}{2}+\ldots\right)}{s\left(1-\frac{s^2}{6}+\ldots\right)}=\frac{1}{s}\left(1-\frac{s^2}{2}+\ldots\right)\left(1+\frac{s^2}{6}+\ldots\right)$$

$$=\frac{1}{s}-\frac{s}{3}+\ldots,$$

and

$$\frac{\varphi(s)}{s}=\frac{1}{s}\left(\frac{1}{s}-\frac{s}{3}+\ldots\right)\left(\frac{z}{s}-\frac{s}{3z}+\ldots\right),$$

and that $\cot s$ is periodic with period π. The residues of $\frac{\varphi(vs)}{s}$ at $s=0$, $s=\frac{\pi k}{v}$, and $s=\frac{\pi k}{v}z$, are respectively

$$-\frac{1}{3}\left(z+\frac{1}{z}\right),\quad \frac{1}{\pi k}\cot\frac{\pi k}{z},\quad \frac{1}{\pi k}\cot\pi kz\,.$$

Let P denote the parallelogram (Fig. 10) in the s-plane with vertices at $A(s=1)$, $B(s=z=iy,\ y>0)$, $C(s=-1)$, and $D(s=-z=-iy)$, and ∂P denote its boundary. Then, since $v=\pi(n+\frac{1}{2})$, we have

$$\frac{1}{2\pi i}\int_{\partial P}\frac{\varphi(vs)}{s}\,ds=-\frac{1}{3}\left(z+\frac{1}{z}\right)+\sum_{\substack{k=-n\\k\neq 0}}^{n}\frac{1}{\pi k}\left(\cot\pi kz+\cot\frac{\pi k}{z}\right),$$

or

$$\int_{\partial P}\frac{\varphi(vs)}{8s}\,ds=-\frac{\pi i}{12}\left(z+\frac{1}{z}\right)+\sum_{\substack{k=-n\\k\neq 0}}^{n}\frac{i}{4k}\left(\cot\pi kz+\cot\frac{\pi k}{z}\right).$$

To prove (3.6) it suffices therefore to prove that

$$\lim_{n\to\infty}\int_{\partial P}\frac{\varphi(vs)}{8s}\,ds=\frac{1}{2}\log\frac{z}{i},\qquad v=\pi\left(n+\frac{1}{2}\right).\tag{3.8}$$

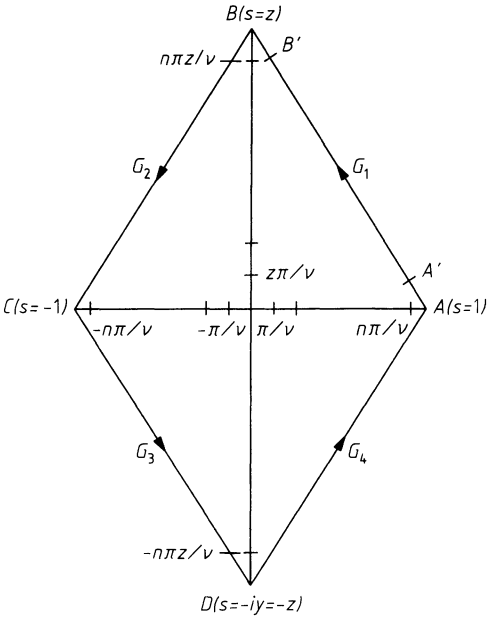


Fig. 10

To show this we consider the behaviour of $\varphi(vs)$ as s describes ∂P. Since $\cot s$ is periodic, with period π, and since $|\cot s| < M_1 < \infty$, for $-\frac{\pi}{2} \leqq \mathrm{Re}\, s \leqq \frac{\pi}{2}$, $|s| \geqq \delta_0 > 0$, we have

$$\left|\frac{\varphi(vs)}{s}\right| < M < \infty\,, \tag{3.9}$$

for $s \in \partial P$, $v = \pi(n+\frac{1}{2})$, $n = 1, 2, \ldots$, where M is independent of s and v. On the other hand, since

$$\cot s = i\frac{(e^{2is}+1)}{(e^{2is}-1)}\,,$$

we have

$$\cot s \to \begin{cases} -i, & \text{as} \quad \mathrm{Im}\, s \to +\infty\,, \\ +i, & \text{as} \quad \mathrm{Im}\, s \to -\infty\,, \end{cases} \tag{3.10}$$

for $-\frac{\pi}{2} \leqq \mathrm{Re}\, s \leqq \frac{\pi}{2}$. By periodicity, this holds in every strip: $-\frac{\pi k}{2} \leqq \mathrm{Re}\, s \leqq$ $\leqq -\frac{\pi k}{2} + \pi$, $k = 1, 2, \ldots$. Hence $\cot vs \to -i$, as $v \to +\infty$, uniformly in every compact set in the *upper* half-plane $\mathrm{Im}\, s > 0$; and $\cot vs \to +i$, as $v \to +\infty$, uniformly in every compact set in the *lower* half-plane $\mathrm{Im}\, s < 0$. Similarly $\cot \frac{vs}{z}$

$\to -i$, as $v\to+\infty$, uniformly in every compact set in the *left* half-plane $\mathrm{Re}\, s<0$; and $\cot\frac{vs}{z}\to+i$, as $v\to+\infty$, uniformly in every compact set in the *right* half-plane $\mathrm{Re}\, s>0$.

Hence we have:

$$\varphi(vs)=\cot(vs)\cdot\cot\left(\frac{vs}{z}\right)\to+1,\quad \text{as}\quad v\to+\infty, \tag{3.11}$$

uniformly for s in G_1 and G_3, where

$$G_1=\{s|0<\theta\leqq\arg s\leqq\tfrac{\pi}{2}-\theta<\tfrac{\pi}{2}\},\qquad G_3=\left\{s|\pi<\pi+\theta\leqq\arg s\leqq\frac{3\pi}{2}-\theta<\frac{3\pi}{2}\right\}$$

and

$$\varphi(vs)=\cot(vs)\cdot\cot\left(\frac{vs}{z}\right)\to-1,\quad \text{as}\quad v\to+\infty, \tag{3.12}$$

uniformly for s in G_2 and G_4, where

$$G_2=\{s|\tfrac{\pi}{2}<\tfrac{\pi}{2}+\theta\leqq\arg s\leqq\pi-\theta<\pi\},$$

$$G_4=\left\{s\left|\frac{3\pi}{2}<\frac{3\pi}{2}+\theta\leqq\arg s\leqq 2\pi-\theta<2\pi\right.\right\}.$$

In order to prove (3.8), we shall show that

$$\lim_{n\to\infty}\int_{\partial P}\frac{\varphi(vs)}{s}\,ds=\left(\int_1^z-\int_z^{-1}+\int_{-1}^{-z}-\int_{-z}^{1}\right)\frac{ds}{s},\qquad v=\pi(n+\tfrac{1}{2}), \tag{3.13}$$

the four integrals on the right-hand side corresponding to the four sides AB, BC, CD, DA of the parallelogram P.

Let us first consider the integral along the path AB, and split it into three parts, namely A to A', A' to B', and B' to B, such that the distance $|AA'|$, and the distance $|BB'|$, equal $\delta>0$, to be chosen sufficiently small.

Given $\varepsilon>0$, there exist a $\delta>0$, and a v_0, such that for $v\geqq v_0$, we have

$$\left|\int_A^B\frac{\{\varphi(vs)-1\}}{s}\,ds\right|\leqq\left|\int_{A'}^{B'}\frac{\varphi(vs)-1}{s}\,ds\right|+\left|\int_A^{A'}\right|+\left|\int_B^{B'}\right|<\varepsilon,$$

since, for any preassigned $\varepsilon'>0$, we have $|\varphi(vs)-1|<\varepsilon'$, for $v\geqq v_0$, $s\in A'B'$, while $|s|>\delta_1>0$ in the first integral on the right-hand side; whereas the second and third integrals are less than $M\delta$ by (3.9). Hence

$$\lim_{n\to\infty}\int_A^B\frac{\varphi(vs)}{s}\,ds=\int_A^B\frac{ds}{s}=\int_1^z\frac{ds}{s}; \tag{3.14}$$

and similar results hold for the other three sides of P, and (3.13) follows. Thus

$$\lim_{n\to\infty}\int_{\partial P}\frac{\varphi(vs)}{s}\,ds=2\left(\int_1^z\frac{ds}{s}+\int_{-1}^{z}\frac{ds}{s}\right).$$

To evaluate the right-hand side, let L be a *simple* curve (in the half-plane $\mathrm{Im}\, s>0$) joining A and C, and containing B in its interior (Fig. 11). Then

$$\lim_{n\to\infty}\int_{\partial P}\frac{\varphi(vs)}{s}\,ds = 4\int_1^z\frac{ds}{s}+2\int_L\frac{ds}{s}$$
$$= 4\log z - 4\log i$$
$$= 4\log\frac{z}{i},$$

(where $\log z$ is real for real, positive z). This proves (3.8), and hence the theorem.

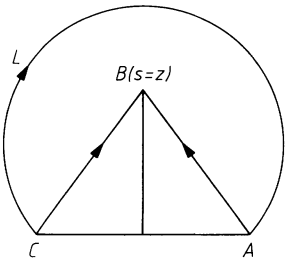


Fig. 11

Theorem 3. *Let a, b, c, d be integers such that $ad-bc=1$. Then*

$$\eta\left(\frac{az+b}{cz+d}\right)=\omega\cdot\sqrt{cz+d}\cdot\eta(z),\quad \textit{for}\quad \mathrm{Im}\, z>0,$$

where ω is some 24th root of unity, depending on a, b, c, d, but not on z.

Proof. Let M, A, B denote respectively the modular transformations

$$z\to\frac{az+b}{cz+d},\qquad z\to z+1,\qquad z\to-\frac{1}{z}.$$

Then the relation

$$\eta^{24}(Mz)=(cz+d)^{12}\eta^{24}(z) \tag{3.15}$$

holds for $M=A^{\pm 1}$ and $M=B$, because of (3.1) and Theorem 2. If it holds for an arbitrary modular transformation M, then it holds also for $MA^{\pm 1}$ and MB, for

$$\eta^{24}(MA^{\pm 1}z)=(cz\pm c+d)^{12}\eta^{24}(z),\quad \text{since}\quad MA^{\pm 1}z=\frac{az\pm a+b}{cz\pm c+d},$$
$$\eta^{24}(MBz)=(dz-c)^{12}\eta^{24}(z),\quad \text{since}\quad MBz=\frac{bz-a}{dz-c}.$$

Since M is generated by A and B, and $B^{-1}=B$, it follows that (3.15) holds in general. Taking the 24th root, and noting that $\sqrt{cz+d}$, with the chosen determination of the square-root, is holomorphic for $\mathrm{Im}\, z>0$, we obtain Theorem 3.

§ 4. Siegel's proof of Theorem 1. We define, for $\mathrm{Im}\, z>0$,

$$g(z)=e^{\pi iz/12}\theta_3\left(\frac{1}{2}+\frac{z}{2}, 3z\right). \tag{4.1}$$

Then the behaviour of $g(z)$ under the transformations: $A(z\to z+1)$, and $B\left(z\to -\frac{1}{z}\right)$ is similar to that of $\eta(z)$, as shown by the following

Theorem 4. *We have the relations*

$$g(z+1)=e^{\pi i/12}g(z), \quad g\left(-\frac{1}{z}\right)=\sqrt{\frac{z}{i}}\,g(z), \tag{4.2}$$

where $\sqrt{\frac{z}{i}}=+1$ *for* $z=i$.

Proof (Siegel). Since

$$g(z)=e^{\pi iz/12}\sum_{n=-\infty}^{\infty}(-1)^n e^{\pi iz(3n^2+n)}, \tag{4.3}$$

by the definition of θ_3 (cf. (3.3), Ch. V), clearly $g(z+1)=e^{\pi i/12}g(z)$.

Let $f(z)=e^{-\pi iz/12}g(z)$, so that

$$f(z)=\theta_3\left(\frac{1}{2}+\frac{z}{2}, 3z\right)=\sum_{n=-\infty}^{\infty}e^{3\pi in^2z+2\pi in\left(\frac{1}{2}+\frac{z}{2}\right)}, \qquad \mathrm{Im}\, z>0,$$

and

$$f\left(-\frac{1}{z}\right)=\theta_3\left(\frac{1}{2}-\frac{1}{2z}, -\frac{3}{z}\right).$$

We have proved in Theorem 9 of Chapter V that

$$\theta_3\left(v, -\frac{1}{\tau}\right)=\sqrt{\frac{\tau}{i}}\sum_{n=-\infty}^{\infty}e^{\pi i\tau(n+v)^2},$$

where v is a complex variable, and $\mathrm{Im}\,\tau>0$. On setting $\tau=z/3$, and $v=\frac{1}{2}-\frac{1}{2z}$, we obtain

$$\begin{aligned} f\left(-\frac{1}{z}\right) &= \sqrt{\frac{z}{3i}}\sum_{n=-\infty}^{\infty}e^{\frac{\pi iz}{3}\left(n+\frac{1}{2}-\frac{1}{2z}\right)^2} \\ &= \sqrt{\frac{z}{3i}}\cdot e^{\frac{\pi i}{12z}}\cdot\sum_{n=-\infty}^{\infty}e^{\frac{\pi iz}{12}(2n+1)^2-\frac{\pi i}{6}(2n+1)} \\ &= \sqrt{\frac{z}{3i}}\cdot e^{\pi i/12z}\sum_{n=-\infty}^{\infty}e^{\pi iz\frac{N^2}{12}-\pi i\frac{N}{6}}, \qquad N=2n+1,\ -\infty<n<+\infty. \end{aligned}$$

The series on the right-hand side is invariant under the substitution $N \to -N$, so that

$$f\left(-\frac{1}{z}\right) = \sqrt{\frac{z}{3i}} \cdot e^{\frac{\pi i}{12z}} \cdot \sum_{N=-\infty}^{\infty} e^{\frac{\pi i z N^2}{12}} \left\{ \frac{e^{-\frac{\pi i N}{6}} + e^{\frac{\pi i N}{6}}}{2} \right\}, \tag{4.4}$$

where N runs through all *odd* integers.

Obviously $N \equiv 1, 3, 5 \pmod 6$. If $N \equiv 3 \pmod 6$, then $N = 3 + 6n = 3(2n+1)$, so that

$$\frac{1}{2}\left(e^{-\frac{\pi i N}{6}} + e^{\frac{\pi i N}{6}}\right) = \cos\left(\frac{\pi N}{6}\right) = \cos(\tfrac{\pi}{2} + n\pi) = 0 .$$

It remains for us only to sum the series in (4.4) over $N \equiv \pm 1 \pmod 6$. The substitution $N \to -N$ carries numbers of the form $N = 6n+1$ into numbers of the form $N = 6n-1$ (as n goes from $-\infty$ to $+\infty$), so that it suffices to carry out the summation over the residue-class $N \equiv 1 \pmod 6$, and then to double that sum. Thus we have

$$f\left(-\frac{1}{z}\right) = \sqrt{\frac{z}{3i}} \cdot e^{\pi i/12z} \cdot 2 \cdot \sum_{n=-\infty}^{\infty} e^{3\pi i z n^2 + \pi i n z + \frac{\pi i z}{12}} \left\{ \cos\frac{\pi}{6}(6n+1) \right\}.$$

Since $2\cos\frac{\pi}{6}(6n+1) = \sqrt{3} \cdot \cos \pi n$, we obtain from this

$$\begin{aligned} f\left(-\frac{1}{z}\right) &= \sqrt{\frac{z}{i}} \cdot e^{\frac{\pi i z}{12} + \frac{\pi i}{12z}} \cdot \sum_{n=-\infty}^{\infty} e^{3\pi i z n^2 + 2\pi i n(\frac{1}{2} + \frac{z}{2})} \\ &= \sqrt{\frac{z}{i}} \cdot e^{\frac{\pi i z}{12} + \frac{\pi i}{12z}} \cdot f(z) , \end{aligned}$$

which implies that $g\left(-\frac{1}{z}\right) = \sqrt{\frac{z}{i}}\, g(z)$, thus proving (4.2).

We shall now use Theorem 4 to prove Theorem 1. Let

$$K(z) = \frac{\eta(z) - g(z)}{\eta(z)}, \qquad \operatorname{Im} z > 0 . \tag{4.5}$$

Since η has no zeros in the half-plane $\operatorname{Im} z > 0$, $K(z)$ is holomorphic there. By Theorems 4 and 2 combined with (3.1), we have

$$g(z+1) = e^{\frac{\pi i}{12}} g(z), \qquad g\left(-\frac{1}{z}\right) = \sqrt{\frac{z}{i}}\, g(z),$$

and

$$\eta(z+1) = e^{\pi i/12} \eta(z), \qquad \eta\left(-\frac{1}{z}\right) = \sqrt{\frac{z}{i}}\, \eta(z) .$$

Hence

$$K(z+1) = K(z), \qquad K\left(-\frac{1}{z}\right) = K(z) .$$

Since the modular group is generated by the two transformations $z \to z+1$, and $z \to -\frac{1}{z}$ (cf. (5.7) of Chapter I), it follows that $K(Mz)=K(z)$, for any modular transformation M. Hence $K(z)$ *is a modular function.* We shall now prove

Theorem 5. *If* $\mathrm{Im}\, z>0$, *then* $K(z)\equiv 0$, *so that* $\eta(z)=g(z)$.

Proof (Siegel). Let $F=\{z|\mathrm{Im}\, z>0,\ |z|\geqq 1,\ |\mathrm{Re}\, z|\leqq \frac{1}{2}\}$. Since

$$\prod_{n=1}^{\infty}(1-e^{2\pi i n z})\to 1,$$

as $\mathrm{Im}\, z\to+\infty$, uniformly in the strip $|\mathrm{Re}\, z|\leqq\frac{1}{2}$, we have

$$K(z)\to 0, \quad \text{as} \quad \mathrm{Im}\, z\to+\infty, \tag{4.6}$$

uniformly for $|\mathrm{Re}\, z|\leqq\frac{1}{2}$. The set $F_0=\{z|\mathrm{Im}\, z>0,\ |z|\geqq 1,\ -\frac{1}{2}\leqq \mathrm{Re}\, z\leqq 0\}$. $\cup\cdot\{z|\mathrm{Im}\, z>0,\ |z|>1,\ 0<\mathrm{Re}\, z<\frac{1}{2}\}$ is a fundamental domain for the modular group Γ (see (5.94) of Ch. I), and $F\supset F_0$.

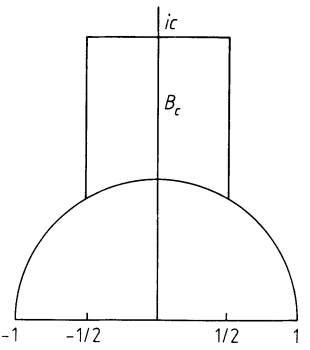


Fig. 12

Since $K(z)$ is holomorphic for $\mathrm{Im}\, z>0$, and invariant under the modular group, we have

$$\sup_{\mathrm{Im}\, z>0}|K(z)|=\sup_{z\in F}|K(z)|=\alpha, \text{ say}.$$

If $K(z)\not\equiv 0$, we have $0<\alpha\leqq\infty$. Since $K(z)\to 0$ as $\mathrm{Im}\, z\to+\infty$, uniformly for $|\mathrm{Re}\, z|\leqq\frac{1}{2}$, and since the set

$$B_c=F\cdot\cap\cdot\{z|\mathrm{Im}\, z\leqq c,\ c>0\}$$

is compact (Fig. 12), we have $\alpha<\infty$. If c is chosen so large that $|K(z)|\leqq\alpha/2$, for $\mathrm{Im}\, z\geqq c$, $z\in F$, then

$$\sup_{z\in B_c}|K(z)|=\alpha.$$

Since B_c is compact, there is a point $z_0 \in B_c$, such that

$$|K(z_0)| = \alpha = \sup_{\mathrm{Im}\, z > 0} |K(z)| .$$

Hence, by the maximum principle, K reduces to a constant, which must be zero, because of (4.6).

§ 5. Connexion between $\eta(z)$ and the modular functions $J(z)$, $\lambda(z)$. Since

$$\eta^{24}(z) = (2\pi)^{-12} \Delta(z), \qquad \mathrm{Im}\, z > 0,$$

by (1.4), and since

$$\Delta(z) \equiv \Delta(1, z) = (2\pi)^4 (\theta'(0, z))^8,$$

by (5.7) of Chapter V, with $\omega_1 = 1$ and $\omega_2 = z$, we have

$$\eta^{24}(z) = \left\{ \frac{\theta'(0, z)}{2\pi} \right\}^8 . \tag{5.1}$$

This enables us to express $\eta^{24}(z)$ in terms of the modular functions $J(z)$ and $\lambda(z)$.

Theorem 6. *For* $\mathrm{Im}\, z > 0$, *we have*

$$\eta^{24}(z) = \frac{1}{(48\pi^2)^3} \cdot \frac{\{J'(z)\}^6}{\{J(z)\}^4 \{1 - J(z)\}^3} . \tag{5.2}$$

Proof. By (8.1) of Chapter VII, we have, for $\mathrm{Im}\, z > 0$,

$$J(z) = \frac{4}{27} \cdot \frac{(1 - \lambda + \lambda^2)^3}{\lambda^2 (\lambda - 1)^2} , \tag{5.3}$$

where

$$\lambda \equiv \lambda(z) = \frac{\theta_1^4(0, z)}{\theta_3^4(0, z)}, \qquad \lambda \neq 0, 1 , \tag{5.4}$$

as in (7.1) of Chapter VII. By logarithmic differentiation, we obtain

$$\frac{J'(z)}{J(z)} = \frac{(2\lambda - 1)(\lambda + 1)(\lambda - 2) \cdot \lambda'}{(\lambda^2 - \lambda + 1) \cdot \lambda(\lambda - 1)} , \tag{5.5}$$

where λ' stands for the derivative of λ. On the other hand, we have

$$1 - J(z) = -\frac{1}{27} \cdot \frac{(\lambda + 1)^2 (\lambda - 2)^2 (2\lambda - 1)^2}{\lambda^2 (\lambda - 1)^2} . \tag{5.6}$$

From (5.5) and (5.6), we obtain

$$\frac{\{J'(z)\}^6}{J^6(z) \cdot (1 - J(z))^3} = -(27)^3 \cdot (\lambda')^6 (\lambda^2 - \lambda + 1)^{-6} , \tag{5.7}$$

or

$$\frac{\{J'(z)\}^6}{J^4(z)\,(1 - J(z))^3} = -(27 \times 4^2) \cdot \frac{(\lambda')^6}{\lambda^4 (\lambda - 1)^4} . \tag{5.8}$$

By logarithmic differentiation of $\lambda(z)$, we obtain

$$\lambda'(z)=\lambda(z)\cdot\left\{\frac{4\dfrac{\partial\theta_1(0,z)}{\partial z}}{\theta_1(0,z)}-\frac{4\dfrac{\partial\theta_3(0,z)}{\partial z}}{\theta_3(0,z)}\right\}.$$

By (5.4) and (5.2) of Chapter V, it follows that

$$\lambda'(z)=\lambda(z)\cdot\frac{1}{\pi i}(e_3-e_1). \tag{5.9}$$

Since

$$\theta_1^4(0,z)+\theta_2^4(0,z)=\theta_3^4(0,z), \tag{5.10}$$

by (3.10) of Chapter VII, and

$$(e_3-e_1)^6=\pi^{12}\cdot\theta_2^{24}(0,z), \tag{5.11}$$

by (5.6) of Chapter V (with $\omega_1=1$), we obtain, from (5.11), (5.10), (5.9), and (5.4),

$$\begin{aligned}\frac{(\lambda')^6}{\lambda^4(\lambda-1)^4}&=\frac{\lambda^2}{(\lambda-1)^4}\cdot(e_3-e_1)^6\cdot(-\pi^{-6})\\&=\frac{\{\theta_1(0,z)/\theta_3(0,z)\}^8}{\{\theta_2(0,z)/\theta_3(0,z)\}^{16}}\cdot(\pi^{12}\cdot\theta_2^{24}(0,z))\cdot(-\pi^{-6})\\&=-\pi^6\cdot\{\theta_1(0,z)\cdot\theta_2(0,z)\cdot\theta_3(0,z)\}^8\\&=-\pi^{-2}\{\theta'(0,z)\}^8,\quad\text{(by Th. 5 of Ch. V)}\\&=-(2\pi)^8\cdot\pi^{-2}\cdot\{\theta'(0,z)\cdot(2\pi)^{-1}\}^8\\&=-4^4\cdot\pi^6\cdot\eta^{24}(z),\quad\text{(by (5.1))}.\end{aligned} \tag{5.12}$$

The theorem follows from (5.12) and (5.8).

Remark. Since $J(z)$ is invariant under all modular transformations, Theorem 3 follows also from Theorem 6.

Notes on Chapter VIII

§ 1. There are two papers by Dedekind that are relevant here. The first one (*Ges. Werke*, I, 159–173) contains his comments on two posthumously published fragments of Riemann's work, (*Bernhard Riemann's gesammelte mathematische Werke und wissenschaftlicher Nachlass*, 2. Aufl., S. 466–478 (1892)), where he notes the central rôle played by the η-function, defined as in (1.3), in the theory of 'modular' functions, recognizes the relation $2\pi\eta^3(z)=\theta'(0,z)$, and deduces the transformation formula for $\eta(z)$ given in Theorem 3 as a consequence of the transformation formula for $\theta'(0,z)$ given in (8.14) of Chapter V. He derives also a transformation formula for $\log\eta(z)$ in which the so-called *Dedekind sums* make their first appearance. (These are sums of the

form

$$s(h,k)=\sum_{\mu=1}^{k-1}\left(\frac{\mu}{k}-\frac{1}{2}\right)\left(\frac{\mu h}{k}-\left[\frac{\mu h}{k}\right]-\frac{1}{2}\right),$$

where h, k are integers, $k>0$, $(h,k)=1$, and $[x]$ denotes the integral part of x). By iterating that formula, Dedekind obtains a reciprocity law for such sums. The method of the paper leads to the determination of the 24th root of unity occurring in Theorem 3. See the comments by R. Fricke following Dedekind's paper, and also Th. Molien, *Bericht über die Verhandlungen der Königlich Sächsischen Gesellschaft der Wissenschaft zu Leipzig, Mathematische-Physische Classe*, 37 (1885), 25–38; L. Kiepert, *Math. Annalen*, 26 (1886), 369–454, particularly Abschnitt III, 404–422. Dedekind is aware of the similarity of the rôle played by the Dedekind sums in the determination of the 24th root of unity in the transformation formula for $\eta(z)$ with the rôle played by Gauss's sums in Hermite's determination of the 8th root of unity in the transformation formula for $\theta'(0,z)$.

For further work on Dedekind sums, see, for instance, H. Rademacher, *J. Indian Math. Soc.*, 19 (1955), 25–30; *Math. Zeit.* 63 (1956), 445–463.

The second paper by Dedekind (*Ges. Werke*, I, 174–201) is the one in which he introduces the function $J(z)$, already cited in the Notes on Chapter VI and on Chapter I. There he proves that the function

$$w(z)=c\cdot J^{-1/3}(z)\cdot(1-J(z))^{-1/4}\cdot(J'(z))^{1/2},$$

where c is a constant, and $\mathrm{Im}\,z>0$, satisfies the hypergeometric differential equation

$$J(1-J)\frac{d^2w}{dJ^2}+\left(\frac{2}{3}-\frac{7}{6}J\right)\frac{dw}{dJ}-\frac{1}{144}w=0,$$

thereby establishing another link with Riemann's work. See Fricke's comments following the paper, p. 201. He then *defines* $\eta(z)$ as the square-root of $w(z)$, and deduces the transformation formula for $\eta(z)$ from the invariance of $J(z)$ under modular transformations. Using the theory of theta-functions, he proves that

$$J(z)=\frac{4}{27}\cdot\frac{(1-\lambda+\lambda^2)^3}{\lambda^2(1-\lambda)^2},\qquad \mathrm{Im}\,z>0,\qquad \lambda=\lambda(z),$$

(formula (8.3), Ch. VII); and using the definition of λ as a quotient of theta-functions ((7.1) of Ch. VII), he shows that $2\pi\eta^3(z)=\theta'(0,z)$; and using the infinite product for $\theta'(0,z)$ ((6.5) and (6.9), Ch. V), he *proves* that

$$\eta(z)=e^{\frac{\pi iz}{12}}\prod_{n=1}^{\infty}(1-e^{2\pi inz}),\qquad \mathrm{Im}\,z>0. \tag{1.3}$$

He adds the remark that he did not succeed in obtaining this product-representation without the aid of the theta-functions.

The significance of Siegel's proof of Theorem 2 is that it has made it possible for us to *define* $\eta(z)$ by the product in (1.3), and to prove the transformation formula of Theorem 3 *without* an appeal to the theory of theta functions.

§ 2. For Euler's theorem (1754/5) see his *Opera Omnia* (1) II, 390–398; (1) VIII, § 323–325, p. 270.

For a combinatorial proof of Corollary 1, due to F. Franklin, *Comptes Rendus*, 92 (1881), 448–50, see Hardy and Wright, *An introduction to the theory of numbers*, § 19.11.

Jacobi proved Euler's theorem in two different ways, first with the help of theta-functions, and then without. For (2.1) see his *Fundamenta Nova*, § 64 (1829); *Ges. Werke*, I, p. 232–234. For Corollary 1, see *ibid.* § 66; *Ges. Werke*, I, p. 237, formula (6); and for the second proof, see *Crelle's J.* 21 (1840), 13–32; *Ges. Werke*, VI, 281–302. For Corollary 4, see formula (5) of *Fundamenta Nova*, § 66; *Ges. Werke*, I, p. 237.

Corollary 3 was proved by Gauss in 1808. See his *Werke*, (ii), p. 20.

Let s_n denote the sum of the first n terms of the arithmetical progression

$$1, 1+(r-2), \ldots, 1+(n-1)(r-2), \ldots,$$

with the first term 1 and common difference $r-2$, where r is an integer, $r \geqq 3$. If $r=3$, then $s_n = \dfrac{n(n+1)}{2}$; if $r=4$, $s_n = n^2$; if $r=5$, $s_n = \dfrac{n(3n-1)}{2}$; these three cases correspond to the *triangular numbers*, *square numbers*, and *pentagonal numbers*, respectively. The reason for this nomenclature will be obvious if one replaces the numbers by dots, and the dots are arranged as figures (Fig. 13):

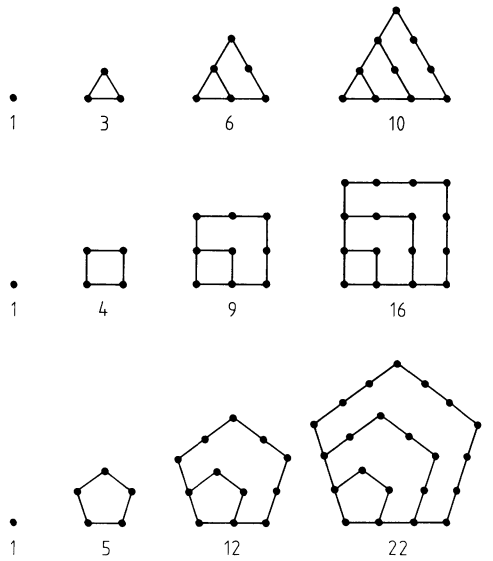


Fig. 13

The 'partition function' $p(n)$ is connected with the η-function by the formula

$$e^{\pi iz/12} \cdot \eta(z) = \sum_{n=0}^{\infty} p(n) e^{2\pi inz}, \quad \operatorname{Im} z > 0, \quad p(0) = 1.$$

The identities of Euler and of Jacobi given in Corollaries 1 and 4 were used by S. Ramanujan to prove the following congruence properties of $p(n)$:

$$p(5m+4) \equiv 0 \pmod 5),$$
$$p(7m+5) \equiv 0 \pmod 7).$$

Proc. Cambridge Phil. Soc. XIX (1919), 207–210; *Collected Papers*, No. 25. The study of the congruence properties of the coefficients (c_n) in the expansion of $J(z)$, referred to in the Notes on Chapter VI was motivated by the results on $p(n)$. For the general theory of partitions, see P. Bachmann's *Niedere Zahlentheorie*, (ii), Ch. 3; Hardy and Wright, loc. cit., Ch. XIX; Hardy's *Ramanujan*, Ch. VI.

For the *asymptotic* formula of Hardy and Ramanujan (and J.V. Uspensky), and the *identity* of Rademacher, for the partition function, see, for instance, the author's *Arithmetical functions*, Ch. VII (Springer Verlag, Grundlehren Bd. 167). The original sources are: G.H. Hardy and S. Ramanujan, *Proc. London Math. Soc.* (2) 17 (1918), 75–115 (= No. 36 of Ramanujan's *Collected Papers*); J.V. Uspensky, *Bull. de l'Acad. des Sciences de l'URSS*, (6) 14 (1920), 199–218; H. Rademacher, *Proc. London Math. Soc.* (2) 43 (1937), 241–254; *Proc. Nat. Acad. Sci.* USA, 23 (1937), 78–84; *Annals of Math.* (2) 44 (1943), 416–422.

The η-function is connected with yet another interesting arithmetical function, namely the τ-function of Ramanujan, by the relation

$$(\eta(z))^{24} = \sum_{n=1}^{\infty} \tau(n) e^{2\pi inz}, \quad \operatorname{Im} z > 0.$$

See S. Ramanujan, *Trans. Cambridge Phil. Soc.* XXII, No. 9 (1916), 159–184 (= No. 18, *Collected Papers*); Hardy's *Ramanujan*, Ch. X. Ramanujan conjectured, and L.J. Mordell proved (*Proc. Cambridge Phil. Soc.* 19 (1917), 117–124), that $\tau(n)$ is multiplicative, and stated, for example, the congruence properties

$$\tau(7m+3) \equiv 0 \pmod 7),$$

and $\tau(23+k) \equiv 0 \pmod{23}$, when k is any *quadratic non-residue* of 23; the first can be proved with the help of Corollary 4, and the second with that of Corollary 1. The congruence properties of $\tau(n)$ are more numerous than those of $p(n)$. See H.P.F. Swinnerton-Dyer, *Modular functions of one variable*, III (1973), 1–55. (Lecture Notes in Mathematics, No. 350, Springer Verlag); J.-P. Serre, *L'Enseignement Mathématique*, XXII (1976), 227–260.

There is also an identity for $\tau(n)$, involving Bessel functions, which is a special case of a more general identity for the Fourier coefficients of certain *modular (cusp) forms*. See, for instance, A. Selberg, *Neuvième Congrès des*

Mathematiciens Scandinaves, Helsingfors (1938), 320–322. See also R.A. Rankin's book, *Modular forms and functions* (Cambridge, 1977), §§ 5.3–5.8.

Ramanujan's conjecture that $\tau(n)=O\left(n^{\frac{11}{2}+\varepsilon}\right)$, for every $\varepsilon>0$, has been proved by algebraic-geometric methods, as a special case of a far more general result, by P. Deligne, Lecture Notes in Mathematics, 179 (1971), 139–172, Springer Verlag; *Inst. Hautes Études Sci. Publ. Math.* 43 (1974), 273–307. A simpler and more direct proof would be of interest.

It is not yet known whether $\tau(n)\neq 0$ for *all* integers $n\geqq 1$. Nor is it known whether *all* the non-real zeros of the entire function $L_\tau(s)$, which has the representation $L_\tau(s)=\sum_{n=1}^{\infty}\tau(n)n^{-s}$, for $\mathrm{Re}\,s>13/2$, lie on the line $\mathrm{Re}\,s=6$ (analogue of the *Riemann hypothesis* for $\zeta(s)=\sum_{n=1}^{\infty}n^{-s}$, $\mathrm{Re}\,s>1$). Nor is it known whether

$$\frac{1}{T}\int_0^T |L_\tau(6+it)|^{2n}dt=O(T^\varepsilon),\qquad \varepsilon>0,$$

for *every* integer $n\geqq 1$ (analogue of the *Lindelöf hypothesis* for $\zeta(s)$). That this is true for $n=1$ follows from a theorem of Anton Good on the mean-values of Dirichlet series associated to modular forms, *Mathematika*, 29 (1982), 278–295.

For mean-value theorems on $\zeta(s)$, see D.R. Heath-Brown, *Recent progress in analytic number theory*, I (1981), 115–119.

§ 3. Siegel's one-page proof is in *Mathematika*, 1 (1954), 4; *Ges. Abhandlungen*, III, 188; and shows him as the consummate artist that he is. Many more pages are devoted to his exposition of the same proof in his Bombay lectures on *Advanced analytic number theory* (Tata Institute of Fundamental Research, Bombay), *Lecture Notes in Mathematics*, No. 23 (1961); reprinted as No. 9, *Studies in Mathematics*, (1980), 15–21. That Siegel's proof extends to *all* modular transformations has been shown by Rademacher, *J. Indian Math. Soc.* 19 (1955), 25–30.

The proof of Theorem 3 given here is *not* different from the proof of (8.14) in Chapter V; it has been repeated for the reader's convenience.

§ 4. Siegel's *analytical* proof of Theorem 4 focuses attention on the rôle of modular functions even at the level of Euler's identity, which clearly inspired the later work by Jacobi, followed by Ramanujan. It is given in his Göttingen lectures (1963/64) on *Analytische Zahlentheorie*, II.

Chapter IX

The law of quadratic reciprocity

§ 1. Reciprocity of generalized Gaussian sums. As a limiting case of the transformation formula connecting the theta-function $\theta_3(v,\tau)$ with $\theta_3\left(v,-\frac{1}{\tau}\right)$, we shall prove a transformation formula for exponential sums (Theorem 1), which yields, as a special case, a reciprocity formula for *generalized Gaussian sums* (Corollary 2) which, in turn, enables us not only to evaluate *Gaussian sums* but to prove the law of quadratic reciprocity.

Theorem 1 (Cauchy-Kronecker). *Let a and b be positive integers, and w a rational number, such that $ab+2aw\equiv 0 \pmod 2$. Then we have*

$$\frac{1}{\sqrt{b}}\sum_{h=1}^{b} e^{\pi i\frac{a}{b}(h+w)^2} = \frac{\varrho}{\sqrt{a}}\sum_{h=1}^{a} e^{-\pi i\frac{b}{a}h^2+2\pi ihw}, \tag{1.1}$$

where $\varrho = e^{\pi i/4}$.

Proof. For complex z, with $\mathrm{Im}\, z>0$, and complex v, we have the formula ((8.1) of Chapter V)

$$\sum_{n=-\infty}^{\infty} e^{\pi iz(n+v)^2} = \sqrt{\frac{i}{z}}\cdot\sum_{n=-\infty}^{\infty} e^{-\pi i\frac{n^2}{z}+2\pi inv}, \tag{1.2}$$

where $\sqrt{\frac{i}{z}}=1$ for $z=i$.

Now let $z=\frac{a}{b}+i\varepsilon$, $\varepsilon>0$, where a and b are integers, $a>0$, $b>0$, and let v be rational and satisfy the congruence $ab+2av\equiv 0 \pmod 2$. We wish to consider the limiting case of formula (1.2), for these values of z and v, as $\varepsilon\to 0+$.

First of all, we note that

$$-\frac{1}{z} = -\frac{1}{\frac{a}{b}+i\varepsilon} = \frac{-b/a}{1+(ib\varepsilon/a)} = \frac{-(b/a)\left(1+\frac{ib\varepsilon}{a}-\frac{ib\varepsilon}{a}\right)}{1+(ib\varepsilon/a)}$$

$$= -\frac{b}{a}+\frac{i\varepsilon(b^2/a^2)}{1+i\left(\frac{b}{a}\right)\varepsilon}, \tag{1.3}$$

so that when z describes the half-line $\mathrm{Re}\, z = \frac{a}{b}$, $\mathrm{Im}\, z > 0$, $z' = -\frac{1}{z}$ describes, in the upper half-plane, the semi-circular arc extending from the origin to the point $-b/a$.

Secondly, the left-hand side of (1.2) equals

$$\sum_{n=-\infty}^{\infty} e^{\pi i \frac{a}{b}(n+v)^2 - \pi\varepsilon(n+v)^2}, \tag{1.4}$$

where the factor $e^{\pi i \frac{a}{b}(n+v)^2}$ is *periodic* in n, with the period b, since

$$\frac{a}{b}\{(n+v+b)^2-(n+v)^2\} = \frac{a}{b}\{b^2+2b(n+v)\} = ab+2an+2av$$
$$\equiv 0 \pmod 2,$$

by assumption. We therefore replace n in (1.4) by $(h+nb)$, where h runs through the integers $1, 2, \ldots, b$, and n runs through *all* integers, and obtain

$$\sum_{h=1}^{b} e^{\pi i \frac{a}{b}(h+v)^2} \cdot \sum_{n=-\infty}^{\infty} e^{-\pi\varepsilon b^2\left(n+\frac{v+h}{b}\right)^2}$$

for the left-hand side of (1.2). If we now use a special case of the theta-formula (1.2), with $i\varepsilon b^2$ in place of z, and $\frac{v+h}{b}$ in place of v, we see that the left-hand side of (1.2) equals

$$\frac{1}{b\sqrt{\varepsilon}} \sum_{n=-\infty}^{\infty} e^{-\frac{\pi n^2}{b^2\varepsilon} + 2\pi i \frac{n}{b}(v+h)} \cdot \sum_{h=1}^{b} e^{\pi i \frac{a}{b}(h+v)^2}. \tag{1.5}$$

Now

$$\lim_{\varepsilon \to 0+} \sum_{n=-\infty}^{\infty} e^{-\frac{\pi n^2}{b^2\varepsilon} + 2\pi i \frac{n}{b}(v+h)} = 1,$$

for

$$\left| \sum_{n=-\infty}^{\infty} e^{-\frac{\pi n^2}{b^2\varepsilon} + 2\pi i \frac{n}{b}(v+h)} - 1 \right| = \left| \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} e^{-\frac{\pi n^2}{b^2\varepsilon} + 2\pi i \frac{n}{b}(v+h)} \right|$$
$$\leqq 2 \sum_{n=1}^{\infty} e^{-\frac{\pi n^2}{b^2\varepsilon}} = 2 \sum_{n=1}^{\infty} e^{-cn^2}, \qquad c = \frac{\pi}{b^2\varepsilon} > 0,$$

and

$$\sum_{n=1}^{\infty} e^{-cn^2} < \sum_{n=1}^{\infty} e^{-cn} = \frac{e^{-c}}{1-e^{-c}} \to 0, \quad \text{as} \quad c \to +\infty.$$

Hence, if $z=\frac{a}{b}+i\varepsilon$, where $\varepsilon>0$, and a and b are integers, and v rational, with $ab+2av\equiv 0 \pmod 2$, then we have (from the left-hand side of (1.2)),

$$\sum_{n=-\infty}^{\infty} e^{\pi i z(n+v)^2} \sim \frac{1}{b\sqrt{\varepsilon}} \sum_{h=1}^{b} e^{\pi i \frac{a}{b}(h+v)^2}, \tag{1.6}$$

as $\varepsilon\to 0+$, because of (1.5).

We now consider the right-hand side of (1.2) for these values of z and v. First of all, we have

$$\lim_{\varepsilon\to 0+} \sqrt{\frac{i}{(a/b)+i\varepsilon}} = \sqrt{\frac{bi}{a}} = \sqrt{\frac{b}{a}}\cdot e^{\frac{\pi i}{4}}, \tag{1.7}$$

and secondly, for $z=\frac{a}{b}+i\varepsilon$, we have

$$\sum_{n=-\infty}^{\infty} e^{-\pi i\frac{n^2}{z}+2\pi i n v} = \sum_{n=-\infty}^{\infty} e^{-\pi i\frac{b}{a}n^2+2\pi i n v-\frac{\pi\cdot\frac{b^2}{a^2}\cdot\varepsilon}{1+i\frac{b}{a}\varepsilon}n^2}, \tag{1.8}$$

because of (1.3). Here the factor $\exp\left(-\frac{\pi i b}{a}n^2+2\pi i n v\right)$ is *periodic* in n with the period a, for

$$\begin{aligned} &2(n+a)v-\frac{b}{a}(n+a)^2-\left(2nv-\frac{b}{a}n^2\right)\\ &\quad=2av-ab-2bn\\ &\quad=2av+ab-2ab-2bn\equiv 0 \pmod 2, \end{aligned}$$

by assumption. On writing $h+an$ instead of n, where h runs through $1,2,\ldots,a$, and n through *all* integers, we see that the sum in (1.8) gives

$$\sum_{h=1}^{a} e^{-\pi i\frac{b}{a}h^2+2\pi i h v}\cdot\sum_{n=-\infty}^{\infty} e^{-\frac{\pi b^2\varepsilon}{1+i\frac{b}{a}\varepsilon}\left(n+\frac{h}{a}\right)^2}. \tag{1.9}$$

If we use the theta-formula (1.2) once more, with $\frac{ib^2\varepsilon}{1+i\frac{b}{a}\varepsilon}$ in place of z, and $\frac{h}{a}$ in place of v, *the last infinite series* equals

$$\frac{\sqrt{1+i\frac{b}{a}\varepsilon}}{b\sqrt{\varepsilon}}\cdot\sum_{n=-\infty}^{\infty} e^{-\frac{\pi n^2}{b^2\varepsilon}-\frac{\pi i n^2}{ab}+2\pi i\cdot\frac{h}{a}n}. \tag{1.10}$$

Now

$$\lim_{\varepsilon\to 0+}\sum_{n=-\infty}^{\infty} e^{-\frac{\pi n^2}{b^2\varepsilon}-\frac{\pi i n^2}{ab}+2\pi i\cdot\frac{h}{a}n}=1.$$

Hence, from the right-hand side of (1.2), we have

$$\sqrt{\frac{i}{z}} \sum_{n=-\infty}^{\infty} e^{-\pi i \frac{n^2}{z} + 2\pi i n v} \sim \frac{1}{b\sqrt{\varepsilon}} \sum_{h=1}^{a} e^{-\pi i \frac{b}{a} h^2 + 2\pi i h v} \cdot \left(\sqrt{\frac{b}{a}} \cdot e^{\pi i/4} \right), \quad (1.11)$$

as $\varepsilon \to 0+$, because of (1.7)–(1.10). Now (1.11) and (1.6) together with (1.2) give the theorem.

If we set $w = \frac{b}{2}$, so that $\frac{a}{b}(h+w)^2 = \frac{a}{b}h^2 + ah + \frac{ab}{4}$, we obtain

Corollary 1. *If a and b are positive integers, then we have*

$$\frac{1}{\sqrt{b}} \sum_{h=1}^{b} e^{\pi i \frac{a}{b} h^2 + \pi i a h} = \frac{\varrho^{1-ab}}{\sqrt{a}} \cdot \sum_{h=1}^{a} e^{-\pi i \frac{b}{a} h^2 - \pi i b h}, \qquad \varrho = e^{\frac{\pi i}{4}},$$

(since $e^{-\pi i b h} = e^{\pi i b h}$).

Corollary 2. *If m and n are any integers, with $m \neq 0$, $n \neq 0$, and we define*

$$G(m,n) = \sum_{h=1}^{|n|} e^{\pi i \frac{m}{n} h^2 + \pi i m h},$$

then

$$\frac{1}{\sqrt{b}} G(a,b) = \frac{1}{\sqrt{a}} G(-b,a) \varrho^{1-ab}, \qquad \varrho = e^{\pi i/4},$$

where a and b are positive integers.

The sums $G(m,n)$ are *the generalized Gaussian sums*, and Corollary 2 gives *a reciprocity formula* for such sums.

Corollary 3. *If b is any positive integer, and $\varrho = e^{\pi i/4}$, then we have*

$$\frac{1}{\sqrt{b}} G(2,b) \equiv \frac{1}{\sqrt{b}} \sum_{h=1}^{b} e^{2\pi i \frac{h^2}{b}} = \frac{\varrho^{1-2b}}{\sqrt{2}} \left(1 + e^{-\pi i \frac{b}{2} + \pi i b}\right).$$

This results from Corollary 2, on setting $a = 2$.

Corollary 4. *Let b be any positive integer, $\sqrt{b}$ the positive square root, and $\xi = e^{2\pi i/b}$. Then we have*

$$\sum_{h \pmod b} \xi^{h^2} = \begin{cases} (1+i)\sqrt{b}, & \text{if } b \equiv 0 \pmod 4; \\ \sqrt{b}, & \text{if } b \equiv 1 \pmod 4; \\ 0, & \text{if } b \equiv 2 \pmod 4; \\ i\sqrt{b}, & \text{if } b \equiv 3 \pmod 4. \end{cases}$$

This results from Corollary 3, for

$$\frac{1}{\sqrt{b}} G(2,b) = \frac{1}{\sqrt{b}} \sum_{h \bmod b} \xi^{h^2} = \varrho(i^{-b}) \cdot \frac{1}{\sqrt{2}} (1+i^b)$$

$$= \frac{(1+i)}{2} \cdot (1+i^{-b}).$$

The sums $\sum_{h (\bmod b)} \xi^{h^2}$, where h runs through a complete system of residues modulo b, are known as *Gaussian sums*. Corollary 4 gives an evaluation of such sums.

Theorem 1 can be used to study the properties of quadratic residues, which we shall next consider.

§ 2. Quadratic residues. Let p be an *odd* prime, and a an integer prime to p, written $(a, p) = 1$. If there exists an integer x, such that $x^2 \equiv a \pmod{p}$, then a is said to be *a quadratic residue* mod p, and we write ${}_aR_p$. If there exists no such integer x, then a is said to be *a quadratic non-residue* mod p, and we write ${}_aN_p$. If $p = 3$ for example, we have ${}_1R_3$, ${}_2N_3$.

We define *the Legendre symbol* $\left(\frac{m}{p}\right)$ as follows:

$$\left(\frac{m}{p}\right) = +1, \quad \text{if} \quad {}_mR_p;$$

$$\left(\frac{m}{p}\right) = -1, \quad \text{if} \quad {}_mN_p;$$

$$\left(\frac{m}{p}\right) = 0, \quad \text{if} \quad p|m;$$

where p is an odd prime, and m an integer. To find out how many of the numbers $1, 2, 3, \ldots, p-1$ are quadratic residues mod p, we have to find out how many solutions the congruence $x^2 \equiv a \pmod{p}$ has, when a runs through the values $1, 2, 3, \ldots, p-1$. For that purpose we consider the sequence of integers 1^2, 2^2, $3^2, \ldots, \left(\frac{p-1}{2}\right)^2$. If r^2 and s^2 are two elements from this sequence, with $r \neq s$, for which $r^2 \equiv s^2 \pmod{p}$, then we must have *either* $r \equiv s \pmod{p}$, *or* $r \equiv -s \pmod{p}$. Neither can be satisfied, since $1 \leqq r,\ s \leqq \frac{p-1}{2}$. On the other hand, we have always: $r^2 \equiv (p-r)^2 \pmod{p}$. Hence when x, in the congruence $x^2 \equiv a \pmod{p}$, runs through the values $1, 2, 3, \ldots, p-1$, then a runs through *exactly* $\frac{1}{2}(p-1)$ *different* values. Therefore *there are exactly* $\frac{1}{2}(p-1)$ *quadratic residues* mod p, *and exactly* $\frac{1}{2}(p-1)$ *quadratic non-residues* mod p. Thus

$$\sum_{m=0}^{p-1} \left(\frac{m}{p}\right) = 0, \tag{2.1}$$

and

$$\left(\frac{m_1}{p}\right)=\left(\frac{m_2}{p}\right), \quad \text{if} \quad m_1 \equiv m_2 \pmod{p}. \tag{2.2}$$

In order to establish *the criterion of Euler* for deciding whether a given integer is, or is not, a quadratic residue modulo a given odd prime, we need two elementary arithmetical propositions.

Proposition A (Euler). *If m is a positive integer, and $(a, m)=1$, then*

$$a^{\varphi(m)} \equiv 1 \pmod{m}, \tag{2.3}$$

where $\varphi(m)$ denotes the number of integers among $1, 2, \ldots, m$ which are prime to m.

Proof. Let $r_1, r_2, \ldots, r_{\varphi(m)}$ be a *complete prime residue system* modulo m, that is to say, $(r_1, m)=1, (r_2, m)=1, \ldots, (r_{\varphi(m)}, m)=1, 1 \leqq r_1, r_2, \ldots, r_{\varphi(m)} \leqq m, r_j \neq r_k$ for $j \neq k$. If $(a, m)=1$, then $ar_1, ar_2, \ldots, ar_{\varphi(m)}$ also form a complete prime residue system modulo m. Hence

$$r_1 \cdot r_2 \ldots r_{\varphi(m)} \equiv ar_1 \cdot ar_2 \ldots ar_{\varphi(m)} \pmod{m},$$

or

$$(a^{\varphi(m)}-1) r_1 \cdot r_2 \ldots r_{\varphi(m)} \equiv 0 \pmod{m}.$$

Since $r_1, r_2, \ldots, r_{\varphi(m)}$ are prime to m, it follows that $a^{\varphi(m)}-1 \equiv 0 \pmod{m}$.

Proposition B (Lagrange). *Let p be a prime. Then the congruence*

$$a_0 x^n + a_1 x^{n-1} + \ldots + a_n \equiv 0 \pmod{p}, \qquad (a_0, p)=1, \tag{2.4}$$

has at most n solutions, n being any positive integer, and $a_0, a_1, \ldots, a_n$ being integers.

Proof. If x is a solution of the congruence, so is any integer congruent to $x \pmod{p}$. When we speak of the number of solutions of a congruence, we mean the number of residue classes (mod p), whose elements satisfy the congruence.

This proposition is true for $n=1$, since $(a_0, p)=1$. Suppose it is true with $n-1$ in place of n, where $n>1$. Then it is trivially true for the degree n, if the corresponding congruence (2.4) has no solution. If it does have a solution, say x_1, then

$$a_0 x_1^n + a_1 x_1^{n-1} + \ldots + a_n \equiv 0 \pmod{p},$$

and by subtraction from (2.4), we have

$$a_0(x^n - x_1^n) + a_1(x^{n-1} - x_1^{n-1}) + \ldots + a_{n-1}(x - x_1) \equiv 0 \pmod{p}, \tag{2.5}$$

and this is obviously satisfied by any solution of (2.4). But (2.5) can be written as

$$(x - x_1)(a_0 x^{n-1} + b_1 x^{n-2} + \ldots + b_{n-1}) \equiv 0 \pmod{p},$$

where $b_1, b_2, \ldots, b_{n-1}$ are integers (which depend on x_1 and on the integers a_0, $a_1, \ldots, a_{n-1}$). Therefore every solution of (2.4) must satisfy either the congruence $x - x_1 \equiv 0 \pmod{p}$, which yields the original solution $x = x_1$, or $a_0 x^{n-1} + b_1 x^{n-2} + \ldots + b_{n-1} \equiv 0 \pmod{p}$, $(a_0, p) = 1$, which is of degree $n-1$, and has, by our assumption, at most $(n-1)$ solutions. In either case, the congruence in (2.4) can have at most n solutions, as claimed in the proposition.

We are now in a position to prove two basic theorems on quadratic residues.

Theorem A (Euler). *If p is an odd prime, and a any integer, then*

$$a^{\frac{1}{2}(p-1)} \equiv 1 \pmod{p}, \tag{2.6}$$

if and only if a is a quadratic residue mod p, *written* ${}_aR_p$.

Proof. Let p be an odd prime and let $(a, p) = 1$. (If p divides a, then (2.6) is impossible). Assuming that ${}_aR_p$, we shall prove (2.6). Obviously there exists an integer x, such that $x^2 \equiv a \pmod{p}$, and $(x, p) = 1$, since $(a, p) = 1$. Since $\frac{1}{2}(p-1)$ is an integer, we can raise this congruence to the power $\frac{1}{2}(p-1)$, and get

$$(x^2)^{\frac{1}{2}(p-1)} = x^{p-1} \equiv a^{\frac{p-1}{2}} \pmod{p}.$$

By proposition A, however, we have: $x^{p-1} \equiv 1 \pmod{p}$, hence (2.6).

Conversely, the congruence $x^{\frac{1}{2}(p-1)} \equiv 1 \pmod{p}$ has, by Proposition B, at most $\frac{1}{2}(p-1)$ solutions. Since there are exactly $\frac{1}{2}(p-1)$ quadratic residues (mod p), and each of them satisfies this congruence, by the first part of the proof, they are all the solutions, so that (2.6) implies that ${}_aR_p$.

Theorem B. *If p is an odd prime, then*

$$m^{\frac{1}{2}(p-1)} \equiv \left(\frac{m}{p}\right) \pmod{p},$$

hence

$$\left(\frac{m}{p}\right) \cdot \left(\frac{n}{p}\right) = \left(\frac{mn}{p}\right).$$

Proof. If $(x, p) = 1$, then $x^{p-1} - 1 \equiv \left(x^{\frac{p-1}{2}} - 1\right)\left(x^{\frac{p-1}{2}} + 1\right) \equiv 0 \pmod{p}$, by Proposition A. Hence we have *either* $x^{\frac{1}{2}(p-1)} \equiv 1 \pmod{p}$, *or* $x^{\frac{1}{2}(p-1)} \equiv -1 \pmod{p}$. The former cannot be satisfied if ${}_xN_p$; hence the latter has to be satisfied if ${}_xN_p$. If we combine this observation with Theorem A, we obtain Theorem B.

§ 3. The law of quadratic reciprocity

Theorem 2 (Gauss). *If m and n are two odd primes, with $m \neq n$, then we have*

$$\left(\frac{m}{n}\right) \cdot \left(\frac{n}{m}\right) = (-1)^{\frac{m-1}{2} \cdot \frac{n-1}{2}}.$$

Proof. For *integers* m, n, with $m \neq 0$, $n \neq 0$, we define, as in § 1,

$$G(m,n) = \sum_{h=1}^{|n|} e^{\pi i \frac{m}{n} h^2 + \pi i m h}.$$

We shall first prove, by means of Theorem 1, that

$$G(m,n) = \sum_{\lambda} e^{\pi i \lambda \cdot \frac{m}{n}(n+1)} - \sum_{\mu} e^{\pi i \mu \frac{m}{n}(n+1)}, \tag{3.1}$$

in case m and n are two *different*, *odd*, *primes*, and λ runs through the quadratic residues mod n, and μ runs through the quadratic non-residues mod n.

Since $h^2 \equiv h \pmod 2$, for any positive integer h, we have, *in case n is an odd prime, and* $(m,n)=1$,

$$\begin{aligned} G(m,n) &= 1 + \sum_{h=1}^{n-1} e^{\pi i h^2 \cdot \frac{m}{n} + \pi i m h^2} \\ &= 1 + \sum_{h=1}^{n-1} e^{\pi i h^2 \frac{m}{n}(n+1)}. \end{aligned}$$

If $h^2 \equiv \lambda \pmod n$, then $e^{\pi i h^2 \cdot \frac{m}{n}(n+1)} = e^{\pi i \lambda \cdot \frac{m}{n}(n+1)}$; if, however, $h^2 \equiv \lambda \pmod n$, *and* $1 \leqq h \leqq n-1$, then λ is a quadratic residue mod n, and $(n-h)^2 \equiv h^2 \equiv \lambda \pmod n$. Hence when h runs through the sequence $1, 2, \ldots, n-1$, then h^2, reduced modulo n, runs *twice* through all the quadratic residues mod n. Thus

$$G(m,n) = 1 + 2 \sum_{\lambda} e^{\pi i \lambda \cdot \frac{m}{n}(n+1)}. \tag{3.2}$$

Let us now consider the sum

$$\sum_{\mu} e^{\pi i \mu \cdot \frac{m}{n}(n+1)}, \quad \text{where } {}_{\mu}N_n.$$

Obviously we have

$$1 + \sum_{\lambda} e^{\pi i \lambda \cdot \frac{m}{n}(n+1)} + \sum_{\mu} e^{\pi i \mu \cdot \frac{m}{n}(n+1)} = \sum_{h=0}^{n-1} e^{\pi i h \frac{m}{n}(n+1)}. \tag{3.3}$$

Since $n+1$ is *even*, $e^{\pi i h \cdot \frac{m}{n}(n+1)}$ is the hth power of an nth root of unity, say ξ, $\xi \neq 1$ (since $n \nmid m$). Hence

$$1 + \sum_{\lambda} e^{\pi i \lambda \cdot \frac{m}{n}(n+1)} + \sum_{\mu} e^{\pi i \mu \cdot \frac{m}{n}(n+1)} = \sum_{h=0}^{n-1} \xi^h = \frac{1-\xi^n}{1-\xi} = 0. \tag{3.4}$$

From (3.2), (3.3), and (3.4), we obtain, *on condition that m and n are integers*, $m \neq 0$, $n \neq 0$, *n is an odd prime, and* $(m,n)=1$,

$$G(m,n) = \sum_{\lambda} e^{\pi i \lambda \cdot \frac{m}{n}(n+1)} - \sum_{\mu} e^{\pi i \mu \cdot \frac{m}{n}(n+1)}. \tag{3.5}$$

Now we have to consider separately the cases ${}_{m}R_n$ and ${}_{m}N_n$.

Case (i). Let ${}_mR_n$. When λ runs through all the quadratic residues mod n, so does (λm), since $(m, n)=1$, and ${}_mR_n$. When μ runs through all quadratic non-residues mod n, so does (μm), by Theorem B. Hence (3.5) gives

$$\begin{aligned} G(m,n) &= \sum_{\lambda} e^{\pi i\lambda\left(\frac{n+1}{n}\right)} - \sum_{\mu} e^{\pi i\mu\left(\frac{n+1}{n}\right)} \\ &= G(1,n), \text{ by (3.5)} \\ &= \left(\frac{m}{n}\right)\cdot G(1,n). \end{aligned} \tag{3.6}$$

Case (ii). Let ${}_mN_n$. Then, as in Case (i), we have by (3.5) and Theorem B,

$$\begin{aligned} G(m,n) &= \sum_{\mu} e^{\pi i\mu(n+1)/n} - \sum_{\lambda} e^{\pi i\lambda(n+1)/n} \\ &= -G(1,n), \text{ by (3.5)} \\ &= \left(\frac{m}{n}\right)\cdot G(1,n). \end{aligned} \tag{3.7}$$

From (3.6) and (3.7), we obtain

$$G(m,n)=\left(\frac{m}{n}\right)\cdot G(1,n), \tag{3.8}$$

on condition that *n is an odd prime, and* $(m, n)=1$.

But by Corollary 2 of Theorem 1, we have

$$\frac{1}{\sqrt{n}}G(1,n)=G(-n,1)\cdot\varrho^{1-n}, \qquad \varrho=e^{\pi i/4}.$$

By definition, we have $G(-n, 1)=1$, hence

$$\frac{1}{\sqrt{n}}G(1,n)=\varrho^{1-n}. \tag{3.9}$$

By combining (3.8) with (3.9), we obtain

$$\frac{1}{\sqrt{n}}G(m,n)=\left(\frac{m}{n}\right)\cdot\varrho^{1-n}, \tag{3.10}$$

where n is an odd prime, and $(m, n)=1$.

Now *let m also be an odd prime*, $m\neq n$. Then (3.10) implies that

$$\frac{1}{\sqrt{m}}G(-n,m)=\left(\frac{-n}{m}\right)\cdot\varrho^{1-m}. \tag{3.11}$$

By Corollary 2 of Theorem 1, together with (3.10) and (3.11), we obtain

$$\left(\frac{m}{n}\right)\cdot\varrho^{1-n}=\left(\frac{-n}{m}\right)\cdot\varrho^{1-m}\cdot\varrho^{1-mn},$$

so that

$$\left(\frac{m}{n}\right)^2 \cdot \varrho^{1-n+mn-2+m} = \left(\frac{m}{n}\right)\left(\frac{-n}{m}\right),$$

or

$$\left(\frac{m}{n}\right)\left(\frac{-n}{m}\right) = \varrho^{(m-1)(n+1)} = \varrho^{4\left(\frac{m-1}{2}\right)\left(\frac{n+1}{2}\right)} = (-1)^{\left(\frac{m-1}{2}\right)\left(\frac{n+1}{2}\right)}.$$

But

$$\left(\frac{m}{n}\right)\left(\frac{-n}{m}\right) = \left(\frac{m}{n}\right)\left(\frac{n}{m}\right)\left(\frac{-1}{m}\right),$$

and

$$\left(\frac{-1}{m}\right) \equiv (-1)^{(m-1)/2} \pmod{m},$$

by Theorem B, while $\left(\frac{-1}{m}\right) = \pm 1$, and m is an odd prime. Hence

$$\left(\frac{-1}{m}\right) = (-1)^{(m-1)/2},$$

and

$$\left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = (-1)^{\{(m-1)/2\}\{(n+1)/2\} - \{(m-1)/2\}} = (-1)^{\{(m-1)/2\}\{(n-1)/2\}},$$

as claimed in Theorem 2.

Theorem 3 (Lagrange). *If n is an odd prime, then*

$$\left(\frac{2}{n}\right) = (-1)^{\frac{(n^2-1)}{8}}.$$

Proof. We have, by (3.10),

$$\left(\frac{2}{n}\right) = \frac{G(2,n)}{\sqrt{n}} \cdot \varrho^{n-1}, \qquad \varrho = e^{\pi i/4}, \tag{3.12}$$

and by Corollary 2 of Theorem 1,

$$\frac{1}{\sqrt{n}} G(2,n) = \frac{1}{\sqrt{2}} G(-n,2) \cdot \varrho^{1-2n}.$$

By definition, however, we have

$$\frac{G(-n,2)}{\sqrt{2}} = \frac{1}{\sqrt{2}}\left(1 + e^{\frac{\pi i n}{2}}\right),$$

hence (3.12) gives

$$\left(\frac{2}{n}\right)=\frac{1}{\sqrt{2}}(1+e^{(\pi in)/2})\cdot\varrho^{1-2n+n-1}=\frac{1}{\sqrt{2}}(1+e^{\pi in/2})\cdot e^{-\pi in/4}$$

$$=\frac{1}{\sqrt{2}}(e^{-\pi in/4}+e^{\pi in/4})=\sqrt{2}\cos\frac{\pi n}{4}=(-1)^{(n^2-1)/8}. \tag{3.13}$$

Remarks. In the proof of Theorem 3 we have *not* used Corollary 3 of Theorem 1 in which $G(2,b)$ is evaluated. If we did, we would get from (3.10), with $m=2$,

$$\frac{1}{\sqrt{n}}G(2,n)=\left(\frac{2}{n}\right)\cdot\varrho^{1-n},\qquad \varrho=e^{\pi i/4},$$

$$=\begin{cases}1, & \text{if } n\equiv 1 \pmod 4,\\ i, & \text{if } n\equiv 3 \pmod 4.\end{cases}$$

Since

$$\varrho^{n-1}=\begin{cases}1, & \text{if } n\equiv 1 \pmod 8,\\ i, & \text{if } n\equiv 3 \pmod 8,\\ -1, & \text{if } n\equiv 5 \pmod 8,\\ -i, & \text{if } n\equiv 7 \pmod 8,\end{cases}$$

we have

$$\left(\frac{2}{n}\right)=\frac{G(2,n)}{\sqrt{n}}\cdot\varrho^{n-1}=\begin{cases}1, & \text{if } n\equiv\pm 1 \pmod 8,\\ -1, & \text{if } n\equiv\pm 3 \pmod 8;\end{cases}$$

$$=(-1)^{(n^2-1)/8}.$$

Notes on Chapter IX

§ 1. As a general reference see P. Bachmann, *Niedere Zahlentheorie*, (i), Ch. 6. (Reprint 1968, New York).

This arrangement of the proof of Theorem 1, *and* the Corollaries, is due to C.L. Siegel. See his lectures on *Analytische Zahlentheorie*, II (Göttingen), 1963–64. In handling the *limiting case* of the transformation formula (1.2), a special case of (1.2) itself is used, once on the left-hand side, and again on the right-hand side. This artifice is Siegel's. The usual way of dealing with the limit

$$\lim_{\varepsilon\to 0}\sum_{n=-\infty}^{\infty}e^{-\pi\varepsilon b^2\left(n+\frac{v+h}{b}\right)^2}$$

is as in Hardy's *Ramanujan*, p. 143, last line. Siegel's proof of

$$\lim_{\varepsilon \to 0} \sum_{n=-\infty}^{\infty} e^{-\frac{\pi n^2}{b^2 \varepsilon} + \frac{2\pi i n b}{v}(v+h)} = 1,$$

however, is *not* what is given here.

The case $w=0$, $a=2$ of Theorem 1 was treated by Dirichlet and Cauchy; Kronecker treated the case in which a is any positive integer. Dirichlet used the method of Fourier series for the evaluation of Gaussian sums, *Abhand. Berliner Akad.* (1835), 391–407; *Werke*, I, 237–256; *Crelle's J.* XVII, 57–67; *Werke*, I, 257–270; *Crelle's J.* XXI (1839/40), 134–155; *Werke*, I, 473–479. At about the same time, and for the same purpose, Cauchy used the transformation formula for $\theta_3(0, z)$, *Liouville's J.*, V, 154–168; *Oeuvres* (i) V, 152–166. Kronecker recognized the common principle inherent in the proofs of Dirichlet and Cauchy, and, using complex integration, obtained a reciprocity formula for Gaussian sums from which the law of quadratic reciprocity could be deduced, *Monatsberichte Kgl. Preuss. Akad. Wiss. Berlin* (1880), 686–698; 854–860; *J. für Math.* 105 (1889), 267–268; *Werke*, IV, 277–300; *Festschrift der Math. Gesellschaft in Hamburg aus dem Jahre* 1890, 32–36; *Werke*, I, 303–307. G. Landsberg gave a method by which *both* the transformation formula for the theta function, and the reciprocity formula for Gaussian sums, could be deduced (including the determination of the eighth root of unity, referred to in the Notes on Chapter V), using complex integration, like Kronecker, but combining it with a skilful use of Euclid's division algorithm, *Crelle's J.* 111 (1893), 234–253. It is needless, perhaps, to say that Gauss's evaluation of the sums preceded all this. See his *Werke*, II, p. 9, and p. 155. E. Lindelöf refers to the work of Schaar (1848) on page 68 of his *Calcul des Résidus*.

The reciprocity formula for *generalized* Gaussian sums can be proved, without the use of the transformation formula for theta-functions, by means of Cauchy's theorem of residues, in the manner of Kronecker and Landsberg. See, for instance, Chapter V of the author's *Introduction to analytic number theory* (1968) where the proof given is, in substance, due to Siegel.

§ 2. This section is closely allied to Chapter II, IV, V of the author's *Introduction to analytic number theory*. (Springer-Verlag, 1968).

For Proposition A, see P. Fermat (1640), *Oeuvres*, (ii), 209; and L. Euler (1760), *Opera Omnia* (1868), (ii), 531. For Proposition B, see J. L. Lagrange, *Oeuvres* (1868), (ii), 667–69. For historical references, see Bachmann, *Niedere Zahlentheorie*, (i), Ch. 6.

§ 3. Euler discovered Theorem 2, in a more complicated form, empirically, so to say, and published it in 1783. In 1785 Legendre rediscovered the theorem, put it in this elegant form by the use of his *symbol*, called it the 'law of reciprocity' because of the symmetry in the rôle of the two (odd) primes, and proved at least a part of it (in the sense, that his proof depended on certain assumptions which could not be proved at that time). Gauss rediscovered the

reciprocity law of Legendre in 1795, again empirically, and proved it a year later. See Section IV of his *Disquisitiones arithmeticae*, *Werke*, I. Gauss presented Theorem 2 in the form: *A prime p is a quadratic residue or non-residue of another prime q, according as* $(-1)^{\frac{q-1}{2}}$ *q is a quadratic residue or non-residue of p.* (Here both p and q are odd). He later found six more totally different proofs. The shortest is his fifth proof, which is based on the so-called *Lemma of Gauss*. See, for instance, Hardy and Wright, loc. cit., §6.11; E. Hlawka and J. Schoiszengeier, *Zahlentheorie* (1979), Ch. XI; J.V. Uspensky and M.A. Heaslet, *Elementary number theory* (1939), Ch. X; P. Bachmann, *Niedere Zahlentheorie*, (i), Ch. 6, which contains many different proofs together with their history.

In later developments, E. Hecke *defined Gaussian sums* in any algebraic number field of degree n, and proved a reciprocity formula for such sums by considering the boundary behaviour of theta-functions in n-variables. From this he derived a proof of the law of quadratic reciprocity, which D. Hilbert was the first to set up in algebraic number fields by the introduction of his norm-residue symbol. See Hecke, *Göttingen Nachrichten*, (1919), 265–278; *Werke*, 235–248; *Vorlesungen über die Theorie der algebraischen Zahlen* (1923), Kap. VIII. Hecke remarks: 'Es ist die Tatsache, daß die genauere Kenntnis des Verhaltens einer analytischen Funktion in der Nähe ihrer singulären Stellen eine Quelle von arithmetischen Sätzen ist'. (The fact of the matter is that a more precise knowledge of the behaviour of an analytic function near its singularities is a source of arithmetical results.) A more explicit form of the law than Hecke's (which contains an undetermined eighth root of unity) is due to Hasse (for the l-th power residues), *J. für Math.* 153 (1924), 192–207. Siegel has given a proof of Hasse's law of reciprocity (for $l=2$) by generalizing the Gauss-Hecke sums, and proving a reciprocity formula for them, in the manner of Kronecker, coupled with an argument of induction on the degree of the field. Siegel's proof avoids the use of theta-functions of several variables, and the use of class-field theory. See *Göttinger Nachrichten* (1960), 1–16; *Ges. Abhandlungen*, III, 334–339.

There is an *approximate* reciprocity formula, due to J.R. Wilton, for exponential sums of the form $\sum_{m \leqq T} d(m)e^{2\pi i x m}$, where $d(m)$ denotes the divisor function, with applications to problems of Diophantine approximation originally considered by Hardy and Littlewood (and depending on the nature of the real number x). See *J. für Math.* 169 (1932), 219–237, (which contains a list of references). Such approximate reciprocity formulae in algebraic number fields have been considered by the author and Raghavan Narasimhan, *Commentarii Math. Helvetici*, 43 (1968), 296–310; 54 (1979), 523–561, Theorem 2.

Theorem 3 of this Chapter was discovered by Lagrange, *Nouv. Mém. de l'Acad. de Berlin*, (1775); *Oeuvres*, III, 759–795; Bachmann (loc. cit.) I, p. 196. As Siegel remarks in his Göttingen lectures on *Analytische Zahlentheorie*, II (1963/64), Theorem 3 can be simply deduced from Euler's criterion, stated as

Theorem B in § 2. For if p is an odd prime, then

$$2^{(p-1)/2} \equiv \left(\frac{2}{p}\right) \pmod{p}.$$

But $\dfrac{(1+i)^2}{i} = 2$, hence

$$\begin{aligned}\left(\frac{2}{p}\right) &\equiv \frac{(1+i)^{p-1}}{i^{(p-1)/2}} \pmod{p} \\ &\equiv \frac{(1+i)^p}{(1+i)i^{(p-1)/2}} \pmod{p} \equiv \frac{1+i^p}{(1+i)i^{(p-1)/2}} \pmod{p} \\ &\equiv \frac{e^{\frac{\pi i p}{4}} + e^{-\frac{\pi i p}{4}}}{e^{\frac{\pi i}{4}} + e^{-\frac{\pi i}{4}}} \pmod{p} \equiv \frac{\cos\frac{\pi p}{4}}{\cos\frac{\pi}{4}} \pmod{p}.\end{aligned}$$

Since $p > 2$, and both sides of the congruence have absolute value 1, it follows that

$$\left(\frac{2}{p}\right) = \frac{\cos\frac{\pi p}{4}}{\cos\frac{\pi}{4}} = \begin{cases} 1, & \text{for} \quad p \equiv \pm 1 \pmod{8}, \\ -1, & \text{for} \quad p \equiv \pm 3 \pmod{8}, \end{cases}$$

which is Theorem 3.

Chapter X

The representation of a number as a sum of four squares

§ 1. The theorems of Lagrange and of Jacobi. We have seen in Chapter VIII that the identity

$$\eta(z)=e^{\pi iz/12}\theta_3\left(\frac{1}{2}+\frac{z}{2},3z\right), \qquad \mathrm{Im}z>0, \tag{1.1}$$

which connects Dedekind's η-function with the theta-function θ_3 implies Euler's theorem on pentagonal numbers. That was proved by analytical methods in two different ways. The first consisted in representing $\theta_3(v,z)$, initially defined by an infinite series, as an infinite product, and identifying the defining product of $\eta(z)$ with that which results from the right-hand side of (1.1). The second consisted in combining the transformation formula for $\theta_3(v,z)$, namely

$$\theta_3\left(0,-\frac{1}{z}\right)=\sqrt{\frac{z}{i}}\cdot\theta_3(0,z), \qquad \mathrm{Im}z>0, \tag{1.2}$$

with the functional equation of $\eta(z)$, namely

$$\eta\left(-\frac{1}{z}\right)=\sqrt{\frac{z}{i}}\eta(z), \qquad \mathrm{Im}z>0, \tag{1.3}$$

so as to construct a modular function which vanishes identically in the upper half-plane $\mathrm{Im}z>0$, and thereby yields (1.1).

We shall now see that, for $\mathrm{Im}z>0$, we have the further identity

$$\theta_3^4(0,z)=f(z), \tag{1.4}$$

where

$$f(z)=1+8\sum_{n=1}^{\infty}\frac{ne^{\pi inz}}{1-e^{\pi inz}}-8\sum_{n=1}^{\infty}\frac{4n\cdot e^{4\pi inz}}{1-e^{4\pi inz}}, \tag{1.5}$$

and this implies the theorem of Lagrange that every integer greater than or equal to one can be represented as a sum of four squares, as well as Jacobi's theorem giving the *number* of such representations.

We shall prove identity (1.4) again by analytical methods in two different ways. The first consists in using the differential equation satisfied by the four theta-functions together with Jacobi's formula

$$\theta'(0,z)=\pi\theta_1(0,z)\cdot\theta_2(0,z)\cdot\theta_3(0,z).$$

The second consists in using the relation

$$f(z)=\frac{8i}{\pi}\cdot\frac{d}{dz}\left(\log\frac{\eta(z/2)}{\eta(2z)}\right),\qquad \mathrm{Im}\,z>0\,, \tag{1.6}$$

and constructing a modular function which vanishes identically in the upper half-plane and thereby yields (1.4).

Whereas in the case of (1.1) we considered transformations belonging to the (full) modular group Γ, in the case of (1.4) we consider transformations belonging to a (proper) sub-group Δ of the modular group (Lemma 2, § 7, Ch. VII).

Let n be an integer, $n\geqq 1$, with the representation

$$n=n_1^2+n_2^2+n_3^2+n_4^2\,,$$

where the (n_k) are integers (inclusive of zero). Representations which differ only in the order, or sign, of the summands n_k are counted as *distinct. Let* $r(n)$ *denote the number of such representations.* Then $r(n)$ is also the number of *lattice-points* (that is to say, points whose coordinates are integers) on the surface of a four-dimensional sphere with the origin as centre and radius $\sqrt{n}$.

For example, $r(3)=32=8(1+3)$, since $3=0^2+(\pm 1)^2+(\pm 1)^2+(\pm 1)^2=$ $=(\pm 1)^2+0^2+(\pm 1)^2+(\pm 1)^2=(\pm 1)^2+(\pm 1)^2+0^2+(\pm 1)^2=(\pm 1)^2+(\pm 1)^2+$ $+(\pm 1)^2+0^2$; and $r(13)=112=8(1+13)$, since $13=0^2+0^2+2^2+3^2=1^2+$ $+2^2+2^2+2^2$; the first representation here gives rise to 48 distinct representations, four through changes of sign of the summands, each of which yields twelve more through changes in the order of the summands, while the second gives rise to 64 distinct representations, sixteen through changes of sign, each of them giving rise to four more through changes in the order.

We shall prove the following

Theorem 1 (Jacobi). *If n is an integer, $n\geqq 1$, then*

$$r(n)=8\sum_{d\mid n,\,4\nmid d} d\,, \tag{1.7}$$

where the summation is over those positive divisors of n, which are not divisible by 4.

As an immediate consequence, we obtain

Corollary 1 (Lagrange). *If n is an integer, $n\geqq 1$, then $r(n)>0$, so that every integer greater than or equal to one can be expressed as a sum of four squares (of integers).*

Theorem 1 also implies

Corollary 2 (Jacobi). *If n is an odd integer, $n\geqq 1$, then*

$$r(n)=8\sum_{d\mid n} d\,, \tag{1.8}$$

while, if n is even, then

$$r(n)=24\sum_{d|n,\ d\,\mathrm{odd}} d. \tag{1.9}$$

If n is *odd*, (1.8) follows from (1.7). To consider the case in which n is *even*, we first note that the function $\frac{r(n)}{8}$, as *defined* by (1.7), is *multiplicative*. For let n_1, n_2 be two integers, with $n_1 \geqq 1$, $n_2 \geqq 1$, *and* $(n_1, n_2)=1$. Then in case $4\nmid n_1$, $4\nmid n_2$, we have trivially

$$\sum_{d|n_1,\,4\nmid d} d\cdot \sum_{d|n_2,\,4\nmid d} d=\sum_{d|n_1\cdot n_2,\,4\nmid d} d.$$

If $4|n_1$, however, and $(n_1, n_2)=1$, then $2\nmid n_2$, and

$$\sum_{d|n_1,\,4\nmid d} d\cdot \sum_{d|n_2} d=\sum_{d|n_1\cdot n_2,\,4\nmid d} d.$$

The multiplicativity of $\frac{r(n)}{8}$ implies that if $n=2m$, where m is *odd*, then $\frac{r(n)}{8}$ $=\frac{r(2)}{8}\cdot\frac{r(m)}{8}$. Since $r(2)=8(1+2)=24$, by direct verification, we have

$$r(n)=3r(m)=3\cdot 8\sum_{d|m,\,4\nmid d} d=3\cdot 8\sum_{d|m} d,$$

since m is odd, and this is (1.9). If $n=2^k$, where k is an integer, $k\geqq 1$, then $r(n)=24$, by (1.7), which confirms (1.9) in this case as well.

§ 2. Proof of Jacobi's theorem by means of theta-functions

Proof of Theorem 1. If $q=e^{\pi i z}$, with $\mathrm{Im}\,z>0$, then we have, by definition ((3.3), Ch. V)

$$\theta_3(0,z)=\sum_{k=-\infty}^{\infty} q^{k^2},$$

and clearly

$$\{\theta_3(0,z)\}^4=\left(\sum_{k=-\infty}^{\infty} q^{k^2}\right)^4=\sum_{n=0}^{\infty} r(n)q^n, \qquad r(0)=1. \tag{2.1}$$

Theorem 1 asserts that

$$r(n)=8\sum_{d|n} d-8\sum_{4d|n} 4d, \tag{2.2}$$

or, equivalently, that

$$\sum_{n=0}^{\infty} r(n)q^n=1+8\sum_{d,m=1}^{\infty} dq^{dm}-8\sum_{d,m=1}^{\infty} 4d\cdot q^{4dm}, \qquad (n=dm), \tag{2.3}$$

where d and m run through all integers greater than or equal to one.

Since $|q|<1$, and

$$\sum_{m=1}^{\infty} q^{md} = \frac{q^d}{1-q^d},$$

we can rewrite (2.3) (with n in place of d) as

$$\sum_{n=0}^{\infty} r(n)q^n = 1+8\sum_{n=1}^{\infty}\frac{nq^n}{1-q^n} - 8\sum_{n=1}^{\infty}\frac{4n\cdot q^{4n}}{1-q^{4n}}. \tag{2.4}$$
$$\equiv f(z),$$

by (1.5).

In order to prove Theorem 1, we have therefore only to show, after (2.1) and (2.4), that

$$\{\theta_3(0,z)\}^4 = f(z). \tag{2.5}$$

In the course of proving Jacobi's formula for $\theta'(0,z)$, we have shown ((5.2), Ch. V) that if (ω_1,ω_2) is a pair of basic periods of Weierstrass's $\wp$-function, and $\wp\left(\frac{\omega_1}{2}\right)=e_1$, $\wp\left(\frac{\omega_2}{2}\right)=e_2$, $\wp\left(\frac{\omega_1+\omega_2}{2}\right)=e_3$, then

$$e_r = \frac{1}{\omega_1^2}\left(\frac{1}{3}\cdot\frac{\theta'''(0,z)}{\theta'(0,z)} - \frac{\theta_r''(0,z)}{\theta_r(0,z)}\right), \qquad r=1,2,3,$$

where

$$\theta'(0,z) = \left[\frac{\partial}{\partial v}\theta(v,z)\right]_{v=0}, \qquad \theta_r''(0,z) = \left[\frac{\partial^2}{\partial v^2}\theta_r(v,z)\right]_{v=0},$$

$$\theta'''(0,z) = \left[\frac{\partial^3}{\partial v^3}\theta(v,z)\right]_{v=0}.$$

We have also seen (§4, Ch. V) that

$$\frac{\partial^2}{\partial v^2}\{\theta_r(v,z)\} = 4\pi i\cdot\frac{\partial}{\partial z}\{\theta_r(v,z)\}, \qquad r=1,2,3,$$

and ((5.4), Ch. V) that

$$\theta'''(0,z) = 4\pi i\cdot\frac{\partial}{\partial z}\{\theta'(0,z)\}.$$

Hence

$$e_r = \frac{4\pi i}{\omega_1^2}\left[\frac{1}{3}\cdot\frac{\partial}{\partial z}\{\log\theta'(0,z)\} - \frac{\partial}{\partial z}\{\log\theta_r(0,z)\}\right], \qquad r=1,2,3. \tag{2.6}$$

On the other hand, we have proved ((5.6), Ch. V) that

$$\{\theta_3(0,z)\}^4 = (e_1-e_2)\left(\frac{\omega_1}{\pi}\right)^2.$$

If we now use (2.6), we get

$$\{\theta_3(0,z)\}^4 = \frac{4i}{\pi}\left[\frac{\partial}{\partial z}\{\log\theta_2(0,z)\} - \frac{\partial}{\partial z}\{\log\theta_1(0,z)\}\right]$$
$$= \frac{4i}{\pi}\left[\frac{\partial}{\partial z}\left\{\log\left(\frac{\theta_2(0,z)}{\theta_1(0,z)}\right)\right\}\right].$$

Since $q = e^{\pi i z}$, and $\frac{dq}{dz} = \pi i q$, this implies that

$$\{\theta_3(0,z)\}^4 = 4q\cdot\frac{\partial}{\partial q}\left(\log\left\{\frac{\theta_1(0,z)}{\theta_2(0,z)}\right\}\right).$$

By Theorem 6 of Chapter V, however, we have

$$\frac{\theta_1(0,z)}{\theta_2(0,z)} = \frac{2q^{1/4}\cdot\prod_{n=1}^{\infty}(1+q^{2n})^2}{\prod_{n=1}^{\infty}(1-q^{2n-1})^2} = \frac{2q^{1/4}\cdot\prod_{n=1}^{\infty}(1-q^{4n})^2}{\prod_{n=1}^{\infty}(1-q^n)^2}.$$

Hence

$$\{\theta_3(0,z)\}^4 = 4q\cdot\frac{\partial}{\partial q}\left\{\log 2 + \frac{1}{4}\log q + \sum_{n=1}^{\infty}\log(1-q^{4n})^2 - \sum_{n=1}^{\infty}\log(1-q^n)^2\right\}$$
$$= 1 + 8\sum_{n=1}^{\infty}\frac{nq^n}{1-q^n} - 8\sum_{n=1}^{\infty}\frac{4nq^{4n}}{1-q^{4n}}$$
$$\equiv f(z),$$

by (1.5), which proves the theorem.

§ 3. Siegel's proof of Jacobi's theorem. We first consider the behaviour of the function $f(z)$, defined in (1.5), under the modular transformation: $z \to -\frac{1}{z}$.

Theorem 2. *If* $\mathrm{Im}\, z > 0$, *then*

$$f\left(-\frac{1}{z}\right) = -z^2 f(z), \tag{3.1}$$

where $f(z)$ *is defined as in* (1.5).

Proof. The Dedekind η-function is defined by the relation

$$\eta(z) = e^{\pi i z/12}\cdot\prod_{n=1}^{\infty}(1-e^{2\pi i n z}), \qquad \mathrm{Im}\, z > 0.$$

Since $\eta(z) \neq 0$, for $\mathrm{Im}\, z > 0$, we have

$$\frac{d}{dz}(\log\eta(z)) = \frac{\pi i}{12} - 2\pi i\sum_{n=1}^{\infty}\frac{ne^{2\pi i n z}}{1-e^{2\pi i n z}}.$$

Replacing z by $\frac{1}{2}z$, and by $2z$, we obtain

$$2\frac{d}{dz}(\log\eta(z/2))=\frac{\pi i}{12}-2\pi i\sum_{n=1}^{\infty}\frac{ne^{\pi inz}}{1-e^{\pi inz}}, \tag{3.2}$$

and

$$\frac{1}{2}\frac{d}{dz}(\log\eta(2z))=\frac{\pi i}{12}-2\pi i\sum_{n=1}^{\infty}\frac{ne^{4\pi inz}}{1-e^{4\pi inz}}. \tag{3.3}$$

On multiplying (3.2) by $\frac{4i}{\pi}$, and (3.3) by $\frac{-16i}{\pi}$, and adding, we get

$$\frac{8i}{\pi}\cdot\frac{d}{dz}\left(\log\frac{\eta(z/2)}{\eta(2z)}\right)=f(z), \tag{3.4}$$

and the transformation: $z\to-\frac{1}{z}$ yields

$$\frac{8i}{\pi}\cdot z^2\cdot\frac{d}{dz}\left(\log\frac{\eta(-1/2z)}{\eta(-2/z)}\right)=f(-1/z). \tag{3.5}$$

But by Theorem 1 of Chapter VIII, we have

$$\eta(-1/z)=\sqrt{\frac{z}{i}}\cdot\eta(z),$$

with $\sqrt{\frac{z}{i}}=1$ for $z=i$. Hence

$$\begin{aligned}f(-1/z)&=\frac{8iz^2}{\pi}\cdot\frac{d}{dz}\left(\log\left\{\frac{\sqrt{2z/i}}{\sqrt{z/2i}}\cdot\frac{\eta(2z)}{\eta(z/2)}\right\}\right)\\&=-z^2\cdot\frac{8i}{\pi}\cdot\frac{d}{dz}\left(\log\left\{\frac{\eta(z/2)}{\eta(2z)}\right\}\right)\\&=-z^2f(z),\end{aligned}$$

by (3.4), which proves the theorem.

Theorem 3. *Let*

$$K(z)=\frac{\theta_3^4(0,z)-f(z)}{\eta^4(z)},\qquad \operatorname{Im}z>0. \tag{3.6}$$

Then we have

$$\begin{aligned}&K(z+2)=\varepsilon^{-8}K(z),\qquad \varepsilon=e^{\pi i/12};\\&K(-1/z)=K(z),\end{aligned} \tag{3.7}$$

so that the function $|K(z)|$ is invariant for the transformations $z \to z+2$, and $z \to -\frac{1}{z}$, and hence invariant under all transformations belonging to the group Δ generated by them.

Proof. That $f(z+2)=f(z)$ follows from the definition of $f(z)$ in (1.5), while Theorem 2 implies that $f(-1/z)=-z^2 f(z)$. Similarly the definition of θ_3 (in (3.3) of Ch. V) gives: $\theta_3^4(0, z+2)=\theta_3^4(0, z)$, while Theorem 9 of Chapter V implies that $\theta_3^4(0, -1/z)=-z^2\theta_3^4(0, z)$. On the other hand, by (3.1) and (3.2) of Chapter VIII, we have

$$\eta(z+2)=e^{2\pi i/12}\cdot\eta(z), \qquad \eta(-1/z)=e^{-\pi i/4}\cdot\sqrt{z}\cdot\eta(z),$$

and the assertions about $K(z)$ in (3.7) follow.

Theorem 4. *For* $\mathrm{Im}\, z>0$, *we have* $\theta_3^4(0, z)=f(z)$.

Proof (Siegel). Since $\eta(z)\neq 0$ for $\mathrm{Im}\, z>0$, it suffices to show that the function $K(z)$ defined in (3.6) is identically zero. Since $|K(z)|$ is invariant under the transformations of the group Δ, which is a proper sub-group of the modular group Γ, it suffices to consider the behaviour of $K(z)$ in a *fundamental domain* F_Δ associated with Δ. We have already seen that the set

$$F_\Delta=\{z|\mathrm{Im}\, z>0,\ |z-\tfrac{1}{2}|\geqq\tfrac{1}{2},\ 0\leqq\mathrm{Re}\, z\leqq 1, \quad \text{with} \quad \tfrac{1}{2}\leqq\mathrm{Re}\, z\leqq 1 \quad \text{if} \quad |z-\tfrac{1}{2}|=\tfrac{1}{2},$$
$$\text{and} \quad \mathrm{Im}\, z\geqq 1 \quad \text{if} \quad \mathrm{Re}\, z=1, \quad \text{and} \quad \mathrm{Im}\, z\leqq 1 \quad \text{if} \quad \mathrm{Re}\, z=0\}$$

is a fundamental domain for Δ (cf. Ch. VII, § 7, after Lemma 3).

Let $S=\{z|\mathrm{Im}\, z>0,\ |z-\frac{1}{2}|\geqq\frac{1}{2},\ 0\leqq\mathrm{Re}\, z\leqq 1\}$, so that $F_\Delta\subset S$. Consider the closed subset $N_c=\{ABCDEF\}$ of S. The boundary ∂N_c of N_c consists of the line-segments DE, EF, FA and the arcs AB, BC, CD. Here FE consists of the points $\{z|0\leqq\mathrm{Re}\, z\leqq 1,\ \mathrm{Im}\, z=c\}$, where c is *sufficiently large*; FA is a segment of the imaginary axis; AB is the inverse of FE relative to the circle $|z|=1$; BC is an arc of the circle $|z-\frac{1}{2}|=\frac{1}{2}$; CD is symmetric to AB relative to the line $\mathrm{Re}\, z=\frac{1}{2}$. As $c \to +\infty$, the radius of the circle of which AB is an arc, as well as its centre, tend to zero; and a similar thing happens with CD (see the following Figure).

We shall first show that

$$K(z)\to 0, \quad \text{as} \quad \mathrm{Im}\, z\to+\infty, \tag{3.8}$$

uniformly for $-1\leqq\mathrm{Re}\, z\leqq 1$. Since

$$\eta^4(z)=e^{\pi iz/3}\cdot\prod_{n=1}^{\infty}(1-e^{2\pi inz})^4, \qquad \mathrm{Im}\, z>0,$$

where the product converges absolutely, and uniformly in every compact set in the upper half-plane $\mathrm{Im}\, z>0$, we have

$$\eta^4(z)\cdot e^{-\pi iz/3}\to 1, \quad \text{as} \quad \mathrm{Im}\, z\to+\infty,$$

uniformly for $-1 \leqq \mathrm{Re} z \leqq 1$. On the other hand, we have

$$\{\theta_3(0,z)\}^4 = \left(1+2\sum_{n=1}^{\infty} e^{\pi i n^2 z}\right)^4 = (1+2e^{\pi i z}+O(e^{-4\pi \mathrm{Im} z}))^4$$
$$= 1+8e^{\pi i z}+O(e^{-4\pi \mathrm{Im} z}), \quad \mathrm{Im} z>0;$$

and

$$f(z) = 1+8\sum_{n=1}^{\infty} \frac{ne^{\pi i n z}}{1-e^{\pi i n z}} - 8\sum_{n=1}^{\infty} \frac{4ne^{4\pi i n z}}{1-e^{4\pi i n z}}$$
$$= 1+8e^{\pi i z}+O(e^{-2\pi \mathrm{Im} z}), \quad \mathrm{Im} z>0.$$

It follows that

$$K(z) = O\left(e^{-\mathrm{Im} z\left(2\pi-\frac{\pi}{3}\right)}\right);$$

hence $K(z)\to 0$ as $\mathrm{Im} z\to\infty$, uniformly for $-1 \leqq \mathrm{Re} z \leqq 1$, which proves (3.8).

We next use (3.8) to consider the behaviour of $K(z)$ as $z\to 0$, $z\in F_\Delta$.

Let F_Δ^* be the image of F_Δ (Fig. 14) under the transformation $z^* = -\frac{1}{z}$. The interior of F_Δ^* is the translate, under $z\to z-1$, of the interior of F_Δ, and $F_\Delta^* = \{z | \mathrm{Im} z>0, |z+\frac{1}{2}| \geqq \frac{1}{2}, -1 \leqq \mathrm{Re} z \leqq 0$, with $-\frac{1}{2} \leqq \mathrm{Re} z \leqq 0$ if $|z+\frac{1}{2}| = \frac{1}{2}$, and $\mathrm{Im} z \leqq 1$ if $\mathrm{Re} z = -1$, while $\mathrm{Im} z \geqq 1$ if $\mathrm{Re} z = 0\}$.

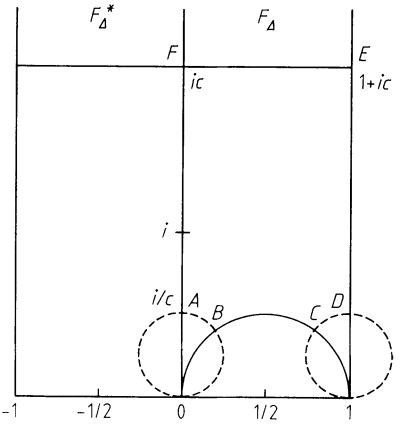


Fig. 14

Clearly $z\to 0$, $z\in F_\Delta$, if and only if $z^*\to\infty$, $z^*\in F_\Delta^*$, while $K(z^*) = K(z)$, by Theorem 3. It follows from (3.8) that $K(z^*)\to 0$ as $\mathrm{Im} z^*\to\infty$, uniformly for $-1 \leqq \mathrm{Re} z^* \leqq 0$. Hence

$$K(z)\to 0, \quad \text{as} \quad z\to 0, \quad z\in F_\Delta, \tag{3.9}$$

the limit being uniform as $z\to 0$, $z\in AB$, and $c\to+\infty$.

To consider the behaviour of $K(z)$ as $z\to 1$, $z\in F_\Delta$, we note that the transformation $z\to z'=1-\frac{1}{z}$ carries F_Δ into itself, and the cusps at $z=\infty$, 0, 1 into $z'=1$, ∞, 0, while the point $z=e^{\pi i/3}$ remains fixed. We shall prove that

$$K\left(1-\frac{1}{z}\right)\to 0, \quad \text{as} \quad \operatorname{Im} z\to\infty,$$

uniformly for $0\leqq \operatorname{Re} z\leqq 1$, which implies that

$$K(z)\to 0, \quad \text{as} \quad z\to 1, \quad z\in F_\Delta, \tag{3.10}$$

the limit being uniform as $z\to 1$, $z\in CD$, and $c\to+\infty$. For this we have to consider the behaviour, in turn, of $\theta_3^4\left(0,1-\frac{1}{z}\right)$, $\eta^4\left(1-\frac{1}{z}\right)$, and $f\left(1-\frac{1}{z}\right)$.

Since $n^2\equiv n \pmod 2$, we have

$$\theta_3\left(0,1-\frac{1}{z}\right)=\sum_{n=-\infty}^{\infty} e^{-\pi i(n^2/z)+\pi i n},$$

and the transformation formula (Theorem 9 of Ch. V)

$$\sqrt{\frac{z}{i}}\cdot\sum_{n=-\infty}^{\infty} e^{\pi i z(n+v)^2}=\sum_{n=-\infty}^{\infty} e^{-\pi i(n^2/z)+2\pi i n v}$$

with $v=\frac{1}{2}$ leads to the relation

$$\begin{aligned}\left\{\theta_3\left(0,1-\frac{1}{z}\right)\right\}^4 &= -z^2(2e^{\pi i z/4}+2e^{9\pi i z/4}+\ldots)^4\\ &= -z^2(16e^{\pi i z}+\ldots).\end{aligned} \tag{3.11}$$

Since

$$\eta\left(1-\frac{1}{z}\right)=\varepsilon\eta\left(-\frac{1}{z}\right)=\varepsilon\cdot\sqrt{\frac{z}{i}}\cdot\eta(z), \quad \text{where} \quad \varepsilon=e^{\pi i/12},$$

by (3.1) and (3.2) of Chapter VIII, we have

$$\eta^4\left(1-\frac{1}{z}\right)=-e^{\pi i/3}\cdot z^2\cdot\eta^4(z). \tag{3.12}$$

From (3.4) we have

$$f\left(1-\frac{1}{z}\right)=\frac{8iz^2}{\pi}\cdot\frac{d}{dz}\left(\log\frac{\eta\left(\frac{z-1}{2z}\right)}{\eta\left(2-\frac{2}{z}\right)}\right),$$

and the right-hand side can be simplified by the repeated application of the functional equation for $\eta(z)$, obtained in Theorem 2 of Chapter VIII, for

$$\begin{aligned}\eta\left(\frac{z-1}{2z}\right) &= \sqrt{\frac{2z}{i(1-z)}}\cdot\eta\left(\frac{2z}{1-z}\right) = \sqrt{\frac{2z}{i(1-z)}}\cdot\eta\left(-2+\frac{2}{1-z}\right)\\ &= \varepsilon^{-2}\sqrt{\frac{2z}{i(1-z)}}\cdot\eta\left(-\frac{2}{z-1}\right), \qquad \varepsilon = e^{\pi i/12},\\ &= \varepsilon^{-2}\sqrt{\frac{2z}{i(1-z)}}\cdot\sqrt{\frac{z-1}{2i}}\cdot\eta\left(\frac{z-1}{2}\right)\\ &= \varepsilon^{-2}\cdot\sqrt{z}\cdot\eta\left(\frac{z-1}{2}\right),\end{aligned}$$

and

$$\eta\left(2-\frac{2}{z}\right) = \varepsilon^2\cdot\eta\left(-\frac{2}{z}\right) = \varepsilon^2\cdot\sqrt{\frac{z}{2i}}\cdot\eta\left(\frac{z}{2}\right).$$

Hence

$$f\left(1-\frac{1}{z}\right) = \frac{8iz^2}{\pi}\cdot\frac{d}{dz}\left(\log\frac{\eta\left(\frac{z-1}{2}\right)}{\eta\left(\frac{z}{2}\right)}\right).$$

Since

$$\log\left(\frac{1}{1-z}\right) = -\log(1-z) = z+\tfrac{1}{2}z^2+\tfrac{1}{3}z^3+\ldots, \quad \text{for} \quad |z|\leqq 1, \quad z\neq 1,$$

we have

$$\begin{aligned}\log\eta(z) &= \frac{\pi i z}{12}+\sum_{n=1}^{\infty}\log(1-e^{2\pi i n z}), \quad \text{for} \quad \operatorname{Im} z>0,\\ &= \frac{\pi i z}{12}-e^{2\pi i z}-\ldots,\end{aligned}$$

and

$$\log\frac{\eta\left(\frac{z-1}{2}\right)}{\eta\left(\frac{z}{2}\right)} = \frac{\pi i(z-1)}{24}-\frac{\pi i z}{24}-e^{\pi i(z-1)}+e^{\pi i z}+\ldots,$$

so that

$$\frac{d}{dz}\left(\log\frac{\eta\left(\frac{z-1}{2}\right)}{\eta\left(\frac{z}{2}\right)}\right) = 2\pi i\cdot e^{\pi i z}+\ldots,$$

the expansion on the right-hand side having no constant term. Hence

$$f\left(1-\frac{1}{z}\right)=\frac{8iz^2}{\pi}(2\pi i\cdot e^{\pi iz}+\ldots)=-16z^2\cdot e^{\pi iz}+\ldots. \tag{3.13}$$

By (3.13), (3.12), and (3.11), we have

$$K\left(1-\frac{1}{z}\right)=\frac{\theta_3^4\left(0,1-\frac{1}{z}\right)-f\left(1-\frac{1}{z}\right)}{\eta^4\left(1-\frac{1}{z}\right)}=\frac{-z^2(16e^{\pi iz}+\ldots)+16z^2e^{\pi iz}((1+\ldots)}{-e^{\pi i/3}\cdot z^2\cdot e^{\pi iz/3}\cdot\prod_{m=1}^{\infty}(1-e^{2\pi imz})^4},$$

so that

$$\left|K\left(1-\frac{1}{z}\right)\right|=\left|\frac{O(e^{-\pi\operatorname{Im}z})}{e^{\pi iz/3}(1+O(e^{-\pi\operatorname{Im}z}))}\right|,$$

$$=O(e^{-\frac{2\pi}{3}\cdot\operatorname{Im}z})\to 0,$$

as $\operatorname{Im}z\to\infty$, uniformly for $0\leqq\operatorname{Re}z\leqq 1$, which proves (3.10).

Reasoning again as in the proof of Theorem 5 of Chapter VIII, we prove that if

$$\alpha=\sup_{\operatorname{Im}z>0}|K(z)|,$$

and $K(z)\not\equiv 0$, then, because of (3.8), (3.9), and (3.10), we have: $0<\alpha<\infty$, since N_c is compact. And if c is sufficiently large,

$$\alpha=\sup_{z\in N_c}|K(z)|,$$

and (since N_c is compact) there is a point $z_0\in N_c$, such that

$$|K(z_0)|=\alpha=\sup_{\operatorname{Im}z>0}|K(z)|,$$

which implies, by the maximum principle, that K reduces to a constant, which must be zero because of (3.8).

Notes on Chapter X

§§ 1–2. It is a classical problem in the theory of numbers to find out *when* a positive integer n can be represented as the sum of k squares, where k is a positive integer given in advance, $k\geqq 2$; and to determine the *exact number* of such representations, taking into account the order, and the sign of the summands. See Hardy's notes on the problem, in his book *Ramanujan*, pp. 157–160; and the remarks on the problem, in the context of modular forms, in Rankin's book, p. 241 (cited in the Notes on Ch. VIII).

It is easy to see that *not* every n is the sum of two squares. Since $(2m)^2\equiv 0 \pmod 4$, and $(2m+1)^2\equiv 1 \pmod 4$, no number of the form $4m+3$ can be the sum of two squares.

Lagrange (1770) proved that *every* positive integer can be represented as the sum of *four* squares (hence also as the sum of five or more squares), *Oeuvres*, iii, p. 189.

If $r_k(n)$ denotes the *number* of representations of n as the sum of k squares, Jacobi (1828) solved the problem of evaluating $r_k(n)$, in terms of simpler arithmetical functions, for $k=2, 4, 6$, and 8, by finding appropriate *identities* for $\{\theta_3(0,z)\}^k$, which yield an expression for $r_k(n)$ on equating coefficients. Typical is identity (1.4), for the case $k=4$. See his *Fundamenta Nova*, §§ 40–42, §§ 65–66; *Ges. Werke*, I, 49–239; *Crelle's J.* III, 191; *Ges. Werke*, I, p. 247.

The arrangement of the proof of Theorem 1 is the same as in Hurwitz's *Funktionentheorie*, and is, in essentials, the same as Jacobi's.

If $k=2$, Jacobi's identity is

$$(*) \qquad \{\theta_3(0,z)\}^2 = 1+4\sum_{n=1}^{\infty}\frac{(-1)^{n-1}q^{2n-1}}{1-q^{2n-1}}, \qquad q=e^{\pi iz}, \qquad \mathrm{Im}\, z>0,$$

which is easier to prove than (1.4), and which gives the result that

$$(**) \qquad r_2(n)=4\sum_{d\,\mathrm{odd},\,d|n}(-1)^{\frac{1}{2}(d-1)}=4\{d_1(n)-d_3(n)\},$$

where $d_1(n)$ and $d_3(n)$ are the numbers of the divisors of n of the form $4m+1$ and $4m+3$ respectively. Gauss stated a result which is equivalent to this in his *Disquisitiones arithmeticae*, § 182. See P. Bachmann, *Niedere Zahlentheorie*, (ii), Ch. VII. The result can be deduced without the use of elliptic functions, from the theory of *Gaussian integers*. See, for instance, Hardy and Wright, *An Introduction to the theory of numbers* (2nd edition, 1945), Ch. XV.

S. Ramanujan gave an elementary deduction of (1.4) from (*). See his *Collected Papers*, No. 18, formula (17) on p. 139. His proof has been reproduced by Hardy and Wright, loc. cit., pp. 311–314.

There is a proof of Theorem 1 by the use of integral quaternions in L.E. Dickson, *Algebren und ihre Zahlentheorie*, Kap. IX, Satz 22, p. 181–182. Landau gives an elementary deduction of (1.7) from (**), in his *Vorlesungen über Zahlentheorie*, (i), 110–113.

Liouville gave the formulae for $r_{10}(n)$ and $r_{12}(n)$, *J. de Math.* (2) 9 (1864), 296–8; 11 (1866), 1–8; Glaisher gave a table of formulae for $2\leqq k\leqq 18$, *Proc. London Math. Soc.* (2) 5 (1907), 479–90, based on previous published work of his; Uspensky developed Liouville's methods to handle the cases $2\leqq k\leqq 12$, and his paper, *Trans. American Math. Soc.* 30 (1928), 385–404, contains references to his earlier work. V. Bulygin (= Boulyguine) gave *general* formulae for $r_k(n)$ for *even* $k>8$, *Izv. Imp. Akad. Nauk*, (6) (1914), 389–404. Uspensky (loc. cit.) states that Liouville's methods enable him to prove Bulygin's formulae. See, for example, J.V. Uspensky and M.A. Heaslet, *Elementary number theory*, (1939), Ch. XIII.

If we consider the problem for k *odd*, and begin with $k=3$, it is easy to see that no number of the form $8m+7$ can be the sum of three squares, since for any integer m, we have $m^2\equiv 0$, 1, or 4 (mod 8). Legendre proved that a number n is

the sum of three squares *if and only if* $n \neq 4^a(8b+7)$, $a \geqq 0$, $b \geqq 0$, *Essai sur la théorie des nombres* (1798), 202; 398–9; 3rd edition (retitled *Théorie des nombres*, I (1830), 393–395). The function $r_3(n)$ has been evaluated by Dirichlet as a finite sum involving symbols of quadratic reciprocity, *Werke*, I, p. 496. Gauss had previously expressed $r_3(n)$ as a multiple of the number of classes, in the principal genus, of the properly primitive, binary quadratic forms of determinant -1, *Disquisitiones arithmeticae*, §291; *Werke*, I, p. 343. See Dirichlet, loc. cit., p. 414; also, for example, Landau's *Vorlesungen über Zahlentheorie*, (i) Sätze 186, 187.

The functions $r_5(n)$, $r_7(n)$ have been evaluated by Eisenstein (*Crelle's J.* 35 (1847), 275–276; *Werke*, I, 483–502; *Crelle's J.* 35 (1847), 368; *Werke*, II, 505), H.J.S. Smith (*Proc. Royal Soc.* 16 (1867), 197–208; *Collected papers*, 510–523; p. 521), and H. Minkowski (*Acta Math.* 7, pp. 201–258; *Werke*, I, 157–202; also I, 3–144, Kap. XXI, p. 117). While Eisenstein gave no proofs, the arithmetical method of Smith and Minkowski is well described by Bachmann in his book *Die Arithmetik von quadratischen Formen*, Ch. X. A different method of proof, which is analytical, and due to Hardy, consists in combining the circle-method (originating in the Hardy-Ramanujan paper on partitions, referred to in the Notes on Ch. VIII) developed by Hardy and Littlewood to tackle a whole range of problems in additive number theory, with the theory of modular functions (first applied by L.J. Mordell in this context). Hardy's method is applicable to both odd and even k. See L.J. Mordell, *Quarterly J. Math.* 48 (1917), 93–104; *Trans. Cambridge Phil. Soc.* 22 (1919), 361–372; G.H. Hardy, *Proc. Nat. Acad. Sci.* 4 (1918), 189–193; *Trans. American Math. Soc.* 21 (1920), 255–284; *Errata*, 29 (1927), 845–847. The method has been presented by L.E. Dickson in his *Studies in the theory of numbers*, Ch. XIII. Hardy has worked out the cases $k=8$ and $k=24$ in his book on *Ramanujan*, Ch. IX.

The classical cases $k=3$, and 4, require special attention, as Hardy himself noticed. In particular, the case $k=3$ involves the handling of double series which are not absolutely convergent. By applying the ideas of E. Hecke, and modifying Hardy's method, $r_3(n)$ can be evaluated, as has been shown by H. Maass, *Abh. Math. Seminar Hansische Univ.* 12 (1938), 133–162; H. Streefkerk, *Thesis, Free Univ. of Amsterdam*, 1943; P.T. Bateman, *Trans. American Math. Soc.* 71 (1951), 70–101. T. Estermann has since given a method for tackling the non-absolutely convergent series without using the ideas of Hecke, *Proc. London Math. Soc.* (3) IX (1959), 573–594.

In addition to the *exact* formulae for $5 \leqq k \leqq 8$, Hardy gave also *asymptotic formulae* for $r_k(n)$, as $n \to \infty$, for arbitrary $k > 4$. A detailed proof of the results can be found in an article by A.Z. Walfisz, *Uspehi Mat. Nauk* (N.S.) 7, no. 6 (52), (1952), 97–178 (American Math. Soc. Translations, Series 2, Vol. 3 (1956), 163–248). For the asymptotic formulae to be exact it is *necessary and sufficient* that $k \leqq 8$. The reasons for this have been shown in a wider view of the problem by C.L. Siegel, (see the Notes on Ch. XI).

§ 3. Siegel's proof of Theorem 1 is similar in method to Hardy's, but his use of $\eta^4(z)$ makes for both simplicity and elegance.

The problem of obtaining an *asymptotic formula* for $r_k(n)$, however, is a very special case of what is known as *Waring's problem*. Given a fixed integer $l>1$, Waring's problem consists in proving the existence of an integer m, such that every integer $N\geqq 1$ can be written as a sum of m l-th powers of non-negative integers. The problem was solved by D. Hilbert, *Math. Annalen*, 67 (1909), 281–300; *Ges. Abhandlungen*, I, 510–527; F. Hausdorff, *Math. Annalen*, 67 (1909), 301–305. Using their circle-method, Hardy and Littlewood proved the deeper result that if $r_{l,m}(N)$ denotes the number of positive integral solutions (n_j), $j=1,\ldots,m$, of the equation $n_1^l+n_2^l+\ldots+n_m^l=N$, then for *any* fixed $m>(l-2)2^{l-1}+5$,

$$(***)\qquad r_{l,m}(N)\sim\frac{\Gamma^m\left(1+\frac{1}{l}\right)}{\Gamma\left(\frac{m}{l}\right)}\cdot\sigma\cdot N^{\frac{m}{l}-1},$$

as $N\to\infty$, where σ, the *singular series*, is a function of N lying between finite, positive, bounds. If $l\geqq 3$, it is sufficient to have $m>l\cdot 2^{l-1}$.

Associated with the validity of the formula (***) is the problem of finding $G(k)$, the *least* integer such that every integer *beyond a certain stage* is the sum of $G(k)$ k-th powers of non-negative integers, and $g(k)$, the least integer, such that *every* positive integer is the sum of $g(k)$ k-th powers of non-negative integers. Clearly $G(k)\leqq g(k)$, and the number $G(k)$ is the more fundamental one, since a finite number of idiosyncratic integers can substantially affect the value of $g(k)$.

We know, after Lagrange (loc. cit.), that $g(2)=G(2)=4$; and, after H. Davenport, that $G(4)=16$, *Annals of Math.* 40 (1939), 731; but the exact value of $G(k)$ for any other k is not yet known. Therefore an upper bound for $G(k)$ is of interest. Building on the foundations laid by Hardy and Littlewood, I.M. Vinogradov has shown, using his powerful method of trigonometric sums (instead of power series) that $G(k)<2k\log k(1+\delta_k)$, where $\delta_k\to 0$, as $k\to\infty$; and also that (***) holds for $m>4k^2\log k(1+\delta_k')$, where $\delta_k'\to 0$, as $k\to\infty$. See his books: *The method of trigonometric sums in the theory of numbers, Trav. Inst. Math. Steklov*, 10 (1937); Second edition, 23 (1947); English translation, Interscience, London-New York, 1954; as well as his second book, *Izdat, Nauka*, Moscow, 1971.

Thanks to the work of S.S. Pillai, L.E. Dickson, R.K. Rubugunday, I. Niven, and Chen Jing-run, the value of $g(k)$ is known for $k\geqq 5$. On the other hand, Wiefrich, and Kempner, showed that $g(3)=9$. The exact value of $g(4)$ is not yet known. An account of the major developments up to 1971 is given in the author's article in the *Proceedings of the International Conference on Number Theory, Moscow, 14–18 September, 1971.* (*Nauka*, Moscow, 1973), 7–26.

These problems have a generalization to algebraic numbers. If K is any algebraic number field of degree n, Siegel proved in 1921 the theorem, first

stated by Hilbert, that every totally positive integer in K can be expressed as a sum of four squares of numbers from the same field, *Math. Zeit.* II (1921), 246; *Ges. Abhandlungen*, I, 47. He later proved that if the field K is totally real, and *every* totally positive integer in K is a sum of squares of *integers* from the same field, then K is either the rational number field Q, or the real quadratic field $Q(\sqrt{5})$; and if the field K is *not* totally real, then *every* totally positive integer is a sum of *integral* squares if and only if the discriminant of K is odd, *Annals of Math.* 46 (1945), 313; *Ges. Abhandlungen*, III, 12.

The analogue of the asymptotic formula of Hardy and Littlewood for $r_{l,m}(N)$ for algebraic number fields has also been proved by Siegel, *American J. Math.* 66 (1944), 122; *Ges. Abhandlungen*, II, 406, using I.M. Vinogradov's method of trigonometrical sums instead of the Hardy-Littlewood method of power series, under the restriction $m > (2^{l-1} + n)nl$, where n is the degree of K. Siegel's question whether this restriction on m can be relaxed, so as to avoid its dependence on n, has been answered in the affirmative by C.P. Ramanujan, and by B.J. Birch, independently of each other. See *Mathematika*, 10 (1963), 137.

For more recent work on problems of Waring's type, see, for instance, C. Hooley, *Recent progress in analytic number theory*, I (1981), 127–191.

Chapter XI

The representation of a number by a quadratic form

§ 1. Positive-definite quadratic forms. The arithmetical function $r(n)$, defined as the number of representations of a positive integer n by the form $n_1^2+n_2^2+n_3^2+n_4^2$, where the (n_k) are integers, is related to the fourth power of the theta-function $\theta_3(0,z)$, which is defined by the *theta-series*

$$\sum_{n_1,n_2,n_3,n_4=-\infty}^{\infty} q^{n_1^2+n_2^2+n_3^2+n_4^2}, \qquad q=e^{\pi iz}, \qquad \mathrm{Im}\, z>0,$$

by the formula

$$\theta_3^4(0,z)=\sum_{n=0}^{\infty} r(n)q^n, \qquad r(0)=1, \qquad q=e^{\pi iz}, \qquad \mathrm{Im}\, z>0,$$

(cf. (2.1), Ch. X) which enabled us to determine $r(n)$ in terms of the divisors of n. We now consider the more general problem of finding the number of representations of a positive integer by a *positive-definite quadratic form*, by constructing more general theta-series, and studying their behaviour under modular transformations.

Let (α_{kl}) be a real, symmetric, $m\times m$ matrix: that is to say, α_{kl} is real for $1\leqq k\leqq m$, $1\leqq l\leqq m$, and $\alpha_{kl}=\alpha_{lk}$. We associate to it the quadratic form $Q(x_1,\ldots,x_m)$ of m real variables $x_1,\ldots,x_m$, defined by

$$\begin{aligned} Q(x_1,\ldots,x_m)&=\sum_{k=1}^{m}\sum_{l=1}^{m}\alpha_{kl}x_kx_l \\ &=\alpha_{11}x_1^2+2\alpha_{12}x_1x_2+\ldots+2\alpha_{1m}x_1x_m+ \\ &\quad+\alpha_{22}x_2^2+2\alpha_{23}x_2x_3+\ldots+2\alpha_{2m}x_2x_m+ \\ &\quad+\ldots+ \\ &\quad+\alpha_{mm}x_m^2. \end{aligned}$$

We denote $Q(x_1,\ldots,x_m)$ by $Q(x)$. We say that $Q(x)$ is *positive-definite*, if $Q(x_1,\ldots,x_m)\geqq 0$ for all real $x_1,\ldots,x_m$, and $Q(x_1,\ldots,x_m)=0$ only for $x_1=x_2=\ldots=x_m=0$.

The determinant of the matrix (α_{kl}), denoted by $\det(\alpha_{kl})$, is called *the determinant of the form* $Q(x)$.

If $Q(x)$ is positive-definite, then $\alpha_{11}>0$. For if $\alpha_{11}\leqq 0$, then $Q(x)\leqq 0$ for $x_1=1$, $x_2=x_3=\ldots=x_m=0$, which contradicts the fact that $Q(x)$ is positive-definite.

We now list some simple properties of positive-definite forms.

(1.1) A positive-definite form $Q(x_1, \ldots, x_m)$, where $m>1$, can be written as

$$Q(x_1, \ldots, x_m)=\frac{1}{\alpha_{11}}(\alpha_{11}x_1+\alpha_{12}x_2+\ldots+\alpha_{1m}x_m)^2+R(x_2, \ldots, x_m),$$

where R is a positive-definite form of the $(m-1)$ variables $x_2, \ldots, x_m$.

For the difference

$$Q(x_1. \ldots, x_m)-\frac{1}{\alpha_{11}}(\alpha_{11}x_1+\ldots+\alpha_{1m}x_m)^2$$

is independent of x_1, and equals

$$R(x_2, \ldots, x_m)=\sum_{k,l=2}^{m}\beta_{kl}x_kx_l, \qquad \beta_{kl}=\beta_{lk}, \quad \text{say},$$

where $\beta_{kl}=\alpha_{kl}-\frac{\alpha_{1k}\alpha_{1l}}{\alpha_{11}}$. If the quadratic form R, of the variables $x_2, \ldots, x_m$, were not positive-definite, there would exist real numbers $x_2, \ldots, x_m$, not all of them zero, such that $R\leqq 0$. If we now define x_1 by the requirement

$$\alpha_{11}x_1+\alpha_{12}x_2+\ldots+\alpha_{1m}x_m=0,$$

while noting that $\alpha_{11}>0$, then $Q(x_1, \ldots, x_m)\leqq 0$, for such a choice of x_1, $x_2, \ldots, x_m$, contradicting the assumption that Q is positive-definite.

(1.2) If a linear transformation $y_k=\sum_{l=1}^{m}\mu_{kl}x_l$, $1\leqq k\leqq m$, carries the quadratic form

$$T(y_1, \ldots, y_m)=\sum_{k,l=1}^{m}\gamma_{kl}y_ky_l, \qquad \gamma_{kl}=\gamma_{lk},$$

into the quadratic form

$$U(x_1, \ldots, x_m)=\sum_{k,l}\xi_{kl}x_kx_l, \qquad \xi_{kl}=\xi_{lk},$$

then the corresponding matrix (ξ_{kl}), where k indicates rows and l columns, satisfies the relation

$$(\xi_{kl})=(\mu_{lk})\cdot(\gamma_{kl})\cdot(\mu_{kl}),$$

where a point ' $\cdot$ ' denotes matrix-multiplication, defined by $(\mu_{kl})\cdot(\gamma_{kl})=(c_{kl})$, $c_{kl}=\sum_{j=1}^{m}\mu_{kj}\cdot\gamma_{jl}$. For

$$T(y_1, \ldots, y_m)=\sum_{q,r=1}^{m}\gamma_{qr}y_qy_r=\sum_{q,r}\gamma_{qr}\sum_{k}\mu_{qk}x_k\sum_{l}\mu_{rl}x_l$$

$$=\sum_{k,l}x_kx_l\sum_{q,r}\mu_{qk}\gamma_{qr}\mu_{rl}.$$

(1.3) If D is the determinant of a positive-definite form Q of m variables, then $D>0$. If $m>1$, and D_1 the determinant of the form R in (1.1), then $D=\alpha_{11}D_1$.

For the form $T(y_1, \ldots, y_m) = \alpha_{11}^{-1} y_1^2 + R(y_2, \ldots, y_m)$, $m > 1$, has the determinant

$$\begin{vmatrix} \alpha_{11}^{-1}, & 0, \ldots, 0 \\ 0, & \beta_{22}, \ldots, \beta_{2m} \\ \cdot & \cdot \cdot \cdot \\ \cdot & \cdot \cdot \cdot \\ \cdot & \cdot \cdot \cdot \\ 0, & \beta_{m2}, \ldots, \beta_{mm} \end{vmatrix} = \alpha_{11}^{-1} \cdot D_1,$$

and the transformation

$$y_1 = \alpha_{11} x_1 + \alpha_{12} x_2 + \ldots + \alpha_{1m} x_m, \quad y_2 = x_2, \ldots, y_m = x_m$$

carries $T(y_1, \ldots, y_m)$ into $Q(x_1, \ldots, x_m)$. By (1.2), we have therefore $D = \alpha_{11} \cdot \frac{1}{\alpha_{11}} D_1 \cdot \alpha_{11} = \alpha_{11} \cdot D_1$.

That $D > 0$ if $m = 1$ was noted before (1.1); if $m > 1$, the same is true by induction.

(1.4) If $Q(x_1, \ldots, x_m)$ is positive-definite, and (α_{kl}) is symmetric, and (A_{kl}) denotes the matrix inverse of (α_{kl}) – so that (A_{kl}) is also symmetric – then the quadratic form

$$Q'(x_1, \ldots, x_m) = \sum_{k,l=1}^{m} A_{kl} x_k x_l$$

is likewise positive-definite, with the determinant $D' = \frac{1}{D}$, and the transformation $y_k = \sum_l \alpha_{kl} x_l$ carries $Q'(y_1, \ldots, y_m)$ into $Q(x_1, \ldots, x_m)$.

This follows from (1.2), since $(\alpha_{lk})(A_{kl})(\alpha_{kl}) = (\alpha_{lk}) = (\alpha_{kl})$. That $Q'(y_1, \ldots, y_m) > 0$, except for $y_1 = \ldots = y_m = 0$, follows from the fact that otherwise there would exist a point $(x_1, \ldots, x_m) \neq (0, \ldots, 0)$, such that $Q(x_1, \ldots, x_m) \leqq 0$ (because $y_k = \sum_l \alpha_{kl} x_l$, and $\det(\alpha_{kl}) \neq 0$).

(1.5) The form Q' in (1.4) is defined to be the *inverse form* of Q. Since (α_{kl}) is the matrix inverse of (A_{kl}), it follows also that Q is the inverse form of Q'.

(1.6) If $m > 1$, and, as in (1.1),

$$Q(x_1, \ldots, x_m) = \alpha_{11}^{-1} (\alpha_{11} x_1 + \ldots + \alpha_{1m} x_m)^2 + R(x_2, \ldots, x_m),$$

and $R'(x_2, \ldots, x_m)$ the inverse form of $R(x_2, \ldots, x_m)$, then

$$(1.61) \qquad Q'(x_1, \ldots, x_m) = \alpha_{11}^{-1} x_1^2 + R'\left(x_2 - \frac{\alpha_{12} x_1}{\alpha_{11}}, \ldots, x_m - \frac{\alpha_{1m} x_1}{\alpha_{11}} \right),$$

where Q' is the inverse form of Q. For the form

$$\alpha_{11}^{-1} y_1^2 + R'\left(y_2 - \frac{\alpha_{12} y_1}{\alpha_{11}}, \ldots, y_m - \frac{\alpha_{1m} y_1}{\alpha_{11}} \right)$$

is carried into

$$\alpha_{11}^{-1}(\alpha_{11}x_1+\ldots+\alpha_{1m}x_m)^2+R'\left(\sum_{l=2}^{m}\beta_{2l}x_l,\ldots,\sum_{l=2}^{m}\beta_{ml}x_l\right) \tag{1.62}$$

by the transformation $y_k=\sum_l \alpha_{kl}x_l$, as in (1.4), since

$$y_k-\frac{\alpha_{1k}y_1}{\alpha_{11}}=\sum_{l=1}^{m}\left(\alpha_{kl}-\frac{\alpha_{1k}}{\alpha_{11}}\cdot\alpha_{1l}\right)x_l=\sum_{l=2}^{m}\beta_{kl}x_l,$$

where $2\leqq k\leqq m$ (cf. (1.1)). Now (1.62) equals

$$\alpha_{11}^{-1}(\alpha_{11}x_1+\ldots+\alpha_{1m}x_m)^2+R(x_2,\ldots,x_m)=Q(x_1,\ldots,x_m),$$

because of (1.4). The right-hand side of (1.61) is therefore the inverse form of Q.

(1.7) If Q is positive-definite, there exists a number $c>0$, which depends only on (α_{kl}), such that

$$Q(x_1,\ldots,x_m)\geqq c\cdot(x_1^2+\ldots+x_m^2).$$

For if $y_1,\ldots,y_m$ are real numbers, such that $y_1^2+\ldots+y_m^2=1$, then $Q(y_1,\ldots,y_m)>0$, since Q is positive-definite. The continuous function Q attains a minimum $c>0$ on $\sum_{j=1}^{m}y_j^2=1$. That is to say, $Q(y_1,\ldots,y_m)\geqq c$, for $y_1^2+\ldots+y_m^2=1$. For arbitrary real numbers x_k, $1\leqq k\leqq m$, not all zero, set

$$y_k=\frac{x_k}{\left(\sum_{j=1}^{m}x_j^2\right)^{1/2}},$$

and the result follows.

§ 2. Multiple theta-series and quadratic forms

Theorem 1. *Let $Q(x_1,\ldots,x_m)$ be a positive-definite quadratic form, and let γ_1, $\gamma_2,\ldots,\gamma_m$ and δ_1, $\delta_2,\ldots,\delta_m$ be arbitrary complex constants. Then the series*

$$\sum_{n_1,n_2,\ldots,n_m=-\infty}^{\infty} e^{-\pi Q(n_1+\gamma_1,\ldots,n_m+\gamma_m)+2\pi i(\delta_1 n_1+\ldots+\delta_m n_m)},$$

where the summation is over all the lattice-points (that is to say, points with integer co-ordinates $(n_1,\ldots,n_m)$ in m-dimensional Euclidean space) is absolutely convergent, and if

$$\theta(\gamma,\delta;Q)=\sum_{n_1,\ldots,n_m=-\infty}^{\infty} e^{-\pi Q(n_1+\gamma_1,\ldots,n_m+\gamma_m)+2\pi i(\delta_1 n_1+\ldots+\delta_m n_m)},$$

then we have

$$\theta(0,\delta;Q)=\frac{1}{\sqrt{D}}\cdot\theta(\delta,0;Q'), \tag{2.1}$$

where D is the determinant of Q, and Q' the inverse form of Q.

Remark. The series

$$\sum_{n_1,n_2,\ldots,n_m=-\infty}^{\infty} e^{-\pi Q(n_1+u_1,\ldots,n_m+u_m)},$$

where $u_1,\ldots,u_m$ are real variables, is known as a *multiple theta-series.*

Proof. The exponent in the general term of the given series can be written in the form

$$-\pi Q(n_1,\ldots,n_m)+\varepsilon_1 n_1+\ldots+\varepsilon_m n_m+\varepsilon,$$

where $\varepsilon_1,\varepsilon_2,\ldots,\varepsilon_m,\varepsilon$ depend only on (α_{kl}), (γ_q), (δ_r), and *not* on the n_k. Here (α_{kl}) is the matrix associated to Q. By (1.7), we have

$$\begin{aligned}&\mathrm{Re}\{-\pi Q(n_1,\ldots,n_m)+\varepsilon_1 n_1+\ldots+\varepsilon_m n_m+\varepsilon\}\\ &\leqq -\pi c(n_1^2+\ldots+n_m^2)+\mathrm{Re}\,\varepsilon_1 n_1+\ldots+\mathrm{Re}\,\varepsilon_m n_m+\mathrm{Re}\,\varepsilon,\quad c>0,\end{aligned}$$

and the series $\sum_{n=-\infty}^{\infty} e^{-\pi cn^2+\varrho n}$, where ϱ is real, converges, since, for sufficiently large $|n|$, we have: $-\pi cn^2+\varrho n<-|n|$, and the geometric series $\sum_{n=0}^{\infty} e^{-n}$ converges.

We shall prove (2.1) by induction on m. Let $m=1$. We have proved in Theorem 1 of Chapter V that

$$\theta_3\left(v,-\frac{1}{\tau}\right)=\sqrt{\frac{\tau}{i}}\sum_{n=-\infty}^{\infty} e^{\pi i\tau(n+v)^2},$$

where

$$\theta_3(v,\tau)=\sum_{n=-\infty}^{\infty} e^{\pi i\tau n^2+2\pi inv},\quad \text{for}\quad \mathrm{Im}\,\tau>0,$$

v complex. If $\tau=iy$, $y>0$, we get

$$\sum_{n=-\infty}^{\infty} e^{-\pi\frac{n^2}{y}+2\pi inv}=\sqrt{y}\sum_{n=-\infty}^{\infty} e^{-\pi y(n+v)^2},$$

or

$$\frac{1}{\sqrt{y}}\sum_{n=-\infty}^{\infty} e^{-\pi\frac{n^2}{y}+2\pi inv+\pi yv^2}=\sum_{n=-\infty}^{\infty} e^{-\pi yn^2-2\pi nyv}.$$

Replacing n by $-n$ on the left-hand side, we get

$$\sum_{n=-\infty}^{\infty} e^{-n^2\pi y-2n\pi vy}=\frac{1}{\sqrt{y}}\sum_{n=-\infty}^{\infty} e^{\pi y\left(v-\frac{ni}{y}\right)^2},$$

for $y>0$, and v complex.

Setting $y=\alpha_{11}$, $v=-\dfrac{i}{\alpha_{11}}\cdot\delta_1$, $n=n_1$, we obtain

$$\sum_{n_1=-\infty}^{\infty} e^{-\pi\alpha_{11}n_1^2+2\pi i\delta_1 n_1}=\frac{1}{\sqrt{\alpha_{11}}}\sum_{n_1=-\infty}^{\infty} e^{\pi\alpha_{11}\left(-\frac{i\delta_1}{\alpha_{11}}-\frac{n_1 i}{\alpha_{11}}\right)^2}$$

$$=\frac{1}{\sqrt{\alpha_{11}}}\sum_{n_1=-\infty}^{\infty} e^{-\frac{\pi}{\alpha_{11}}(n_1+\delta_1)^2}.$$

Since for $m=1$, we have $D=\alpha_{11}$, and $Q'(n_1)=\dfrac{1}{\alpha_{11}}n_1^2$, formula (2.1) holds for $m=1$.

If $m>1$, we assume that (2.1) holds with $m-1$ in place of m, and prove it then for m.

From (1.1) we have

$$Q(0,x_2,\ldots,x_m)=\frac{1}{\alpha_{11}}(\alpha_{12}x_2+\ldots+\alpha_{1m}x_m)^2+R(x_2,\ldots,x_m),$$

so that

$$\theta(0,\delta;Q)=\sum_{n_1,\ldots,n_m=-\infty}^{\infty} e^{-\pi Q(n_1,\ldots,n_m)+2\pi i(\delta_1 n_1+\ldots+\delta_m n_m)}$$

$$=\sum_{n_2,\ldots,n_m=-\infty}^{\infty} e^{-\frac{\pi L^2}{\alpha_{11}}-\pi R(n_2,\ldots,n_m)+2\pi i(\delta_2 n_2+\ldots+\delta_m n_m)}\times$$

$$\times\sum_{n_1=-\infty}^{\infty} e^{-\pi\alpha_{11}n_1^2-2\pi L n_1+2\pi i\delta_1 n_1},$$

where $L=\alpha_{12}n_2+\ldots+\alpha_{1m}n_m$. The last sum equals $\theta(0,\delta_1+Li;U)$, where $U(x_1)=\alpha_{11}x_1^2$, so that, on using formula (2.1) with $m=1$, we get

$$\theta(0,\delta_1+Li;U)=\frac{1}{\sqrt{\alpha_{11}}}\sum_{n_1=-\infty}^{\infty} e^{-\frac{\pi}{\alpha_{11}}(n_1+\delta_1+Li)^2}.$$

Now the multiple series

$$\sum_{n_1,\ldots,n_m=-\infty}^{\infty} e^{-\frac{\pi L^2}{\alpha_{11}}-\pi R(n_2,\ldots,n_m)-\frac{\pi}{\alpha_{11}}(n_1+\delta_1)^2+\frac{\pi}{\alpha_{11}}L^2-\frac{2\pi i}{\alpha_{11}}(n_1+\delta_1)L+2\pi i(\delta_2 n_2+\ldots+\delta_m n_m)}$$

converges absolutely, for the real part of the quadratic terms in the exponent, namely

$$-\frac{\pi}{\alpha_{11}}L^2-\pi R(n_2,\ldots,n_m)-\frac{\pi}{\alpha_{11}}n_1^2+\frac{\pi}{\alpha_{11}}L^2=-\pi\left(\frac{n_1^2}{\alpha_{11}}+R(n_2,\ldots,n_m)\right),$$

equals (-1) times a positive-definite form of m variables. Hence

$$\theta(0,\delta;Q)=\frac{1}{\sqrt{\alpha_{11}}}\sum_{n_1=-\infty}^{\infty} e^{-\frac{\pi}{\alpha_{11}}(n_1+\delta_1)^2}\times$$

$$\cdot\sum_{n_2,\ldots,n_m=-\infty}^{\infty} e^{-\pi R(n_2,\ldots,n_m)+2\pi i(\delta_2 n_2+\ldots+\delta_m n_m)-\frac{2\pi i}{\alpha_{11}}(n_1+\delta_1)L}.$$

For fixed n_1, the inner sum equals

$$\theta\left(0, \delta_k - \frac{(n_1+\delta_1)\alpha_{1k}}{\alpha_{11}}; R\right), \quad (k=2, \ldots, m),$$

where the form R has the determinant D/α_{11}. By the induction hypothesis, we have, however

$$\theta\left(0, \delta_k - \frac{(n_1+\delta_1)\alpha_{1k}}{\alpha_{11}}; R\right) = \frac{\sqrt{\alpha_{11}}}{\sqrt{D}} \cdot \sum_{n_2, \ldots, n_m = -\infty}^{\infty} e^{-\pi R'\left(n_k + \delta_k - \frac{(n_1+\delta_1)}{\alpha_{11}}\alpha_{1k}\right)},$$

so that

$$\theta(0, \delta; Q) = \frac{1}{\sqrt{\alpha_{11}}} \cdot \frac{\sqrt{\alpha_{11}}}{\sqrt{D}} \cdot \sum_{n_1, \ldots, n_m = -\infty}^{\infty} e^{-\frac{\pi}{\alpha_{11}}(n_1+\delta_1)^2 - \pi R'\left(n_k + \delta_k - \frac{(n_1+\delta_1)}{\alpha_{11}}\alpha_{1k}\right)},$$

$$2 \leqq k \leqq m.$$

Since we have, by (1.61),

$$\frac{1}{\alpha_{11}} x_1^2 + R'\left(x_2 - \frac{x_1}{\alpha_{11}} \cdot \alpha_{12}, \ldots, x_m - \frac{x_1}{\alpha_{11}} \cdot \alpha_{1m}\right) = Q'(x_1, \ldots, x_m)$$

identically in $x_1, x_2, \ldots, x_m$, the exponent in the last sum is

$$-\frac{\pi}{\alpha_{11}}(n_1+\delta_1)^2 - \pi R'\left(n_k + \delta_k - \frac{(n_1+\delta_1)}{\alpha_{11}}\alpha_{1k}\right) = -\pi Q'(n_1+\delta_1, \ldots, n_m+\delta_m),$$

$2 \leqq k \leqq m$, hence

$$\theta(0, \delta; Q) = \frac{1}{\sqrt{D}}\, \theta(\delta, 0; Q')$$

as claimed.

§ 3. Theta-functions associated to positive-definite forms. Let z be a complex number, with $\operatorname{Im} z > 0$. Let (α_{kl}) be a real, symmetric, $m \times m$ matrix, and let the associated quadratic form $Q(x) = \sum_{k,l=1}^{m} \alpha_{kl} x_k x_l$ be positive-definite. After Theorem 1, the multiple series

$$\sum_{n_1, \ldots, n_m = -\infty}^{\infty} e^{\pi i z Q(n_1, \ldots, n_m)}$$

converges absolutely, and uniformly in every compact set, in the upper half-plane $\operatorname{Im} z > 0$. The *theta-function associated to* Q is defined to be

$$\theta(z, Q) = \sum_{n_1, \ldots, n_m = -\infty}^{\infty} e^{\pi i z Q(n_1, \ldots, n_m)}. \tag{3.1}$$

In case (α_{kl}) is the unit matrix, $Q(n_1, \ldots, n_m) = n_1^2 + \ldots + n_m^2$, and $\theta(z, Q)$ reduces, in the notation of Chapter V, to $\{\theta_3(0, z)\}^m$.

If we replace α_{kl} by $\frac{i}{z}\alpha_{kl}$, where $z=it$, $t>0$, then the form associated to the matrix $\left(\frac{i}{z}\alpha_{kl}\right)$ is positive-definite, and for *such* z, we have by Theorem 1, with $\delta=(0,\ldots,0)$,

$$\sum_{(k)} e^{-\frac{\pi i}{z}Q(k)} = \frac{(\sqrt{z/i})^m}{\sqrt{D}} \cdot \sum_{(l)} e^{\pi i z Q'(l)},$$

where (k), (l) run over all lattice-points in m-dimensional Euclidean space. By analytic continuation, this formula holds for all complex z with $\operatorname{Im} z>0$. Thus we have

Theorem 2. *Let (α_{kl}) be a real, symmetric, $m\times m$ matrix. Let the associated quadratic form $Q(x)=\sum_{k,l=1}^{m}\alpha_{kl}x_kx_l$ be positive-definite, with determinant D, and with Q' as its inverse. Then*

$$\theta\left(-\frac{1}{z}, Q\right) = \left(\sqrt{\frac{z}{i}}\right)^m \cdot D^{-1/2} \cdot \theta(z, Q'), \tag{3.2}$$

for all complex z with $\operatorname{Im} z>0$.

Let us now consider the special case in which all the numbers α_{kl} are *integers*, and $\alpha_{11}, \alpha_{22}, \ldots, \alpha_{mm}$ are *even*. Then the definition of $\theta(z, Q)$ in (3.1) yields the relation

$$\theta(z+1, Q)=\theta(z, Q). \tag{3.3}$$

We have noticed in (1.4) that the inverse form of Q, namely $Q'(y)=\sum_{k,l=1}^{m}A_{kl}y_ky_l$, where (A_{kl}) is the matrix inverse of (α_{kl}), is carried into $Q(x)$ by the substitution $y_k=\sum_{l=1}^{m}\alpha_{kl}x_l$. In case *all* the numbers α_{kl} are *integers*, and $D=1$, then (x) runs over all lattice-points in R^m when (y) does. Hence we have

Theorem 3. *Let (α_{kl}) be a symmetric, $m\times m$ matrix of integers, where α_{kk} are all even for $k=1,\ldots,m$, and let the determinant of (α_{kl}) be equal to 1. Let the associated quadratic form Q be positive-definite. Then we have*

$$\theta(z+1, Q)=\theta(z, Q), \quad \theta\left(-\frac{1}{z}, Q\right) = \left(\sqrt{\frac{z}{i}}\right)^m \cdot \theta(z, Q), \tag{3.4}$$

for $\operatorname{Im} z>0$.

Remarks. If $m=2$, the matrix (α_{kl}) of Theorem 3 is of the form

$$\begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix},$$

where a, b, and c are integers, with $a>0$, and with $|(\alpha_{kl})|=4ac-b^2=1$. Since $4ac \equiv 0 \pmod 4$, and $b^2 \equiv 0, 1 \pmod 4$, there is *no* solution in integers of the equation $4ac-b^2=1$. Hence there exists no 2×2 matrix which satisfies the conditions of Theorem 3. The following theorem gives a necessary and sufficient condition for the existence of matrices of the type specified in Theorem 3.

Theorem 4 (Minkowski). *A necessary and sufficient condition for the existence of a matrix of the type specified in Theorem 3 is that m be divisible by 8.*

Proof (Witt). If such a matrix (α_{kl}) exists, then m *is* divisible by 8, for on writing $\theta(z)\equiv\theta(z,Q)$, for $\mathrm{Im}\, z>0$, we have $\theta(z+1)=\theta(z)$, $\theta(-1/z)=(\sqrt{z/i})^m\theta(z)$, by Theorem 3. Now

$$\theta\left(1-\frac{1}{z}\right)=\theta\left(-\frac{1}{z}\right)=(\sqrt{z/i})^m\theta(z),$$

and, on the other hand,

$$\theta\left(1-\frac{1}{z}\right)=\theta\left(\frac{z-1}{z}\right)=\left(\sqrt{\frac{z}{i(1-z)}}\right)^m\theta\left(\frac{z}{1-z}\right)=\left(\sqrt{\frac{z}{i(1-z)}}\right)^m\theta\left(\frac{z-1+1}{1-z}\right)$$

$$=\left(\sqrt{\frac{z}{i(1-z)}}\right)^m\theta\left(\frac{1}{1-z}\right)$$

$$=\left(\sqrt{\frac{z}{i(1-z)}}\right)^m\cdot\left(\sqrt{\frac{z-1}{i}}\right)^m\theta(z).$$

Hence

$$\left(\sqrt{\frac{z}{i}}\right)^m=\left(\sqrt{\frac{z}{i(1-z)}}\right)^m\cdot\left(\sqrt{\frac{z-1}{i}}\right)^m.$$

$\left(\text{Here we define } \sqrt{\frac{z}{i}}=1 \text{ for } z=i\right)$. On setting $z=i$, we get

$$1=\left(\sqrt{\frac{1}{1-i}}\right)^m(\sqrt{1+i})^m=\left(\sqrt{\frac{\sqrt{2}}{1-i}}\right)^m\left(\sqrt{\frac{1+i}{\sqrt{2}}}\right)^m$$

$$=\left(\sqrt{\frac{1+i}{\sqrt{2}}}\right)^{2m}=(e^{\pi i/8})^{2m}=e^{2\pi i\cdot\frac{m}{8}},$$

whence 8 divides m.

Conversely, if m is divisible by 8, one can construct a matrix of the type specified, as follows.

Let $x_1, x_2, \ldots, x_m$ be arbitrary real numbers, and let

$$u_1=\frac{x_1}{2},\quad u_2=\frac{x_1}{2}+x_2,\quad \ldots,\quad u_{m-1}=\frac{x_1}{2}+x_{m-1},$$

$$u_m=\frac{x_1}{2}+x_2+\ldots+x_{m-1}+2x_m.$$

If we set

$$U=\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{pmatrix}, \quad L=\begin{pmatrix} \frac{1}{2}, 0, 0, \ldots, 0, 0 \\ \frac{1}{2}, 1, 0, \ldots, 0, 0 \\ \frac{1}{2}, 0, 1, \ldots, 0, 0 \\ \vdots\ \vdots\ \vdots\ \vdots\ \vdots\ \vdots \\ \frac{1}{2}, 0, 0, \ldots, 1, 0 \\ \frac{1}{2}, 1, 1, \ldots, 1, 2 \end{pmatrix}, \quad X=\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}, \tag{3.5}$$

then $U=LX$ (where L has m rows and m columns), and

$$U=\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \text{ if and only if } X=\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}; \tag{3.6}$$

further

$$\sum_{k=1}^{m} u_k^2=U^tU=X^tL^tLX=\sum_{k,l=1}^{m} \alpha_{kl}x_kx_l, \tag{3.7}$$

where

$$(\alpha_{kl})=L^t\cdot L, \tag{3.8}$$

(t denoting *transpose*), giving

$$(\alpha_{kl})=\begin{pmatrix} \frac{m}{4}, 1, 1, \ldots, 1, 1, 1 \\ 1, 2, 1, \ldots, 1, 1, 2 \\ 1, 1, 2, \ldots, 1, 1, 2 \\ \vdots\ \vdots\ \vdots\ \vdots\ \vdots\ \vdots\ \vdots \\ 1, 1, 1, \ldots, 2, 1, 2 \\ 1, 1, 1, \ldots, 1, 2, 2 \\ 1, 2, 2, \ldots, 2, 2, 4 \end{pmatrix}. \tag{3.9}$$

Since $8|m$, α_{kk} is *even* for $k=1, \ldots, m$; (α_{kl}) is symmetric by (3.8); since $|L|=1$, we have $|(\alpha_{kl})|=|L|^2=1$; by (3.6) and (3.7) the quadratic form $\sum_{k,l=1}^{m} \alpha_{kl}x_kx_l$ associated to (α_{kl}) is positive-definite.

§ 4. Representation of an even integer by a positive-definite form. The analytic method which has been used to prove Euler's theorem in Chapter VIII, and Jacobi's theorem in Chapter X, will now be used to prove the following

Theorem 5 (Hecke). *Let (α_{kl}) be a symmetric, $m\times m$ matrix of integers, in which $\alpha_{11}, \alpha_{22}, \ldots, \alpha_{mm}$ are all even, and having determinant 1. Let the associated quadratic form $\sum_{k,l=1}^{m} \alpha_{kl}x_kx_l$ be positive-definite. For any positive integer n, let a_n denote the number of solutions $(x_1, x_2, \ldots, x_m)$, with x_k integral for every k such*

that $1 \leqq k \leqq m$, *of the equation*

$$\sum_{k,l=1}^{m} \alpha_{kl} x_k x_l = 2n .$$

Then we have

$$a_n = 240\sigma_3(n), \quad \textit{for} \quad m=8,$$

and

$$a_n = 480\sigma_7(n), \quad \textit{for} \quad m=16,$$

where $\sigma_k(n) = \sum_{d|n, d>0} d^k$, $k=0,1,2,\ldots$.

Proof (Siegel). By Theorem 4, the existence of (α_{kl}) requires that m be divisible by 8.

Let $Q_m(n) = \sum_{k,l=1}^{m} \alpha_{kl} n_k n_l$, and

$$f_m(z) = \sum_{n_1,\ldots,n_m=-\infty}^{\infty} e^{\pi i z Q_m(n)}, \qquad \operatorname{Im} z > 0, \qquad (n) = (n_1, \ldots, n_m).$$

By Theorem 3, we have the relations

$$f_m(z+1) = f_m(z); \qquad f_m\left(-\frac{1}{z}\right) = z^{m/2} f_m(z); \qquad \operatorname{Im} z > 0. \tag{4.1}$$

It follows from them that

$$f_m\left(\frac{az+b}{cz+d}\right) = (cz+d)^{m/2} f_m(z), \tag{4.2}$$

where a, b, c, d are integers, with $ad-bc=1$. For the modular group is generated by the two transformations: $A: z \to z+1$, and $B: z \to -1/z$, for which (4.2) is valid because of (4.1); *and* if (4.2) holds for any modular transformation $M: z \to \dfrac{az+b}{cz+d}$, then it holds also for $A^{\pm 1}M$, $B^{\pm 1}M$. (§ 5, Ch. I).

Since $Q_m(n_1, \ldots, n_m)$ is an *even integer*, and $f_m(z)$ is holomorphic for $\operatorname{Im} z > 0$, we have an expansion of $f_m(z)$ in powers of $e^{2\pi i z}$, given by

$$f_m(z) = 1 + \sum_{n=1}^{\infty} a_n e^{2\pi i n z}, \qquad \operatorname{Im} z > 0, \tag{4.3}$$

where a_n is the number of solutions $(n_1, \ldots, n_m)$ of the equation

$$Q_m(n_1, \ldots, n_m) = 2n .$$

Given two complex numbers ω_1, ω_2, with $\operatorname{Im} \dfrac{\omega_2}{\omega_1} > 0$, we have seen that the series

$$E_\varrho(\omega_1, \omega_2) = \sum_{\substack{k,l=-\infty \\ (k,l) \neq (0,0)}}^{\infty} (k\omega_1 + l\omega_2)^{-\varrho} \tag{4.4}$$

converges absolutely for any integer $\varrho > 2$ (Theorem 1, Ch. III). If ϱ is *odd*, the terms cancel in pairs (Remarks after Theorem 4, Ch. III), so that $E_\varrho(\omega_1, \omega_2) = 0$, for $\varrho = 3, 5, 7, \ldots$. We therefore consider only the case in which ϱ is *even*.

If we write, as before, $\tau = \omega_2/\omega_1$, then we have $E_\varrho(\omega_1, \omega_2) = \omega_1^{-\varrho} E_\varrho(\tau)$, where

$$E_\varrho(\tau) = \sum_{\substack{k,l=-\infty \\ (k,l)\neq(0,0)}}^{\infty} (l\tau + k)^{-\varrho}. \tag{4.5}$$

The function $E_\varrho(\omega_1, \omega_2)$ is left invariant, when ω_1 and ω_2 are replaced by ω_1^* and ω_2^*, where $\omega_1^* = d\omega_1 + c\omega_2$, $\omega_2^* = b\omega_1 + a\omega_2$, with integers a, b, c, d satisfying the condition $ad - bc = 1$ (Proof of Theorem 1, Ch. VI), which is equivalent to replacing τ by $\tau^* = \dfrac{\omega_2^*}{\omega_1^*} = \dfrac{a\tau + b}{c\tau + d}$. Hence we have

$$E_\varrho\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^\varrho E_\varrho(\tau). \tag{4.6}$$

We have proved (Lemma 1, Ch. VI) that if ϱ is an even integer, $\varrho \geqq 4$, then

$$\frac{E_\varrho(\tau)}{2\zeta(\varrho)} = 1 + \frac{(-2\pi i)^\varrho}{(\varrho-1)!\,\zeta(\varrho)} \cdot \sum_{N=1}^{\infty} \sigma_{\varrho-1}(N)\, e^{2\pi i N\tau} = g_\varrho(\tau), \tag{4.7}$$

say. From (4.6) we have

$$g_\varrho\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^\varrho g_\varrho(\tau), \qquad ad - bc = 1. \tag{4.8}$$

By (4.7), $g_\varrho(\tau) \to 1$, as $\operatorname{Im}\tau \to +\infty$, uniformly for $|\operatorname{Re}\tau| \leqq \frac{1}{2}$. On the other hand, because of (4.3), $f_m(\tau) \to 1$, as $\operatorname{Im}\tau \to +\infty$, uniformly for $|\operatorname{Re}\tau| \leqq \frac{1}{2}$. Hence

$$h_\varrho(\tau) \equiv f_m(\tau) - g_\varrho(\tau) \to 0, \quad \text{as} \quad \operatorname{Im}\tau \to +\infty, \tag{4.9}$$

uniformly for $|\operatorname{Re}\tau| \leqq \frac{1}{2}$. By (4.2) and (4.8) we have

$$h_\varrho\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^\varrho h_\varrho(\tau), \qquad \varrho = \frac{m}{2},\ 8 \mid m. \tag{4.10}$$

On the other hand, the Dedekind η-function satisfies the relation (Theorem 3, Ch. VIII)

$$\eta\left(\frac{a\tau + b}{c\tau + d}\right) = \omega \cdot (c\tau + d)^{1/2} \eta(\tau),$$

where ω is some 24th root of unity. It follows that the function

$$\left|\frac{h_\varrho(\tau)}{\eta^{2\varrho}(\tau)}\right|$$

is invariant under all transformations of the modular group.

Now

$$|h_\varrho(\tau)| = O(|e^{2\pi i\tau}|) = O(e^{-2\pi \operatorname{Im}\tau}), \quad \text{as} \quad \operatorname{Im}\tau \to +\infty,$$

uniformly for $|\mathrm{Re}\,\tau| \leqq \frac{1}{2}$, because of (4.3) and (4.7), while

$$\eta^{2\varrho}(\tau) = e^{\frac{\pi i \tau}{12} \cdot 2\varrho}\{1 + o(1)\}, \quad \text{as} \quad \mathrm{Im}\,\tau \to +\infty,$$

uniformly for $|\mathrm{Re}\,\tau| \leqq \frac{1}{2}$, so that

$$|\eta^{2\varrho}(\tau)| > c \cdot e^{-\frac{2\pi\varrho}{12} \cdot \mathrm{Im}\,\tau}, \quad c > 0, \quad \text{as} \quad \mathrm{Im}\,\tau \to +\infty.$$

Hence

$$\left|\frac{h_\varrho(\tau)}{\eta^{2\varrho}(\tau)}\right| \to 0, \quad \text{as} \quad \mathrm{Im}\,\tau \to \infty,$$

uniformly for $|\mathrm{Re}\,\tau| \leqq \frac{1}{2}$, for $\varrho < 12$, where ϱ is an even integer and $\varrho \geqq 4$.

Reasoning as we did in the proof of Theorem 5 of Chapter VIII, we conclude that

$$\frac{h_\varrho(\tau)}{\eta^{2\varrho}(\tau)} \equiv 0, \quad \text{for} \quad \varrho < 12, \quad \varrho \geqq 4.$$

Since $\eta(\tau) \neq 0$ for $\mathrm{Im}\,\tau > 0$, it follows that $h_\varrho(\tau) \equiv 0$, or $f_m(\tau) = g_\varrho(\tau)$ for $4 \leqq \varrho < 12$, or $8 \leqq m < 24$, or $m = 8$, 16 (since $\varrho = m/2$ and $8|m$).

If $\varrho = 4$, or $m = 8$, we have from (4.7),

$$g_4(\tau) = 1 + \frac{(-2\pi i)^4}{3!\zeta(4)} \cdot \sum_{N=1}^{\infty} \sigma_3(N)\, e^{2\pi i N\tau}, \tag{4.11}$$

and if $\varrho = 8$, or $m = 16$,

$$g_8(\tau) = 1 + \frac{(-2\pi i)^8}{7!\zeta(8)} \cdot \sum_{N=1}^{\infty} \sigma_7(N)\, e^{2\pi i N\tau}. \tag{4.12}$$

Since $\zeta(4) = \dfrac{\pi^4}{90}$, and $\zeta(8) = \dfrac{\pi^8}{9450}$, (Lemma 2, Ch. VI), we obtain from the equality

$$f_m(\tau) = g_{m/2}(\tau), \quad \text{for} \quad m = 8, 16,$$

on comparing the coefficients in (4.3), (4.11), and (4.12), the relations

$$a_n = 240\sigma_3(n), \quad \text{for} \quad m = 8,$$

and

$$a_n = 480\sigma_7(n), \quad \text{for} \quad m = 16,$$

as claimed in the theorem.

Remark. If $\varrho = 12$, so that $m = 24$, then $a_1 = c \cdot \sigma_{11}(1)$, where $c = \dfrac{24 \cdot 2730}{691}$, which is not an integer.

Notes on Chapter XI

§§ 1–2. For generalities on positive-definite quadratic forms and multiple theta-series see, for instance, E. Landau, *Einführung in die elementare und analytische Theorie der algebraischen Zahlen und der Ideale*, 1918.

§ 3. For Theorem 4, see E. Witt's paper, *Abhandlungen Math. Seminar Hamburg*, 14 (1941), 323–337, and the Göttingen lectures by C.L. Siegel on *Analytische Zahlentheorie*, II (1963/64) (Notes by K.F. Kürten).

§ 4. Theorem 5 is obtained as a special case of a more general result by E. Hecke in his remarkable paper, *Analytische Arithmetik der positiven Quadratischen Formen, Kgl. Danska Vidensk. Selskab. Mathematisk-fysisk Meddelelser*, XIII, 12 (1940), s. 134; *Math. Werke* (1959), Nr. 41, 789–918; see § 7, p. 868.

The proof given here is due to C.L. Siegel, and is expounded by him in his Göttingen lectures, loc. cit. § 3. It is of a piece with his proof of Euler's theorem in Chapter VIII, and of Jacobi's theorem in Chapter X.

The arithmetical result of Theorem 5 establishes a connexion between theta-series on the one hand, and Eisenstein series on the other. Note that $E_\varrho(\tau)$ is a modular form of weight $\varrho(\varrho \geqq 4, \varrho$ even), while $f_m(\tau)$ is a modular form of weight $m/2$ (where $8|m$). The division by $\eta^{2\varrho}$, in the proof, is an artifice of Siegel's.

The interest of this proof will be evident to those acquainted with Siegel's papers on the analytic theory of quadratic forms, particularly *indefinite forms*. See No. 20 (1935) of his *Ges. Abhandlungen*, Bd. I, p. 329, where he writes: "Diese zeigt dann, daß der bekannten Relation, aus der Jacobi seinen Satz über die Zerlegungen einer Zahl in 4 Quadrate entnahm, ein auf beliebige quadratischen Formen bezüglicher analytischer Zusammenhang zugrundeliegt." (Free translation: This then shows that in the case of quadratic forms in general there is an underlying analytical relation of the same kind as the well-known one on which Jacobi's theorem on the representation of a number as the sum of 4 squares is based). See also papers No. 22 (1936), p. 410, and No. 27 (1937), Bd. II, p. 7, where he says: "Die Bedeutung des Hauptsatzes für die Analysis ist nun darin zu sehen, daß durch ihn eine Verbindung hergestellt wird zwischen den aus Thetareihen gebildeten Funktionen $F(s, z)$ und den einfachsten Bausteinen der allgemeinen Modulfunktionen, nämlich den Eisensteinschen Reihen". (The analytical significance of the main theorem is to be seen in the fact that it establishes a connexion between the functions $F(s, z)$ constructed out of theta-functions and the simplest building blocks of modular functions in general, namely the Eisenstein series). In paper No. 30 (1938), Bd. II, p. 41, he says: "Die analytische Theorie der quadratischen Formen liefert nun einen Zusammenhang zwischen Thetareihen und Eisensteinschen Reihen, und zwar auch für indefinite Formen". (The analytic theory of quadratic forms now provides a connexion between theta-series and Eisenstein series even in the case of indefinite forms). See also papers No. 42 (1943), Bd. II, p. 363; No. 45 (1944), Bd. II, p. 421; No. 58 (1951), Bd. III, p. 105; No. 60 (1952), Bd. III, p. 154; No. 66 (1956), Bd. III, p. 228, where he begins by saying: "Die Eisensteinschen Reihen

für die vierten bis achten Potenzen der Funktion

$$\theta(\tau)=\sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} \quad (\tau=\xi+i\eta, \eta>0)$$

sind zuerst von Hardy unter Benutzung der Kreismethode gewonnen worden. Bei Annäherung an einen rationalen Randpunkt a/c der oberen τ-Halbebene gilt nämlich

$$(c\tau-a)^{1/2}\theta(\tau)\to\gamma\left(\frac{a}{c}\right) \quad (c>0,(a,c)=1),$$

wobei $\gamma(a/c)$ eine achte Einheitswurzel oder 0 ist, und als Eisensteinsche Entwicklung der Potenzen von $\theta(\tau)$ ergibt sich dann mittels der Theorie der Modulfunktionen

$$\theta^m(\tau)=1+\sum_{a/c}\gamma^m(a/c)\,(c\tau-a)^{-m/2} \quad (m=4,\ldots,8). \tag{1}$$

Hierin durchläuft a/c alle rationale Zahlen, und im Falle $m=4$ hat man wegen der bedingten Konvergenz zuerst über a bei festen c und sodann über c zu summieren. Die Hardysche Formel war der Ausgangspunkt für die Untersuchungen des Verfassers zur analytischen Theorie der quadratischen Formen, wodurch (1) als Spezialfall eines allgemeineren Satzes erkannt wurde". (The Eisenstein series for the kth powers of the function $\theta(\tau)=\sum_{n=-\infty}^{\infty}\exp(\pi i n^2\tau)$, where $\tau=\xi+i\eta$, $\eta>0$, and $4\leqq k\leqq 8$, were first obtained by Hardy using the circle-method. When we approach a rational boundary-point a/c of the upper half-plane of τ, we have: $(c\tau-a)^{1/2}\theta(\tau)\to\gamma(a/c)$, where $c>0$, $(a,c)=1$, and $\gamma(a/c)$ is an eighth root of unity or zero, and it follows from the theory of modular functions that the Eisenstein series of the powers of $\theta(\tau)$ are given by $\theta^m(\tau)=1+\sum_{a/c}\gamma^m(a/c)\ (c\tau-a)^{-m/2}$, $m=4,\ldots,8$. Here a/c runs through all rational numbers, and in the case $m=4$, because of the non-absolute convergence, one has to sum first over a for fixed c, and then over c. Hardy's formula was the starting point of the author's investigations in the analytic theory of quadratic forms, as a result of which (1) came to be recognized as a special case of a more general theorem). See also pp. 184–186 of Siegel's *Lectures on quadratic forms*, Bombay (1957), (Notes by K.G. Ramanathan).

It is these papers of Siegel, in which the methods of Minkowski and of Hardy are combined and adapted, about which Hermann Weyl (in his *Ges. Abhandlungen*, No. 136 (1944/45), Bd. IV, p. 232) wrote: "In a series of publications which continue the great tradition of Gauss, Dirichlet, Hermite, Minkowski and which I am sure will be admired by generations of mathematicians to come Siegel has vastly added to our knowledge in the theory of definite and indefinite quadratic forms and of arithmetical reduction."

Chronological table

Fagnano, Conte G.C. di	1682–1766	Riemann, B.	1826–1866
Euler, L.	1707–1783	Dedekind, R.	1831–1916
Lagrange, J. L.	1736–1813	Weber, H.	1842–1913
Legendre, A. M.	1752–1833	Klein, F.	1849–1925
Fourier, J.	1768–1830	Hurwitz, A.	1859–1919
Gauss, C. F.	1777–1855	Minkowski, H.	1864–1909
Cauchy, A.	1789–1857	Hardy, G. H.	1877–1947
Abel, N. H.	1802–1829	Fueter, R.	1880–1950
Jacobi, C. G. J.	1804–1851	Littlewood, J. E.	1885–1977
Dirichlet, G. Lejeune	1805–1859	Weyl, H.	1885–1955
Liouville, J.	1809–1882	Hecke, E.	1887–1947
Weierstrass, K.	1815–1897	Ramanujan, S.	1887–1920
Hermite, C.	1822–1901	Mordell, L. J.	1888–1972
Eisenstein, G.	1823–1852	Vinogradov, I. M.	1891–1983
Kronecker, L.	1823–1891	Siegel, C. L.	1896–1981

The enduring is not a substitute for the transient,
Neither the one for the other.

T. S. Eliot

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